

Split Zero and the Arithmetic Trace

A reconstruction from the complete support geometry

Split-Zero research programme

23 September 2026; full-support purity continuation

This manuscript constructs the programme from its coefficient operations, all relevant prime spectra, and the maps connecting them. Every assertion below has its stated mathematical scope. The manuscript is being written cumulatively; it does not assert completion of the RH argument or of the entire requested reconstruction.

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1 The coefficient operations and their prime spectra

Definition 1.1. For a nonzero commutative unital ring R , let $G(R) = R \sqcup \{\tau\}$. Supported elements have their original ring sum and product; $x + \tau = x$ and $x\tau = \tau$. Write $e = 0_R$ for the supported zero. Thus $e \neq \tau$, $e^2 = e$, and $er = e$ for every supported r .

The injection $\tau \mapsto (0, 0)$, $r \mapsto (r, 1)$ into $R \times \mathbb{B}$ proves all semiring laws, including distributivity. Here addition in $\mathbb{B}$ is join and multiplication is meet. The two projections are

$$p : G(R) \rightarrow R, \quad p(\tau) = 0, \quad p(r) = r; \quad \chi : G(R) \rightarrow \mathbb{B}, \quad \chi(\tau) = 0, \quad \chi(r) = 1. \quad (1)$$

They jointly retain the entire original element. Every map to a ring kills e since $e + e = e$, and then factors uniquely through p . The supported inclusion of R does not preserve the specified additive zero and is not a semiring section of p .

Theorem 1.2 (The scalar spectrum and its stalks). *The ideals of $G(R)$ are $\{\tau\}$ and $J \cup \{\tau\}$ for ring ideals J . Its primes are*

$$P_\tau = \{\tau\}, \quad P_{\mathfrak{p}} = \mathfrak{p} \cup \{\tau\} \quad (\mathfrak{p} \in \text{Spec } R).$$

The map induced by p identifies $\text{Spec } R$ with $V(e)$, while $D(e) = \{P_\tau\}$ is open and dense. The localization stalks are

$$G(R)_{P_\tau} = \mathbb{B}, \quad G(R)_{P_{\mathfrak{p}}} = G(R_{\mathfrak{p}}).$$

For $R = \mathbb{Z}$, all strict maximal prime chains are

$$\{\tau\} \subsetneq \{\tau, e\} \subsetneq p\mathbb{Z} \cup \{\tau\}. \quad (2)$$

In particular its dimension is two.

Proof. Every ideal contains τ . If an ideal contains a supported element, multiplication by e places e in the ideal. Its supported intersection is a ring ideal because multiplication by -1 supplies additive inverses. This proves the ideal classification and its converse. A product equals τ only if a factor does, proving primality of P_τ . Testing supported products proves primality of $J \cup \{\tau\}$ exactly when J is prime.

For supported a , $D(a) = \{P_\tau\} \cup D_R(a)$, whereas $D(\tau)$ is empty. These formulas prove the topology assertions, and P_τ is contained in every prime. At P_τ , every supported element is inverted, including e . Its idempotence forces $e = 1$, and $er = e$ then forces every supported r to equal 1. The character χ factors through the localization and distinguishes 1 from τ , proving that the result is exactly $\mathbb{B}$. At $P_{\mathfrak{p}}$ the denominators are $R \setminus \mathfrak{p}$. The fraction map $r/s \mapsto r/s$, $\tau/s \mapsto \tau$ is an isomorphism: equality of supported fractions is precisely the usual localization relation; an absent fraction cannot equal a supported fraction since denominators and the extra multiplier remain supported. The list of ring primes of $\mathbb{Z}$ now proves (2) and excludes longer chains. $\square$

Proposition 1.3 (The direction of the absolute-point maps). *The maps in (1) induce*

$$\text{Spec } R \xrightarrow{\text{closed}} \text{Spec } G(R) \xleftarrow{\text{open}} \text{Spec } \mathbb{B}.$$

There is no unital semiring map $\mathbb{B} \rightarrow G(R)$. The pointed multiplicative monoid $\{0, 1\}$ does map to the multiplicative monoid of $G(R)$ by $0 \mapsto \tau$, $1 \mapsto 1$.

Proof. The first assertion is the localization and prime contraction already proved. A map from $\mathbb{B}$ would send $1 + 1 = 1$ to $1_R + 1_R = 1_R$, which contradicts $1_R \neq 0_R$. The two-element pointed monoid has no such additive relation, and its product table verifies the last map. Thus the two arrows and the pointed-monoid map retain their exact categories. $\square$

1.1 Every multiplicative prime, before imposing addition

Let M be the pointed multiplicative monoid underlying $G(\mathbb{Z})$ and let $\mathcal{P}$ be the rational primes. A monoid ideal is absorbing under multiplication; no additive closure is required.

Theorem 1.4 (The complete monoid spectrum). *For $\Sigma \subseteq \mathcal{P}$, put*

$$J_\Sigma = \{\tau, e\} \cup \{n \in \mathbb{Z} \setminus \{0\} : p \mid n \text{ for some } p \in \Sigma\}.$$

Then

$$\text{Spec}_{\text{mon}} M = \{\{\tau\}\} \sqcup \{J_\Sigma : \Sigma \subseteq \mathcal{P}\}, \quad J_\Sigma \subseteq J_{\Sigma'} \iff \Sigma \subseteq \Sigma'.$$

This spectrum has infinite dimension. The map

$$j : \text{Spec } G(\mathbb{Z}) \longrightarrow \text{Spec}_{\text{mon}} M \tag{3}$$

that keeps each prime ideal as a subset is a continuous topological embedding. Its image is exactly $\{\tau\}$, $J_\emptyset$, and the $J_{\{p\}}$.

Proof. The displayed sets are proper absorbing ideals. Products in J_Σ either involve τ , vanish in the integral domain $\mathbb{Z}$, or have a prime factor in Σ , so they satisfy primality. If a prime monoid ideal contains a supported element it contains e . It contains neither unit $1, -1$. Every nonzero element in it has an integer prime factor in it, by repeated primality and integer factorization. Conversely it contains every multiple of every such prime factor. This proves exhaustion, including $J_\emptyset = \{\tau, e\}$. Membership of a rational prime p tests whether $p \in \Sigma$, proving the inclusion statement. Arbitrarily long strictly nested finite subsets give arbitrarily long prime chains.

The scalar theorem identifies the image of j . For every $a \in M$, the inverse image under j of the monoid basic open $D_M(a)$ is the semiring basic open $D_{G(\mathbb{Z})}(a)$. These basics also generate the topology on the source, proving that the injection is an embedding. Finally if Σ contains distinct primes p, q , Bézout integers u, v give $up + vq = 1$. Both summands belong to J_Σ , so additive closure would force 1 into it. Such a monoid prime therefore cannot be a semiring prime. This also proves exactly which additive relation removes it. $\square$

Proposition 1.5 (Reconstruction retains the missing relations). *Let $\mathbb{N}_0[M]$ be the monoid semiring with the basis symbol $[\tau]$ identified with its additive zero. Then the quotient by*

$$[a] + [b] \sim [a + b] \quad (a, b \in \mathbb{Z})$$

is canonically $G(\mathbb{Z})$.

Proof. Evaluation $[a] \mapsto a$ respects all displayed relations. Every nonempty supported sum reduces to the single basis symbol of its integer sum, while the empty sum remains the symbol τ . Conversely $a \mapsto [a]$, $\tau \mapsto 0$ preserves multiplication, every supported addition, and addition with the specified zero. These maps are inverse. The contraction of $[\tau]$ is essential: in the uncontracted free semiring, the relations $[x] + [\tau] = [x]$ alone do not identify $[\tau]$ with the empty sum. Indeed they hold in the semiring obtained by adjoining another external zero below the already specified zero of $G(\mathbb{Z})$, where the two remain distinct. This gives an explicit separating model for the missing relation. $\square$

1.2 The complete lattice of zero-amplitude support

Let L be a nontrivial bounded distributive lattice and put

$$S_L = G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L.$$

Operations are $(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu)$ and $(r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu)$. Set $z_\lambda = (0, \lambda)$, $\tau_L = z_0$, $e_L = z_1$, and $Z_L = \{z_\lambda\}$.

Theorem 1.6 (All support and arithmetic primes). *The primes of S_L are exactly*

$$P_{\mathfrak{a}} = \{z_\lambda : \lambda \in \mathfrak{a}\}, \quad \mathfrak{a} \in \text{Spec}_{\text{lat}} L; \quad Q_{\mathfrak{p}} = Z_L \cup \{(r, 1) : r \in \mathfrak{p}\}, \quad \mathfrak{p} \in \text{Spec } R.$$

Within each family, inclusions retain the original prime order; every $P_{\mathfrak{a}}$ is strictly contained in every $Q_{\mathfrak{p}}$. Topologically

$$\text{Spec } S_L = \text{Spec}_{\text{lat}} L \blacktriangleright \text{Spec } R, \quad D(e_L) = \text{Spec}_{\text{lat}} L, \quad V(e_L) = \text{Spec } R, \quad (4)$$

where opens are opens of the first stratum and sets consisting of that entire stratum together with an open of the second. If both dimensions are finite, their dimensions add with one further step between the strata.

Proof. An ideal containing a supported element contains e_L by multiplication by it and therefore all of Z_L , since $e_L z_\lambda = z_\lambda$. Its supported elements form a ring ideal. Every remaining ideal is $\{z_\lambda : \lambda \in \mathfrak{a}\}$ for a proper lattice ideal: closure under join follows from addition, and downward closure follows from multiplication by z_μ for $\mu \leq \lambda$. Conversely these are all ideals. Primality in the first family is the meet-prime condition on $\mathfrak{a}$; in the second it is the ring-prime condition. Products involving a zero-fibre element obey exactly these tests. The inclusions follow from the displayed sets. Finally

$$D(z_\lambda) = \{P_{\mathfrak{a}} : \lambda \notin \mathfrak{a}\}, \quad D((r, 1)) = \text{Spec}_{\text{lat}} L \cup D_R(r)$$

prove the topology and the two strata. Every chain is a support chain followed by an arithmetic chain, and maximal finite chains concatenate, proving the dimension formula. $\square$

Proposition 1.7 (Two support channels and their exact comparison). *For $L = \mathcal{P}(\{1, 2\})$, $\{\tau_L\}$ is not prime, since $z_{\{1\}} z_{\{2\}} = \tau_L$ although neither factor is absent. There are two support primes*

$$P_1 = \{\tau_L, z_{\{2\}}\}, \quad P_2 = \{\tau_L, z_{\{1\}}\}.$$

For $R = \mathbb{Z}$, each specializes to the common $Q_{(0)}$ and then to every $Q_{(p)}$. Moreover

$$G_{\mathbb{B}^2}(R) \simeq G(R) \times_R G(R), \quad (5)$$

where the two maps to R are ring reflections. The two factor-generic primes contract to P_1, P_2 , while their arithmetic primes above the same $\mathfrak{p}$ both contract to $Q_{\mathfrak{p}}$.

Proof. The two coordinate maps $L \rightarrow \mathbb{B}$ have precisely the displayed prime kernels; a proper prime ideal must contain exactly one of the two atoms, proving exhaustion. For (5), send z_A to the pair with entry e on A and τ off A , and send $(r, \{1, 2\})$ to (r, r) . Equal nonzero amplitudes force both coordinates to be that supported value, whereas common amplitude zero allows exactly the four masks. Hence the map is onto the fibre product and injective; the operations verify it is an isomorphism. Taking inverse images of the factor primes proves their contractions. $\square$

Proposition 1.8 (What a Boolean branch retains). *A bounded lattice homomorphism $\epsilon : L \rightarrow \mathbb{B}$ induces $\beta_\epsilon : G_L(R) \rightarrow G_{\mathbb{B}}(R)$ by $z_\lambda \mapsto z_{\epsilon(\lambda)}$ and $(r, 1) \mapsto (r, 1)$. Its spectral image is*

$$\{P_{\epsilon^{-1}(0)}\} \blacktriangleright \text{Spec } R.$$

Proof. The map preserves joins, meets and supported arithmetic, hence is a semiring homomorphism. The inverse image of the scalar absent prime is $\{z_\lambda : \epsilon(\lambda) = 0\}$, and the inverse image of each arithmetic prime contains the full zero fibre and exactly its original amplitudes. This proves the asserted image. In particular a branch selects one support prime; the formula does not recover the other primes. $\square$

These proofs reconstruct the scalar and multiplicative results of [1, 2] and the lattice-support results of [3]. They give explicit maps whenever their spectra differ. The following section uses the multiplicative prime faces to recover the finite sets of places occurring in the arithmetic trace, retaining the support geometry of (4) for the cohomological coefficient construction.

2 Supported zero as the central arithmetic prime

The nonzero prime element e is part of the starting geometry, not a notation for the semiring zero. The assertion already appears explicitly in [1, thm:zero-prime-not-irr]. Here its ideal, height, localization, quotient, and power filtration are reconstructed on the full lattice-support base.

Theorem 2.1 (The supported-zero prime and the arithmetic closed stratum). *Let R be a nonzero integral domain and L a nontrivial bounded distributive lattice. In $S = G_L(R)$ the supported zero $e = z_1$ is a nonzero nonunit prime element. Precisely,*

$$eS = Z_L = \ker p_L =: E, \quad e \mid x \iff p_L(x) = 0, \quad e \mid xy \implies e \mid x \text{ or } e \mid y. \quad (6)$$

Its principal ideal E is prime, $E^2 = E$, and it is not irreducible because $e = e \cdot e$ with neither factor a unit. Every support prime lies strictly below E . The primes above E are exactly the arithmetic primes Q_p , and

$$V(E) \simeq \operatorname{Spec} R.$$

For finite-dimensional $\operatorname{Spec}_{\text{lat}} L$,

$$\operatorname{ht}(E) = \dim \operatorname{Spec}_{\text{lat}} L + 1.$$

For $R = \mathbb{Z}$ and finite Boolean $L = \mathbb{B}^I$ with $I \neq \emptyset$, E has height one and each $Q_{(p)}$ has height two.

Proof. For $z_\lambda \in Z_L$, $ez_\lambda = z_\lambda$, so every zero-amplitude element belongs to eS . Conversely a product with e has zero amplitude, proving the first equality and the divisibility equivalence. Since R is a domain, $p_L(x)p_L(y) = 0$ implies that one factor has zero amplitude; this proves the prime-element condition and primality of its ideal. The element e differs from $\tau = z_0$, and it is not a unit since every product with it has amplitude zero, whereas 1 has nonzero amplitude. Its idempotence proves both $E^2 = E$ and the failure of irreducibility.

The complete prime classification places every support prime below $Q_{(0)} = E$, and a domain has no arithmetic prime strictly below (0). Thus every chain ending at E consists of a support chain followed by this final step. Taking suprema of lengths proves the height formula. A finite Boolean lattice has only its coordinate prime ideals, which are pairwise incomparable: primality and disjointness of atoms force exactly one atom outside a prime ideal. Its lattice spectrum has dimension zero, proving the last assertions. $\square$

Theorem 2.2 (The stalk and residue at the supported-zero prime). *If $K = \operatorname{Frac} R$, then*

$$S_E \simeq G_L(K), \quad S_E/ES_E \simeq K. \quad (7)$$

The first map retains every support element z_λ ; the second is the ring reflection and uses the ideal's Bourne congruence. Inverting the element e , in contrast, gives

$$S[e^{-1}] \simeq L. \quad (8)$$

Proof. The complement of E consists exactly of supported nonzero amplitudes. Inverting them creates their original field fractions. For a zero-fibre element, multiplication by any such denominator fixes z_λ , so its fraction still equals the same z_λ . Different zero-fibre labels remain distinct: the support map $S \rightarrow L$ sends every denominator to 1 and therefore distinguishes them in the localization. A nonzero field fraction cannot become a zero-fibre element because the ring-reflection map to K distinguishes their amplitudes. These observations prove the first isomorphism, including all fraction identifications.

The Bourne relation for E identifies x, y when $x + i = y + j$ for some $i, j \in E$. It preserves amplitudes. Conversely if amplitudes agree, adding e to either element gives the same supported

amplitude; hence the relation identifies exactly equal amplitudes. This proves the quotient assertion. For (8), invertibility of the idempotent e forces $e = 1$. Each supported r then satisfies $r = er = e = 1$. Every z_λ survives, and its inherited join and meet give L with unit e . The support map realizes this identification, proving its universal property. $\square$

Proposition 2.3 (The complete power filtration of the prime E). *For every $n \geq 1$, $E^n = E$. The ideal-congruence quotients and their inverse limit are*

$$S/E^n \simeq R, \quad \varprojlim_n S/E^n \simeq R. \quad (9)$$

Every transition map is the identity on R , and the completion map is p_L . The support-only ideal E is isomorphic to L with its own multiplicative identity e ; its inclusion in S is not unit preserving.

Proof. Idempotence $E^2 = E$ proves the power identity by induction. The Bourne-quotient argument in the preceding proof applies before localization and identifies S/E exactly with R . Thus all levels and transition maps are as stated. The last assertion follows from $z_\lambda z_\mu = z_{\lambda \wedge \mu}$, $z_\lambda + z_\mu = z_{\lambda \vee \mu}$, and $ez_\lambda = z_\lambda$. Since $e \neq 1_S$, its internal unit is not the unit of S . $\square$

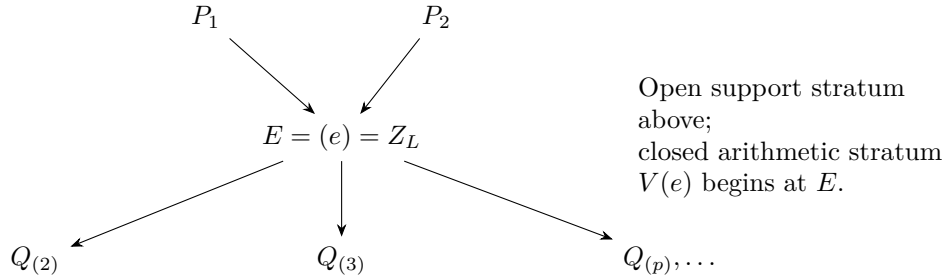


Figure 1: Exact specialization diagram for $G_{\mathbb{B}^2}(\mathbb{Z})$, with arrows from a smaller prime to a larger one. The two minimal support primes specialize to the supported-zero prime E ; every finite arithmetic prime lies one step below it in this drawing. Theorem 2.1 proves that their heights are 0, 1, 2. The two branches share the arithmetic stratum.

The coordinate E is simultaneously a nonzero prime ideal of S and the generic point of its closed arithmetic subspace. These facts describe its actual position; calling it the old zero prime alone loses its nonzero principal generator and its height in the new space. The arithmetic constructions transported through p_L live over $V(E)$. Extending their conclusions to the whole support spectrum requires the support diagram and its connecting maps, which the rebuilt cohomology retains.

3 The absolute-base maps and the role of every integer

This section specifies the actual arrows used by the proposed geometry over τ . It retains supported integer zero throughout. Monoid schemes and blueprints in the original source are attributed to Deitmar and Lorscheid [11, 12]; the elementary maps needed here are proved below.

Proposition 3.1 (Integer coefficients and their vanishing loci). *Let $S = G(\mathbb{Z})$ and $X = \text{Spec } S$. The supported inclusion $j : \mathbb{Z} \hookrightarrow S$, $n \mapsto n$, preserves addition, multiplication and one. Its zero is $j(0) = e$. The ring reflection satisfies $pj = I_{\mathbb{Z}}$. The initial unital semiring map $\iota : \mathbb{N} \rightarrow S$ is instead*

$$\iota(0) = \tau, \quad \iota(n) = j(n) \quad (n > 0). \quad (\text{AZ1})$$

Writing $T = (\tau)$, $E = (e)$, and $P_p = \{\tau\} \cup p\mathbb{Z}$, one has

$$\iota^{-1}(T) = \iota^{-1}(E) = \{0\}, \quad \iota^{-1}(P_p) = p\mathbb{N}. \quad (\text{AZ2})$$

Every supported integer defines a function on the semiring spectrum and its exact vanishing set is

$$V_X(j(0)) = V_X(E) \simeq \text{Spec } \mathbb{Z}, \quad V_X(j(n)) = \{P_p : p \mid n\} \quad (n \neq 0). \quad (\text{AZ3})$$

For $n = \pm 1$ the latter set is empty. The zero integer has its own nonzero prime principal ideal E , at height one.

Proof. On supported elements the two operations are exactly integer operations, proving the claims about j and p . A unital map from $\mathbb{N}$ sends its zero to the target additive identity and sends a positive n to a sum of n copies of one. This proves (AZ1), including preservation of the two operations when an input is zero. Testing its values against the explicit prime sets proves (AZ2). The ideals T contain no supported integers, E contains exactly the supported integer zero, and P_p contains exactly the integers divisible by p together with τ . This proves every case of (AZ3). The height of E is the already proved foundational theorem. $\square$

Thus the two zero conventions have an exact relation: j keeps the integer zero as an element with support, and ι is the unital semiring structure map. Neither substitutes one zero for the other in a calculation.

Proposition 3.2 (Generic localization and the monoidal base). *The support morphism*

$$\chi : S \longrightarrow \mathbb{B}, \quad \chi(\tau) = 0, \quad \chi(j(n)) = 1 \quad (n \in \mathbb{Z}) \quad (\text{AZ4})$$

is localization at e . Its spectral arrow is the open immersion

$$\text{Spec } \mathbb{B} \longrightarrow X, \quad \text{image} = D_X(e) = \{T\}. \quad (\text{AZ5})$$

There is no unital semiring homomorphism $\mathbb{B} \rightarrow S$. There is a unital pointed-monoid map

$$\mathbb{F}_1^{\text{mon}} = \{0, 1\} \longrightarrow (S, \cdot), \quad 0 \longmapsto \tau, \quad 1 \longmapsto 1, \quad (\text{AZ6})$$

and hence a structure morphism from the entire multiplicative spectrum to the one-point monoidal base. Give $(S, \cdot)$ its actual additive relations: a formal sum is declared equivalent to another exactly when their sums are equal in S , and the empty sum equals τ . Then (AZ6) preserves the relations of the initial blueprint and gives a blueprint structure map. The subset of multiplicative primes closed under these additive relations is exactly $\text{Spec } S$; its induced map to the one-point base still includes every support and arithmetic prime.

Proof. Support of a sum is the join of its supports, even when supported amplitudes cancel to e . Support of a product is their meet. Hence χ is a unital semiring morphism. Inverting the idempotent e makes $e = 1$; then $ej(n) = e$ makes every supported integer one. The resulting two elements are still distinguished by χ . This proves localization and (AZ5). For a unital semiring map from $\mathbb{B}$, its relation $1 + 1 = 1$ would become $j(2) = j(1)$ in S , contradicting the injectivity of the supported inclusion. This proves the specific obstruction to a reverse semiring arrow.

For (AZ6), the multiplication tables agree, including absorption by zero, so the indicated map exists and is unique. The only proper prime of $\{0, 1\}$ is $\{0\}$, proving the base assertion. The initial blueprint has no additional nontrivial additive identifications; the empty-sum relation is respected because it is sent to τ . A multiplicative ideal closed under the declared additive relations is closed under addition, and is thus a semiring ideal. Conversely every semiring ideal has precisely that closure. The multiplicative prime condition is the same product test in both settings, proving the asserted spectral identification and the restriction of the structure map. $\square$

Equations (AZ4)–(AZ6) retain two different useful arrows with their directions proved: generic localization inside the full semiring spectrum, and the absolute monoidal/blueprint structure map of the whole object. The full-support cohomology diagram in this manuscript is attached to that whole object. It is not defined by replacing it with the stalk at T .

4 Exact localizations and contraction maps on the full support spectrum

22 September 2026. This derivation starts with the full bounded distributive support lattice. It does not select a Boolean branch before localizing. Its original provider is **The Clankers**, *Split Support Geometry*, version 11, `split_support_geometry_arithmetic_curve_v11.tex`: `def:lattice-split` and `thm:lattice-normal-form` at lines 1210–1321. The localization theorem `thm:lattice-localization` is at lines 1457–1487; the complete ideal/prime/ordinal-sum proofs at lines 1607–1771; and the branch construction and inverse-prime map at lines 1875–1987. Those sections were read, and the stalks, finite Boolean fibre product, and all contractions below are derived explicitly.

4.1 1. Full coefficient object and its primes

Let $R \neq 0$ be a commutative unital ring, and let L be a nontrivial bounded distributive lattice. Set

$$S = G_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{r} = (r, 1_L) : r \in R\} \subset R \times L. \quad (\text{TL1})$$

The two parts meet exactly at $e_L = z_{1_L} = \widehat{0}$. The semiring zero is $\tau_L = z_{0_L}$, and its one is $\widehat{1}$. The operations are

$$\begin{aligned} z_\lambda + z_\mu &= z_{\lambda \vee \mu}, & z_\lambda z_\mu &= z_{\lambda \wedge \mu}, & \widehat{r} + z_\lambda &= \widehat{r}, & \widehat{r} z_\lambda &= z_\lambda, \\ \widehat{r} + \widehat{s} &= \widehat{r + s}, & \widehat{r} \widehat{s} &= \widehat{rs}. \end{aligned} \quad (\text{TL2})$$

The formulas agree at the overlap. Distributivity follows from that of the ring and lattice. Define $Z_L = \{z_\lambda : \lambda \in L\}$.

A lattice ideal is lower, contains 0_L , and is closed under finite joins. It is prime if it is proper and $\lambda \wedge \mu$ in the ideal forces one factor to be in the ideal. For a prime lattice ideal $\mathfrak{a}$ and a ring prime $\mathfrak{p}$, put

$$P_{\mathfrak{a}} = \{z_\lambda : \lambda \in \mathfrak{a}\}, \quad Q_{\mathfrak{p}} = Z_L \cup \{\widehat{r} : r \in \mathfrak{p}\}. \quad (\text{TL3})$$

These are every prime of S . Here is a complete verification of the needed classification. An ideal containing a supported $\widehat{r}$ contains e_L : multiply by -1 and add when $r \neq 0$, while $r = 0$ already gives it. Multiplication of e_L by z_λ then places every z_λ in the ideal. Its supported amplitudes form a ring ideal, by TL2 and multiplication by $\widehat{-1}$. An ideal containing no supported element is contained in Z_L , and its support labels form a proper lattice ideal by TL2 and the identity $z_\mu z_\lambda = z_\mu$ for $\mu \leq \lambda$. Conversely these two constructions give ideals. For the second family, primality is precisely lattice primality; no product of supported elements belongs to a proper pure-support ideal. For the first family, primality reduces on supported factors to ring primality; every product with a zero-fibre factor is already in the amplitude ideal. This proves the assertion.

The complete inclusions are

$$P_{\mathfrak{a}} \subseteq P_{\mathfrak{b}} \iff \mathfrak{a} \subseteq \mathfrak{b}, \quad Q_{\mathfrak{p}} \subseteq Q_{\mathfrak{q}} \iff \mathfrak{p} \subseteq \mathfrak{q}, \quad P_{\mathfrak{a}} \subsetneq Q_{\mathfrak{p}} \text{ for every pair.} \quad (\text{TL4})$$

Indeed the pure-support ideals omit e_L , while every amplitude ideal contains it. The basic opens are

$$D(z_\lambda) = \{P_{\mathfrak{a}} : \lambda \notin \mathfrak{a}\}, \quad D(\widehat{r}) = \text{SpecLat } L \sqcup D_R(r). \quad (\text{TL5})$$

These formulas retain the whole support-prime space as an open stratum and the whole arithmetic spectrum as a closed stratum. They also prove the stated spectral ordinal-sum topology directly.

4.2 2. The exact lattice localization at a prime

Fix a prime lattice ideal $\mathfrak{a}$, and let $F = L \setminus \mathfrak{a}$. This is an upward-closed meet filter containing 1_L and excluding 0_L . Define

$$\lambda \sim_F \mu \iff \exists f \in F : f \wedge \lambda = f \wedge \mu. \quad (\text{TL6})$$

This is a bounded-lattice congruence. Reflexivity uses $f = 1_L$, and symmetry is immediate. If witnesses f, g give two consecutive equalities, their meet $f \wedge g \in F$ witnesses transitivity. For congruence under join and meet, replace two witnesses by their meet. Distributivity then gives equality after meeting either join with that common witness; associativity gives the corresponding equality for meets. Therefore the quotient

$$L_{\mathfrak{a}} = L / \sim_F, \quad \ell_{\mathfrak{a}}(\lambda) = [\lambda] \quad (\text{TL7})$$

is a bounded distributive lattice, regarded as a semiring with join and meet. Its bottom and top are distinct: equality $[0_L] = [1_L]$ would force some $f \in F$ to equal zero.

The map $\ell_{\mathfrak{a}}$ is the semiring localization inverting F . Every $f \in F$ becomes one because $f \wedge f = f \wedge 1_L$. Conversely, let $u : L \rightarrow T$ be a semiring map to any semiring, inverting every $f \in F$. Each $u(f)$ is multiplicatively idempotent and invertible, hence equals one. Equality in TL6 therefore implies $u(\lambda) = u(\mu)$, so u factors uniquely through TL7. This proves the full universal property, not only a description of representatives.

The top class has the exact preimage

$$[\lambda] = 1_{L_{\mathfrak{a}}} \iff \lambda \in F. \quad (\text{TL8})$$

For the nontrivial direction, $f \wedge \lambda = f$ implies $f \leq \lambda$, and upward closure gives $\lambda \in F$. For the reverse direction use witness $f = \lambda$. In any nontrivial lattice semiring the only unit is the top: $x \wedge y = 1$ forces $x = y = 1$. Hence $L_{\mathfrak{a}}$ is local, with unique maximal ideal

$$\mathfrak{m}_{\mathfrak{a}} = \{[\lambda] : \lambda \in \mathfrak{a}\} = L_{\mathfrak{a}} \setminus \{1\}. \quad (\text{TL9})$$

It is an ideal because $\mathfrak{a}$ is closed under joins and downward containment, and all proper ideals must omit the unit. The branch character

$$\epsilon_{\mathfrak{a}}(\lambda) = \begin{cases} 0, & \lambda \in \mathfrak{a}, \\ 1, & \lambda \notin \mathfrak{a} \end{cases} \quad (\text{TL10})$$

is a bounded lattice homomorphism: join lies in $\mathfrak{a}$ exactly when both factors do, while meet lies there exactly when at least one factor does. It factors through $L_{\mathfrak{a}}$, and its zero fibre is TL9. Quotienting the local lattice by that maximal ideal gives $\mathbb{B}$. Explicitly every nonunit is identified with zero by the ideal quotient, while one cannot be identified with zero because an equality $1 \vee i = j$, with $i, j \in \mathfrak{m}_{\mathfrak{a}}$, would force $j = 1$. Thus the Boolean object is the local residue of this support stalk; the entire stalk need not be Boolean.

4.3 3. Pure-support prime stalks, including all surviving levels

The complement $S \setminus P_{\mathfrak{a}}$ contains every supported $\hat{r}$, including e_L , and the elements z_f for $f \in F$. Inverting e_L first gives L by the exact support map

$$\chi_L : S \longrightarrow L, \quad \hat{r} \mapsto 1_L, \quad z_{\lambda} \mapsto \lambda. \quad (\text{TL11})$$

Indeed an invertible idempotent image of e_L equals one; then $e_L \hat{r} = e_L$ forces every supported amplitude to map to one. The images of the z_{λ} preserve join, meet, and both endpoints. This proves the universal property of TL11 as localization at e_L .

After TL11, the remaining denominators are precisely F . Combining its universal property with TL7 proves

$$\boxed{S_{P_a} \cong L_a, \quad \widehat{r} \mapsto 1, \quad z_\lambda \mapsto [\lambda].} \quad (\text{TL12})$$

Every denominator maps to one. The numerator fraction has the concrete value

$$\widehat{r}/t \mapsto 1, \quad z_\lambda/t \mapsto [\lambda] \quad (t \notin P_a),$$

because every inverted denominator is idempotent after TL11 and therefore equal to one. Conversely any map inverting the original complement must factor first through TL11 and then through TL7, which proves that no additional support identifications have been omitted.

When L is finite, define

$$f_a = \bigwedge_{f \in F} f \in F. \quad (\text{TL13})$$

Then TL6 is equivalent to $f_a \wedge \lambda = f_a \wedge \mu$. One implication follows by meeting a witnessing equality with $f_a \leq f$, and the reverse uses f_a itself as witness. Therefore

$$\boxed{L_a \xrightarrow{\sim} \downarrow f_a, \quad [\lambda] \mapsto f_a \wedge \lambda.} \quad (\text{TL14})$$

This map is onto because any $\nu \leq f_a$ is its own image. Its bottom is 0_L , its unit is f_a , and its operations are the original join and meet restricted to that principal interval.

For the complete chain $C_n = \{0 < 1 < \dots < n\}$, every prime ideal has the form $\mathfrak{a}_j = \{0, \dots, j-1\}$, $1 \leq j \leq n$. Lowerness forces every proper ideal to be such an initial segment, and primality follows because a minimum lies below j exactly when one input does. Its complement has least element j , so

$$\boxed{G_{C_n}(R)_{P_{\mathfrak{a}_j}} \cong C_j, \quad z_\lambda \mapsto \min(\lambda, j), \quad \widehat{r} \mapsto j.} \quad (\text{TL15})$$

For $j > 1$, this stalk retains the $j-1$ distinct support elements strictly between absence and its unit. The residue map to $\mathbb{B}$ sends all of them to zero and sends j to one. This is an explicit counterexample to replacing every pure-support prime stalk by $\mathbb{B}$.

Every ring-valued realization of L_a is the zero ring, because $1 \vee 1 = 1$ forces $1 + 1 = 1$ in the target. This arithmetic reflection statement does not identify the distinct support elements of the stalk itself.

4.4 4. Amplitude prime stalks retain the entire lattice

Fix $\mathfrak{p} \in \text{Spec } R$. The complement of $Q_{\mathfrak{p}}$ consists exactly of supported $\widehat{s}$ with $s \notin \mathfrak{p}$. In particular it contains no element of Z_L . The exact localization is

$$\boxed{S_{Q_{\mathfrak{p}}} \cong G_L(R_{\mathfrak{p}}), \quad \widehat{r}/\widehat{s} \mapsto \widehat{r/s}, \quad z_\lambda/\widehat{s} \mapsto z_\lambda.} \quad (\text{TL16})$$

The second formula follows from $z_\lambda \widehat{s} = z_\lambda$, after $\widehat{s}$ becomes invertible. For the first, if $r/s = r'/s'$, there is $u \notin \mathfrak{p}$ with $us'r = usr'$. Any semiring map inverting the denominators gives equal images of these fractions after cancellation of the invertible factors. Thus the formulas are well-defined. Their compatibility with TL2 proves the mixed addition and multiplication cases. Conversely a map from S that inverts $R \setminus \mathfrak{p}$ extends by those same formulas, uniquely, so TL16 has the required universal property.

No two distinct support labels become equal: in the fractional equivalence relation, multiplication of z_λ and z_μ by any supported denominator leaves both unchanged, so equality forces $\lambda = \mu$. A supported fraction can coincide with a zero-fibre element only at the top label 1_L , when its ring fraction equals zero. This retains precisely the original overlap in TL1.

The unit group of $G_L(A)$, for any nonzero ring A , is $A^\times$: a product equal to $\widehat{1}$ must have nonzero amplitude and hence both factors are supported, where invertibility is ordinary ring invertibility. Thus TL16 is local with maximal ideal

$$Z_L \cup \{\widehat{a} : a \in \mathfrak{p}R_{\mathfrak{p}}\}. \quad (\text{TL17})$$

The support-preserving residue and its ordinary ring shadow are

$$G_L(R_{\mathfrak{p}}) \longrightarrow G_L(\kappa(\mathfrak{p})) \xrightarrow{p_L} \kappa(\mathfrak{p}), \quad \kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}. \quad (\text{TL18})$$

The first map keeps every z_λ and reduces the amplitude; the second kills every z_λ . The composite is the ideal quotient by TL17. Indeed killing the ideal kills e_L , and $e_L + z_\lambda = e_L$ then kills every support element; the surviving supported ring quotient is exactly $\kappa(\mathfrak{p})$. Conversely TL18 has that exact zero fibre. Thus ordinary residue-field norms remain available without discarding the full support residue $G_L(\kappa(\mathfrak{p}))$.

4.5 5. Every generalization and localization contraction

For lattice primes $\mathfrak{a} \subseteq \mathfrak{b}$, the filters satisfy $F_{\mathfrak{b}} \subseteq F_{\mathfrak{a}}$, so the exact generalization map is

$$S_{P_{\mathfrak{b}}} = L_{\mathfrak{b}} \longrightarrow L_{\mathfrak{a}} = S_{P_{\mathfrak{a}}}, \quad [\lambda]_{\mathfrak{b}} \mapsto [\lambda]_{\mathfrak{a}}. \quad (\text{TL19})$$

It is well-defined because every equality witness from the smaller filter remains a witness in the larger filter. If L is finite, $f_{\mathfrak{a}} \leq f_{\mathfrak{b}}$, and in TL14 coordinates this is the explicit map

$$\downarrow f_{\mathfrak{b}} \longrightarrow \downarrow f_{\mathfrak{a}}, \quad \lambda \mapsto \lambda \wedge f_{\mathfrak{a}}. \quad (\text{TL20})$$

For ring primes $\mathfrak{q} \subseteq \mathfrak{p}$, the generalization map is $G_L(R_{\mathfrak{p}}) \rightarrow G_L(R_{\mathfrak{q}})$, obtained by the ring fraction map and the identity on all support labels. For every pure-support/amplitude pair, TL4 gives the generalization

$$S_{Q_{\mathfrak{p}}} = G_L(R_{\mathfrak{p}}) \xrightarrow{\chi_L} L \xrightarrow{\ell_{\mathfrak{a}}} L_{\mathfrak{a}} = S_{P_{\mathfrak{a}}}. \quad (\text{TL21})$$

It sends $\widehat{r/s}$ to one and z_λ to $[\lambda]$. It inverts every element in the larger complement, so the universal property proves it is exactly the stalk map. These maps compose according to the inclusions in TL4, as is also immediate from their formulas.

The prime contractions of TL12 are exactly

$$\text{Spec } L_{\mathfrak{a}} \longleftrightarrow \{\mathfrak{b} \in \text{SpecLat } L : \mathfrak{b} \subseteq \mathfrak{a}\}, \quad \mathfrak{b} \longmapsto \ell_{\mathfrak{a}}(\mathfrak{b}), \quad (\text{TL22})$$

and the corresponding contraction to S is $P_{\mathfrak{b}}$. For a direct proof, if $\mathfrak{b} \subseteq \mathfrak{a}$, it is disjoint from $F_{\mathfrak{a}}$. If $\lambda \sim_F \mu$ and $\lambda \in \mathfrak{b}$, then a witnessing $f \notin \mathfrak{b}$ satisfies $f \wedge \mu = f \wedge \lambda \in \mathfrak{b}$. Primality gives $\mu \in \mathfrak{b}$. Thus membership in $\mathfrak{b}$ is well-defined on equivalence classes, and its image is prime. Conversely the inverse image of any proper prime in the quotient is prime in L , omits every $f \in F$ because they map to one, and is therefore contained in $\mathfrak{a}$. The constructions are inverse by surjectivity of $\ell_{\mathfrak{a}}$.

The prime contractions of TL16 are exactly all the pure-support primes and the amplitude primes below $Q_{\mathfrak{p}}$:

$$\begin{aligned} P_{\mathfrak{b}} \subset G_L(R_{\mathfrak{p}}) &\longmapsto P_{\mathfrak{b}} \subset S, \\ Q_{\mathfrak{q}R_{\mathfrak{p}}} &\longmapsto Q_{\mathfrak{q}}, \quad \mathfrak{q} \in \text{Spec } R, \quad \mathfrak{q} \subseteq \mathfrak{p}. \end{aligned} \quad (\text{TL23})$$

The support contraction follows because the labels are unchanged and no supported element belongs to a pure-support ideal. For the ring part, a prime ideal $\mathfrak{q} \subseteq \mathfrak{p}$ has extension consisting of fractions with numerator in $\mathfrak{q}$. Membership is well-defined because a denominator outside $\mathfrak{p}$ lies outside $\mathfrak{q}$, and primality cancels it from the fractional equality. That extension is prime and contracts to $\mathfrak{q}$. Conversely a prime of $R_{\mathfrak{p}}$ contracts to a ring prime disjoint from $R \setminus \mathfrak{p}$, and every fraction is a numerator times an invertible denominator, proving extension and contraction inverse. Combined with TL3, this proves that TL23 lists every contraction.

4.6 6. Scalar inclusion, branch maps, and their local forms

The scalar inclusion is

$$\iota : G(R) = G_{\mathbb{B}}(R) \hookrightarrow G_L(R), \quad \tau \mapsto z_{0_L}, \quad r \mapsto \hat{r}. \quad (\text{TL24})$$

For a lattice prime $\mathfrak{a}$, use its branch TL10 to define $\beta_{\mathfrak{a}} : G_L(R) \rightarrow G(R)$: supported amplitudes remain unchanged, while z_{λ} maps to τ for $\lambda \in \mathfrak{a}$ and to e otherwise. TL2 and the lattice homomorphism property prove it is a zero- and one-preserving homomorphism, and $\beta_{\mathfrak{a}}\iota = \text{id}$. Its complete prime inverse images and the inclusion contractions are

$$\begin{aligned} \beta_{\mathfrak{a}}^{-1}(P_{\tau}) &= P_{\mathfrak{a}}, & \beta_{\mathfrak{a}}^{-1}(P_{\mathfrak{p}}) &= Q_{\mathfrak{p}}, \\ \iota^{-1}(P_{\mathfrak{b}}) &= P_{\tau} \text{ for every } \mathfrak{b}, & \iota^{-1}(Q_{\mathfrak{p}}) &= P_{\mathfrak{p}}. \end{aligned} \quad (\text{TL25})$$

These equalities follow by evaluating the maps on supported elements, bottom, and top. Every pure-support prime omits the top and contains the bottom. At the chosen pure-support point the induced local map is

$$L_{\mathfrak{a}} \longrightarrow \mathbb{B}, \quad [\lambda] \mapsto \epsilon_{\mathfrak{a}}(\lambda). \quad (\text{TL26})$$

It is exactly the residue map of TL9–10, and can have a nontrivial kernel containing several distinct support levels. At an amplitude point it is $G_L(R_{\mathfrak{p}}) \rightarrow G(R_{\mathfrak{p}})$ with the same chosen branch. Thus the scalar stalk computation is recovered through a displayed local morphism; it is not substituted for the entire full-support stalk.

4.7 7. Finite Boolean support as an exact fibre product

Let I be a finite nonempty set and $L = 2^I$, with union and intersection. For every $i \in I$ let $\pi_R : G(R) \rightarrow R$ be the ring reflection. Form the fibre product in the category of unital zero-preserving semirings,

$$\mathcal{F}_I(R) = \prod_{i \in I, R} G(R) = \{(x_i) \in G(R)^I : \pi_R(x_i) = \pi_R(x_j) \text{ for all } i, j\}. \quad (\text{TL27})$$

It is a subsemiring because equal amplitudes remain equal under coordinatewise operations. There is an exact isomorphism

$$\boxed{\Phi : G_{2^I}(R) \xrightarrow{\sim} \mathcal{F}_I(R), \quad z_A \mapsto (x_i), \quad x_i = \begin{cases} e, & i \in A, \\ \tau, & i \notin A, \end{cases} \quad \hat{r} \mapsto (r)_{i \in I}.} \quad (\text{TL28})$$

To prove surjectivity, let r be the common amplitude of a fibre-product tuple. If $r \neq 0$, each coordinate must be the supported r , since the reflection fibre of a nonzero amplitude is a singleton. If $r = 0$, every coordinate is either τ or e , so it determines a unique subset A . These cases also prove injectivity, including the overlap $r = 0, A = I$. Coordinatewise union/intersection on zero amplitudes and the supported rules TL2 prove preservation of every operation and both identities.

The coordinate maps of TL28 are exactly the branch maps β_i for $\epsilon_i(A) = \mathbf{1}_{i \in A}$. Its scalar inclusion TL24 becomes the diagonal $G(R) \rightarrow G(R)^I$. Thus one has the actual diagram of semiring maps

$$G(R) \xrightarrow{\iota} G_{2^I}(R) \xrightarrow{\Phi} \mathcal{F}_I(R) \hookrightarrow G(R)^I \xrightarrow{\text{pr}_i} G(R), \quad \text{pr}_i \Phi \iota = \text{id}. \quad (\text{TL29})$$

This is the amplitude fibre product. The different synchronized-support identity $G(R^I) = \prod_{i \in I, \mathbb{B}} G(R)$ is verified by the same coordinate reasoning: equal support means either all coordinates absent, or all present with independent amplitudes. The independent product $G(R)^I$

contains both subobjects. In TL28 amplitudes are equal and the whole zero-support cube is retained; in $G(R^I)$ support is equal and amplitudes are independent. These exact fibre-product maps state their relation without identifying them.

Every lattice prime of 2^I is

$$\mathfrak{a}_i = \{A \subseteq I : i \notin A\}, \quad i \in I. \quad (\text{TL30})$$

Indeed the top is the finite join of singleton atoms, so a prime ideal must omit at least one atom. It cannot omit two distinct atoms, because their meet is bottom and primality would fail. If it omits $\{i\}$, every subset not containing i is a join of included atoms and lies in the ideal, while no subset containing i can lie in it by lowerness. This proves TL30. Its complement has least element $\{i\}$, so every pure-support stalk in this finite Boolean case is $\downarrow \{i\} \cong \mathbb{B}$. This conclusion depends on the Boolean atoms and does not apply to TL15's longer chains.

4.8 8. Complete contractions from the independent finite product

Every prime of the independent product $G(R)^I$ is $\text{pr}_i^{-1}(\mathfrak{P})$ for exactly one $i \in I$ and $\mathfrak{P} \in \text{Spec } G(R)$. To prove this, let d_i have supported one at i and absent zero elsewhere. Then $\sum_i d_i = 1$ and $d_i d_j = 0$ for $i \neq j$. A proper prime must omit some d_i and can omit at most one. Every x decomposes as $\sum_j d_j x$; all terms except the omitted coordinate lie in the prime. Hence x lies in the prime exactly when $d_i x$ does: one direction uses multiplication by d_i , and the other uses the displayed finite sum. The values at that coordinate form a proper prime of $G(R)$, proving existence and uniqueness of the claimed description.

Contract these primes along the embedding in TL29. For all $i \in I$ and ring primes $\mathfrak{p}$, the complete formulas are

$$\boxed{\Phi^{-1}(\text{pr}_i^{-1}(P_\tau)) = P_{\mathfrak{a}_i}, \quad \Phi^{-1}(\text{pr}_i^{-1}(P_{\mathfrak{p}})) = Q_{\mathfrak{p}}.} \quad (\text{TL31})$$

For the first, a zero-mask coordinate equals τ precisely when $i \notin A$, and no supported diagonal amplitude equals τ . For the second, both τ and supported e belong to $P_{\mathfrak{p}}$, so every zero mask belongs to the contraction, while $\hat{r}$ belongs precisely when $r \in \mathfrak{p}$. TL3 and TL30 show that these formulas exhaust every prime of $G_{2^I}(R)$.

Thus the spectral map from the independent product onto the amplitude fibre product has singleton fibres over each pure-support prime $P_{\mathfrak{a}_i}$, and exactly $|I|$ points over each amplitude prime $Q_{\mathfrak{p}}$, one from each coordinate. The map to the scalar subobject then contracts all $P_{\mathfrak{a}_i}$ to P_τ while keeping the ring-prime labels, as in TL25. These are contractions of prime ideals under the actual displayed homomorphisms, with every support branch present.

At a product prime $\text{pr}_i^{-1}(\mathfrak{P})$, localization inverts d_i , whose principal factor kills every other coordinate. Its stalk is therefore $G(R)_{\mathfrak{P}}$. The local maps induced by TL31 are the identity $\mathbb{B} \rightarrow \mathbb{B}$ at $P_{\mathfrak{a}_i}$, and

$$G_{2^I}(R_{\mathfrak{p}}) \xrightarrow{\beta_i} G(R_{\mathfrak{p}}) \quad (\text{TL32})$$

at an amplitude prime. In the latter map the full zero-support cube is mapped through its specified i -th branch; the full source stalk remains TL16. This completes the local and spectral contraction diagram without prematurely replacing the full coefficient object by a branch.

The classical ideal arithmetic is reproved in the following chapter; its human-source context is Neukirch [10], retained from the foundational bibliography.

5 The full ideal arithmetic, divisor coordinates, and norms

Let K be a number field, let $A = \mathcal{O}_K$, and retain

$$S = G(A) = A \sqcup \{\tau\}, \quad e = 0_A, \quad T = \{\tau\}, \quad E = \{\tau, e\}.$$

Thus the zero ideal of the semiring is T , whereas $E = eS$ is a nonzero prime ideal. The supported-zero prime theorem is prior work: [1, `thm:zero-prime-not-irr`] states it explicitly. The programme's originator attributes its original derivation to Gemini 2.5 Pro. The present section calculates the complete fractional ideal structure and the exact class quotients using that theorem. Ideal products mean finite sums of elementwise products; an empty sum is τ . The integral ideal classification and Dedekind zeta identity are also retained prior results of [1, `thm:OK-ideals-spectrum`, `thm:Dedekind-zeta`]. The arguments below give their full fractional-ideal setting and the comparison for the exact displayed class relation.

5.1 Cancellation and the localization retaining both zeros

Proposition 5.1 (The exact regular elements). *Multiplication $x \mapsto ax$ on S is injective exactly for $a \in A \setminus \{0_A\}$. The units of S are $A^\times$. The only element whose addition map is injective is τ . Nevertheless every $a \neq \tau$, including e , satisfies*

$$ax = \tau \implies x = \tau.$$

Thus absence of a nontrivial product equal to τ is a weaker condition here than multiplicative cancellation.

Proof. For supported nonzero a , a supported product cannot equal an absent product, and supported equality cancels in the domain A . Multiplication by e identifies 1 and e ; multiplication by τ is constant. A product is 1 exactly when its factors are inverse elements of A , proving the unit assertion. For supported a , $a + \tau = a = a + e$, whereas addition by τ is the identity. Finally a product equals τ exactly when one factor equals τ . $\square$

Call injective multiplication scalars *regular*. Their localization is

$$Q_{\text{reg}}(S) = (A \setminus \{0_A\})^{-1}S = G(K). \quad (10)$$

Indeed τ/d maps to τ and supported a/d to the corresponding field fraction. The localization relation identifies supported fractions precisely by equality in K , and cannot identify one with τ . The units of $G(K)$ are $K^\times$. Localization at all of $S \setminus \{\tau\} = A$ instead inverts e and yields $\mathbb{B}$, as in [1, `thm:frac-collapse`]. Every fractional ideal below uses exactly (10).

5.2 The number-field ideal calculation used in the comparison

Lemma 5.2 (Integral lattice and local valuation rings). *The ring A is free abelian of rank $n = [K : \mathbb{Q}]$, is Noetherian and integrally closed, and every nonzero prime $\mathfrak{p}$ is maximal with finite residue field. For each such $\mathfrak{p}$, $A_{\mathfrak{p}}$ is a discrete valuation ring. Its valuation $v_{\mathfrak{p}} : K^\times \rightarrow \mathbb{Z}$ has $v_{\mathfrak{p}}(\pi_{\mathfrak{p}}) = 1$ for a generator of its maximal ideal.*

Proof. Choose a $\mathbb{Q}$ -basis $\alpha_1, \dots, \alpha_n$ of K of algebraic integers by scaling any basis by suitable nonzero integers. The trace matrix $B = (\text{Tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j))$ has integer entries and nonzero determinant. To check nondegeneracy, choose a primitive element θ of the separable extension. The matrix of its n distinct embeddings on $1, \theta, \dots, \theta^{n-1}$ is an invertible Vandermonde matrix; the trace matrix is its transpose times itself. Change of basis preserves nondegeneracy. For $x \in A$, each $\text{Tr}(x \alpha_j)$ is integral, being rational and a sum of algebraic integers. Thus if $x = \sum_i c_i \alpha_i$, then $c_i \in (\det B)^{-1} \mathbb{Z}$. Consequently

$$\sum_i \mathbb{Z} \alpha_i \subseteq A \subseteq (\det B)^{-1} \sum_i \mathbb{Z} \alpha_i.$$

It follows that A is free of rank n . Each ideal is a subgroup of this lattice, hence has finitely many abelian generators, which also generate it as an A -module. This proves Noetherianity.

If $x \in K$ is integral over A , adjoin to $\mathbb{Z}$ the finitely many coefficients of its monic equation and then x . The resulting ring is a finitely generated $\mathbb{Z}$ -module: powers of each successive integral generator reduce using its monic equation. Multiplication by x , applied to a finite set of module generators, gives a matrix; the determinant of $xI - C$ annihilates that module and hence its element 1. This is a monic integer equation for x , proving $x \in A$. For $0 \neq a \in A$, the nonzero constant term of its monic minimal polynomial belongs to aA . Every nonzero ideal therefore contains a nonzero integer d , and its quotient is a quotient of the finite group A/dA . A finite domain is a field. Hence every nonzero prime is maximal with finite residue field.

Fix $\mathfrak{p} \neq 0$, and write $B_{\mathfrak{p}} = A_{\mathfrak{p}}$ and $\mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}$. This is a Noetherian local domain with only primes $0, \mathfrak{m}$. It is integrally closed: given an integral equation for x over $A_{\mathfrak{p}}$, choose $s \notin \mathfrak{p}$ clearing every coefficient. Substituting $y = sx$ gives a monic equation over A , since its j -th coefficient is s^j times the old one. Thus $sx \in A$ and $x \in A_{\mathfrak{p}}$.

For $0 \neq t \in \mathfrak{m}$, $\sqrt{tB_{\mathfrak{p}}} = \mathfrak{m}$. Finite generation implies $\mathfrak{m}^r \subseteq tB_{\mathfrak{p}}$ for some r : a power of each generator lies in $tB_{\mathfrak{p}}$, and a sufficiently long product includes one of those powers. Choose the least r . If $r = 1$, then $\mathfrak{m} = tB_{\mathfrak{p}}$. Otherwise choose $y \in \mathfrak{m}^{r-1} \setminus tB_{\mathfrak{p}}$ and set $z = y/t$. Then $z \notin B_{\mathfrak{p}}$ and $z\mathfrak{m} \subseteq B_{\mathfrak{p}}$. If $z\mathfrak{m} \subseteq \mathfrak{m}$, express multiplication by z on finitely many generators of the nonzero ideal $\mathfrak{m}$. The determinant of $zI - C$ annihilates them. Cancellation of a nonzero generator in the field makes this determinant zero, forcing z integral over $B_{\mathfrak{p}}$, a contradiction. Thus the ideal $z\mathfrak{m}$ contains a unit and equals $B_{\mathfrak{p}}$. Some $\pi \in \mathfrak{m}$ satisfies $z\pi = 1$, and $u = \pi(zu)$ for every $u \in \mathfrak{m}$; therefore $\mathfrak{m} = \pi B_{\mathfrak{p}}$.

A nonzero $b \in B_{\mathfrak{p}}$ cannot lie in every $\pi^j B_{\mathfrak{p}}$. Otherwise the ascending sequence (b/π^j) of ideals stabilizes, forcing π to be a unit after cancellation of b . Thus $b = \pi^j u$ for a unique $j \geq 0$ and a unit u . Taking the least exponent occurring in a nonzero ideal proves that the ideal is a power of $\mathfrak{m}$. Extending exponents to fractions gives the claimed discrete valuation. $\square$

Lemma 5.3 (Fractional factorization and norm). *Let $\mathcal{I}(A)$ be the nonzero fractional ideals in K . Every such ideal is invertible and has a unique expression*

$$J = \prod_{\mathfrak{p} \neq 0} \mathfrak{p}^{v_{\mathfrak{p}}(J)}, \quad v_{\mathfrak{p}}(J) \in \mathbb{Z}, \quad (11)$$

with finite support. Its ideal operations satisfy

$$\begin{aligned} J \subseteq H &\iff v_{\mathfrak{p}}(J) \geq v_{\mathfrak{p}}(H) \quad \text{for every } \mathfrak{p}, \\ v_{\mathfrak{p}}(J + H) &= \min(v_{\mathfrak{p}}(J), v_{\mathfrak{p}}(H)), \\ v_{\mathfrak{p}}(J \cap H) &= \max(v_{\mathfrak{p}}(J), v_{\mathfrak{p}}(H)), \\ v_{\mathfrak{p}}(JH) &= v_{\mathfrak{p}}(J) + v_{\mathfrak{p}}(H). \end{aligned}$$

For integral $J \neq 0$,

$$|A/J| = \prod_{\mathfrak{p}} |A/\mathfrak{p}|^{v_{\mathfrak{p}}(J)}. \quad (12)$$

Proof. Every bounded A -submodule of K is finitely generated, since a common denominator embeds it in the Noetherian module A . Localizing J gives a power of the maximal ideal in each discrete valuation ring. For integral J , its exponents are nonnegative and vanish outside the finitely many primes containing J ; there are finitely many because A/J is finite. Clearing a denominator proves finite support for fractional J .

Fractional ideals are determined by these localizations. If $x \notin J$, the proper ideal $(J : x) = \{a \in A : ax \in J\}$ lies in a maximal $\mathfrak{p}$. Membership $x \in J_{\mathfrak{p}}$ would give an $s \notin \mathfrak{p}$ with $sx \in J$, a contradiction. Define $\mathfrak{p}^{-1} = \{x \in K : x\mathfrak{p} \subseteq A\}$. For $0 \neq a \in \mathfrak{p}$, this module lies in $a^{-1}A$, so it is fractional. At any maximal $\mathfrak{q}$, its localization equals the inverse of $\mathfrak{p}A_{\mathfrak{q}}$: to prove the reverse inclusion, clear denominators outside $\mathfrak{q}$ for finitely many generators of $\mathfrak{p}$. At $\mathfrak{q} \neq \mathfrak{p}$, $\mathfrak{p}A_{\mathfrak{q}} = A_{\mathfrak{q}}$; at $\mathfrak{p}$ it is the principal maximal ideal. Thus $(\mathfrak{p}\mathfrak{p}^{-1})_{\mathfrak{q}} = A_{\mathfrak{q}}$ for every $\mathfrak{q}$. The equality test proves

$\mathfrak{p}\mathfrak{p}^{-1} = A$. The product in (11) now has the same localizations as J , which proves existence and uniqueness. The same local test with powers in each valuation ring proves the formulas for inclusion, sum, intersection, and product.

Put $q_{\mathfrak{p}} = |A/\mathfrak{p}|$. Each module $\mathfrak{p}^j/\mathfrak{p}^{j+1}$ is killed by $\mathfrak{p}$, so localizing it at $\mathfrak{p}$ changes nothing: every scalar outside $\mathfrak{p}$ already acts invertibly. Its localization is $\pi^j A_{\mathfrak{p}}/\pi^{j+1} A_{\mathfrak{p}}$, one-dimensional over $A/\mathfrak{p}$. Hence $|A/\mathfrak{p}^r| = q_{\mathfrak{p}}^r$. Powers of distinct maximal ideals are comaximal, because a maximal ideal containing their sum would contain both radicals. For comaximal ideals, writing $1 = u + v$ with u, v in the two ideals proves the surjectivity of the Chinese remainder map and that intersection equals product. Induction gives

$$A/\prod_{\mathfrak{p}} \mathfrak{p}^{r_{\mathfrak{p}}} \simeq \prod_{\mathfrak{p}} A/\mathfrak{p}^{r_{\mathfrak{p}}}.$$

Cardinalities prove (12). □

5.3 Every fractional ideal, with both zero ideals retained

Definition 5.4. A fractional S -ideal in $G(K)$ is an S -subsemimodule $F \subseteq G(K)$ with $dF \subseteq S$ for some $d \in A \setminus \{0_A\}$. It contains τ and is closed under addition and multiplication by S . No condition removes E or T . Let $\mathcal{F}(S)$ be all such ideals and put $I_J = J \cup \{\tau\}$ for $J \in \mathcal{I}(A)$. A fractional ideal is regular when it contains a regular element.

Theorem 5.5 (The entire fractional ideal structure). *Every fractional S -ideal is finitely generated, and*

$$\mathcal{F}(S) = \{T, E\} \sqcup \{I_J : J \in \mathcal{I}(A)\}. \quad (13)$$

Conversely, every finitely generated S -subsemimodule of $G(K)$ is a fractional ideal in this list. Its regular ideals are exactly the I_J , and its operations are

$$I_J I_H = I_{JH}, \quad I_J + I_H = I_{J+H}, \quad I_J \cap I_H = I_{J \cap H}, \quad (14)$$

$$TF = T, \quad T + F = F, \quad T \cap F = T, \quad (15)$$

$$EF = E \ (F \neq T), \quad E + F = F \ (F \neq T), \quad E \cap F = E \ (F \neq T). \quad (16)$$

Thus $E^r = E$ for every $r \geq 1$. The unit group of the multiplicative monoid $\mathcal{F}(S)$ consists exactly of the regular ideals, with $I_J^{-1} = I_{J^{-1}}$. Multiplication by F on $\mathcal{F}(S)$ is injective exactly when F is regular.

Proof. If $F \cap K = \emptyset$, then $F = T$. Otherwise multiplication by e puts e in F , and multiplication by -1 supplies additive inverses on $F \cap K$. Thus $J = F \cap K$ is an A -submodule. A common denominator makes dJ an ideal of A . If $J = 0$, then $F = E$; otherwise $J \in \mathcal{I}(A)$, so $F = I_J$. Conversely every listed set has the required closure and denominator. The ideals T, E are generated by τ, e . If $a_1, \dots, a_r$ generate J over A , their S -span is I_J : supported sums give the A -linear combinations and the absent sum gives τ . Hence all fractional ideals are finitely generated. Conversely finitely many elements of $G(K)$ have finitely many supported field denominators. Their product is a nonzero common denominator for their entire S -span.

Every supported product from I_J, I_H lies in JH , and every element of JH is a finite sum of such products. This proves the product formula, including supported zero. Sum and intersection follow from their subsets. The relations involving T follow from its being the specified zero subsemimodule. Every $F \neq T$ contains E , and multiplication by e sends its supported elements to e ; hence (16). By Lemma 5.3, $I_J I_{J^{-1}} = S$. Neither E nor T is invertible, because its products stay in E . Invertible ideals act injectively; E identifies E and S under multiplication, and T acts constantly. □

The entire ideal monoid therefore has an exact divisor presentation. Set

$$D_K = \bigoplus_{\mathfrak{p} \neq 0} \mathbb{Z}[\mathfrak{p}], \quad \widehat{D}_K = D_K \sqcup \{\epsilon, \perp\},$$

with extended addition

$$\epsilon + D = \epsilon, \quad \epsilon + \epsilon = \epsilon, \quad \perp + x = \perp \quad (x \in \widehat{D}_K).$$

Unique ideal factorization and the theorem give a monoid isomorphism

$$\mathcal{F}(S) \xrightarrow{\sim} \widehat{D}_K, \quad I_J \mapsto \sum_{\mathfrak{p}} v_{\mathfrak{p}}(J)[\mathfrak{p}], \quad E \mapsto \epsilon, \quad T \mapsto \perp. \quad (17)$$

Integral ideals correspond to effective divisors together with $\epsilon, \perp$. The inclusion order keeps the strict levels

$$T \subsetneq E \subsetneq I_J, \quad I_J \subseteq I_H \iff v_{\mathfrak{p}}(J) \geq v_{\mathfrak{p}}(H) \text{ for all } \mathfrak{p}.$$

Every homomorphism from the full monoid to an abelian group is trivial: $E I_J = E$ forces each I_J to the identity, and idempotence forces E, T there too. The object $\widehat{D}_K$ retains all three kinds of ideals.

Proposition 5.6 (Extended coordinates and support). *Define*

$$\tilde{v}_{\mathfrak{p}}(I_J) = v_{\mathfrak{p}}(J), \quad \tilde{v}_{\mathfrak{p}}(E) = \tilde{v}_{\mathfrak{p}}(T) = +\infty.$$

With $m + \infty = \infty$, these coordinates send product to addition, sum to minimum, and intersection to maximum. Together with $\sigma(T) = 0$, $\sigma(F) = 1$ for $F \neq T$, they determine every fractional ideal. Also $\sigma(FH) = \sigma(F)\sigma(H)$.

Proof. For regular ideals this is Lemma 5.3; the two other cases follow from (15)–(16). Regular ideals have integer exponent tuples with finite support. Only E, T have the all-infinite tuple, and σ distinguishes them. A product of supported ideals contains e , whereas a product with T is T . $\square$

5.4 The original class relation and its exact comparison

Write $\text{Cl}(A) = \mathcal{I}(A)/\{cA : c \in K^\times\}$ for the quotient of the fractional ideal group by its principal subgroup. The source [1, definition before `thm:class-semigroup`] defines, on nonzero semiring ideals,

$$F \sim_{\text{src}} H \iff \exists a, b \in S \setminus \{\tau\} : aF = bH. \quad (18)$$

Its displayed scalar set is A , so it contains e . We retain this relation exactly as written.

Theorem 5.7 (Both class quotients and their map). *On nonzero integral ideals, or on $\mathcal{F}(S) \setminus \{T\}$, the source quotient is the one-element monoid. Define a distinct relation with regular scalars:*

$$F \sim_{\text{reg}} H \iff \exists a, b \in A \setminus \{0_A\} : aF = bH. \quad (19)$$

On either of those two domains its quotient is

$$\text{Cl}_{\text{reg}}^+(S) \simeq \text{Cl}(A) \sqcup \{\epsilon\}, \quad \epsilon c = \epsilon, \quad \epsilon^2 = \epsilon. \quad (20)$$

The isomorphism is $[I_J]_{\text{reg}} \mapsto [J]$, $[E]_{\text{reg}} \mapsto \epsilon$. On regular ideals alone it is a group isomorphism $\text{Cl}_{\text{reg}}(S) \simeq \text{Cl}(A)$. The identity on ideals induces the exact comparison

$$q : \text{Cl}(A) \sqcup \{\epsilon\} \longrightarrow \text{Cl}_{\text{src}}(S) = \{*\}, \quad q([J]) = q(\epsilon) = *. \quad (21)$$

If T is included, the regular quotient adds a second absorbing level $\perp$, while the source quotient has exactly two classes, represented by T, E , with the Boolean multiplicative table.

Proof. For every $F \neq T$, $eF = E$. Choosing $a = b = e$ in (18) relates every two such ideals, including E, S , and proves the first assertion. For regular scalars, $aI_J = bI_H$ is equivalent to $aJ = bH$, or $J = (b/a)H$ in K . Conversely every multiplier in $K^\times$ has this form with $a, b \in A \setminus \{0\}$. Thus on regular ideals the relation is exactly ordinary ideal-class equivalence. Every fractional ideal scales to an integral one, so integral ideals have the same class set. Regular scalars fix E and cannot send a regular ideal to E . Its class is therefore a distinct absorber, and all other classes form the ordinary ideal class group. The implication $\sim_{\text{reg}} \Rightarrow \sim_{\text{src}}$ gives the map q with the displayed values. Finally every supported scalar sends T to T and a supported ideal to a supported ideal, so T is a separate class in both relations. The product table follows from $TF = T$. The explicit descriptions also verify transitivity and compatibility with multiplication on each stated domain. $\square$

Thus the regular-scalar class group is an exact comparison object within the complete ideal calculation, and q records the precise quotient imposed by the original scalars. The comparison construction in [3, cor:dedekind-preservation] must likewise specify its regular ideals and regular principal scalars; it is not used to rename (18).

5.5 Codimension and the divisor map

Theorem 5.8 (The complete codimension filtration). *Put $X = \text{Spec } S$, $Y = V(E)$. Every point, height, closure, stalk, and residue object is in this table; residue objects use the Bourne congruence of the stalk's maximal ideal.*

point	height in X	closure	$\mathcal{O}_{X,x}$	$\kappa(x)$
$\eta = T$	0	X	$\mathbb{B}$	$\mathbb{B}$
$\xi = E$	1	Y	$G(K)$	K
$x_{\mathfrak{p}} = I_{\mathfrak{p}}$	2	$\{x_{\mathfrak{p}}\}$	$G(A_{\mathfrak{p}})$	$A/\mathfrak{p}$

The reflection induces a closed immersion $i : \text{Spec } A \rightarrow X$ with image Y , sending the ordinary generic point to E and increasing every codimension by one. Precisely,

$$X^{(0)} = \{\eta\}, \quad X^{(1)} = \{E\}, \quad X^{(2)} = \{x_{\mathfrak{p}} : \mathfrak{p} \neq 0\}, \quad X^{(r)} = \emptyset \quad (r > 2). \quad (22)$$

Hence $X^{\geq 0} = X$, $X^{\geq 1} = Y$, $X^{\geq 2} = \{x_{\mathfrak{p}}\}$, and $X^{\geq 3} = \emptyset$. The subset $X^{\geq 2}$ is dense in Y and is not closed in X .

Proof. Every ideal containing a supported element contains e , and its supported intersection is a ring ideal. Thus the integral ideals are T and $J \cup \{\tau\}$ for integral ring ideals J , including 0. Product testing shows that T is prime and $J \cup \{\tau\}$ is prime exactly when J is prime. By Lemma 5.2, all maximal strict chains are $T \subsetneq E \subsetneq I_{\mathfrak{p}}$. Closures consist of primes containing the given prime. This proves the heights, closures, and filtration.

At T , localization inverts e ; its idempotence forces $e = 1$, and then $a = ea = e = 1$ for every supported a . The support map to $\mathbb{B}$ distinguishes these from τ , proving the first stalk. At E , localization is (10). At $I_{\mathfrak{p}}$, the same fraction calculation with denominators $A \setminus \mathfrak{p}$ gives $G(A_{\mathfrak{p}})$.

For any ring R and ideal J , the Bourne relation of $J \cup \{\tau\} \subseteq G(R)$ is $x \sim y$ when $x + u = y + v$ for some $u, v \in J \cup \{\tau\}$. It implies equality of amplitudes in R/J . Conversely adding e makes any absent element supported; if the two amplitudes differ by an element of J , add that difference to one side. This proves that the quotient is exactly R/J . For T , the relation is equality. Applying these facts at the three stalks proves the residues in the table. Contraction along reflection sends J to $J \cup \{\tau\}$; the formula $D_X(a) \cap Y = D_A(a)$ proves its asserted topology.

For every rational prime ℓ , the nonzero ring $A/\ell A$ has ℓ^n elements and a maximal ideal. Its inverse image is a prime above ℓ ; distinct rational primes give distinct maximal ideals of A . A fixed nonzero a belongs to only finitely many such ideals, since A/aA is finite. Thus every nonempty $D_A(a)$ contains a closed point. The closed points are dense in Y but omit E , so their set is not closed. $\square$

Proposition 5.9 (Arithmetic divisor sequence). *The free cycle groups on points of codimension r are*

$$Z^0(X) = \mathbb{Z}[\eta], \quad Z^1(X) = \mathbb{Z}[E], \quad Z^2(X) = \bigoplus_{\mathfrak{p} \neq 0} \mathbb{Z}[x_{\mathfrak{p}}].$$

The inclusion i identifies $Z^0(Y)$ with $Z^1(X)$, and $Z^1(Y)$ with $Z^2(X)$, point by point. There is an exact sequence

$$1 \longrightarrow A^\times \longrightarrow K^\times \xrightarrow{\text{div}_{\text{ar}}} Z^2(X) \longrightarrow \text{Cl}_{\text{reg}}(S) \longrightarrow 0, \quad \text{div}_{\text{ar}}(a) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(a)[x_{\mathfrak{p}}]. \quad (23)$$

A regular fractional ideal I_J has the unique divisor with coefficients $v_{\mathfrak{p}}(J)$. Its extension at an arithmetic point is the fractional ideal

$$I_J G(A_{\mathfrak{p}}) = \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}(J)} A_{\mathfrak{p}} \cup \{\tau\} \subseteq G(K).$$

At E , it localizes to $G(K)$, while E itself localizes to the proper ideal $\{\tau, e\}$. At T , every nonzero ideal localizes to the whole stalk $\mathbb{B}$.

Proof. The cycle groups and i 's maps follow from the point classification. Unique factorization identifies regular fractional ideals with $Z^2(X)$, carrying $I_{aA} = aS$ to the displayed principal divisor. The quotient by these principal ideals is exactly $\text{Cl}_{\text{reg}}(S)$. If every $v_{\mathfrak{p}}(a)$ is zero, then $aA = A$, equivalent to $a \in A^\times$, proving exactness. At $I_{\mathfrak{p}}$, adjoining τ to the ordinary equality $JA_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}(J)} A_{\mathfrak{p}}$ gives the displayed local ideal. Every nonzero fractional ring ideal localizes to K at its generic point, whereas E keeps its two elements at the stalk $G(K)$. At T , a supported ideal contains e , which becomes 1, so its extension is the whole stalk. $\square$

The point E is therefore a codimension-one prime with an idempotent principal ideal. Any group-valued homomorphism on ideals sends E to the group identity because $E^2 = E$; the full divisor object $\widehat{D}_K$ retains it as ϵ . A regular a is a unit of $G(K)$, and direct prime membership gives

$$V_X(a) = \{x_{\mathfrak{p}} : v_{\mathfrak{p}}(a) > 0\} \quad (0 \neq a \in A), \quad V_X(e) = Y.$$

Thus the vanishing set of a regular principal nonunit ideal lies in codimension two, whereas the codimension-one ideal E has the nonregular generator e .

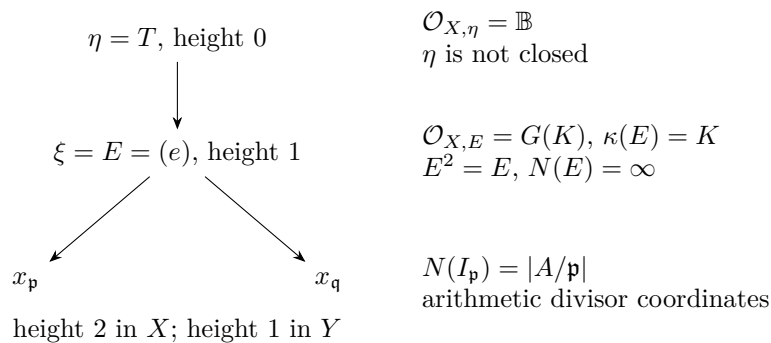


Figure 2: Complete types of specialization in $X = \text{Spec } G(\mathcal{O}_K)$. Arrows mean strict inclusion of primes. The bottom points represent arbitrary distinct finite primes, not a two-point spectrum. Theorems 5.8 and 5.10 prove heights and norms; Proposition 5.9 proves the divisor map. The supported-zero prime is the prior result [1, thm:zero-prime-not-irr].

5.6 Finite norms and the ideal zeta function

Theorem 5.10 (All integral ideal norms and closed points). *For an integral $F \subseteq S$, let $N_B(F)$ be the cardinality of its Bourne-congruence quotient. Then*

$$N_B(T) = \infty, \quad N_B(E) = \infty, \quad N_B(I_J) = |A/J| \quad (0 \neq J \subseteq A). \quad (24)$$

The finite-norm ideals are exactly the regular integral ideals. The closed points are exactly $I_{\mathfrak{p}}$ for nonzero primes $\mathfrak{p}$, all with finite norm $|A/\mathfrak{p}|$. The point E is not closed in X , has infinite quotient A , and infinite residue field K . It is closed in its local spectrum $\text{Spec } G(K)$, still with infinite residue field. The generic point T has residue object $\mathbb{B}$ of size two, but is not closed and has infinite global ideal quotient.

Proof. The Bourne relation of T is equality, giving quotient S . The quotient calculation in Theorem 5.8 gives $S/E = A$ and $S/I_J = A/J$. The first two quotients are infinite, and nonzero ring ideals have finite quotient by Lemma 5.2. Closed points are maximal primes, as already classified. The spectrum of $G(K)$ consists of $T \subsetneq E$, so E is closed there, with residue K . The remaining assertions follow from the stalk and closure table. $\square$

Proposition 5.11 (Norm homomorphism and Euler product). *On regular fractional ideals the norm is*

$$N(I_J) = \prod_{\mathfrak{p}} |A/\mathfrak{p}|^{v_{\mathfrak{p}}(J)} \in \mathbb{Q}_{>0}.$$

It is multiplicative, with $N(cS) = |\text{Norm}_{K/\mathbb{Q}}(c)|$ for $c \in K^\times$. The sum over all finite Bourne-norm ideals is

$$\zeta_{S,B}(s) = \sum_{N_B(F) < \infty} N_B(F)^{-s} = \sum_{0 \neq J \subseteq A} |A/J|^{-s} = \prod_{\mathfrak{p} \neq 0} (1 - |A/\mathfrak{p}|^{-s})^{-1} = \zeta_K(s) \quad (\text{Re } s > 1). \quad (25)$$

Equation (24) accounts explicitly for both zero ideals in this summation rule.

Proof. The ideal norm formula is Lemma 5.3. For $0 \neq c \in A$, multiplication by c on a $\mathbb{Z}$ -basis of A is a full-rank integer matrix with image cA . Its determinant's absolute value is the lattice index $[A : cA]$: integer elementary operations put the matrix in Smith diagonal form, where each quantity is the product of the positive diagonal entries. The rational determinant is the field norm by definition. For $c = a/b \in K^\times$, multiplicativity gives the quotient of these two identities.

Theorem 5.10 proves the termwise equality of the sums. Unique ideal factorization gives the Euler product as a sum of nonnegative terms for real $s > 1$. To prove convergence, let $n = [K : \mathbb{Q}]$. Primes above a rational prime ℓ have residue sizes $\ell^{f_{\mathfrak{p}}}$ with $f_{\mathfrak{p}} \geq 1$. There are at most n such primes: the Chinese remainder surjection

$$A/\ell A \longrightarrow \prod_{\mathfrak{p}|\ell} A/\mathfrak{p}$$

gives $\sum_{\mathfrak{p}|\ell} f_{\mathfrak{p}} \leq n$. Every nonzero prime lies over a rational prime, since its residue field is finite. Hence for real $\sigma > 1$,

$$\prod_{\mathfrak{p}} (1 - |A/\mathfrak{p}|^{-\sigma})^{-1} \leq \prod_{\ell \text{ prime}} (1 - \ell^{-\sigma})^{-n} = \left(\sum_{m \geq 1} m^{-\sigma} \right)^n < \infty.$$

The last sum converges by the integral test; its product expression follows from integer unique factorization and increasing finite products. Thus all rearrangements are valid for complex s with real part greater than one. The final ideal sum is the defining Dedekind zeta series, proving (25). $\square$

The quotient retaining a separate absent element has its own exact comparison map. For $0 \neq J \subseteq A$, set

$$x \equiv_J^{\text{sp}} y \iff (x = y = \tau) \text{ or } (x, y \in A, x - y \in J).$$

Its quotient is $G(A/J)$, of size $|A/J| + 1$, with zero class T , rather than I_J . The natural morphism

$$G(A/J) \longrightarrow A/J, \quad \tau \mapsto 0, \quad [a] \mapsto [a],$$

identifies exactly its two zero-amplitude elements and has kernel $\{\tau, 0_{A/J}\}$. This proves the exact relation between the two quotient constructions. Their weights obey

$$(N(J) + 1)(N(H) + 1) - (N(JH) + 1) = N(J) + N(H).$$

The support-preserving cardinality therefore has its stated counting rule, while the Bourne norm retains the multiplicative formula (25).

6 The full adelic valuation space and its derived geometry

6.1 Objects and source scope

Throughout, a ring is commutative and has an identity. For a nonzero ring A , define the semiring

$$G(A) = \{\tau\} \sqcup A.$$

The symbol τ is the additive identity and multiplicative absorber. Addition and multiplication of two supported elements of A are their original ring operations, with their results remaining in the supported copy of A . We write e_A for its element 0_A . Thus

$$\tau \neq e_A, \quad e_A + e_A = e_A, \quad a + (-a) = e_A, \quad e_A a = e_A \quad (a \in A). \quad (\text{AV1.1})$$

The last formula does not apply to $a = \tau$: $e_A \tau = \tau$. All semiring homomorphisms below preserve the additive identity and the multiplicative identity. Associativity and commutativity follow on supported inputs from those of A and otherwise from the rules for τ . Distributivity follows from that of A when all three inputs are supported; if the multiplier is τ , both sides are τ ; if either summand is τ , the identity reduces to $ab = ab + \tau$. These checks prove that $G(A)$ is a semiring. A ring homomorphism $f : A \rightarrow B$ gives a semiring homomorphism $G(f)$ by $\tau \mapsto \tau$ and $a \mapsto f(a)$, including $e_A \mapsto e_B$.

The human source used here is Antoine Sédillot, *Topological adelic curves: Zariski-Riemann spaces, algebraic coverings, Harder-Narasimhan filtrations and heights*, arXiv:2503.20156v2. The exact author file is `topological_adelic_curves_V2.tex`. Its definition of pseudo-absolute values is at source lines 623–629; integral structures at lines 674–689, label `subsub:prelim_integral_structures`; the algebraic Zariski–Riemann construction at lines 819–848, labels `subsub:Zariski-Riemann_space_algebraic` and `example:ZR_algebraic`; the general coherence and Tor assertion at lines 942–985, label `prop:coherence_algebraic_ZR_sheaf`. The spelling of the latter label is retained so the locator matches the source. The statements about split objects below are proved here. The general Tor assertion of the source is examined in Section 6.7; it is not used as a substitute for the proofs in this note.

6.2 The entire space of split pseudo-absolute values

Equip $[0, \infty]$ with its compact order topology. Extended addition has $r + \infty = \infty$. Extended multiplication is used only away from the two ordered pairs $(0, \infty)$ and $(\infty, 0)$.

Definition 6.1. For a field K , let $\mathcal{P}(\mathbf{G}(K))$ consist of maps $w : \mathbf{G}(K) \rightarrow [0, \infty]$ such that

$$w(\tau) = 0, \quad w(1) = 1, \quad w(a + b) \leq w(a) + w(b), \quad (\text{AV2.1})$$

and

$$w(ab) = w(a)w(b) \quad \text{if } \{w(a), w(b)\} \neq \{0, \infty\}. \quad (\text{AV2.2})$$

Give this set its topology of pointwise convergence. Let M_K be the space of pseudo-absolute values in Sédillot's definition: the same conditions on K , with $v(0_K) = 0$ and $v(1) = 1$.

Theorem 6.2 (Exact component decomposition). *Define*

$$\chi_K(\tau) = 0, \quad \chi_K(a) = 1 \ (a \in K), \quad j_K(v)(\tau) = 0, \quad j_K(v)(a) = v(a) \ (a \in K).$$

Then

$$\mathcal{P}(\mathbf{G}(K)) = \{\chi_K\} \sqcup j_K(M_K) \quad (\text{AV2.3})$$

is a homeomorphism of a disjoint union of two compact Hausdorff spaces. The two components are respectively $w(e_K) = 1$ and $w(e_K) = 0$ and are both open and closed. In particular, the full space is not identified with M_K by restriction to the supported coefficients.

Proof. Since $(-1)^2 = 1$, multiplicativity applies to the repeated value $w(-1)$ and gives $w(-1)^2 = 1$. Neither zero nor infinity has square one, so $w(-1) = 1$. The equality $1 + (-1) = e_K$ implies $w(e_K) \leq 2$. The equality $e_K^2 = e_K$ then gives $w(e_K)^2 = w(e_K)$, whence $w(e_K) \in \{0, 1\}$.

If $w(e_K) = 1$, then for every supported a the pair $(w(e_K), w(a))$ is allowed in (AV2.2); hence $1 = w(e_K a) = w(a)$. Thus $w = \chi_K$. This map satisfies the triangle inequality: the sum of two supported elements is supported, so its value one is at most two; a sum involving τ is immediate. It is multiplicative because a product is supported exactly when both factors are supported.

If $w(e_K) = 0$, its restriction to K satisfies precisely Sédillot's three axioms. Conversely, any such v extends by $v(\tau) = 0$: the new triangle inequalities are equalities and each newly required product identity has value zero on both sides. This proves the bijection. Both restriction and extension are continuous in the product topologies. The continuous evaluation at e_K takes only the values zero and one, which have disjoint open neighbourhoods. It therefore makes both components open and closed.

For completeness, M_K is a closed subset of $[0, \infty]^K$. The two value constraints and every triangle inequality are closed conditions. For a convergent net satisfying multiplicativity, if its limiting pair is allowed, it has a neighbourhood avoiding the two forbidden pairs; extended multiplication is continuous at that pair, so the limiting identity holds. Thus its multiplicativity condition is closed as well. The product is compact Hausdorff and the trivial absolute value supplies a point. This proves the compactness and Hausdorff assertions, and then those of (AV2.3). $\square$

Proposition 6.3 (Field maps preserve both components). *A field embedding $i : K \hookrightarrow L$ induces a continuous restriction $\mathcal{P}(\mathbf{G}(L)) \rightarrow \mathcal{P}(\mathbf{G}(K))$. It sends χ_L to χ_K and $j_L(v)$ to $j_K(v|_K)$. Every extension of χ_K is χ_L ; extensions of $j_K(v)$ correspond exactly to extensions of v to L .*

Proof. Precomposition preserves every displayed axiom and is continuous in the pointwise topology. It preserves e , so preserves its value zero or one. Theorem 6.2 then gives all assertions. $\square$

6.3 Finiteness, residue, and the complete prime space

Proposition 6.4 (Finiteness and residue proved from the axioms). *For $v \in M_K$ put*

$$A_v = \{a \in K : v(a) < \infty\}, \quad \mathfrak{m}_v = \{a \in K : v(a) = 0\}, \quad \kappa_v = A_v / \mathfrak{m}_v.$$

Then A_v is a valuation ring of K , its unique maximal ideal is $\mathfrak{m}_v$, and v induces an absolute value $\bar{v}$ on κ_v . The finite locus of $j_K(v)$ is exactly $\mathbf{G}(A_v)$.

Proof. As above, $v(-1) = 1$, so $v(-a) = v(a)$. The triangle and product axioms make A_v a subring and $\mathfrak{m}_v$ an ideal in A_v . For $a \neq 0$, the identity $aa^{-1} = 1$ gives the following exhaustive possibilities. If $0 < v(a) < \infty$, then $v(a^{-1}) = v(a)^{-1}$: all pairs with the positive finite value $v(a)$ are allowed. If $v(a) = 0$, then $v(a^{-1})$ must be infinity, since every finite alternative would give product value zero. If $v(a) = \infty$, then $v(a^{-1})$ must be zero, since every nonzero alternative would give product value infinity. Consequently a or a^{-1} belongs to A_v , proving the valuation property, and its units are precisely the elements of positive finite value. This proves the assertion about the maximal ideal.

If $a - b \in \mathfrak{m}_v$, the triangle inequality gives $v(a) \leq v(b)$ and $v(b) \leq v(a)$; hence the value depends only on the residue class. It vanishes precisely on the zero class, is finite there, and satisfies the ordinary triangle and multiplication laws. Thus it is an absolute value of the residue field. The final assertion follows from the value zero of τ . $\square$

Proposition 6.5 (All ideals and all primes). *For every nonzero ring A , the ideals of $\mathbf{G}(A)$ are $\{\tau\}$ and*

$$I^+ = \{\tau\} \sqcup I \quad (I \text{ an ideal of } A). \quad (\text{AV3.1})$$

Its proper prime ideals are $\{\tau\}$ and $\mathfrak{p}^+$ for $\mathfrak{p} \in \text{Spec } A$. The map $\mathfrak{p} \mapsto \mathfrak{p}^+$ identifies $\text{Spec } A$ with the closed subset $V(e_A)$ of $\text{Spec } \mathbf{G}(A)$. The remaining point $\{\tau\}$ is open and is a generalization of every point. For supported a ,

$$D_{\mathbf{G}(A)}(a) = \{\{\tau\}\} \sqcup D_A(a), \quad D_{\mathbf{G}(A)}(e_A) = \{\{\tau\}\}. \quad (\text{AV3.2})$$

If A is a domain and $\mathfrak{p}$ a nonzero prime, its original chain is retained as

$$(\tau) = \{\tau\} \subsetneq (e_A) = \{\tau, e_A\} \subsetneq \mathfrak{p}^+. \quad (\text{AV3.3})$$

Proof. An ideal containing a supported element contains e_A by multiplication by e_A . Its supported elements are closed under addition, multiplication by every element of A , and negatives by multiplication by -1 ; hence they form a ring ideal. The converse follows directly from the operations. A product of two supported elements is always supported, even when its coefficient is zero, proving primality of $\{\tau\}$. Primality of $\mathfrak{p}^+$ is exactly primality of $\mathfrak{p}$ in A . The formulas for the basic opens follow by membership in these ideals; they prove the topological assertions and the strict chain. $\square$

Theorem 6.6 (Residue map with supported zero). *For $v \in M_K$ there is a commuting diagram of semirings*

$$\begin{array}{ccc} \mathbf{G}(A_v) & \xrightarrow{\mathbf{G}(\pi_v)} & \mathbf{G}(\kappa_v) \\ q_{A_v} \downarrow & & \downarrow q_{\kappa_v} \\ A_v & \xrightarrow{\pi_v} & \kappa_v. \end{array} \quad (\text{AV3.4})$$

Here $q_A(\tau) = q_A(e_A) = 0_A$ and $q_A(a) = a$ on supported coefficients. The top map is surjective and local: its inverse image of the unique maximal ideal $\{\tau, e_{\kappa_v}\}$ is $\mathfrak{m}_v^+$. Its fibre over the additive identity is only $\{\tau\}$. Its full congruence is

$$\tau \sim \tau, \quad a \sim b \iff a - b \in \mathfrak{m}_v \quad (a, b \in A_v), \quad (\text{AV3.5})$$

with no equivalence of τ and a supported element. On $\mathbf{G}(A_v)$ one has

$$j_K(v) = j_{\kappa_v}(\bar{v}) \circ \mathbf{G}(\pi_v). \quad (\text{AV3.6})$$

Proof. Every map in (AV3.4) respects its stated operations; on a supported input both composites are its residue coefficient and on τ both are zero. The assertions concerning fibres, congruence, and localness follow respectively from preservation of support, equality of residue coefficients, and Proposition 6.5. The last formula is Proposition 6.4 on supported elements and the value zero on τ . $\square$

The restriction to the finite locus is essential. If $0 \neq a \in \mathfrak{m}_v$, then $v(a) = 0$, $v(a^{-1}) = \infty$, and $v(aa^{-1}) = 1$. Thus $\{w = 0\}$ is generally not an ideal of $G(K)$; it becomes the prime ideal $\mathfrak{m}_v^+$ of $G(A_v)$. The precise defect is the set of forbidden products

$$\mathcal{D}_v = \{(a, b) \in K^2 : v(a) = 0, v(b) = \infty\} \cup \{(a, b) : v(a) = \infty, v(b) = 0\}.$$

The restriction map to $G(A_v)$ removes every such pair and gives the exact residue factorization (AV3.6); it does not identify the two zeros.

6.4 Integral structures and the support point

Let A be a Prüfer domain with fraction field K . Here this means that every $A_{\mathfrak{p}}$ is a valuation ring. Let $\|\cdot\|$ be a Banach ring norm as in Sédillot's integral-structure definition. A bounded multiplicative seminorm α on A means a finite map to $[0, \infty)$ with $\alpha(0) = 0$, $\alpha(1) = 1$, triangle inequality, multiplicativity, and $\alpha(a) \leq \|a\|$.

Proposition 6.7 (Exact extension from the integral spectrum). *Restriction is a bijection between such α and the subset*

$$M_{K,A} = \{v \in M_K : v(a) \leq \|a\| \text{ for all } a \in A\}. \quad (\text{AV4.1})$$

For $\mathfrak{p} = \ker \alpha$, the inverse map has finiteness ring $A_{\mathfrak{p}}$ and is given by

$$v_{\alpha}(x) = \begin{cases} \bar{\alpha}(x \bmod \mathfrak{p}A_{\mathfrak{p}}), & x \in A_{\mathfrak{p}}, \\ \infty, & x \notin A_{\mathfrak{p}}, \end{cases} \quad (\text{AV4.2})$$

where $\bar{\alpha}$ is the absolute value on $\kappa(\mathfrak{p}) = \text{Frac}(A/\mathfrak{p})$ defined by $\bar{\alpha}(\bar{a}/\bar{b}) = \alpha(a)/\alpha(b)$. The subset (AV4.1) is closed in M_K , and this bijection is a homeomorphism for the pointwise topologies.

Proof. Multiplicativity makes $\mathfrak{p}$ prime. The triangle inequality proves that α depends only on a residue class, as in Proposition 6.4, and its quotient norm extends uniquely by the indicated fraction formula. The residue field of $A_{\mathfrak{p}}$ is exactly $\text{Frac}(A/\mathfrak{p})$ because its nonzero residue denominators are the elements outside $\mathfrak{p}$.

For any valuation ring $V \subset K$ with maximal ideal $\mathfrak{m}$ and residue absolute value β , the map equal to β on residues of V and infinity outside V satisfies the pseudo-absolute-value axioms. To verify this, sums of two finite inputs use the residue triangle inequality and other sums have infinite right side. Products of finite inputs use the residue multiplication law. An element outside V has inverse in $\mathfrak{m}$. Multiplication by a unit of V preserves its being outside V , and the product of two outside elements is outside because its inverse is a product in $\mathfrak{m}$. The sole remaining case is one residue-zero factor and one infinite factor, for which no identity is imposed. This proves existence of (AV4.2).

For uniqueness let v restrict to α . Since every element of A is finite, $A \subset A_v$. All elements of $A \setminus \mathfrak{p}$ are units of A_v , giving $A_{\mathfrak{p}} \subset A_v$ with the displayed residue values. If $x \in A_v \setminus A_{\mathfrak{p}}$, the valuation property implies $x^{-1} \in \mathfrak{p}A_{\mathfrak{p}}$, whose value is zero. But then multiplicativity for the finite pair (x, x^{-1}) would give $v(1) = 0$, a contradiction. Thus the finiteness ring and every value are forced.

Each inequality in (AV4.1) is closed in the pointwise topology. This subset is compact by Theorem 6.2. Restriction is a continuous bijection into a Hausdorff product space, so its inverse is continuous: a closed subset of a compact domain has compact, hence closed, image in a Hausdorff space. $\square$

Corollary 6.8 (What a bound does to the split point). Extend the norm bound by $\|\tau\| = 0$ and $\|e_A\| = \|0_A\| = 0$. The maps in $\mathcal{P}(G(K))$ obeying this bound on $G(A)$ are precisely $j_K(M_{K,A})$. In particular χ_K does not obey it. The full split extension of the integral space is instead the explicit disjoint union $\{\chi_K\} \sqcup j_K(M_{K,A})$.

Proof. The character has $\chi_K(e_A) = 1 > 0$, whereas every lifted valuation has value zero there. On every other supported input the bound is exactly (AV4.1). Apply Theorem 6.2. $\square$

Proposition 6.9 (The center map includes the additional prime). *For the explicit space $\widetilde{M}_{K,A} = \{\chi_K\} \sqcup j_K(M_{K,A})$, there is a continuous map*

$$c_A : \widetilde{M}_{K,A} \longrightarrow \text{Spec } G(A), \quad c_A(\chi_K) = \{\tau\}, \quad c_A(j_K(v)) = (\mathfrak{m}_v \cap A)^+. \quad (\text{AV4.3})$$

At $j_K(v)$, writing $\mathfrak{p} = \mathfrak{m}_v \cap A$, the associated local semiring is $G(A)_{\mathfrak{p}^+} = G(A_{\mathfrak{p}}) = G(A_v)$. At the additional prime one instead has $G(A)_{\{\tau\}} \cong \mathbb{B}$, where $\mathbb{B} = \{0, 1\}$ has $1 + 1 = 1$.

Proof. The values of v are finite on A , so its zero locus there is a prime ideal, by multiplication of finite nonnegative values. For a supported $a \in A$, the inverse image of the basic open $D(a)$ is $\{\chi_K\} \sqcup \{j_K(v) : v(a) > 0\}$, which is open in the stated pointwise topology. The inverse image of $D(\tau)$ is empty. These opens prove continuity.

For the local semiring at $\mathfrak{p}^+$, denominators are exactly supported $s \in A \setminus \mathfrak{p}$. Send a supported fraction a/s to its supported coefficient in $A_{\mathfrak{p}}$ and the fraction τ/s to τ . The usual cross-multiplication criterion proves equality of two supported fractions exactly when their ring fractions agree; no supported fraction equals a τ fraction since multiplication of supported elements stays supported. This proves the first localization isomorphism. The equality with A_v was proved in Proposition 6.7.

At $\{\tau\}$ every supported element, including e_A , is inverted. An invertible multiplicative idempotent equals one, so $e_A = 1$ in that localization. Then $e_A a = e_A$ forces every supported a to equal one. The two classes remain distinct because the support map to $\mathbb{B}$ sends all supported denominators to a unit, and therefore factors through the localization while distinguishing their class from τ . Its addition is $1 + 1 = 1$ by the image of $e_A + e_A = e_A$. $\square$

6.5 The exact additive bridge and its limits

Theorem 6.10 (Universal ring recovery). *The map $q_A : G(A) \rightarrow A$ is universal among semiring homomorphisms from $G(A)$ to rings. It identifies exactly τ and e_A and keeps all other coefficients distinct. Consequently the category of $G(A)$ -semimodules whose additive monoid is an abelian group is canonically equivalent to the category of A -modules, by restriction along q_A . Under this equivalence tensor products, resolutions, and Tor are exactly the ordinary A -module constructions.*

Proof. If $f : G(A) \rightarrow R$ is a semiring homomorphism with R a ring, then $f(e_A) + f(e_A) = f(e_A)$ implies $f(e_A) = 0$. It already sends τ to zero. Its restriction to supported A preserves addition, multiplication, and identities, hence is a ring homomorphism. It is uniquely determined and factors f through q_A .

On an abelian-group semimodule M , the relation $e_A m + e_A m = e_A m$ similarly gives $e_A m = 0_M$ for all m . The action therefore factors through A , and all semimodule homomorphisms are precisely A -linear maps. Conversely every A -module gives such an action. These are inverse functors and preserve each addition and scalar action. A balanced biadditive map for one description is exactly one for the other, so the tensor universal properties agree. Free and projective objects, exact sequences, and tensor resolutions in this abelian category therefore agree, proving the Tor assertion. $\square$

There is also a way to retain the two zeros on modules. For an A -module M , put $G(M) = \{\tau_M\} \sqcup M$, with supported addition inherited from M , and the scalar action induced from A , with either τ input absorbing. The semimodule laws follow by the same cases as in (AV1.1).

Proposition 6.11 (Support-preserving module category). *A homomorphism $h : G(M) \rightarrow G(N)$ is either the constant map to τ_N , or is $G(f)$ for a unique A -linear map $f : M \rightarrow N$. The morphisms preserving supported elements form a category equivalent to A -modules. Its additive*

zero morphism is the supported map $m \mapsto 0_N$, extended by $\tau_M \mapsto \tau_N$; it differs from the constant- τ_N homomorphism in the ambient semimodule category. For every linear f ,

$$(\mathbf{G}(f))^{-1}(\{\tau_N\}) = \{\tau_M\}, \quad \mathbf{G}(\ker f) = \{\tau_M\} \sqcup \{m : f(m) = 0_N\}. \quad (\text{AV5.1})$$

The second expression, with its inclusion, is the kernel in the support-preserving category; the first is the categorical zero fibre in the ambient category.

Proof. The only additive idempotents in $\mathbf{G}(N)$ are τ_N and 0_N : a supported idempotent satisfies $n + n = n$ in a group, so $n = 0_N$. Hence $h(0_M)$ is one of these. If $h(0_M) = \tau_N$, then $h(m) + h(-m) = \tau_N$ forces both summands to be τ_N , since any supported summand makes a sum supported. This is the constant map. If $h(0_M) = 0_N$, the identity $0_M + m = m$ rules out $h(m) = \tau_N$, because $0_N + \tau_N = 0_N \neq \tau_N$. Thus all supported elements have supported images. The homomorphism laws become exactly linearity of $f = h|_M$. Conversely every linear map extends, proving the categorical claim. The two kernel statements now follow by direct inverse-image calculation, including the distinct images of 0_M and τ_M . $\square$

In particular, no assertion about derived functors in the entire nonadditive semimodule category follows merely by renaming ordinary module Tor . Theorem 6.10 and Proposition 6.11 specify two exact functors and the exact categories in which the asserted transfer holds.

6.6 Valuation stalks and the sheaf that retains support

Proposition 6.12 (Tor bound over a valuation ring). *Let V be a valuation ring with fraction field F . Every torsion-free V -module is flat. Every V -module has a flat resolution of length at most one. Every finitely presented V -module has a finite free resolution of length at most one. Consequently*

$$\text{Tor}_i^V(M, N) = 0 \quad (i \geq 2) \quad (\text{AV6.1})$$

for all V -modules M, N .

Proof. First a finitely generated torsion-free module is free. Given generators $m_1, \dots, m_n$, if they are dependent choose a nonzero relation $\sum a_i m_i = 0$. Among its nonzero coefficients choose a_j dividing all the others; this is possible because principal ideals in a valuation ring are totally ordered. Write $a_i = a_j b_i$, with $b_j = 1$. Torsion-freeness gives $\sum b_i m_i = 0$, expressing m_j in the other generators. Repeating decreases the number of generators until an independent generating set remains. This is a free basis.

An arbitrary torsion-free module is the filtered union of its finitely generated submodules, all free by the preceding argument. It is flat: tensor commutes with filtered colimits, and filtered colimits preserve injections and exactness of modules. Explicitly, a finite tensor relation already lies in some finitely generated submodule, and if its image vanishes in the colimit it vanishes at a later stage; exactness there gives exactness in the colimit. Thus no relation obstructing flatness appears in the union.

Choose a free surjection $P \rightarrow M$. Its kernel is a submodule of a free module over a domain and so is torsion-free, hence flat. This is the claimed length-one flat resolution. If M is finitely presented, choose P finite free and its kernel finitely generated; that kernel is then finite free by the first paragraph. Tensoring either resolution proves (AV6.1). $\square$

If $(Z, \mathcal{O}_Z)$ is a locally ringed space with valuation-ring stalks, then for all sheaves of modules $\mathcal{F}, \mathcal{H}$,

$$\mathcal{T}or_i^{\mathcal{O}_Z}(\mathcal{F}, \mathcal{H}) = 0 \quad (i \geq 2). \quad (\text{AV6.2})$$

Indeed stalks commute with tensor and homology; applying a flat resolution and taking stalks identifies the stalk of this sheaf with $\text{Tor}_i^{\mathcal{O}_{Z,z}}(\mathcal{F}_z, \mathcal{H}_z)$, which vanishes by Proposition 6.12. A sheaf

with all stalks zero is zero, since every section whose germs vanish vanishes on a neighbourhood of each point and hence globally by the sheaf axiom.

There is a small sheaf issue that matters for retaining support. The assignment $U \mapsto \mathbf{G}(\mathcal{O}_Z(U))$ need not itself be a sheaf: on two disjoint nonempty opens, a supported section on one and τ on the other must be able to glue. Define instead

$$\mathcal{S}_Z(U) = \{(C, s) : C \subset U \text{ is open and closed, } s \in \mathcal{O}_Z(C)\}. \quad (\text{AV6.3})$$

For $C = \emptyset$ the unique pair is τ . Addition has support $C \cup D$ and is the original sum on $C \cap D$, the first section on $C \setminus D$, and the second on $D \setminus C$. Multiplication has support $C \cap D$ and is the original product there. These pieces are open and closed in the union, so the sheaf axiom for $\mathcal{O}_Z$ defines each operation. In particular cancellation on $C \cap D$ leaves a supported zero.

Proposition 6.13 (Sheaf-level split and recovery). *The construction (AV6.3) is a sheaf of semirings with stalk $\mathcal{S}_{Z,z} = \mathbf{G}(\mathcal{O}_{Z,z})$. Extension by zero on $U \setminus C$ gives a morphism $q : \mathcal{S}_Z \rightarrow \mathcal{O}_Z$ which has the universal ring-recovery property stalkwise and as a sheaf. Sheaves of $\mathcal{S}_Z$ -semimodules with abelian-group addition are exactly sheaves of $\mathcal{O}_Z$ -modules. If the stalks of $\mathcal{O}_Z$ are valuation rings, their Tor sheaves in this abelian category obey (AV6.2).*

Proof. Compatible supports on an open cover glue to a subset whose membership is locally constant, hence to an open-and-closed subset. The supported sections then glue uniquely by the original sheaf axiom. All semiring identities hold locally according to whether each support is present, and there reduce to the verified identities of $\mathbf{G}$. A germ either has absent support on a neighbourhood or has full support on a neighbourhood. These alternatives give respectively τ and a unique supported coefficient germ, proving the stalk identification. The extension-by-zero map is compatible with restrictions and operations by the same local cases. Any map to a sheaf of rings factors uniquely on stalks by Theorem 6.10; equivalently, a local section splits on its support and its complement, where the factorizations are unique and so glue. The action on every additive-group semimodule likewise factors locally and uniquely, giving the stated equivalence. Its tensor products agree on stalks by Theorem 6.10, and hence as sheaves; the Tor conclusion follows from (AV6.2). $\square$

6.6.1 The full spectral extension, including the extra generic point

Let $(Z, \mathcal{O}_Z)$ be a nonempty irreducible locally ringed space whose local rings are nonzero. Form the space

$$Z^+ = Z \sqcup \{\eta_\tau\} \quad (\text{AV6.4})$$

whose opens are the empty set, $\{\eta_\tau\}$, and $U_W = \{\eta_\tau\} \cup W$ for nonempty open $W \subset Z$. These sets form a topology: arbitrary unions are of the same form, and a finite intersection is obtained by intersecting the corresponding opens in Z , with empty intersection there giving $\{\eta_\tau\}$. Define

$$\mathcal{O}_Z^+(U_W) = \mathbf{G}(\mathcal{O}_Z(W)), \quad \mathcal{O}_Z^+(\{\eta_\tau\}) = \mathbb{B}. \quad (\text{AV6.5})$$

Restriction between two opens with nonempty arithmetic parts is the split extension of the original restriction map. Restriction to $\{\eta_\tau\}$ sends τ to 0 and every supported section to 1. This is a semiring homomorphism to $\mathbb{B}$: supported sums and products remain supported, including supported zero coefficients. On the empty open take the one-element semiring. The restriction maps compose because ordinary restriction never changes the support tag.

Theorem 6.14 (Full spectral gluing and stalks). *The construction (AV6.5) is a sheaf of local semirings. Its stalks are*

$$\mathcal{O}_{Z, \eta_\tau}^+ = \mathbb{B}, \quad \mathcal{O}_{Z, z}^+ = \mathbf{G}(\mathcal{O}_{Z, z}) \quad (z \in Z). \quad (\text{AV6.6})$$

The inclusion $i : Z \hookrightarrow Z^+$ is a closed embedding of spaces. The inverse-image sheaf $i^{-1}\mathcal{O}_Z^+$ is the split sheaf $\mathcal{S}_Z$ of Proposition 6.13; its universal ring quotient is $\mathcal{O}_Z$. If $Z = \operatorname{Spec} A$ for a domain A , this construction is canonically the entire semiring spectrum $\operatorname{Spec} \mathbf{G}(A)$ with its localization sheaf, not only its closed arithmetic subset.

Proof. Every nonempty open of Z^+ contains η_τ . On any cover of such an open, compatible sections must therefore have the same image in the stalk $\mathbb{B}$ at η_τ . If that image is zero, each section on an open with arithmetic part is τ , and every section on the singleton is zero; they glue uniquely to the absent section. If that image is one, the sections on the arithmetic parts are supported. Their coefficients agree wherever the arithmetic parts overlap, because supported equality in $\mathbf{G}$ is coefficient equality. The original sheaf axiom glues these coefficients to one section on W , giving the unique supported glue on U_W . Sections on a singleton member of the cover must be one and impose exactly that support condition. This also proves the sheaf axiom for a cover of the singleton. In the irreducible case, nonempty arithmetic opens intersect, so the coefficient compatibility can equivalently be checked directly on their nonempty intersections.

The singleton is a smallest neighbourhood of η_τ , yielding its stalk. Neighbourhoods of an arithmetic point z are exactly U_W for neighbourhoods W of z . In their colimit no restriction identifies τ with a supported section, and supported germs are precisely $\mathcal{O}_{Z,z}$. This proves (AV6.6). If $(R, \mathfrak{m})$ is a local ring, Proposition 6.5 shows that $\mathbf{G}(R)$ has the unique maximal ideal $\mathfrak{m}^+$. The Boolean stalk has the unique maximal ideal $\{0\}$. Thus all stalks are local semirings.

The complement of $i(Z)$ is the open singleton. Its relative topology is the original topology of Z . The presheaf computing the inverse image of $\mathcal{O}_Z^+$ on an open $W \subset Z$ has the restriction of $\mathbf{G}(\mathcal{O}_Z(W))$ as a cofinal representative, because U_W is the smallest open of Z^+ containing $i(W)$. Sheafifying this presheaf allows the support tag to vary locally, giving exactly the open-and-closed supported subsets and coefficient sections of (AV6.3). The construction in Proposition 6.13 proves the ring quotient assertion.

For a domain A , Proposition 6.5 identifies the points and opens with Z^+ , taking η_τ to the prime $\{\tau\}$. For a nonzero supported a , the basic open $D_{\mathbf{G}(A)}(a)$ has arithmetic part $D_A(a)$, and its local-fraction semiring is $\mathbf{G}(A)_a \cong \mathbf{G}(A_a)$, by the same cross-multiplication calculation as Proposition 6.9. This agrees with $\mathbf{G}(\mathcal{O}_Z(D_A(a)))$. The remaining nonempty basic open $D_{\mathbf{G}(A)}(e_A) = \{\eta_\tau\}$ has localization $\mathbb{B}$ by that proposition. All restriction maps agree: between arithmetic charts they are localization maps, and to the singleton they retain exactly the support. Since these form a basis, gluing identifies their localization sheaf with (AV6.5). $\square$

Proposition 6.15 (Every local morphism extends with its support point). *Let $f : (Z, \mathcal{O}_Z) \rightarrow (X, \mathcal{O}_X)$ be a morphism of nonempty irreducible locally ringed spaces with nonzero local rings. Then*

$$f^+ : Z^+ \longrightarrow X^+, \quad f^+(z) = f(z), \quad f^+(\eta_{\tau,Z}) = \eta_{\tau,X} \quad (\text{AV6.7})$$

is canonically a morphism of locally semiringed spaces. At an arithmetic point its stalk map is $\mathbf{G}(f_z^\#)$, and at the extra point it is the identity of $\mathbb{B}$. The construction respects identities and composition, and its restriction followed by coefficient recovery is the original morphism f .

Proof. The inverse image of U_W is $\{\eta_{\tau,Z}\} \cup f^{-1}(W)$, an open of Z^+ even when the arithmetic inverse image is empty. If it is nonempty, use $\mathbf{G}(f^\#) : \mathbf{G}(\mathcal{O}_X(W)) \rightarrow \mathbf{G}(\mathcal{O}_Z(f^{-1}W))$ on sections. If it is empty, use the support homomorphism to $\mathbb{B}$. On the singleton use its identity. These homomorphisms respect restrictions, including restrictions that make the arithmetic part empty, because coefficient maps preserve the support tag. They therefore define a sheaf morphism and give the stated stalk maps. A local ring map has inverse image of the maximal ideal equal to the maximal ideal. Formula (AV3.1) shows the same for its split extension. The Boolean identity is local as well. Composition and recovery hold on every supported coefficient and on τ , proving every remaining assertion. $\square$

6.7 The exact scope of the Tor assertion and its refinement

6.7.1 The whole-scheme submodel limit does not have the claimed bound

In the supplied source a projective submodel of a projective K -scheme X over $k \subset K$ is a factorization $X \rightarrow \mathcal{X} \rightarrow \operatorname{Spec} k$ with schematically dominant first map and projective second map. Its Zariski–Riemann space is the inverse limit of these submodels.

Proposition 6.16 (An explicit counterexample to the unrestricted claim). *If $k = K$ and X is a projective K -scheme, this submodel limit is X . For $X = \mathbf{P}_K^2$, a coherent point sheaf has nonzero Tor_2 with itself. Thus the source’s assertion of Tor-dimension at most one for all coherent sheaves on its general $\operatorname{ZR}(X/k)$, with only its stated hypotheses, fails in this case.*

Proof. The identity factorization makes X itself a submodel, and the defining map $X \rightarrow \mathcal{X}$ is the unique arrow from this object to each submodel compatible with the factorization. For any compatible cone, its map to the X component determines every other component. Therefore the inverse limit is X , also as a locally ringed space.

Take a K -rational point with affine coordinates $x = y = 0$. Its local ring is $R = K[x, y]_{(x, y)}$, with residue field K . The sequence

$$0 \longrightarrow R \xrightarrow{c \mapsto (-yc, xc)} R^2 \xrightarrow{(a, b) \mapsto xa + yb} R \longrightarrow K \longrightarrow 0 \quad (\text{AV7.1})$$

is exact. The left map is injective because R is a domain. If $xa + yb = 0$, reduction modulo x in the domain $K[y]_{(y)}$ gives $b \equiv 0 \pmod{x}$, so $b = xc$, and then $a = -yc$. The image at R is the ideal (x, y) , giving the last exactness. Tensoring with K makes both differentials zero, so $\operatorname{Tor}_2^R(K, K) = K \neq 0$. The coherent skyscraper sheaf at the point therefore has a nonzero second Tor stalk with itself. All the stated hypotheses hold: K is stably coherent, and $\mathbf{P}_K^2$ is projective and integral. $\square$

6.7.2 The valuation refinement is a new space with an exact map

Let X be an integral separated scheme of finite type over a field and $F = K(X)$ its actual function field. Consider integral schemes Y with a projective birational morphism $Y \rightarrow X$, and compatible morphisms over X . Their category is cofiltered: the closure of the common generic point in $Y_1 \times_X Y_2$ is another such integral model dominating both. Define

$$\mathfrak{Z}_X = \varprojlim_{Y \rightarrow X} Y, \quad \mathcal{O}_{\mathfrak{Z}_X} = \varprojlim_{Y \rightarrow X} p_Y^{-1} \mathcal{O}_Y. \quad (\text{AV7.2})$$

The limit topology and the indicated sheaf are used. Stalks are the filtered colimits of the local rings at compatible centers; these colimits are local because all transition maps between these local rings are local. The projection to the identity model is a morphism of locally ringed spaces $\rho : \mathfrak{Z}_X \rightarrow X$.

Theorem 6.17 (Valuation refinement and the proved amplitude bound). *Points of $\mathfrak{Z}_X$ are in bijection with pairs (V, c) where V is a valuation ring of F and $c : \operatorname{Spec} V \rightarrow X$ extends the generic point $\operatorname{Spec} F \rightarrow X$. At the corresponding point z , the stalk is exactly V . If $x = c(\mathfrak{m}_V)$, the stalk map of ρ is the inclusion $\mathcal{O}_{X, x} \hookrightarrow V$. All sheaves of modules on $\mathfrak{Z}_X$ satisfy*

$$\operatorname{Tor}_i^{\mathcal{O}_{\mathfrak{Z}_X}}(\mathcal{F}, \mathcal{H}) = 0 \quad (i \geq 2). \quad (\text{AV7.3})$$

The split sheaf from (AV6.3) has stalks $\mathbf{G}(V)$ and maps to this ringed space by the exact coefficient quotient; its abelian-group module category has the same bound.

Proof. Given (V, c) and a model $Y \rightarrow X$, its generic map lifts uniquely to $\text{Spec } V \rightarrow Y$. This is the valuative property of a projective map; it can be checked here using homogeneous coordinates: write the generic lift as $[a_0 : \cdots : a_n]$ in a projective embedding of Y , divide by an a_j whose value is least among these finitely many nonzero coordinates, and obtain coordinates in V with one equal to one. The homogeneous equations hold over V because they hold over F . Uniqueness follows from separatedness, since two maps agreeing at the generic point agree on the closed equations of the diagonal. These unique lifts give compatible centers z_Y and therefore z . Each local ring $\mathcal{O}_{Y, z_Y}$ embeds locally into V .

Conversely, a compatible family gives a union $R_z = \varinjlim_Y \mathcal{O}_{Y, z_Y} \subset F$. For each $f \in F^*$, take the closure of the graph of its rational map to $\mathbf{P}^1$ inside $X \times \mathbf{P}^1$. It is an integral projective birational model. Its center lies in at least one of the two standard affine charts, so either f or f^{-1} belongs to its local ring and hence to R_z . Therefore R_z is a valuation ring. Each center map is local, and the inclusion of the X local ring gives the required center on X .

For the family constructed from V , the union of center rings equals V , rather than just being contained in it. If $f \in V$, the graph model just used has center in its finite-coordinate chart: its unique V -lift has coordinate f , which is defined at the closed point of $\text{Spec } V$. Thus f lies in that center's local ring. Every element of V is obtained. The construction is inverse in the other direction because the inclusions of center rings into the union give the unique lifts to each separated model. This proves both the bijection and the stalk and morphism formulas. The Tor assertions follow from (AV6.2) and Proposition 6.13. $\square$

The distinction between (AV7.2) and the source's whole-scheme submodels is an explicit morphism ρ and a change of modification problem: the former models the generic map $\text{Spec } F \rightarrow X$, whereas the latter requires the whole map $X \rightarrow \mathcal{X}$ to survive. For $X = \text{Spec } K$ over a subring, valuation centers recover the valuation stalk setting recalled in Sédillot's `example:ZR_algebraic`, source lines 831–835. The bound is therefore proved on the valuation objects needed by that field-level route without asserting it on every whole-scheme submodel limit.

Corollary 6.18 (The entire spectrum survives valuation refinement). The space $\mathfrak{Z}_X$ in Theorem 6.17 is irreducible with generic point represented by the valuation ring F . The refinement induces the exact morphism

$$\rho^+ : \mathfrak{Z}_X^+ \longrightarrow X^+. \quad (\text{AV7.3a})$$

Its stalk map at a valuation center is $\mathbf{G}(\mathcal{O}_{X, x}) \rightarrow \mathbf{G}(V)$, induced by the original inclusion; its stalk map at the new generic point is the Boolean identity. The ordinary generic point ξ remains in the closed arithmetic stratum and remains distinct from η_τ . At ξ , the stalk is $\mathbf{G}(F)$; the two primes that record the two generic points are $\{\tau\}$ and $\{\tau, e_F\}$. The latter is the supported-zero prime of the ordinary generic point.

Proof. The compatible generic points of all integral birational models form a point of $\mathfrak{Z}_X$, with stalk F by Theorem 6.17. Every nonempty basic open in the inverse-limit topology is a finite intersection of inverse images of opens on models. A common dominating model replaces these finitely many conditions by one open condition. If the intersection is nonempty, that open is nonempty in an integral scheme, so contains its generic point. Thus every nonempty open contains the compatible generic point, proving irreducibility. Proposition 6.15 applies to ρ and gives the stated map and stalk formulas. The arithmetic stratum is closed and omits the new point by Theorem 6.14, so the two are distinct. At ξ , Proposition 6.5 for the field F gives exactly the two listed primes; their localizations are respectively $\mathbb{B}$ and $\mathbf{G}(F)$, as proved in Proposition 6.9. This identifies their roles without identifying e_F and τ . $\square$

6.7.3 Where the derived contribution goes under a blowup

Theorem 6.19 (Complete blowup-chart calculation). Let $A = K[x, y]$, $M = A/(x, y)$, and let

$$B = K[u, t], \quad A \longrightarrow B, \quad x \longmapsto u, \quad y \longmapsto ut \quad (\text{AV7.4})$$

be the first chart of the blowup of (x, y) . Ordinary pullback is $B \otimes_A M = B/(u)$, whose projective dimension over B is one. Derived pullback is represented, in homological degrees 2, 1, 0, by

$$0 \longrightarrow B \xrightarrow{c \mapsto (-utc, uc)} B^2 \xrightarrow{(a, b) \mapsto u(a+tb)} B \longrightarrow 0. \quad (\text{AV7.5})$$

Its homology is

$$H_0 = B/(u), \quad H_1 = (B/(u)) [(-t, 1)], \quad H_2 = 0. \quad (\text{AV7.6})$$

Consequently the kernel and cokernel relevant to the middle map are

$$\ker d_1 = B(-t, 1), \quad \text{im } d_2 = uB(-t, 1), \quad \text{im } d_1 = uB, \quad \text{coker } d_1 = B/(u). \quad (\text{AV7.7})$$

On the whole blowup with exceptional curve $E \cong \mathbf{P}_K^1$, the two homology sheaves are $\mathcal{O}_E$ in degree zero and $\mathcal{O}_E(-1)$ in degree one. These are the complete contributions; no degree-two term remains on either chart.

Proof. The polynomial-ring version of the exact Koszul sequence (AV7.1) has the same proof: reducing modulo x in $K[y]$ shows that a relation $xa + yb = 0$ has $(a, b) = (-yc, xc)$. It is a free A -resolution of M . Tensoring it with the specified A -algebra B gives (AV7.5), which therefore computes derived pullback.

Because u is not a zero divisor, d_2 is injective. The equation $u(a + tb) = 0$ gives $a = -tb$, so $\ker d_1 = B(-t, 1)$. The image of d_2 is precisely u times this free rank-one kernel. The image of d_1 is uB because a can be arbitrary. This proves all four formulas in (AV7.7) and their homology quotients. The sequence $0 \rightarrow B \xrightarrow{u} B \rightarrow B/(u) \rightarrow 0$ is a length-one free resolution. Its cokernel is not projective: after tensoring this resolution with $B/(u)$ the degree-one homology is $B/(u) \neq 0$, whereas a projective module has vanishing positive Tor.

On the second chart $B' = K[v, s]$ with $y = v, x = vs$, the same calculation gives a degree-one generator $(1, -s)$ and quotient $B'/(v)$. On the overlap $s = t^{-1}$ and $v = ut$, and the first generator satisfies

$$(-t, 1) = -t(1, -s). \quad (\text{AV7.8})$$

The tautological line on $E = \mathbf{P}^1$ has local frames $(1, t)$ and $(s, 1)$ with first frame t times the second. Multiplying one frame by -1 changes no line bundle. Thus (AV7.8) is exactly the transition of $\mathcal{O}_E(-1)$. Degree-zero homology is the structure sheaf of the exceptional divisor on both charts, and degree-two homology is zero on this open cover, proving the global assertions. $\square$

$$\begin{array}{ccc} 3\mathbf{A}_K^2 & \xrightarrow{\quad} & \text{Bl}_{(x,y)} \mathbf{A}_K^2 \\ & \searrow \rho & \downarrow (u,t) \mapsto (u,ut) \\ & & \mathbf{A}_K^2 \end{array}$$

Figure 3: Exact maps for the valuation refinement and one blowup chart. The exceptional equation is $u = 0$. The supported point module pulls back to $B/(u)$, while its derived pullback also retains $(B/(u)) [(-t, 1)]$ in degree one. Theorems 6.17 and 6.19 prove every map and homology label. The source comparison is Sédillot, arXiv:2503.20156v2, author-TeX labels lemma:Tor-dimension-pullback and lemma:Tor-dimension-flattening, lines 950–982.

For clarity, even a valuation target keeps the derived defect rather than erasing it. Let $A \rightarrow V$ be a map to a valuation ring contained in $K(x, y)$, with $u = x \neq 0, t = y/x \in V$, and u noninvertible. The same calculation over the domain V gives

$$V \otimes_A M = V/(u), \quad \text{Tor}_1^A(V, M) = V/(u), \quad \text{Tor}_i^A(V, M) = 0 \ (i \geq 2). \quad (\text{AV7.9})$$

The nonzero first group is precisely the defect of flatness on this module. It is compatible with (AV6.1), whose base ring is V : that latter formula bounds derived tensor over the valuation ring, while (AV7.9) computes base change from the surface ring A . The connecting operation is the explicit base-change complex (AV7.5), with B replaced by V . This is the exact relation between the two calculations, including its nonzero preserved term.

6.8 Adelic data transported without changing the product formula

Proposition 6.20 (Exact transport of a topological adelic curve). *Let $(K, \phi : \Omega \rightarrow M_K, \nu)$ satisfy Sédillot's definition: Ω is locally compact Hausdorff, ϕ is continuous, and for each $a \in K^*$, $\omega \mapsto \log \phi(\omega)(a)$ is integrable. Then $\tilde{\phi} = j_K \circ \phi$ is continuous into $\mathcal{P}(\mathcal{G}(K))$, and for every supported unit $a \in K^*$,*

$$\int_{\Omega} \log \tilde{\phi}(\omega)(a) d\nu(\omega) = \int_{\Omega} \log \phi(\omega)(a) d\nu(\omega). \quad (\text{AV8.1})$$

Thus the original product formula is preserved exactly. Conversely, a continuous map into the component $w(e_K) = 0$ with these integrability conditions uniquely recovers the original adelic data. An additional isolated point mapped to χ_K , with any finite nonnegative mass, contributes zero to every integral in (AV8.1).

Proof. Continuity and the inverse assertion are Theorem 6.2. The two integrands in (AV8.1) agree pointwise on their exact original domain K^* , proving integrability and equality without alteration of any values. At the added point, $\chi_K(a) = 1$ for every such a , so $\log \chi_K(a) = 0$. A topological disjoint union with one isolated point is locally compact Hausdorff, and adjoining its finite atomic measure has the asserted integral. No product formula for e_K or τ is claimed, since those elements are not units and their logarithms do not enter the original definition. $\square$

The final construction describes exactly what the support point can and cannot add at this stage: it retains a distinct character and an independent atomic mass, but the logarithms of supported units vanish there. No assertion of a Riemann-hypothesis criterion, a functional equation, or smoothness follows from this transport alone.

7 Prime faces and the complete Weil functional

7.1 The exact semilocalization associated with a monoid prime

For $\Sigma \subseteq \mathcal{P}$, define

$$T_{\Sigma} = \{n \in \mathbb{Z} \setminus \{0\} : p \nmid n \text{ for every } p \in \Sigma\}, \quad R_{\Sigma} = T_{\Sigma}^{-1}\mathbb{Z}.$$

Theorem 7.1. *The complement of the monoid prime J_{Σ} is T_{Σ} . Localization followed by the original supported addition gives exactly $G(R_{\Sigma})$. Its arithmetic spectrum consists of (0) and the primes $p \in \Sigma$, each with its original residue field. If $F \subset \mathcal{P}$ is finite, then*

$$R_{\mathcal{P} \setminus F} = \mathbb{Z}[p^{-1} : p \in F], \quad R_{\mathcal{P} \setminus F}^{\times} = \{\pm \prod_{p \in F} p^{n_p} : n_p \in \mathbb{Z}\} = \Gamma_F. \quad (26)$$

Proof. The complement statement follows directly from the definition of J_{Σ} . A supported fraction a/t obeys the original fraction operations and $\tau/t = \tau$, giving the localization isomorphism in Theorem 1.2. A ring prime survives localization exactly when it misses T_{Σ} : (p) misses it precisely when $p \in \Sigma$, and (0) always misses it. Its localized residue is still $\mathbb{F}_p$, since the denominators are nonzero modulo p . For the complement of F , the allowed denominators have no prime factors outside F , yielding the first equality in (26). A rational number and its inverse belong to that localization exactly when numerator and denominator have no prime factors outside F , proving the unit formula. $\square$

The arithmetic trace space associated with these units is

$$\mathbb{A}_F = \mathbb{R} \times \prod_{p \in F} \mathbb{Q}_p, \quad Y_F = \mathbb{A}_F / \Gamma_F.$$

This is the precise set-of-places and scaling-group construction used by Connes [5, labels YQS, GL1QS]. Equation (26) identifies its group with the original multiplicative-prime localization, without replacing that spectrum by the scalar semiring spectrum.

7.2 Every prime-power, Gamma and pole term

Use the multiplicative convention

$$\widehat{h}(s) = \int_0^\infty h(x) x^{-is} \frac{dx}{x}, \quad g^*(x) = \overline{g(x^{-1})}, \quad (g_1 * g_2)(x) = \int_0^\infty g_1(y) g_2(x/y) \frac{dy}{y}.$$

For $h \in C_c^\infty(\mathbb{R}_+^*)$, set

$$W_p(h) = (\log p) \sum_{m \geq 1} p^{-m/2} (h(p^m) + h(p^{-m})), \quad (27)$$

$$\begin{aligned} W_\infty(h) &= (\log(4\pi) + \gamma_E) h(1) \\ &\quad + \int_1^\infty \left(h(x) + h(x^{-1}) - 2x^{-1/2} h(1) \right) \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}, \end{aligned} \quad (28)$$

$$D_F(h) = \widehat{h}(i/2) + \widehat{h}(-i/2) - W_\infty(h) - \sum_{p \in F} W_p(h). \quad (29)$$

The cancellation in the integrand at $x = 1$ is retained; beyond the support of h , its subtraction term is also retained. These are the explicit-formula conventions in [5, wp, arch, sectweilpos]. In this notation W_∞ is that source's $W_{\mathbb{R}}$, without changing its sign. The full Weil functional is $D_{\mathcal{P}}$, whose prime sum is finite for each such h .

Theorem 7.2 (The entire prime-face valuation). *For finite sets of rational primes F, E ,*

$$D_{F \cup E}(h) + D_{F \cap E}(h) = D_F(h) + D_E(h), \quad (30)$$

$$D_{F \cup \{p\}}(h) - D_F(h) = -W_p(h) \quad (p \notin F). \quad (31)$$

*If $\text{supp } g \subseteq [\lambda^{-1}, \lambda]$ and $h = g * g^*$, every prime term with $p > \lambda^2$ vanishes. Thus for $F_\lambda = \{p : p \leq \lambda^2\}$,*

$$D_{F_\lambda}(g * g^*) = D_{\mathcal{P}}(g * g^*). \quad (32)$$

The full Weil positivity criterion therefore concerns these compatible complete finite-place returns. Equations (30)–(31) specify exactly what changes when a prime face is omitted.

Proof. In (29), both pole terms and the archimedean term are independent of F . Each prime occurs with the indicator $1_F(p)$, and $1_{F \cup E} + 1_{F \cap E} = 1_F + 1_E$, proving the first formula. Subtraction gives the second with its minus sign. In a nonzero convolution integrand for $h(x)$, both y and x/y belong to $[\lambda^{-1}, \lambda]$, since the involution preserves that interval. Hence $x \in [\lambda^{-2}, \lambda^2]$. For $p > \lambda^2$ and $m \geq 1$, neither p^m nor p^{-m} belongs to that interval, so every summand in (27) is zero. This proves (32). The classical equivalence of positivity of this full functional and RH is Weil's criterion as stated in the cited source; the finite-face identities themselves have been proved here without assuming RH. $\square$

Theorem 7.3 (Prime-face increments have both signs, even with both poles killed). *Fix a rational prime p , put $a = \log p$, and choose a nonzero real $\phi \in C_c^\infty((-\eta, \eta))$ with $0 < 4\eta < a$. Set*

$$\psi = \phi'' - \frac{1}{4}\phi, \quad f_\theta(u) = \psi(u) + e^{i\theta}\psi(u - a), \quad g_\theta(x) = f_\theta(\log x).$$

Then $\widehat{g}_\theta(i/2) = \widehat{g}_\theta(-i/2) = 0$, and for $h_\theta = g_\theta * g_\theta^*$,

$$W_p(h_\theta) = 2(\log p)p^{-1/2} \cos \theta \|\psi\|_{L^2(\mathbb{R})}^2. \quad (33)$$

In particular the change (31) is strictly negative at $\theta = 0$ and strictly positive at $\theta = \pi$. Retaining residue norm p does not make prime-face removal a sign-neutral operation.

Proof. If $\psi = 0$, then $\phi'' = \phi/4$. Multiplying by ϕ and integrating gives $-\int |\phi'|^2 = \frac{1}{4} \int |\phi|^2$, forcing $\phi = 0$, a contradiction. Thus $\|\psi\|^2 > 0$. Integration by parts twice gives $\int \psi(u)e^{\pm u/2} du = 0$. Translation multiplies these integrals by $e^{\pm a/2}$, so both Mellin values remain zero for g_θ .

In logarithmic coordinates, $h_\theta(e^v) = \int f_\theta(u) \overline{f_\theta(u-v)} du$. At $v = a$, only the right bump of the first factor and the left bump of the second overlap; the other three products vanish since $2\eta < a$. Their integral is $e^{i\theta} \|\psi\|^2$. At $v = -a$ it is its conjugate. For $|m| \geq 2$, the shifts ma exceed the difference support $[-a - 2\eta, a + 2\eta]$, so the correlations are zero. Substitution in the full sum (27) proves (33). The pole conditions hold exactly, without deleting either pole from the definition of D_F . $\square$

This calculation is a signed arithmetic consequence of the prime-face map. It does not assign positivity to the complete functional: the other prime terms and the archimedean term in (29) remain part of that calculation. It supplies an explicit reason that no argument may infer the full Weil sign from an unchanged list of residue cardinalities.

8 The support defect of the Fourier map

Let K be $\mathbb{R}$ or $\mathbb{Q}_p$, with its self-dual additive Haar measure and the same basic character in both Fourier transforms. Write

$$(\mathcal{F}f)(\xi) = \int_K f(x) \psi_K(x\xi) dx, \quad \mathcal{F}^2 f(x) = f(-x).$$

Then $\mathcal{F}f(0) = \int_K f$ and $\int_K \mathcal{F}f = f(0)$ for $f \in \mathcal{S}(K)$. Give $G(K)$ the disjoint-union topology $K \sqcup \{\tau\}$ and define its test space as

$$\mathcal{S}_\tau(K) = \mathcal{S}(K) \oplus \mathbb{C}\delta_\tau.$$

The pullback of the ring reflection and the support-preserving Fourier operator are respectively

$$Jf = (f, f(0)), \quad \mathcal{F}_\tau(f, c) = (\mathcal{F}f, c).$$

These are explicitly defined operators on test spaces; the scalar on the absent point is not confused with the value at supported zero.

Theorem 8.1 (Exact Fourier defect and its stable kernel). *For every original test function f ,*

$$(\mathcal{F}_\tau J - J\mathcal{F})f = \left(0, f(0) - \int_K f(x) dx\right). \quad (34)$$

For the original unweighted scaling $U_a f(x) = f(a^{-1}x)$, its defect scalar is

$$(U_a f)(0) - \int_K U_a f = f(0) - |a|_K \int_K f. \quad (35)$$

The largest scaling-invariant vector subspace on which (34) vanishes is exactly

$$\mathcal{S}(K)_{00} = \{f : f(0) = 0, \int_K f = 0\}. \quad (36)$$

It is also Fourier invariant. The quotient by this subspace is the two-dimensional scaling representation $\mathbb{C} \oplus \mathbb{C}(| \cdot |_K)$, with quotient map $f \mapsto (f(0), \int_K f)$.

Proof. The first components of the two sides of the commutator agree, and their absent components are $f(0)$ and $\mathcal{F}f(0) = \int f$, proving (34). Change variables in the integral to prove (35). If a scaling-invariant subspace lies in the defect kernel, every f in it satisfies $f(0) = \int f$ and $f(0) = |a|_K \int f$ for every a . Choosing $|a|_K \neq 1$ forces both to vanish. Conversely both conditions are preserved by scaling, and Fourier exchanges the two, so (36) has all claimed invariances.

The map to $\mathbb{C}^2$ is onto. Over $\mathbb{R}$, a smooth bump supported away from zero has zero evaluation and may be chosen to have nonzero integral; an independent bump with value one at zero supplies the other coordinate. Over $\mathbb{Q}_p$, use indicators of a compact open ball away from zero and of a compact open ball containing zero. Subtract a suitable multiple of the first function from the second to set its integral to zero while leaving value one. Thus the kernel is exactly (36) and the quotient is two-dimensional. The transformation rule is $(b, m) \mapsto (b, |a|_K m)$, proving its asserted characters. $\square$

The same two maps $f(0)$ and $\int f$ and the characters 1 and $|a|$ occur in Connes's original adelic construction [4, Section III, (6)–(11)]. Here they have arisen by an explicit commutator with the split-zero ring reflection. This proves the connecting map; it does not identify a local construction with the complete global spectral quotient.

8.1 All mixed support faces

For a finite set V of local fields, use the algebraic tensor test space $\bigotimes_{v \in V} \mathcal{S}(K_v)$; every claim below is on this explicitly specified domain. Tensoring the disjoint-sum test spaces gives

$$\bigotimes_{v \in V} \mathcal{S}_\tau(K_v) = \bigoplus_{A \subseteq V} \bigotimes_{v \in A} \mathcal{S}(K_v).$$

The A component means supported coordinates on A and absent coordinates off A . The ring-reflection pullback J_V has component

$$(J_V f)_A = \text{ev}_{V \setminus A} f,$$

where evaluation sets precisely the missing coordinates equal to their supported zeros. Let $\mathcal{F}_V = \bigotimes_v \mathcal{F}_v$ and let $\mathcal{F}_\tau$ act on face A by $\mathcal{F}_A$.

Theorem 8.2 (Every mixed Fourier defect). *For every face $A \subseteq V$,*

$$((\mathcal{F}_\tau J_V - J_V \mathcal{F}_V) f)_A = \mathcal{F}_A \left(\text{ev}_{V \setminus A} f - \int_{\prod_{v \notin A} K_v} f \right). \quad (37)$$

The largest subspace invariant under independent local scalings on which all these defects vanish is

$$\bigcap_{v \in V} \ker(\text{ev}_v) \cap \ker(\text{int}_v). \quad (38)$$

All conditions here are function-valued in the coordinates not removed.

Proof. For a pure tensor the left term is Fourier transform in A of evaluations in its complement. In the right term, evaluation of Fourier transform in a missing coordinate is integration in that coordinate. These operations commute with Fourier transforms in A , proving the formula for pure tensors and then finite sums. No mixed face is omitted.

Taking $A = V \setminus \{v\}$ and using invertibility of $\mathcal{F}_A$ gives $\text{ev}_v f = \text{int}_v f$. Scaling only coordinate v leaves the former unchanged and multiplies the latter by $|a_v|_v$. Invariance for a scale of nonunit absolute value forces both functions to vanish. This proves necessity for each v . Conversely these separate vanishing conditions imply both terms in every proper-face difference vanish, and the full-face difference is zero identically. They are preserved by scaling. Fourier in coordinate v exchanges its evaluation and integration; Fourier in every other coordinate commutes with them. Thus the subspace is also Fourier invariant. $\square$

The independent-scaling domain in Theorem 8.2 is explicit. Restriction to a diagonal rational action has fewer independent scales and can retain additional defect vectors; that restriction must use its actual character relations. The full mixed-face formula (37) remains exact under such a restriction and provides those relations without collapsing the support lattice to one Boolean branch.

9 All adelic boundary faces and the two pole terms

We now restrict the preceding Fourier construction to the actual diagonal rational action. The calculation keeps every mixed face before taking a quotient. The quotient and its contracting homotopy will be computed, rather than inferred from a support count. The local zero and integral maps are those in Connes's original construction [4, Section III, (6)–(11)]. The product-formula defect is also the defect in Sédillot's definition of a topological adelic curve [8, `def:topological_adelic_curve`].

9.1 The exact finite boundary representation

Fix a nonempty finite set F of rational primes, write $d = |F|$, and put

$$V = \{\infty\} \sqcup F, \quad K_\infty = \mathbb{R}, \quad K_p = \mathbb{Q}_p, \quad \Gamma_F = \mathbb{Z}[F^{-1}]^\times = \{\pm \prod_{p \in F} p^{n_p} : n_p \in \mathbb{Z}\}. \quad (\text{AB1})$$

Use the unchanged absolute values $|p|_p = p^{-1}$ and $|a|_\infty = |a|$, and self-dual additive Haar measures. All tensor products of Schwartz spaces in this section are algebraic tensor products over $\mathbb{C}$. Set $\mathcal{T}_V = \bigotimes_{v \in V} \mathcal{S}(K_v)$ and

$$b_v : \mathcal{S}(K_v) \longrightarrow \mathbb{C}^2, \quad f \longmapsto (f(0), \int f), \quad B_V = \bigotimes_{v \in V} \mathbb{C}^2 = \bigoplus_{A \subseteq V} \mathbb{C} e_A. \quad (\text{AB2})$$

The symbol e_A is a basis vector, not the scalar $e = 0_{\mathbb{Z}}$. The coordinate of $b = \bigotimes b_v$ indexed by A integrates the coordinates in A and evaluates at their supported zeros the coordinates in $V \setminus A$:

$$b_A(f) = \text{int}_A \text{ev}_{V \setminus A} f. \quad (\text{AB3})$$

In terms of the preceding split test space it is exactly the integral of the A face of $J_V f$. Thus these scalar boundary coordinates are explicit receivers of the original function-valued faces, with no identification of an absent coordinate with a supported zero.

Theorem 9.1 (Complete diagonal character calculation). *The map b is onto and its kernel is*

$$\ker b = \sum_{v \in V} \left(\mathcal{S}(K_v)_{00} \otimes \bigotimes_{w \neq v} \mathcal{S}(K_w) \right), \quad \mathcal{S}(K_v)_{00} = \ker b_v. \quad (\text{AB4})$$

For the original unweighted action $(U_\gamma f)(x) = f(\gamma^{-1}x)$, $bU_\gamma = D_\gamma b$, where

$$D_\gamma e_A = \chi_A(\gamma) e_A, \quad \chi_A(\gamma) = \prod_{v \in A} |\gamma|_v = \prod_{p \in F} p^{n_p(\mathbf{1}_{\infty \in A} - \mathbf{1}_{p \in A})}. \quad (\text{AB5})$$

The characters are distinct except for $\chi_\emptyset = \chi_V = 1$. Every nonempty proper face has a nontrivial character. Fourier transform induces the map $Ce_A = e_{V \setminus A}$ and $CD_\gamma C = D_\gamma^{-1}$.

Proof. Each b_v is onto by the explicit two-bump construction in Theorem 8.1. Choose functions $h_{v,0}, h_{v,1}$ with $b_v h_{v,j}$ equal to the j th coordinate vector. Then $\mathcal{S}(K_v) = \ker b_v \oplus \text{span}(h_{v,0}, h_{v,1})$. Tensoring these direct sums proves surjectivity and (AB4): every summand having at least

one kernel factor maps to zero and the remaining summand maps isomorphically to B_V . This also proves that (AB4) is a sum, with its stated domain; it is not a replacement of the earlier intersection of function-valued conditions.

Evaluation at zero is fixed by U_γ and change of variables multiplies integration at v by $|\gamma|_v$. This proves the first part of (AB5). Since γ is an F -unit, its absolute value at every finite place outside F is one. Its exponents at infinity and at p give the last expression. If $\chi_A = \chi_B$, evaluating at each generator p gives

$$\mathbf{1}_{\infty \in A} - \mathbf{1}_{p \in A} = \mathbf{1}_{\infty \in B} - \mathbf{1}_{p \in B}. \quad (\text{AB6})$$

If the two infinite-place indicators agree, all finite indicators agree and $A = B$. If they differ, say by 1, all finite indicators must differ by 1, so $A = V$ and $B = \emptyset$. The other sign gives the reverse pair. This proves the entire character classification, including nontriviality. Fourier exchanges local evaluation and integration, hence complements A . Finally the product formula gives $\chi_A \chi_{V \setminus A} = \prod_{v \in V} |\gamma|_v = 1$, proving the operator identity. $\square$

The logarithm of (AB5) is the exact restricted product-formula defect

$$\log \chi_A(\gamma) = \sum_{v \in A} \log |\gamma|_v. \quad (\text{AB7})$$

The finite places and the Archimedean place are both present. In particular, the complement is the opposite defect, not a separately discarded boundary.

9.2 Original generators, projectors and higher cohomology

For $p \in F$ set $T_p = D_p$. With respect to the original face basis,

$$\det(I - tT_p) = (1 - pt)^{2^{d-1}} (1 - p^{-1}t)^{2^{d-1}} (1 - t)^{2^d}. \quad (\text{AB8})$$

Indeed, the indicators at ∞, p are respectively $(1, 0), (0, 1)$ or equal; the other $d - 1$ indicators are unrestricted. This proves every multiplicity. Define the following polynomial in the actual generators:

$$\Pi = \prod_{p \in F} \frac{(T_p - pI)(T_p - p^{-1}I)}{(1 - p)(1 - p^{-1})}. \quad (\text{AB9})$$

All denominators are nonzero. On a character line the p factor is one exactly when its p eigenvalue is one, and otherwise is zero. Theorem 9.1 therefore proves

$$\Pi B_V = \mathbb{C}e_\emptyset \oplus \mathbb{C}e_V = B_V^{\Gamma_F}, \quad B_V / \text{span}\{(D_\gamma - I)c\} = \Pi B_V. \quad (\text{AB10})$$

The quotient is identified by $[c] \mapsto \Pi c$, with inverse $c \mapsto [c]$ on ΠB_V . For each proper nonempty face, choosing p where its character differs from one expresses e_A as $(\chi_A(p) - 1)^{-1}(T_p - I)e_A$. This proves the kernel assertion in (AB10). In the Hermitian norm making the original face vectors orthonormal, Π is an orthogonal projector of norm one.

There is also an exact calculation in every cohomological degree. Order $F = \{p_1, \dots, p_d\}$ and let $\theta_1, \dots, \theta_d$ be the dual coordinate basis of $\mathbb{C}^d$. The Koszul cochain complex is

$$\mathcal{C}^j = B_V \otimes \Lambda^j \mathbb{C}^d, \quad \delta = \sum_{i=1}^d (T_{p_i} - I) \otimes (\theta_i \wedge -). \quad (\text{AB11})$$

It computes cohomology of $\mathbb{Z}^d$ on B_V : the group algebra is $\mathbb{C}[t_1^{\pm 1}, \dots, t_d^{\pm 1}]$, and its Koszul resolution at $t_i - 1$ is free and exact. To verify exactness, quotienting successively by the first j of these elements leaves a Laurent polynomial ring in $d - j$ variables, where the next $t_{j+1} - 1$ is not

a zero divisor. The one-variable resolution is injective multiplication followed by evaluation at 1; induction by its tensor mapping cone proves the asserted Koszul resolution. Applying module Hom gives (AB11).

On a proper face A , choose $i(A)$ with $a_{i(A),A} = \chi_A(p_{i(A)}) - 1 \neq 0$. Let ι_i be contraction by the vector dual to θ_i and set

$$h_A = a_{i(A),A}^{-1} \iota_{i(A)}. \quad (\text{AB12})$$

The exterior identity $\iota_i(\theta_j \wedge \eta) + \theta_j \wedge \iota_i \eta = \delta_{ij} \eta$ proves $h_A \delta + \delta h_A = I$ on that face. Its coefficient has modulus at most 2, since its denominator is either $p-1 \geq 1$ or $1-p^{-1} \geq 1/2$. The direct-sum homotopy, set equal to zero on the two trivial faces, has norm at most 2 and satisfies

$$h\delta + \delta h = I - \Pi \otimes I. \quad (\text{AB13})$$

On the remaining two faces $\delta = 0$. The sign subgroup $\{\pm 1\}$ acts trivially on the entire boundary space, and its invariants functor on complex representations is exact by the averaging projector $(I + D_{-1})/2$. Consequently

$$H^j(\Gamma_F, B_V) \simeq (\mathbb{C}e_\emptyset \oplus \mathbb{C}e_V) \otimes \Lambda^j \mathbb{C}^d, \quad 0 \leq j \leq d, \quad (\text{AB14})$$

with zero groups above d . Equations (AB9)–(AB14) compute how every mixed scalar boundary face reaches rational-invariant cohomology, including an explicit uniformly bounded homotopy for the other faces.

9.3 Pairing, Fourier duality and the actual pole receiver

Define the Hermitian form

$$\mathcal{B}(c, d) = \sum_{A \subseteq V} \overline{c_A} d_{V \setminus A}. \quad (\text{AB15})$$

Complementation is an involution without fixed subsets because V is nonempty. Each pair has matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so the signature is $(2^d, 2^d)$. Formula (AB5) and the product formula prove $\mathcal{B}(D_\gamma c, D_\gamma d) = \mathcal{B}(c, d)$. Moreover $\mathcal{B}(c, d) = \langle c, Cd \rangle$ in the original face norm. The invariant two-dimensional subspace is orthogonal for $\mathcal{B}$ to the sum of proper face lines, and its restriction has signature $(1, 1)$. This is an explicitly constructed boundary form; its positivity is not being identified with positivity of the full arithmetic spectral form.

For the unchanged multiplicative test space $\mathcal{D} = C_c^\infty(\mathbb{R}_{>0})$ use $d^\times x = dx/x$, involution $g^*(x) = \overline{g(x^{-1})}$, and multiplicative convolution. Put

$$\widehat{g}(s) = \int_0^\infty g(x) x^{-is} d^\times x, \quad P(g) = \widehat{g}(-i/2)e_\emptyset + \widehat{g}(i/2)e_V. \quad (\text{AB16})$$

Then the two pole terms in the symmetric explicit formula are exactly

$$\widehat{g * g^*}(i/2) + \widehat{g * g^*}(-i/2) = \mathcal{B}(Pg, Pg) = 2 \operatorname{Re} \left(\overline{\widehat{g}(-i/2)} \widehat{g}(i/2) \right). \quad (\text{AB17})$$

Indeed Fubini's theorem applies to compact supports, $\widehat{g * g^*}(s) = \widehat{g}(s) \overline{\widehat{g}(\overline{s})}$, and substitution of the two imaginary arguments gives (AB17). These are precisely the pole conventions used in [5, sectweilpos, wp].

The map P is onto. Choose a nonnegative nonzero $g_0 \in \mathcal{D}$. Its two moments $a_- = \widehat{g}_0(-i/2)$ and $a_+ = \widehat{g}_0(i/2)$ are positive. For $t > 0$, $t \neq 1$, let $g_t(x) = g_0(x/t)$. The two moment columns of g_0, g_t have determinant $a_- a_+ (t^{1/2} - t^{-1/2}) \neq 0$. This proves surjectivity and the exact sequence

$$0 \longrightarrow \{g : \widehat{g}(-i/2) = \widehat{g}(i/2) = 0\} \longrightarrow \mathcal{D} \xrightarrow{P} \Pi B_V \longrightarrow 0. \quad (\text{AB18})$$

The relation to the original unweighted adelic scaling can also be stated without changing any test function: define the explicitly different representation $M_t g(x) = t^{1/2} g(x/t)$ on $\mathcal{D}$. Direct

change of variables gives $P(M_t g) = \text{diag}(1, t)P(g)$. On ΠB_V , an idele of modulus t acts by exactly $\text{diag}(1, t)$. Thus P is an intertwiner for these specified representations. The $t^{1/2}$ is part of this stated intertwining map, not an omitted factor in U_γ or in (AB17).

Finally, the chosen $h_{v,j}$ define a concrete section $j_B : B_V \rightarrow \mathcal{T}_V$ by $j_B(e_A) = \bigotimes_{v \in A} h_{v,1} \otimes \bigotimes_{v \notin A} h_{v,0}$, with factors in their fixed order. Thus $b j_B = I$ and

$$b(j_B P(g)) = P(g), \quad \Pi b(j_B P(g)) = P(g). \quad (\text{AB19})$$

This supplies the exact lift from the pole receiver to original local test functions. The section depends on its explicitly chosen functions and is not asserted to be equivariant. The quotient, its characters and its pairing are independent of that choice.

$$\begin{array}{ccc}
 \begin{array}{c} \mathcal{T}_V \\ \text{original local test functions} \end{array} & \xrightarrow{b \text{ (AB3)}} & \begin{array}{c} B_V = \bigoplus_{A \subseteq V} \mathbb{C} e_A \\ \text{all } 2^{d+1} \text{ mixed faces} \end{array} \\
 \uparrow j_B P & & \downarrow \Pi \text{ (AB9)} \\
 \begin{array}{c} C_c^\infty(\mathbb{R}_{>0}) \\ \text{original multiplicative tests} \end{array} & \xrightarrow{P \text{ (AB16)}} & \begin{array}{c} \Pi B_V = \mathbb{C} e_\emptyset \oplus \mathbb{C} e_V \\ \text{two pole characters} \end{array}
 \end{array}$$

Figure 4: The mixed boundary calculation, with its actual maps. The square commutes by (AB19). Every proper face has the character (AB5) and the contracting homotopy (AB12); both surviving faces enter the pole pairing (AB17). This diagram concerns the computed boundary quotient, not an asserted replacement for the full zeta cohomology.

9.4 Passing to every prime with the original reference functions

The finite-place calculation has a specific global continuation. If $F \subset F'$ and $V' = \{\infty\} \sqcup F'$, the original restricted Schwartz tensor product uses $\mathbf{1}_{\mathbb{Z}_p}$ at $p \in F' \setminus F$. Its evaluation and integral are both one for the fixed Haar measure. The induced injective boundary map is therefore

$$j_{FF'} : B_V \longrightarrow B_{V'}, \quad c \longmapsto c \otimes \bigotimes_{p \in F' \setminus F} (e_0 + e_1). \quad (\text{AB20})$$

There is no omitted scalar in this formula. The ordering of the added tensor factors is the ordering of the places. These maps compose exactly and commute with the original Fourier involutions. They are equivariant for Γ_F , since its elements have absolute value one at each added prime. Let

$$B_{\text{all}} = \varinjlim_F B_{\{\infty\} \cup F}. \quad (\text{AB21})$$

The nonempty finite F form a cofinal set, so the preceding results apply. The whole group $\mathbb{Q}^\times$ acts on this direct limit: for any γ , first enlarge F to include its prime factors and then use (AB5). Equivariance at every further enlargement proves independence of choices and the group law. The restricted algebraic Schwartz space

$$\mathcal{T}_{\text{all}} = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_p^I (\mathcal{S}(\mathbb{Q}_p), \mathbf{1}_{\mathbb{Z}_p}) \quad (\text{AB22})$$

maps onto B_{all} by the b_v , because every finite-level map is onto and their transition maps agree by (AB20). This is the original reference-function convention of [4, Section III].

Theorem 9.2 (The global boundary quotient and its invariant vectors). *Let $\pi_F : B_V \rightarrow \mathbb{C}^2$ take the empty and full coordinates, in that order. The maps π_F induce a surjective $\mathbb{Q}^\times$ -invariant map $\pi_{\text{all}} : B_{\text{all}} \rightarrow \mathbb{C}^2$, and*

$$(B_{\text{all}})_{\mathbb{Q}^\times} \xrightarrow[\pi_{\text{all}}]{\sim} \mathbb{C}^2, \quad (B_{\text{all}})^{\mathbb{Q}^\times} = 0. \quad (\text{AB23})$$

The composite from $\mathcal{T}_{\text{all}}$ is exactly $f \mapsto (f(0), \int f)$ with its original adelic measure. The two-dimensional quotient consequently has no equivariant linear section into B_{all} .

Proof. Each endpoint coordinate of (AB20) is the same endpoint coordinate of c , so $\pi_{F'} j_{FF'} = \pi_F$. An element $\gamma \in \mathbb{Q}^\times$ can be tested in a stage containing all its prime factors. Both endpoint characters there equal one by the product formula, proving invariance. Surjectivity holds at any finite stage.

If $\pi_{\text{all}} c = 0$, represent c at a finite stage. Its two endpoint coordinates are zero, so it is a sum of proper-face basis vectors. For each such vector the explicit identity following (AB10) expresses it as $(D_p - I)$ of a vector at that same stage. Hence c is in the span of the coinvariant relations. Conversely invariance of π_{all} kills every such relation. This proves the quotient assertion.

For the invariant-vector assertion, let a nonzero c be represented in B_V . Choose a prime $p \notin F$ and view it in the next stage through $j_{F, F \cup \{p\}}$. For each nonzero coefficient c_A , the two extended faces $(A, 0)$ and $(A, 1)$ both have that same nonzero coefficient. Under p , their eigenvalues are respectively $p^{1_{\infty \in A}}$ and $p^{1_{\infty \in A} - 1}$. Exactly one equals one; the other is p or p^{-1} , and therefore differs from one. Different A yield different coordinate positions, so these nonzero changes cannot cancel. Thus $(D_p - I) j_{F, F \cup \{p\}} c \neq 0$. All later maps are injective, so this remains nonzero in the direct limit. No nonzero vector is invariant. If an equivariant section from the trivial representation $\mathbb{C}^2$ existed, its image would consist of invariant vectors, contradicting this result. Finally both evaluation and integral multiply by one on each reference factor, proving the formula on pure tensors and then on finite sums. $\square$

Both pairings can be tracked through these same maps. From (AB15), for $r = |F' \setminus F|$ one obtains

$$\mathcal{B}_{V'}(j_{FF'} c, j_{FF'} d) = 2^r \mathcal{B}_V(c, d), \quad \mathcal{B}_{V'}(\Pi_{F'} j_{FF'} c, \Pi_{F'} j_{FF'} d) = \mathcal{B}_V(\Pi_F c, \Pi_F d). \quad (\text{AB24})$$

For the first equality each added local factor $(1, 1)$ has value two under its complement pairing, so tensoring gives exactly 2^r . For the second equality both sides equal $\overline{c_\partial} d_V + \overline{c_V} d_\partial$. Consequently the pole pairing (AB17) is compatible at every stage, whereas the full mixed pairing has the explicitly calculated factor 2^r . No unrecorded change of measure or scalar has been made.

This establishes a global quotient from the actual local test spaces. It also identifies exactly what passage to all primes changes: the finite invariant endpoint vectors do not survive as global invariant vectors, while the two pole coordinates survive as a quotient. The kernel of that quotient, including its original arithmetic cohomology, is retained as a separate mathematical object throughout the construction.

10 The mixed-support extension and its arithmetic determinant

The full mixed boundary constructed above contains more information than its two-dimensional quotient. We now compute a nonzero extension class, its local differentials, and their exact contribution to the prime terms of Weil's formula. The primes and their absolute values are those of the original arithmetic stratum $V(e) \subset \text{Spec } G(\mathbb{Z})$. All tests, Haar measures and the prime terms W_p remain those of Section 7. The classical trace-formula source is Connes [4, Section III and Section V], and the exact explicit-formula conventions are those read in [5, **wp**, **arch**]. The extension and determinant calculations below are proved here.

10.1 A nonzero class carried by all the mixed faces

For a finite set F of primes, including $F = \emptyset$, write $B_F = \bigotimes_{v \in \{\infty\} \sqcup F} \mathbb{C}^2$ as in (AB2). Use exactly the bonding maps

$$j_{FF'}c = c \otimes \bigotimes_{p \in F' \setminus F} (e_0 + e_1). \quad (\text{ME1})$$

They come from the original reference function $\mathbf{1}_{\mathbb{Z}_p}$, whose value at zero and integral both equal one. Put $B = \varinjlim_F B_F$. For $\gamma \in \mathbb{Q}^\times$, choose F containing every prime factor of its numerator and denominator. The action on B_F is $D_\gamma e_A = \chi_A(\gamma) e_A$ with (AB5), and enlarging F makes this an action on B .

There is a canonical complex-parameter deformation of this particular action: because all its eigenvalues are positive real numbers, define

$$D_\gamma(s) e_A = \exp(s \log \chi_A(\gamma)) e_A, \quad s \in \mathbb{C}. \quad (\text{ME2})$$

The logarithm here is the real logarithm of a positive number. Thus $D_\gamma(1) = D_\gamma$, no eigenvalue or absolute value is changed, and $D_{\gamma\eta}(s) = D_\gamma(s) D_\eta(s)$ follows from the real logarithm's additivity. The new coordinates in (ME1) have character one on Γ_F , proving compatibility for every s . Denote this module by $B(s)$. Fourier complementation satisfies $CD_\gamma(s)C = D_{\gamma^{-1}}(s)$.

The two endpoint coordinates give a surjection $\epsilon : B(s) \rightarrow \mathbb{C}^2$, independent of F and s . Define $K(s) = \ker \epsilon$. The target has the trivial rational action, by the product formula. Hence there is an exact sequence of actual modules

$$0 \longrightarrow K(s) \longrightarrow B(s) \xrightarrow{\epsilon} \mathbb{C}^2 \longrightarrow 0. \quad (\text{ME3})$$

Let $b_0, b_1 \in B(s)$ be the two basis vectors at $F = \emptyset$; they give the specified vector-space section of ϵ .

Theorem 10.1 (Explicit global mixed-support extension). *For $\Re s > 0$, $B(s)^{\mathbb{Q}^\times} = 0$, and (ME3) has no equivariant section. Its two connecting cocycles are*

$$c_j(\gamma; s) = D_\gamma(s) b_j - b_j \in K(s), \quad j = 0, 1. \quad (\text{ME4})$$

They determine linearly independent elements of $H^1(\mathbb{Q}^\times, K(s))$. For γ with prime factors in F , their complete coordinates are

$$c_0(\gamma; s) = \sum_{A \subseteq F} \left(\prod_{p \in A} |\gamma|_p^s - 1 \right) e_A, \quad (\text{ME5})$$

$$c_1(\gamma; s) = \sum_{A \subseteq F} \left(|\gamma|_\infty^s \prod_{p \in A} |\gamma|_p^s - 1 \right) e_{\{\infty\} \cup A}. \quad (\text{ME6})$$

In particular, on the original four faces at $\{\infty, p\}$,

$$c_0(p; s) = (p^{-s} - 1) e_{\{p\}}, \quad c_1(p; s) = (p^s - 1) e_{\{\infty\}}. \quad (\text{ME7})$$

These are nonzero mixed-face coordinates, not endpoint coordinates.

Proof. At stage F , (ME1) sends b_0 to $\sum_{A \subseteq F} e_A$ and b_1 to $\sum_{A \subseteq F} e_{\{\infty\} \cup A}$. Acting diagonally and subtracting gives (ME5)–(ME6). Their empty and full coordinates vanish, the latter by the full product formula, so they lie in $K(s)$.

If $z \neq 0$ is represented in B_F , choose a prime $p \notin F$. Its image in $B_{F \cup \{p\}}$ has equal coefficients z_A on A and $A \cup \{p\}$. If $\infty \notin A$, the first has eigenvalue one under $D_p(s)$ and the second has eigenvalue p^{-s} . If $\infty \in A$, the eigenvalues are p^s and one. Since $\Re s > 0$, the nonunit eigenvalue

never equals one. At least one nonzero coefficient is therefore changed. Every bonding map is injective, since tensoring by the nonzero vector $e_0 + e_1$ is injective. Thus z cannot become invariant at a later stage either. This proves the assertion about invariants and excludes an equivariant section.

Multiplying the operators gives $c_j(\gamma\eta) = D_\gamma(s)c_j(\eta) + c_j(\gamma)$, the cocycle identity. Suppose $a_0c_0 + a_1c_1$ is a coboundary $D_\gamma(s)k - k$ with $k \in K(s)$. Then $a_0b_0 + a_1b_1 - k$ is invariant. The preceding argument forces it to vanish, and applying ϵ gives $(a_0, a_1) = 0$. This proves linear independence, using the definition of group cohomology rather than an assumed vanishing or comparison theorem. Specializing (ME5)–(ME6) to $F = \{p\}$ proves (ME7). $\square$

At the original value $s = 1$, the complement pairing of (AB15) gives

$$\mathcal{B}(c_0(p; 1), c_1(p; 1)) = (p^{-1} - 1)(p - 1) = -(\sqrt{p} - 1/\sqrt{p})^2. \quad (\text{ME8})$$

This follows by pairing the two complementary face vectors, each with coefficient one in the original pairing. It is an evaluated pairing of the extension, not an assignment of a Weil sign to every test.

10.2 The determinant of the actual local differential

Restrict to the infinite cyclic subgroup generated by p and the invariant mixed-face subspace

$$M_p = \mathbb{C}e_{\{p\}} \oplus \mathbb{C}e_{\{\infty\}} \subset B_{\{p\}}. \quad (\text{ME9})$$

The standard two-term cochain complex in this case is obtained directly from the free resolution $0 \rightarrow \mathbb{C}[t, t^{-1}] \xrightarrow{t-1} \mathbb{C}[t, t^{-1}] \rightarrow \mathbb{C} \rightarrow 0$. The first map is injective because a Laurent polynomial ring is a domain; the cokernel is evaluation at one. Applying module Hom proves, in the displayed order of basis vectors, that the differential is

$$d_p(s) = D_p(s) - I = \begin{pmatrix} p^{-s} - 1 & 0 \\ 0 & p^s - 1 \end{pmatrix} : M_p \longrightarrow M_p. \quad (\text{ME10})$$

It is the same matrix that evaluates the two cocycles (ME7) on the generator. It is invertible for $\Re s > 0$. Its determinant, with both orientations and the original ordered basis retained, is

$$\det d_p(s) = (p^{-s} - 1)(p^s - 1) = -p^s(1 - p^{-s})^2. \quad (\text{ME11})$$

For clarity, the determinant-line calculation can be specified without a torsion convention: in a one-dimensional complex $L^0 \xrightarrow{a} L^1$ with named bases, the map $\det L^0 \otimes (\det L^1)^* \rightarrow \mathbb{C}$ induced by the differential sends the named tensor to a . Its inverse scalar is a^{-1} . For the finite-place line in (ME10) this scalar is $(p^{-s} - 1)^{-1} = -(1 - p^{-s})^{-1}$. Thus the exact arithmetic Euler factor is

$$Z_p(s) = \frac{1}{1 - p^{-s}} = -(\det d_p(s)|_{\mathbb{C}e_{\{p\}}})^{-1}. \quad (\text{ME12})$$

The minus sign is displayed, not absorbed into a basis change. The complementary line has inverse determinant $(p^s - 1)^{-1} = p^{-s}Z_p(s)$. Their product is therefore $-p^{-s}Z_p(s)^2$, in agreement with (ME11).

The extraction of this local line has an exact chain map from all the finite-place data. Fix $p \in F$. Define $r_{F,p} : B_F(s) \rightarrow B_{\{p\}}(s)$ by retaining the coefficients of e_A only when $A \subseteq \{\infty, p\}$ and setting every spectator coordinate to the evaluation component. Then

$$r_{F,p}D_p(s) = D_p(s)r_{F,p}, \quad r_{F,p}j_{\{p\},F} = I. \quad (\text{ME13})$$

Indeed $|p|_q = 1$ at every spectator prime q , and exactly the empty spectator choice survives the projection of the tensor in (ME1). Projection of the full Koszul complex (AB11) to degrees

zero and one along the p direction, followed by $r_{F,p}$, is a cochain map to the cyclic complex: all other exterior generators project to zero. Finally projection onto the two mixed faces and then onto the finite-place line commutes with $D_p(s) - I$. These identities prove the whole chain map that extracts (ME12). They do not identify this determinant with the torsion of the entire higher-rank Koszul complex.

Proposition 10.2 (The full prime product in its domain of convergence). *For $\Re s > 1$, the product of the factors (ME12), one for each original finite prime, converges absolutely and equals*

$$\prod_p Z_p(s) = \sum_{n \geq 1} n^{-s} = \zeta(s). \quad (\text{ME14})$$

Proof. Write $\sigma = \Re s > 1$. Then $\sum_p \sum_{m \geq 1} p^{-m\sigma}/m \leq (1 - 2^{-\sigma})^{-1} \sum_{n \geq 2} n^{-\sigma} < \infty$, so the logarithms of the product converge absolutely. For a finite set of primes the geometric-series expansion is absolutely convergent and enumerates exactly the positive integers with prime factors in that set, once each by unique prime factorization. Its omitted terms are dominated by the summable series $\sum n^{-\sigma}$. Passing to all primes proves (ME14). This derivation concerns the convergent Euler product; it makes no assertion about zeros in its meromorphic continuation. $\square$

10.3 An exact receiver in the original Weil prime term

Define the logarithmic derivative of the same finite-place differential by

$$L_p(s) = -\frac{d}{ds} \log Z_p(s) = \frac{(\log p)p^{-s}}{1 - p^{-s}} = (\log p) \sum_{m \geq 1} p^{-ms}, \quad \Re s > 0. \quad (\text{ME15})$$

There is no branch issue: the first expression means $-Z'_p/Z_p$, and the last equality is an absolutely convergent geometric series. Let $g \in C_c^\infty(\mathbb{R}_{>0})$, let $h = g * g^*$ use the original multiplicative Haar measure dx/x , and put $\widehat{g}(t) = \int_0^\infty g(x)x^{-it}dx/x$. Thus $\widehat{h}(t) = |\widehat{g}(t)|^2$ for real t .

Theorem 10.3 (Mixed-differential formula for every prime-power term). *The exact prime term of the original explicit formula is*

$$\begin{aligned} W_p(h) &= (\log p) \sum_{m \geq 1} p^{-m/2} (h(p^m) + h(p^{-m})) \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 (L_p(\tfrac{1}{2} + it) + L_p(\tfrac{1}{2} - it)) dt. \end{aligned} \quad (\text{ME16})$$

Consequently, with $r = p^{-1/2}$ and $P_r(\theta) = (1 - r^2)/(1 - 2r \cos \theta + r^2)$,

$$W_p(g * g^*) = \frac{\log p}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 (P_{p^{-1/2}}(t \log p) - 1) dt. \quad (\text{ME17})$$

Both pole terms, every prime power and the Archimedean term are retained in the resulting finite-place Weil functional:

$$\begin{aligned} D_F(g * g^*) &= \mathcal{B}(Pg, Pg) - W_\infty(g * g^*) \\ &\quad + \sum_{p \in F} (\log p) \left(\|g\|_{L^2(dx/x)}^2 - \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 P_{p^{-1/2}}(t \log p) dt \right). \end{aligned} \quad (\text{ME18})$$

Here P is precisely the two-pole map (AB16); no new source or metric has been substituted for the original test.

Proof. In logarithmic coordinates the Fourier inversion formula is $h(e^u) = (2\pi)^{-1} \int \widehat{h}(t) e^{itu} dt$. The two series in (ME15), evaluated at $1/2 \pm it$, have total absolute sum at most $2(\log p)p^{-1/2}/(1 - p^{-1/2})$ uniformly in t . Since $\widehat{h}$ is a Schwartz function, absolute integrability justifies interchanging the integrals and the series. Fourier inversion at $u = \pm m \log p$ proves (ME16), including the factor $1/(2\pi)$. For $z = re^{-i\theta}$ direct multiplication gives

$$\frac{z}{1-z} + \frac{\bar{z}}{1-\bar{z}} = \frac{2r \cos \theta - 2r^2}{1 - 2r \cos \theta + r^2} = P_r(\theta) - 1. \quad (\text{ME19})$$

This proves (ME17). Finally the full pole pairing was computed in (AB17), D_F is the original pole term minus the Archimedean term and minus $\sum_{p \in F} W_p$, and Fourier Plancherel in the same logarithmic coordinate gives $(2\pi)^{-1} \int |\widehat{g}|^2 = \|g\|^2$. Substitution proves (ME18). $\square$

Each Poisson energy in (ME18) is positive for $g \neq 0$: its integrand is nonnegative, P_r is strictly positive, and Plancherel excludes a zero Fourier transform for nonzero g . Its subtraction from $\|g\|^2$ has the exact sign already tested by the two-bump examples of Theorem 7.3. Thus the formula retains the competing energies; it does not replace their difference by a positivity assertion. If $\text{supp } g \subset [\lambda^{-1}, \lambda]$ and $p > \lambda^2$, every $h(p^{\pm m})$ vanishes, so (ME16) proves that the corresponding Poisson energy is *exactly* $\|g\|^2$. Consequently (ME18) becomes the full prime return at $F = \{p : p \leq \lambda^2\}$, without summing separately divergent norm and Poisson-energy series.

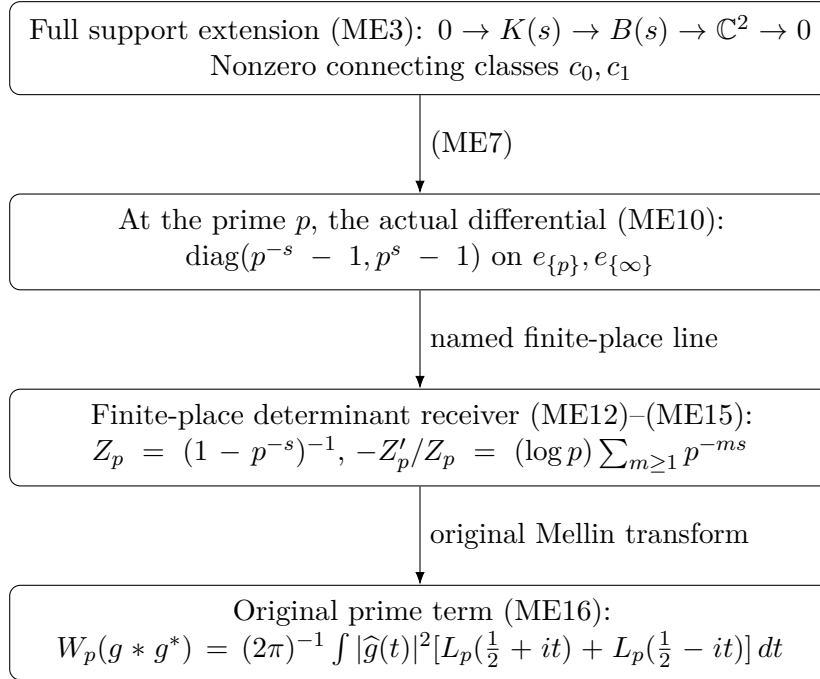


Figure 5: The calculated route from all support faces to each arithmetic prime-power contribution. The complementary line and its minus sign remain in (ME11)–(ME12); the local extraction is the explicit chain map (ME13). Theorem 10.3 proves the last arrow in the conventions of Connes [4, 5]. These are exact maps and equalities, not a diagram asserting the still-unproved full Weil positivity.

The use of the new support geometry in this section is specific: a global nonsplit mixed extension supplies a local differential; its determinant and logarithmic derivative supply the Euler factor and the complete prime-power term. Ordinary endpoint coinvariants alone would not retain that differential. The remaining calculation for the RH programme is the pairing of the full interior cohomology with these retained relative terms and the Archimedean term in (ME18). No coefficient of an earlier large-degree determinant has been used to stand in for that pairing.

11 Relative cohomology at the full supported prime

11.1 The full space and the receiving categories

Let X be a nonempty irreducible topological space. Put $Y = X^+ = X \sqcup \{\eta_\tau\}$. Its nonempty opens are the singleton $\{\eta_\tau\}$ and $U^+ = U \cup \{\eta_\tau\}$ for nonempty open $U \subset X$. Write $i : X \hookrightarrow Y$ and $j : \{\eta_\tau\} \hookrightarrow Y$. All nonempty opens of X intersect. In the first four parts of this chapter, sheaves have values in modules over a fixed commutative coefficient ring Λ ; taking $\Lambda = \mathbb{Z}$ gives abelian sheaves. These are sheaves on the entire underlying prime space. They are not automatically additive-group modules over its semiring structure sheaf. Section 11.4 proves the exact distinction and the resulting morphisms.

For a module B , let cB be the sheaf on X with value B on every nonempty open and identity restriction between such opens. This is a sheaf because any two members of a nonempty open cover intersect, so compatible locally constant values agree. Define

$$g\mathcal{A} = \varinjlim_{\emptyset \neq U \subset X} \mathcal{A}(U), \quad \epsilon_{\mathcal{A}} : \mathcal{A} \longrightarrow c(g\mathcal{A}). \quad (\text{RP1})$$

The indexing order is restriction to smaller nonempty opens. It is filtered by irreducibility. If X has a generic point ξ , then every nonempty open contains ξ , and $g\mathcal{A} = \mathcal{A}_\xi$.

Lemma 11.1 (Exact generic-section adjunction). *The functors g and c are exact, g is left adjoint to c , and $gcB = B$. In particular c preserves injective modules as injective sheaves, and*

$$R\Gamma(X, cB) = B[0]. \quad (\text{RP2})$$

Proof. A morphism $\mathcal{A} \rightarrow cB$ is a family of maps $\mathcal{A}(U) \rightarrow B$ agreeing under all restrictions. The filtered colimit is exactly their universal receiving module, proving the adjunction. The assertions $gcB = B$ and exactness of c follow from its displayed sections or stalks. To check surjectivity of g on an epimorphism of sheaves, represent a generic section by a section on a nonempty U . Sheaf surjectivity supplies a lift on an open cover of U ; choose a nonempty member and take its generic germ. For exactness in the middle, if a section has zero image as a generic germ, restrict to a nonempty smaller open on which its image is zero, then use the kernel sheaf. This proves exactness of g .

The adjunction and exactness of g imply that c takes injectives to injectives: applying $\text{Hom}(-, cI) = \text{Hom}(g(-), I)$ to an exact sequence is exact if I is injective. Apply c to an injective resolution of B . It remains an injective resolution, and its sections on X are the original resolution. This proves (RP2). $\square$

Theorem 11.2 (All sheaves and all gluing maps). *The category $\text{Sh}(Y, \Lambda)$ is equivalent to the category of triples*

$$\mathcal{F} = (\mathcal{A}, B, \phi), \quad \phi : \mathcal{A} \longrightarrow cB, \quad (\text{RP3})$$

where $\mathcal{A} \in \text{Sh}(X, \Lambda)$ and $B \in \Lambda\text{-Mod}$. Its sections are $\mathcal{F}(U^+) = \mathcal{A}(U)$ and $\mathcal{F}(\{\eta_\tau\}) = B$; restriction to the singleton is ϕ_U . A morphism is a pair (a, b) satisfying $\phi' a = c(b)\phi$. Kernels and cokernels are computed componentwise. Equivalently, the gluing is an arbitrary map $\bar{\phi} : g\mathcal{A} \rightarrow B$.

Proof. For a sheaf $\mathcal{F}$ on Y put $B = \mathcal{F}(\{\eta_\tau\})$ and $\mathcal{A}(U) = \mathcal{F}(U^+)$ on nonempty opens. If U is covered by nonempty opens U_a , their pairwise intersections are nonempty, and $(U_a \cap U_b)^+ = U_a^+ \cap U_b^+$. Thus the sheaf axiom for $\mathcal{F}$ is exactly the sheaf axiom for these values on X . Restriction to the singleton gives ϕ . This $\mathcal{A}$ is $i^*\mathcal{F}$, since U^+ is the smallest open of Y containing U and no further sheafification changes the displayed sheaf.

Conversely the displayed formulas for a triple define a sheaf. On a cover of U^+ , sections on nonempty arithmetic parts glue by the sheaf axiom for $\mathcal{A}$; their images in B agree, and hence also agree with any section on a singleton member of the cover. The singleton case is immediate. The

two constructions are inverse. The compatibility equation is precisely restriction compatibility. Since c is exact, both a componentwise kernel and a componentwise cokernel inherit the indicated arrow and satisfy the categorical universal property. The last assertion is Lemma 11.1. $\square$

11.2 The six elementary functors and their derived terms

Theorem 11.3 (Explicit recollement). *In triple notation the functors are*

$$\begin{aligned} i^*(\mathcal{A}, B, \phi) &= \mathcal{A}, & i_*\mathcal{A} &= (\mathcal{A}, 0, 0), & i^!(\mathcal{A}, B, \phi) &= \ker \phi, \\ j^*(\mathcal{A}, B, \phi) &= B, & j_!B &= (0, B, 0), & j_*B &= (cB, B, \text{id}). \end{aligned} \quad (\text{RP4})$$

The adjunctions are $i^ \dashv i_* \dashv i^!$ and $j_! \dashv j^* \dashv j_*$. All displayed functors except possibly $i^!$ are exact. There is in addition an exact left adjoint to i^* ,*

$$U\mathcal{A} = (\mathcal{A}, g\mathcal{A}, \epsilon_{\mathcal{A}}), \quad U \dashv i^*. \quad (\text{RP5})$$

For any triple, there is a canonical short exact sequence

$$0 \longrightarrow j_!B \longrightarrow \mathcal{F} \longrightarrow i_*\mathcal{A} \longrightarrow 0. \quad (\text{RP6})$$

Proof. A map $\mathcal{F} \rightarrow i_*\mathcal{C}$ is precisely a map $\mathcal{A} \rightarrow \mathcal{C}$. A map $i_*\mathcal{C} \rightarrow \mathcal{F}$ is a map $\mathcal{C} \rightarrow \mathcal{A}$ whose composite with ϕ is zero, hence a map to $\ker \phi$. A map $j_!D \rightarrow \mathcal{F}$ is any map $D \rightarrow B$. A map $\mathcal{F} \rightarrow j_*D$ has its arithmetic component forced to be $c(b)\phi$, so is any map $b : B \rightarrow D$. These calculations prove the four adjunctions and all formulas. Exactness follows from componentwise exactness and exactness of c .

A map $U\mathcal{C} \rightarrow \mathcal{F}$ has arithmetic part $a : \mathcal{C} \rightarrow \mathcal{A}$ and must have generic part the unique map $g\mathcal{C} \rightarrow B$ corresponding to ϕa under the adjunction of Lemma 11.1. This proves $U \dashv i^*$; its exactness follows from exactness of g . Finally the maps in (RP6) have components $(0, \text{id}_B)$ and $(\text{id}_{\mathcal{A}}, 0)$, proving its exactness. $\square$

Complexes use cohomological degrees: $B[-1]$ has B in degree one. All derived statements below can be read in the bounded-below derived categories. No assertion requires an unbounded resolution convention.

Theorem 11.4 (Derived closed restriction and relative cohomology). *For $\mathcal{F} = (\mathcal{A}, B, \phi)$ one has canonical identifications*

$$Ri^!\mathcal{F} \simeq [\mathcal{A} \xrightarrow{\phi} cB] = \text{Cone}(\phi)[-1], \quad (\text{RP7})$$

where the two entries are in degrees zero and one. Thus $i^!\mathcal{F} = \ker \phi$, $R^1i^!\mathcal{F} = \text{coker } \phi$, and $R^n i^!\mathcal{F} = 0$ for $n \geq 2$. Furthermore

$$R\Gamma(Y, \mathcal{F}) = R\Gamma(X, \mathcal{A}), \quad R\Gamma_X(Y, \mathcal{F}) \simeq \text{Cone}(R\Gamma(X, \mathcal{A}) \rightarrow B)[-1]. \quad (\text{RP8})$$

Here $R\Gamma_X(Y, -)$ means $R\Gamma(X, Ri^!(-))$, the derived functor of sections with support in the closed subset X . The corresponding localization triangle is

$$i_* Ri^!\mathcal{F} \longrightarrow \mathcal{F} \xrightarrow{(\phi, \text{id})} j_* B \longrightarrow i_* Ri^!\mathcal{F}[1]. \quad (\text{RP9})$$

Proof. There are enough injectives in the triple category. Indeed embed $\mathcal{A}$ into an injective sheaf I and B into an injective module J . The map

$$(\mathcal{A}, B, \phi) \longrightarrow i_* I \oplus j_* J$$

with arithmetic component $(\mathcal{A} \hookrightarrow I, c(B \hookrightarrow J)\phi)$ and generic component $B \hookrightarrow J$ is injective. The target is injective because i_* and j_* are right adjoints of exact functors.

If $\mathcal{I} = (\mathcal{A}_I, B_I, \phi_I)$ is injective, then ϕ_I is split surjective. To see this, the monomorphism $j_! B_I \hookrightarrow j_* B_I$ and the map $j_! B_I \rightarrow \mathcal{I}$ with generic part the identity extend to a map $j_* B_I \rightarrow \mathcal{I}$. Its arithmetic part is a right inverse of ϕ_I . Also $\ker \phi_I$ is injective on X because $i^!$ is right adjoint to the exact functor i_* . The functors i^* and j^* preserve injectives: their respective left adjoints U and $j_!$ are exact.

For an injective resolution of $\mathcal{F}$, therefore, there is a termwise split exact sequence of complexes $0 \rightarrow i^! \mathcal{I}^\bullet \rightarrow i^* \mathcal{I}^\bullet \rightarrow c(j^* \mathcal{I}^\bullet) \rightarrow 0$. The latter two complexes resolve $\mathcal{A}$ and cB by injectives. Its kernel computes $Ri^!$ and is the indicated shifted cone. This proves (RP7). The same termwise calculation in the triple category proves (RP9).

Ordinary global sections equal $\Gamma(X, \mathcal{A})$, and i^* preserves injectives, proving the first identification in (RP8). Apply $R\Gamma(X, -)$ to (RP7) and use (RP2) for the second. At degree zero the kernel is exactly the sections whose restriction to the open point is zero, which proves the stated support interpretation. $\square$

Corollary 11.5 (A nonzero relative object and its connecting map). For every nonzero B , the sheaf $j_! B$ is nonzero and satisfies

$$R\Gamma(Y, j_! B) = 0, \quad Ri^! j_! B = cB[-1], \quad R\Gamma_X(Y, j_! B) = B[-1]. \quad (\text{RP10})$$

The sequence

$$0 \longrightarrow j_! B \longrightarrow j_* B \longrightarrow i_* cB \longrightarrow 0 \quad (\text{RP11})$$

does not split. Its localization boundary identifies the actual module B at the new point with $H_X^1(Y, j_! B)$, with the sign convention fixed by the order of maps in this exact sequence.

Proof. The three calculations are substitutions of $(0, B, 0)$ into Theorem 11.4; its stalk at the new point proves it is nonzero. A section of the surjection in (RP11) would be a triple morphism $(cB, 0, 0) \rightarrow (cB, B, \text{id})$ with arithmetic part the identity. The compatibility equation would instead force that part to be zero. This contradicts $B \neq 0$. In the localization long exact sequence, both global cohomology groups of $j_! B$ adjoining this boundary vanish, so the boundary $B \rightarrow H_X^1(Y, j_! B)$ is the stated isomorphism. $\square$

11.3 Cross extensions, relative traces, and exact pairing maps

Theorem 11.6 (The new cross-extension groups). For $n \geq 1$,

$$\text{Ext}_Y^n(i_* \mathcal{A}, j_! B) = \text{Ext}_X^{n-1}(\mathcal{A}, cB) = \text{Ext}_\Lambda^{n-1}(g\mathcal{A}, B), \quad \text{Hom}_Y(i_* \mathcal{A}, j_! B) = 0. \quad (\text{RP12})$$

All groups $\text{Ext}_Y^n(j_! B, i_* \mathcal{A})$ vanish, including $n = 0$. Under the degree-one identification in (RP12), the class of (RP6) is exactly $\bar{\phi}$. In particular the universal coupling $U\mathcal{A}$ represents $\text{id}_{g\mathcal{A}}$ and is nonsplit if $g\mathcal{A} \neq 0$.

Proof. The exact functor i_* and its right adjoint $i^!$ give, using injective resolutions,

$$\text{RHom}_Y(i_* \mathcal{A}, j_! B) = \text{RHom}_X(\mathcal{A}, Ri^! j_! B).$$

Substitute $Ri^! j_! B = cB[-1]$. The exact adjunction $g \dashv c$, with c preserving injectives, then gives $\text{RHom}_X(\mathcal{A}, cB) = \text{RHom}_\Lambda(g\mathcal{A}, B)$. This proves (RP12). For the reverse direction, use $j_! \dashv j^*$, exactness of $j_!$, and $j^* i_* = 0$.

The extension (RP6) is the pullback of (RP11) along $i_* \phi : i_* \mathcal{A} \rightarrow i_* cB$: the fibre product has exactly the triple $(\mathcal{A}, B, \phi)$. The universal sequence corresponds to id_{cB} under the derived adjunction, fixing its sign; pullback therefore corresponds to ϕ , and then to $\bar{\phi}$. For $U\mathcal{A}$ that map is the identity. $\square$

For a scheme with generic point, (RP12) says exactly that this direct cross-extension channel receives the generic stalk. A sheaf whose generic stalk is zero has no such cross-extension. This fact does not remove its other supported-prime data. The universal coupling in Section 11.5 gives the exact relative object in which the quotients by those prime stalks occur.

Proposition 11.7 (Full mapping complex). *For $\mathcal{F} = (\mathcal{A}, B, \phi)$ and $\mathcal{F}' = (\mathcal{A}', B', \phi')$, their derived mapping complex is the fibre of*

$$\mathrm{RHom}_X(\mathcal{A}, \mathcal{A}') \oplus \mathrm{RHom}_\Lambda(B, B') \longrightarrow \mathrm{RHom}_\Lambda(g\mathcal{A}, B'), \quad (a, b) \longmapsto \bar{\phi}' g(a) - b\bar{\phi}. \quad (\text{RP13})$$

Proof. Resolve the target by injective triples. Their arithmetic and generic components are injective, as proved in Theorem 11.4, and their gluing arrows split. In each degree, the map of ordinary Hom groups in (RP13) is surjective: lift any map $\mathcal{A} \rightarrow cB'_I$ through a splitting of ϕ'_I and take $b = 0$. Its kernel is exactly the compatible pairs defining a triple morphism. Thus the complex of triple morphisms is the kernel of a surjective map of complexes computing the displayed derived Homs. It is therefore their homotopy fibre. Lemma 11.1 identifies the last mapping complex with the stated module-valued one. $\square$

Now take a coefficient field k . If V, W are finite-dimensional k -vector spaces, define

$$E(V, W) = \mathrm{Ext}_Y^1(i_*cV, j_!W) = \mathrm{Hom}_k(V, W). \quad (\text{RP14})$$

There is a canonical perfect pairing

$$E(V, W) \times E(W, V) \longrightarrow k, \quad (\phi, \psi) \longmapsto \mathrm{tr}_V(\psi\phi) = \mathrm{tr}_W(\phi\psi). \quad (\text{RP15})$$

Indeed matrices give both equal traces as $\sum_{a,b} \phi_{ba}\psi_{ab}$; pairing a matrix unit with its transpose shows nondegeneracy. This is a pairing of two explicitly identified extension spaces with exchanged coefficient spaces. It is not a Yoneda product through a nonexistent reverse cross-extension: those reverse groups vanish by Theorem 11.6. For $k = \mathbb{C}$ and specified positive Hermitian metrics, $\mathrm{tr}(\phi^*\psi)$ gives the associated Hermitian form and $\mathrm{tr}(\phi^*\phi) = \sum_{a,b} |\phi_{ba}|^2$ in orthonormal bases. The metrics are additional data; topology alone has not supplied them.

If $T : B \rightarrow B$ is an endomorphism of a finite-dimensional coefficient space, its relative alternating trace is

$$\sum_n (-1)^n \mathrm{tr}(T \mid H_X^n(Y, j_!B)) = -\mathrm{tr}(T). \quad (\text{RP16})$$

This follows from (RP10), with its entire cohomology in degree one. For two such sheaves the relative cup product on their degree-one groups is zero: its target is $H_X^2(Y, j_!(B \otimes_k D)) = 0$. Their tensor sheaf is $j_!(B \otimes D)$ because tensor products are computed on stalks and both arithmetic stalks vanish. Thus the nonzero relative degree-one object and the trace pairing (RP15) have precise different maps; identifying that trace pairing with a degree-two cup pairing would give a false result.

11.4 What the actual split structure sheaf retains

Let X now be an irreducible locally ringed space with nonzero local rings. For a ring R , $\mathbf{G}(R) = \{\tau\} \sqcup R$, with τ the additive identity and multiplicative absorber; write e_R for its supported coefficient zero. Supported addition and multiplication are those of R , so $e_R + e_R = e_R$ and $e_R a = e_R$ for every supported a . The full semiring sheaf is

$$\mathcal{S}(U^+) = \mathbf{G}(\mathcal{O}_X(U)), \quad \mathcal{S}(\{\eta_\tau\}) = \mathbb{B} = \{0, 1\}, \quad 1 + 1 = 1, \quad (\text{RP17})$$

with support restriction to the singleton. Gluing is exact because compatible sections agree in support at the common singleton; absent sections glue to τ , and supported coefficients glue in $\mathcal{O}_X$.

Its stalks are $\mathbf{G}(\mathcal{O}_{X,x})$ and $\mathbb{B}$. The global section e has arithmetic germs $e_{\mathcal{O}_{X,x}}$ and generic germ 1 at the additional point. For an affine domain A , the prime ideals of $\mathbf{G}(A)$ are $\{\tau\}$ and $\{\tau\} \sqcup \mathfrak{p}$, $\mathfrak{p} \in \text{Spec } A$: an ideal containing a supported element contains e_A and its supported elements form a ring ideal by multiplication by -1 . Products of supported elements remain supported, proving primality of $\{\tau\}$. This identifies $V(e)$ with the entire closed arithmetic stratum.

Theorem 11.8 (Exact additive loss and joint recovery). *The additive group completion of $\mathcal{S}$ is $i_*\mathcal{O}_X$. Every sheaf of $\mathcal{S}$ -semimodules with abelian-group addition is uniquely an $i_*\mathcal{O}_X$ -module; its generic stalk at η_τ is zero. In particular neither nonzero $j_!B$ nor the universal coupling $U\mathcal{O}_X$ with nonzero generic fibre is an additive-group module over $\mathcal{S}$ with its stated unit action.*

Nevertheless the joint map of semiring sheaves

$$(\chi, q) : \mathcal{S} \longrightarrow j_*\mathbb{B} \times i_*\mathcal{O}_X \quad (\text{RP18})$$

is injective. On U^+ it identifies the sections with $(0, 0)$ and all $(1, a)$, $a \in \mathcal{O}_X(U)$. Here χ retains support and q retains the exact coefficient, including zero.

Proof. An additive map from $\mathbf{G}(R)$ to an abelian group sends e_R to zero, since $f(e_R) + f(e_R) = f(e_R)$. Its supported restriction is then a group homomorphism from R , and the image of τ is zero. This proves the universal group-completion property with quotient R . At the Boolean stalk the relation $1 + 1 = 1$ kills its sole nonzero generator, so its group completion is zero. Sheafification commutes with this local universal property, yielding $i_*\mathcal{O}_X$.

For an additive-group semimodule, the relation $em + em = em$ gives $em = 0$. Thus its scalar action factors through the indicated ring quotient. At the extra stalk $e = 1$, so this says $m = 0$ for every element. This proves the module and nonexistence assertions. In (RP18), the absent section gives $(0, 0)$ and the supported zero gives $(1, 0)$, which are distinct. All other coefficients are distinguished by their second coordinate. The same checks on operations prove it is a morphism and injective. $\square$

Proposition 11.9 (A multiplicative linearization that retains support). *Let $\mathcal{M}$ be the sheaf of supported multiplicative elements of $\mathcal{S}$: its values on U^+ are the multiplicative monoid of $\mathcal{O}_X(U)$, including its supported zero, and its value on the singleton is $\{1\}$. Let $\mathcal{L} = \mathbb{Z}[\mathcal{M}]$ be the sheafification of its integral monoid algebra. Write $E = [e]$. Then E is a central idempotent, and*

$$\mathcal{L} = E\mathcal{L} \oplus (1 - E)\mathcal{L}, \quad E\mathcal{L} \cong j_*\mathbb{Z}, \quad (1 - E)\mathcal{L} \cong i_*\mathcal{C} \quad (\text{RP19})$$

as rings with the two factor identities E and $1 - E$. Here $\mathcal{C}_x = (1 - [0])\mathbb{Z}[(\mathcal{O}_{X,x}, \cdot)]$. There is a surjection $\mathcal{L} \rightarrow i_\mathcal{O}_X$, $[a] \mapsto a$, whose kernel is the sheaf of ideals generated by $[a + b] - [a] - [b]$. It kills the whole factor $E\mathcal{L}$. The factor $E\mathcal{L}$ contains the canonical nonzero submodule $j_!\mathbb{Z}$, and its quotient is $i_*\mathbb{Z}_X$ as in (RP11).*

Proof. Multiplicatively $e^2 = e$ and $ea = e$ for every supported element. Consequently $E^2 = E$ and $E[a] = E$. The augmentation $\varepsilon : \mathcal{L} \rightarrow j_*\mathbb{Z}$, $[a] \mapsto 1$, is a ring map. Multiplication by E sends a section to its augmentation times E , so its image is the free rank-one ring sheaf with identity E . This proves its identification with $j_*\mathbb{Z}$. The complementary factor has zero stalk at the added point, since $E = 1$ there; the triple description therefore places it on the closed arithmetic stratum and gives its stated stalks. The central idempotent identity proves the direct-product ring decomposition.

The quotient by the displayed additive relations has exactly the universal property of a ring receiving the original supported addition and multiplication. At an arithmetic stalk this is the original ring; at the singleton the relation $[1] - [1] - [1] = -1$ makes the quotient zero. Thus the quotient is precisely $i_*\mathcal{O}_X$. Taking $a = b = 0$ gives $-E$ among these relations, so its entire factor is killed. The last assertion is (RP11) on that factor. All computations commute with restrictions and so prove the sheaf assertions. $\square$

This construction specifies both what survives and what is lost. After tensoring with a coefficient field k , the E factor is the constant ring sheaf j_*k on the full space Y . Its module category is the category of all k -sheaves used above. The complementary factor is supported on X . Every $\mathcal{L} \otimes k$ -module splits as $EM \oplus (1 - E)M$, since these orthogonal idempotents sum to one; the same decomposition of every map makes the module category a product. Its cross-factor Hom and Ext groups are zero, by resolving within each product component. Each supported scalar $[a]$ acts as the identity on the E factor. Thus its rank-one relative trace for any such scalar is -1 , by (RP16); the support factor alone has not encoded a prime norm.

More specifically, if arithmetic coefficients are required to act through q and the additional fibre through ε , equivariance of a gluing map would give

$$\phi(ea) = e\phi(a), \quad \text{hence } 0 = \phi(a). \quad (\text{RP20})$$

The exact space of couplings in the underlying abelian-sheaf category is still $\text{Hom}(g\mathcal{A}, B)$ by (RP12); the specified module compatibility selects its zero subspace. The universal coupling (RP5) retains the entire arithmetic sheaf and gives its identity generic coupling in that explicit receiving category. The next calculation uses this object and does not assert the incompatible semiring-module action.

11.5 Principal parts at the supported primes

Theorem 11.10 (Principal parts are the relative object of the new point). *Let X be an integral scheme with function field F , viewed with its actual structure sheaf. Let $F_X = cF$ be its sheaf of rational functions and define $\mathcal{P}_X = F_X/\mathcal{O}_X$. For the canonical coupled abelian sheaf*

$$\mathcal{U}_X = U\mathcal{O}_X = (\mathcal{O}_X, F, \mathcal{O}_X \hookrightarrow F_X) \quad (\text{RP21})$$

one has

$$Ri^!\mathcal{U}_X = \mathcal{P}_X[-1], \quad (R^1i^!\mathcal{U}_X)_x = F/\mathcal{O}_{X,x}. \quad (\text{RP22})$$

In particular a supported valuation-ring stalk V contributes exactly F/V . The ordinary generic stalk gives zero, while nonfield valuation stalks give nonzero quotients. The maps are the original inclusions $\mathcal{O}_{X,x} \hookrightarrow F$ and their exact cokernels.

Proof. An integral scheme embeds its structure sheaf in its function field: on an affine domain chart, each local ring and each regular section maps injectively into its fraction field, and these maps agree on overlaps. Its generic stalk is F . Hence (RP5) gives exactly (RP21). Substitute this injective arrow into (RP7). The two-term complex has zero degree-zero cohomology and cokernel $\mathcal{P}_X$ in degree one, proving its indicated quasi-isomorphism and every stalk formula. If $V \neq F$, an element of $F \setminus V$ gives a nonzero quotient class; for $V = F$ the quotient is zero. $\square$

Theorem 11.11 (The complete integral prime receiver). *For $X = \text{Spec } \mathbb{Z}$, one has*

$$\mathcal{P}_X(U) = \bigoplus_{p \in U \text{ closed}} \mathbb{Q}/\mathbb{Z}_{(p)} \quad (\text{RP23})$$

for every nonempty open U , with restriction the projection discarding the primes removed from the open. At the generic point the stalk is zero. Each closed-prime stalk admits the canonical isomorphism

$$\mathbb{Q}/\mathbb{Z}_{(p)} \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p. \quad (\text{RP24})$$

The resulting relative cohomology is exactly

$$H_X^1(Y, \mathcal{U}_X) = \mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Q}_p/\mathbb{Z}_p, \quad H_X^n(Y, \mathcal{U}_X) = 0 \quad (n \neq 1). \quad (\text{RP25})$$

Proof. A nonempty open of $\text{Spec } \mathbb{Z}$ omits a finite set S of closed primes and has ring of regular functions $\mathbb{Z}[S^{-1}]$. Indeed it contains a basic open $D(n)$, so its complement is contained in the finitely many prime divisors of n ; a rational number is regular there exactly when its denominator has no prime factors outside S . The map

$$\mathbb{Q} \longrightarrow \bigoplus_{p \notin S} \mathbb{Q}/\mathbb{Z}_{(p)} \quad (\text{RP26})$$

sends a rational number to its indicated quotient classes. It has finite support, since its denominator has finitely many prime factors, and kernel $\mathbb{Z}[S^{-1}]$. It is surjective. For a prescribed finite family of classes, represent its p component by a_p/p^{n_p} with $a_p \in \mathbb{Z}$: any denominator prime to p is invertible modulo p^{n_p} , which supplies this representative. Then the sum $\sum_p a_p/p^{n_p}$ has the desired component at each p , because all the other summands are p -integral. This proves surjectivity.

The direct-sum formula with projection restrictions defines a sheaf. Every open here is quasi-compact: a cover by basic opens has a finite subcover because the ideal generated by their defining integers has a finite set of generators. Compatible families therefore have a finite-subcover union of finite supports, and glue to a finite-support family. They glue uniquely componentwise. Its stalk at p is the displayed p group, because the finitely many other support primes can be excluded by a smaller neighbourhood. Its generic stalk is zero by excluding its finite support. Thus the stalk maps from the quotient sheaf $\mathbb{Q}_X/\mathcal{O}_X$ identify it with the displayed sheaf, proving (RP23).

The inclusion $\mathbb{Q} \subset \mathbb{Q}_p$ induces (RP24): its kernel is the rational numbers with nonnegative p valuation, namely $\mathbb{Z}_{(p)}$. For surjectivity, write a p -adic element as $p^{-n}a$ with $a \in \mathbb{Z}_p$ and choose an integer b with $a - b \in p^n\mathbb{Z}_p$. Then b/p^n has the same class modulo $\mathbb{Z}_p$.

The sheaf in (RP23) is flasque: every restriction projection is surjective, by extension by zero at the omitted primes. Therefore its higher sheaf cohomology vanishes. This elementary acyclicity criterion follows, for example, by the standard injective resolution argument: injective sheaves are flasque, a quotient of a flasque sheaf by a flasque subsheaf is flasque, and global sections of such a quotient lift by extending compatible local lifts; the extension is obtained by adjoining one open set at a time and taking unions of compatible extensions. Induction on a resolution then makes the global-sections complex exact in positive degrees. Applying (RP22) now leaves only its degree-one global sections. Formula (RP26) for $S = \emptyset$ identifies those with $\mathbb{Q}/\mathbb{Z}$ and with the displayed direct sum, proving (RP25). $\square$

11.6 The exact filtration, finite pairings, and length weights

Fix a prime p . Put $C_p = \mathbb{Q}_p/\mathbb{Z}_p$ and

$$C_{p,n} = p^{-n}\mathbb{Z}_p/\mathbb{Z}_p \quad (n \geq 0), \quad C_{p,0} = 0. \quad (\text{RP27})$$

These are the original subquotients at the supported prime, not their numerical replacements.

Theorem 11.12 (All filtration maps and dual maps). *The map $a \bmod p^n \mapsto a/p^n \bmod \mathbb{Z}_p$ identifies $\mathbb{Z}/p^n\mathbb{Z}$ with $C_{p,n}$. In these coordinates the inclusion $C_{p,n} \hookrightarrow C_{p,n+1}$ is $a \mapsto pa$. For $n \geq 1$ there is the exact sequence*

$$0 \longrightarrow C_{p,n-1} \longrightarrow C_{p,n} \xrightarrow{\rho_{p,n}} \mathbb{F}_p \longrightarrow 0, \quad \rho_{p,n}(a/p^n) = a \bmod p. \quad (\text{RP28})$$

Thus $C_p = \bigcup_n C_{p,n}$, its n th layer has length one and cardinality p , and $\text{length}_{\mathbb{Z}_p} C_{p,n} = n$, $|C_{p,n}| = p^n$.

The finite pairing

$$C_{p,n} \times \mathbb{Z}/p^n\mathbb{Z} \longrightarrow \mathbb{C}^*, \quad (a/p^n, b) \longmapsto \exp(2\pi iab/p^n) \quad (\text{RP29})$$

is perfect. Dual to the filtration inclusion is reduction $\mathbb{Z}/p^{n+1} \rightarrow \mathbb{Z}/p^n$. Hence the character group of C_p , with its discrete topology, is $\mathbb{Z}_p$. Its annihilator of $C_{p,n}$ is $p^n\mathbb{Z}_p$, and the induced perfect layer pairing is

$$(C_{p,n}/C_{p,n-1}) \times (p^{n-1}\mathbb{Z}_p/p^n\mathbb{Z}_p) \longrightarrow \mathbb{C}^*, \quad (a/p^n, p^{n-1}b) \longmapsto \exp(2\pi iab/p). \quad (\text{RP30})$$

Proof. Changing a by a multiple of p^n changes a/p^n by an integer; these are exactly the changes giving zero in the quotient. Every $a \in \mathbb{Z}_p$ has an integer representative modulo p^n , proving the first isomorphism. Expressing the same fraction with denominator p^{n+1} multiplies its numerator by p , proving the inclusion map. Reduction of the numerator modulo p is well defined for $n \geq 1$. Its kernel consists of $a = pb$, which gives exactly the preceding filtration group, and it is surjective. This proves the exact sequence, the layer, and, by induction, both length and cardinality. Every element of $\mathbb{Q}_p$ has a finite negative valuation, proving the union.

The exponential is unchanged by changing either residue representative. If $a \not\equiv 0 \pmod{p^n}$, pairing it with $b = 1$ is not one; the same argument holds in the other variable. A character of the cyclic group is determined by its value on $1/p^n$, an arbitrary p^n th root of unity, and these are exactly the displayed values. This proves perfectness, not merely nondegeneracy. Substituting pa at level $n+1$ gives the level- n formula with b reduced modulo p^n ; thus compatible characters are exactly the inverse limit $\mathbb{Z}_p$. A p -adic integer kills every element of $C_{p,n}$ exactly when it is zero modulo p^n . Restricting that assertion to successive annihilators gives precisely (RP30), whose perfection also follows from the same cyclic argument at order p . $\square$

Corollary 11.13 (The same residue layer is the exact degree of multiplication). Multiplication by p gives the exact sequences

$$0 \longrightarrow \mathbb{F}_p \xrightarrow{a \mapsto a/p} C_{p,n+1} \xrightarrow{p} C_{p,n} \longrightarrow 0 \quad (n \geq 0), \quad 0 \longrightarrow \mathbb{F}_p \longrightarrow C_p \xrightarrow{p} C_p \longrightarrow 0. \quad (\text{RP30a})$$

In finite numerator coordinates the middle map is reduction $\mathbb{Z}/p^{n+1} \rightarrow \mathbb{Z}/p^n$. Its dual is multiplication by p , and the dual map on $\mathbb{Z}_p$ is multiplication by p . Every fibre on C_p has exactly p elements. Consequently on $\ell^2(C_p)$ with counting measure the actual pullback $A_p f(d) = f(pd)$ obeys

$$\|A_p f\|^2 = p\|f\|^2. \quad (\text{RP30b})$$

Thus the same one-layer cardinality produces both the logarithmic weight $\log p$ below and the norm factor $\sqrt{p}$ in the companion degree-operator calculation (PD7)–(PD10).

Proof. One has $p(a/p^{n+1}) = a/p^n$, giving the coordinate reduction and its surjectivity. Its kernel consists of the fractions a/p modulo $\mathbb{Z}_p$, exactly the stated copy of $\mathbb{F}_p$. Every element of C_p lies at a finite level, so passing to the union gives the second exact sequence and all fibre sizes. Substitution in (RP29) proves that the dual finite map is $b \mapsto pb$; equivalently on the inverse limit, $\langle pd, z \rangle = \langle d, pz \rangle$. For finite-support f , group the sum $\sum_d |f(pd)|^2$ by its image pd . Each image has exactly p preimages, so the sum is $p \sum_u |f(u)|^2$. Density of finite-support functions extends the pullback as a bounded operator and proves the same equality on its whole Hilbert domain. This uses the original group map and its exact kernel, not a substituted scalar map on a residue field. $\square$

Taking the direct sum of the prime groups in (RP25), a character is independently determined on each summand. Thus its character group is $\prod_p \mathbb{Z}_p = \widehat{\mathbb{Z}}$. Equivalently, for $a/n \bmod \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ and $z \in \widehat{\mathbb{Z}}$ the global pairing is

$$\langle a/n, z \rangle = \exp(2\pi ia(z \bmod n)/n). \quad (\text{RP31})$$

It is independent of the representative because the reductions $z \bmod n$ are compatible under divisibility. Perfection follows from the cyclic finite-level calculation and its inverse limit. At any

finite level $N = p^n$, the exact orthogonality identity is

$$\frac{1}{N} \sum_{b \bmod N} \left| \sum_{a \bmod N} f(a) \exp(2\pi i ab/N) \right|^2 = \sum_{a \bmod N} |f(a)|^2. \quad (\text{RP32})$$

Expanding the square reduces this to $\sum_b \exp(2\pi i(a - a')b/N) = N$ for $a = a'$ and zero otherwise; the latter is a finite geometric series with ratio different from one. Thus the original residue labels and their finite pairing are retained, in addition to the length receiver constructed next.

Theorem 11.14 (The weighted filtration receiver is the prime logarithmic derivative). *Let $K_0^{\text{fl}}(\mathbb{Z}_p)$ be the Grothendieck group of finite length $\mathbb{Z}_p$ -modules, with the relations $[M] = [M'] + [M'']$ from short exact sequences. The length map is an isomorphism $K_0^{\text{fl}}(\mathbb{Z}_p) \cong \mathbb{Z}$, taking $[\mathbb{F}_p]$ to one. The canonical additive weight is*

$$w_p([M]) = \log |M| = (\log p) \text{length}_{\mathbb{Z}_p} M. \quad (\text{RP33})$$

For the exact original filtration (RP27), its successive-layer series therefore satisfies

$$w_p \left(\sum_{m \geq 1} [C_{p,m}/C_{p,m-1}] T^m \right) = (\log p) \frac{T}{1-T}. \quad (\text{RP34})$$

For $\Re s > 0$, put $T = p^{-s}$. Then this convergent series is exactly

$$(\log p) \sum_{m \geq 1} p^{-ms} = -\frac{d}{ds} \log \frac{1}{1-p^{-s}} = L_p(s). \quad (\text{RP35})$$

It equals the expression (ME15), label `eq:ME-log-derivative`, in the companion complete proof `11_mixed_extension_prime_trace.tex`, lines 210–215 of the 22 September 2026 working edition.

Proof. Every simple module over the local ring $\mathbb{Z}_p$ is its residue field: the annihilator of a nonzero generator is a maximal ideal, and the unique maximal ideal is (p) . A composition series of a finite length module therefore gives $[M] = \text{length}(M)[\mathbb{F}_p]$. Length is additive on a short exact sequence, as one sees by adjoining a composition series of the submodule to the inverse images of a composition series of the quotient. Consequently it gives a homomorphism back to $\mathbb{Z}$, inverse to $1 \mapsto [\mathbb{F}_p]$. Each simple factor has p elements. Cardinalities multiply in a short exact sequence of finite groups, since all fibres of the quotient have the size of its kernel. Induction on the composition series proves $|M| = p^{\text{length}(M)}$, establishing (RP33).

By (RP28), every layer has class $[\mathbb{F}_p]$. Summing their formal series gives (RP34). For $\Re s > 0$, $|p^{-s}| < 1$, so it is absolutely convergent and is the ordinary geometric series. With $q = p^{-s}$ one has $dq/ds = -(\log p)q$ and $d \log(1-q)^{-1}/ds = -(\log p)q/(1-q)$, proving the exact sign in (RP35). Thus the receiving weight per layer is $\log p$, while the cumulative weight through level n is $n \log p$; no factor m has been lost or added. $\square$

The receiver in (RP34) is a Grothendieck-group receiver. It does not tensor the actual finite residue group with $\mathbb{C}$: $\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{C} = 0$, since multiplication by p is invertible on $\mathbb{C}$. The full finite groups and their pairings remain in (RP27)–(RP32). The map through K_0 records their exact length and cardinality and is separately identified as such.

11.7 A trace operator receiving the original prime-power Weil term

Let $\mathcal{H}_p = \ell^2(\mathbb{N}_{\geq 1})$ with distinguished orthonormal vectors $\lambda_{p,m}$ representing the length generator $[C_{p,m}/C_{p,m-1}] = [\mathbb{F}_p]$ in grade m . Define

$$D_p(s) \lambda_{p,m} = p^{-ms} \lambda_{p,m} \quad (\Re s > 0). \quad (\text{RP36})$$

The exact domain is this complex length receiver; it is not the $\mathbb{F}_p$ -module on which a nonintegral complex scalar would be undefined. Its trace norm is $\sum_{m \geq 1} p^{-m\Re s} < \infty$, because finite diagonal truncations have that trace norm and form a Cauchy sequence in trace norm. Therefore

$$(\log p) \operatorname{tr} D_p(s) = L_p(s). \quad (\text{RP37})$$

This proves an exact map from the arithmetic principal-parts filtration through its additive logarithmic-cardinality weight to the logarithmic derivative of the mixed prime differential.

For completeness the resulting receiver in the original prime-power test can be displayed without changing its measure or coefficients. Let $g \in C_c^\infty(\mathbb{R}_{>0})$, let $g^*(x) = \overline{g(x^{-1})}$, use convolution for dx/x , and put $h = g * g^*$ and $\widehat{g}(t) = \int_0^\infty g(x) x^{-it} dx/x$. Then

$$\begin{aligned} W_p(h) &= (\log p) \sum_{m \geq 1} p^{-m/2} (h(p^m) + h(p^{-m})) \\ &= \frac{\log p}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 \operatorname{tr} (D_p(\tfrac{1}{2} + it) + D_p(\tfrac{1}{2} - it)) dt. \end{aligned} \quad (\text{RP38})$$

The first line is the unchanged prime-power expression (ME16) in the same companion proof, and the second line substitutes (RP37) into its exact Mellin receiver. One can also verify the substitution directly: in the coordinate $u = \log x$, Fourier inversion gives $h(e^u) = (2\pi)^{-1} \int |\widehat{g}(t)|^2 e^{itu} dt$. The sum of absolute diagonal coefficients in the two traces is $2p^{-1/2}/(1 - p^{-1/2})$, uniformly in t . Since $\widehat{g}$ is rapidly decreasing, this bound justifies exchanging the integral and the trace series. Substituting $u = \pm m \log p$ gives exactly the first line, including $1/(2\pi)$ and every $p^{-m/2}$ coefficient.

Thus the extra-point universal coupling gives the relative principal parts; the supported prime gives its entire $p^{-n}\mathbb{Z}_p$ filtration; its successive residue layers give the exact $\log p$ weight in the original Weil prime term. The independent finite residue pairing is retained alongside this trace receiver. This computation does not assert positivity of the complete Weil functional: the prime term in the explicit formula retains its original sign, and its pole and Archimedean contributions remain the ones in the companion calculation. Nor does it assert an additive-group module action that contradicts (RP20).

$$\begin{array}{ccc} \mathcal{U}_X & \xrightarrow{Ri^!} & \mathcal{P}_X[-1] \xrightarrow{\text{stalk at } (p)} (\mathbb{Q}_p/\mathbb{Z}_p)[-1] \\ & & \uparrow \\ & & C_{p,n}[-1] \\ C_{p,n}/C_{p,n-1} \cong \mathbb{F}_p & \xrightarrow{\text{class}} & [\mathbb{F}_p] \in K_0^{\text{fl}}(\mathbb{Z}_p) \xrightarrow{w_p} \log p \end{array}$$

Figure 6: Exact routes from the full prime space to its arithmetic receiver. The first row uses the universal coupling in the explicitly specified abelian-sheaf category. Its supported-prime stalk retains every filtration group and the finite pairing (RP29)–(RP30). The lower row records the separate length receiver; summing its successive-layer weights with p^{-ns} gives (RP35), exactly (ME15), and hence the unchanged prime-power Weil expression (RP38). Theorems 11.10, 11.11, 11.12, and 11.14 prove every object and map shown.

11.8 Provenance and exact dependence

The earlier complete derivation `split_zero_adelic_bridge.tex`, Theorem `thm:fullspectral`, Proposition `prop:fullspectralfunctor`, and Corollary `cor:fullrefinement`, constructs the full prime space, its local semiring sheaf, and the valuation-refinement morphism. The construction

needed here is stated and checked again in (RP17). The original-author source ledger in `WORKLOG.md` was read before this derivation. Antoine Sédillot's arXiv:2503.20156v2, original author-TeX lines 623–629 and 831–835, supplies the prior valuation and residue-field context. The recollement, universal coupling, principal-parts receiver, finite pairings, and filtration trace maps in this note are proved here; they are not attributed to that paper. No external literature claim is used as a substitute for these proofs. The companion (ME15)–(ME16) source is included in the same working collection; its use is bounded to the displayed prime logarithmic derivative and exact prime-power test, whose connecting calculation has also been proved in (RP35)–(RP38).

12 The prime-degree operator and its exact Weil pairing

We now use the relative principal-parts object produced by the full spectral extension, rather than only comparing it with older determinant bounds. At $\text{Spec } \mathbb{Z}$ its arithmetic part is $\mathbb{Q}/\mathbb{Z}$, and its p -primary part is

$$D_p = \mathbb{Z}[1/p]/\mathbb{Z} \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p. \quad (\text{PD1})$$

The relative cohomological construction and its receiving category are proved in the preceding recollement chapter. The operators and pairings on (PD1) will all be calculated here. The absolute value remains $|p|_p = p^{-1}$, and the additive Haar measure remains $\mu(\mathbb{Z}_p) = 1$. The use of local Fourier transforms follows the same convention as Connes [4, Section III]. No spectral interpretation of the zeros is assumed.

12.1 The actual filtration, degree and finite character pairing

Put $D_{p,n} = p^{-n}\mathbb{Z}/\mathbb{Z}$, for $n \geq 0$. This is a subgroup of D_p of order p^n , and $D_p = \bigcup_n D_{p,n}$. Mapping rational classes into $\mathbb{Q}_p/\mathbb{Z}_p$ is injective: an element a/p^n lying in $\mathbb{Z}_p$ has $p^n \mid a$ after cancellation. It is onto: the finite negative part of any p -adic expansion differs from the number by an element of $\mathbb{Z}_p$. This proves (PD1), including its filtration. Multiplication by p gives

$$0 \longrightarrow D_{p,1} \longrightarrow D_p \xrightarrow{p} D_p \longrightarrow 0, \quad |D_{p,1}| = p. \quad (\text{PD2})$$

Surjectivity follows by replacing a/p^n by a/p^{n+1} ; the kernel consists of the p classes a/p modulo integers. Every fibre therefore has exactly p elements. The exact layer maps are

$$D_{p,n}/D_{p,n-1} \xrightarrow{\sim} \mathbb{F}_p, \quad [a/p^n] \mapsto a \pmod{p}, \quad n \geq 1. \quad (\text{PD3})$$

They are well defined because changing the representative modulo $D_{p,n-1}$ changes a by a multiple of p . They are injective and onto by the same formula. Thus each layer has module length one and logarithmic cardinality $\log p$.

For $d = a/p^n \pmod{\mathbb{Z}}$ and $x \in \mathbb{Z}_p$, define

$$\chi_d(x) = \exp\left(2\pi i \frac{a(x \bmod p^n)}{p^n}\right). \quad (\text{PD4})$$

Changing representatives of either residue changes the exponent by an integer; passing to denominator p^{n+1} gives the same character. At finite level this is a perfect pairing of $D_{p,n}$ and $\mathbb{Z}/p^n\mathbb{Z}$: a nonzero a is detected by $x = 1$, and a nonzero x is detected by $a = 1$. The geometric-series identity gives

$$p^{-n} \sum_{x \bmod p^n} \chi_d(x) \overline{\chi_{d'}(x)} = \mathbf{1}_{d=d'} \quad (d, d' \in D_{p,n}). \quad (\text{PD5})$$

The Haar measure of each residue class is p^{-n} , so these are also the integrals over $\mathbb{Z}_p$.

Let $\ell^2(D_p)$ use counting measure. Formula (PD5) implies that

$$\mathcal{F}_p : \ell^2(D_p) \longrightarrow L^2(\mathbb{Z}_p, dx), \quad \mathcal{F}_p \delta_d = \chi_d \quad (\text{PD6})$$

extends isometrically from finite sums. It is onto: the p^n orthogonal characters at level n form a basis of the p^n -dimensional space of functions constant on residue classes modulo p^n . The union of these spaces is dense in $L^2(\mathbb{Z}_p)$. One way to see this last assertion is to approximate a continuous function uniformly using uniform continuity on the compact space $\mathbb{Z}_p$ and then use the density of continuous functions for the finite regular Haar measure. In particular $\mathcal{F}_p \delta_0 = \mathbf{1}_{\mathbb{Z}_p}$, with norm exactly one.

12.2 The square-root factor is fixed by the prime degree

The pullback of the actual map (PD2) on finite-support functions is

$$(A_p f)(d) = f(pd), \quad d \in D_p. \quad (\text{PD7})$$

Each fibre has p elements, so direct summation proves $\|A_p f\|^2 = p\|f\|^2$. Consequently its bounded extension has norm exactly $\sqrt{p}$. Define $V_p = p^{-1/2} A_p$ and keep both operators distinguished. The factor in V_p is forced by this degree calculation. Their adjoints and the Fourier transforms are

$$(V_p^* f)(u) = p^{-1/2} \sum_{pd=u} f(d), \quad (\text{PD8})$$

$$(\mathcal{F}_p V_p \mathcal{F}_p^{-1} F)(x) = \sqrt{p} \mathbf{1}_{p\mathbb{Z}_p}(x) F(x/p), \quad (\text{PD9})$$

$$(\mathcal{F}_p V_p^* \mathcal{F}_p^{-1} F)(x) = p^{-1/2} F(px). \quad (\text{PD10})$$

Proof. Taking the counting-measure inner product in (PD7) and collecting terms over each fibre proves (PD8). For finite f , the Fourier transform of $A_p f$ is $\sum_u f(u) \sum_{pd=u} \chi_d(x)$. The inner character sum is zero unless $x \in p\mathbb{Z}_p$: translation of the fibre by $D_{p,1}$ multiplies it by a nontrivial p th root of unity otherwise. For $x = py$ it is $p\chi_u(y)$. This proves (PD9) on the dense finite-support subspace and hence on the whole Hilbert space. Finally changing variables $x = py$, with $dx = p^{-1}dy$, in the inner product for (PD9) proves (PD10). No counting measure or Haar factor has been omitted. $\square$

In particular $V_p^* V_p = I$. In the Fourier representation $V_p^n (V_p^*)^n$ is multiplication by $\mathbf{1}_{p^n \mathbb{Z}_p}$. Their decreasing intersection has measure zero, since $\mu(p^n \mathbb{Z}_p) = p^{-n}$. Thus $\bigcap_n V_p^n \ell^2(D_p) = 0$. The distinction between this isometry and an invertible evolution is resolved by the following explicit extension, not by an assumed inverse.

Theorem 12.1 (Exact unitary dilation and its original vector). *On the actual local space $L^2(\mathbb{Q}_p, dx)$ define*

$$(\mathcal{U}_p F)(x) = \sqrt{p} F(x/p), \quad (\mathcal{U}_p^{-1} F)(x) = p^{-1/2} F(px). \quad (\text{PD11})$$

This is unitary. Extension by zero embeds $L^2(\mathbb{Z}_p)$ isometrically, and $\mathcal{U}_p$ restricts there to (PD9). The union of $\mathcal{U}_p^{-n} L^2(\mathbb{Z}_p)$ is dense in $L^2(\mathbb{Q}_p)$. For the original vector $\xi_p = \mathbf{1}_{\mathbb{Z}_p}$,

$$\langle \mathcal{U}_p^n \xi_p, \xi_p \rangle = p^{-|n|/2}, \quad n \in \mathbb{Z}. \quad (\text{PD12})$$

Its scalar spectral measure on the unit circle is

$$d\nu_p(\theta) = \frac{1}{2\pi} \frac{1 - p^{-1}}{1 - 2p^{-1/2} \cos \theta + p^{-1}} d\theta. \quad (\text{PD13})$$

Proof. Change of variables gives $\int p|F(x/p)|^2 dx = \int |F(y)|^2 dy$, and the two formulas (PD11) compose to the identity. A function supported in $\mathbb{Z}_p$ is carried to one supported in $p\mathbb{Z}_p$, proving the restriction. The inverse images are exactly the spaces supported in $p^{-n}\mathbb{Z}_p$. These sets exhaust $\mathbb{Q}_p$, so truncation of any L^2 function proves density. For $n \geq 0$, $\mathcal{U}_p^n \xi_p = p^{n/2} \mathbf{1}_{p^n \mathbb{Z}_p}$; pairing with $\mathbf{1}_{\mathbb{Z}_p}$ gives $p^{n/2} p^{-n} = p^{-n/2}$. Unitarity gives the complex conjugate value for $-n$, here the same real number. This proves (PD12).

For $0 < r < 1$, absolutely convergent geometric series give $\sum_{n \in \mathbb{Z}} r^{|n|} e^{in\theta} = (1 - r^2)/(1 - 2r \cos \theta + r^2)$. Its constant coefficient is one, and its density is positive. Setting $r = p^{-1/2}$ shows that (PD13) has exactly the moments (PD12). On any Laurent polynomial a , direct expansion then gives $\|a(\mathcal{U}_p)\xi_p\|^2 = \int |a(e^{i\theta})|^2 d\nu_p(\theta)$. Density of Laurent polynomials identifies the cyclic Hilbert space with $L^2(\nu_p)$ and $\mathcal{U}_p$ with multiplication by $e^{i\theta}$ there. This constructs the asserted scalar spectral measure directly. $\square$

12.3 The original prime term is the energy of this operator

Write $a_p = \log p$ and $f(u) = g(e^u)$ for the original multiplicative test g . Define a vector-valued function on the interval $[0, a_p)$ by

$$(\mathcal{E}_p g)(x) = \sum_{n \in \mathbb{Z}} f(x + na_p) \mathcal{U}_p^n \xi_p \in L^2(\mathbb{Q}_p), \quad 0 \leq x < a_p. \quad (\text{PD14})$$

For compact support of f , only finitely many n occur, uniformly in this bounded interval. Thus this is an explicitly defined map into $L^2([0, a_p), dx; L^2(\mathbb{Q}_p))$.

Theorem 12.2 (The relative-prime Hilbert pairing in Weil's formula). *For every $g \in C_c^\infty(\mathbb{R}_{>0})$, with $h = g * g^*$,*

$$\|\mathcal{E}_p g\|^2 = \sum_{j \in \mathbb{Z}} p^{-|j|/2} h(p^j) \quad (\text{PD15})$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 P_{p^{-1/2}}(t \log p) dt. \quad (\text{PD16})$$

In particular the complete arithmetic prime term, including every prime power, is exactly

$$W_p(g * g^*) = (\log p) \left(\|\mathcal{E}_p g\|^2 - \|g\|_{L^2(dx/x)}^2 \right). \quad (\text{PD17})$$

The full finite-place return therefore becomes

$$D_F(g * g^*) = \mathcal{B}(Pg, Pg) - W_\infty(g * g^*) + \sum_{p \in F} (\log p) \left(\|g\|_{L^2(dx/x)}^2 - \|\mathcal{E}_p g\|^2 \right). \quad (\text{PD18})$$

Proof. Use the inner product linear in its first variable. Expanding (PD14) and using (PD12) gives

$$\sum_{n,m} \int_0^{a_p} f(x + na_p) \overline{f(x + ma_p)} p^{-|n-m|/2} dx.$$

Set $j = n - m$ and $u = x + na_p$. For fixed j , the half-open intervals $[na_p, (n+1)a_p)$ partition $\mathbb{R}$ up to measure zero, so the sum is $\sum_j p^{-|j|/2} \int_{\mathbb{R}} f(u) \overline{f(u - ja_p)} du$. The integral is precisely $h(p^j)$ with the original multiplicative convolution, proving (PD15). There are only finitely many nonzero correlations for this test. Fourier inversion of the correlations and the absolutely convergent Poisson series prove (PD16); the dominating sum is $(1+r)/(1-r) |\widehat{g}(t)|^2$ with $r = p^{-1/2} < 1$, which is integrable. The $j = 0$ term of (PD15) is $h(1) = \|g\|^2$. Multiplying the remaining terms by $\log p$ gives the original $W_p(h)$, proving (PD17). Substitution into the unmodified explicit formula proves (PD18). $\square$

12.4 The same determinant and the exact finite support guard

The scalar resolvent on this original vector is also explicit:

$$\langle (I - z\mathcal{U}_p)^{-1}\xi_p, \xi_p \rangle = \sum_{n \geq 0} z^n p^{-n/2} = \frac{1}{1 - zp^{-1/2}}, \quad |z| < 1. \quad (\text{PD19})$$

The operator series is norm convergent in this domain. The scalar series and its last expression continue to $|z| < \sqrt{p}$ without asserting operator resolvent invertibility there. Substituting $z = p^{1/2-s}$ for $\Re s > 1/2$ identifies the scalar with the local determinant receiver $Z_p(s)$ of (ME12); the resulting scalar identity continues to $\Re s > 0$ by the displayed convergent scalar series. This is the exact comparison between the noninvertible prime-degree map, its unitary extension, and the mixed-face determinant.

If $\text{supp } g \subset [\lambda^{-1}, \lambda]$ and $p > \lambda^2$, then $h(p^j) = 0$ for all $j \neq 0$. Formula (PD15) gives $\|\mathcal{E}_p g\|^2 = \|g\|^2$ exactly, so (PD18) agrees with the full prime return once F contains the primes at most λ^2 . It is essential to take the differences before enlarging F ; neither of the two infinite sums of positive energies is asserted to converge separately.

12.5 The precise local weight identity

The unweighted prime pullback A_p has the invertible extension

$$T_p^{\text{rel}} = \sqrt{p} \mathcal{U}_p, \quad (T_p^{\text{rel}} F)(x) = pF(x/p) \quad \text{on } L^2(\mathbb{Q}_p). \quad (\text{PD20})$$

The superscript distinguishes this relative-prime operator from the finite mixed-boundary matrix $T_p = D_p$ of (AB8); the connecting scalar resolvent was proved in (PD19).

Theorem 12.3 (Weight-one norm identity on the computed relative space). *For the actual operator (PD20),*

$$(T_p^{\text{rel}})^* T_p^{\text{rel}} = T_p^{\text{rel}} (T_p^{\text{rel}})^* = pI, \quad \sigma(T_p^{\text{rel}}) = \{z \in \mathbb{C} : |z| = \sqrt{p}\}. \quad (\text{PD21})$$

It has no nonzero Hilbert-space eigenvectors. The spectrum assertion concerns the entire Hilbert spectrum, not a list of zeta eigenvalues.

Proof. The two adjoint identities follow from the proved unitarity of $\mathcal{U}_p$. For the exact spectrum, put $\mathcal{A}_p = \mathbb{Z}_p^\times$ and $H_0 = L^2(\mathcal{A}_p) \subset L^2(\mathbb{Q}_p)$. The measurable disjoint sets $p^n \mathcal{A}_p$, $n \in \mathbb{Z}$, partition $\mathbb{Q}_p \setminus \{0\}$. The missing singleton has Haar measure zero. The summands $\mathcal{U}_p^n H_0$ are therefore mutually orthogonal and their Hilbert direct sum is the whole space. Choose a nonzero unit vector $h \in H_0$; then $h_n = \mathcal{U}_p^n h$ is orthonormal. For $|\lambda| = 1$ put $v_N = (2N+1)^{-1/2} \sum_{n=-N}^N \lambda^{-n} h_n$. Cancellation at the interior indices gives $\|(\mathcal{U}_p - \lambda I)v_N\|^2 = 2/(2N+1)$, while $\|v_N\| = 1$. Hence every such λ is in the spectrum. For $|\lambda| > 1$, the Neumann series for $-\lambda^{-1}(I - \lambda^{-1}\mathcal{U}_p)^{-1}$ inverts $\mathcal{U}_p - \lambda I$; for $|\lambda| < 1$ use $(I - \lambda\mathcal{U}_p^{-1})^{-1}\mathcal{U}_p^{-1}$. Thus there is no other spectrum. Scaling proves (PD21).

If $\mathcal{U}_p F = \lambda F$, then $|\lambda| = 1$ follows by taking norms unless $F = 0$. The orthogonal components in $\mathcal{U}_p^n H_0$ consequently have equal norms for all n . Square summability forces them all to be zero. Scaling again proves the assertion for T_p^{rel} . $\square$

For comparison, Deligne's actual purity result in Weil II, (3.3.6), concerns the image of compactly supported cohomology in ordinary cohomology, using the upper-weight theorem and its dual lower-weight theorem [9]. Formula (PD21) supplies a complex weight-one norm identity on the relative object explicitly constructed here. The remaining programme step is to calculate the corresponding image for the full split cohomology and its trace, with the maps (ME13), (PD14) and the complete expression (PD18). No identification of that image with $L^2(\mathbb{Q}_p)$, and no full cohomological purity, is asserted by the local calculation.

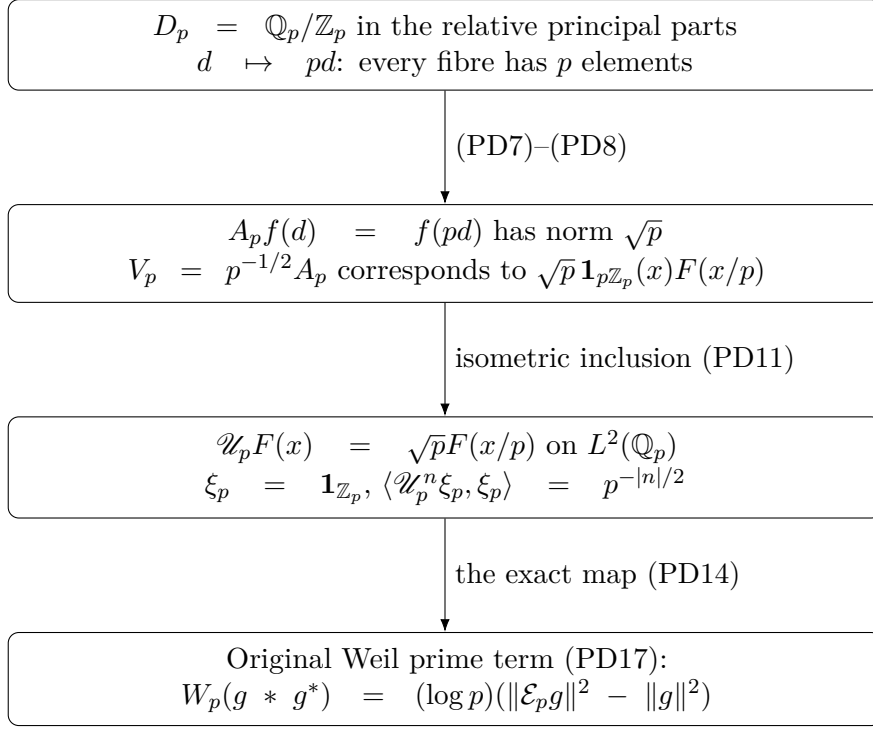


Figure 7: Use of the relative prime data in the original arithmetic pairing. The prime degree fixes every factor of $p^{1/2}$; the original Haar measure fixes every norm. Theorems 12.1 and 12.2 prove the arrows. The formula retains the subtracted diagonal energy, both poles and the Archimedean contribution in (PD18), in the conventions of [4, 5].

These calculations establish a positive local Hilbert representation constructed from the relative prime object and the exact signed map from it into the Weil functional. The full inequality for (PD18) has not been proved: positivity of each individual squared norm does not assign the sign of their displayed difference. The new task supplied by this construction is the full pairing of these relative-prime operators with the interior and Archimedean objects, with the subtraction and the two endpoint terms retained.

13 All integers, including supported zero, on the relative object

The local prime construction extends to the whole relative group $D = \mathbb{Q}/\mathbb{Z}$. We keep its supported zero 0_D and the newly absent element τ_D distinct. In particular the integer $e = 0_{\mathbb{Z}}$ is included in the scalar action, with its exact domain and codomain.

13.1 Simultaneous arithmetic action and the original integral vector

For $n \in \mathbb{Z} \setminus \{0\}$ the map $d \mapsto nd$ on D is onto with kernel $n^{-1}\mathbb{Z}/\mathbb{Z}$, of cardinality $|n|$. Therefore

$$(A_n f)(d) = f(nd), \quad \|A_n f\|^2 = |n| \|f\|^2, \quad V_n = |n|^{-1/2} A_n \quad (\text{IA1})$$

on $\ell^2(D)$ with counting measure. These formulas follow by counting each fibre, as in (PD7), and $A_m A_n = A_{mn}$ follows by composition. The finite character pairings identify $\ell^2(D)$ with $L^2(\widehat{\mathbb{Z}})$, where $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ has Haar measure one. Indeed every $d = a/N$ gives the character $x \mapsto \exp(2\pi i a(x \bmod N)/N)$; the finite cyclic orthogonality proof (PD5) applies with N in place of p^n . The functions factoring through the finite quotients form a dense subspace, by compactness and the same approximation argument.

Let $\mathbb{A}_f$ denote the restricted product of the $\mathbb{Q}_p$ with respect to $\mathbb{Z}_p$, with its original additive Haar measure $\mu(\widehat{\mathbb{Z}}) = 1$. For a positive rational a define

$$(\mathcal{U}_a F)(x) = \sqrt{a} F(x/a), \quad F \in L^2(\mathbb{A}_f). \quad (\text{IA2})$$

The finite-adelic modulus is $|a|_f = a^{-1}$: this follows first for a prime from $|p|_p = p^{-1}$ and $|p|_q = 1$ for $q \neq p$, and then for all positive rationals by prime factorization. Changing variables in (IA2) proves unitarity, and composition proves $\mathcal{U}_a \mathcal{U}_b = \mathcal{U}_{ab}$. For a positive integer n its restriction to $L^2(\widehat{\mathbb{Z}})$ is the Fourier transform of V_n . The proof is the finite character sum over the kernel of multiplication by n : it is zero off $n\widehat{\mathbb{Z}}$, and equals n times the original character on that subgroup. This gives the formula $\sqrt{n} \mathbf{1}_{n\widehat{\mathbb{Z}}}(x) F(x/n)$, including its factor.

Put $\xi = \mathbf{1}_{\widehat{\mathbb{Z}}}$, of norm one. For $a = m/n > 0$ with $\gcd(m, n) = 1$,

$$\langle \mathcal{U}_a \xi, \xi \rangle = \frac{1}{\sqrt{mn}}. \quad (\text{IA3})$$

In fact $\widehat{\mathbb{Z}} \cap a\widehat{\mathbb{Z}} = m\widehat{\mathbb{Z}}$, as checked at each valuation, and its measure is $1/m$. Multiplication by $\sqrt{m/n}$ proves the displayed equality. In particular, for all positive integers m, n ,

$$\langle \mathcal{U}_m \xi, \mathcal{U}_n \xi \rangle = \frac{\gcd(m, n)}{\sqrt{mn}}. \quad (\text{IA4})$$

This retains every integral scalar, not only primes or prime powers. For a finite set of primes the joint scalar spectral measure is exactly the product of the local measures (PD13): its joint Fourier coefficient at (k_p) is $\prod_p p^{-|k_p|/2}$ by (IA3). Expanding Laurent polynomials proves the equality of their pairings with the actual operators, just as in the proof of (PD13).

The determinant of the matrix (IA4) for $1 \leq m, n \leq N$ is

$$\det \left(\frac{\gcd(m, n)}{\sqrt{mn}} \right)_{1 \leq m, n \leq N} = \prod_{d=1}^N \frac{\varphi(d)}{d} > 0. \quad (\text{IA5})$$

Here $\varphi(d)$ counts the units of $\mathbb{Z}/d\mathbb{Z}$, with $\varphi(1) = 1$. For completeness, sorting the integers $1, \dots, k$ by the value of $\gcd(a, k)$ proves $\sum_{d|k} \varphi(d) = k$: those with $\gcd(a, k) = k/d$ correspond bijectively to residues coprime to d . Thus, for the lower-triangular matrix $E_{md} = \mathbf{1}_{d|m}$,

$$(\gcd(m, n))_{m, n} = E \operatorname{diag}(\varphi(1), \dots, \varphi(N)) E^t.$$

The diagonal of E consists of ones, so its determinant is one. Taking determinants and retaining the row and column factors $m^{-1/2}, n^{-1/2}$ proves (IA5). This determinant identity is classical, associated with H. J. S. Smith [13]; the complete proof is given here. The calculation specific to this construction is its realization as the Gram matrix of the relative-prime operators on the original integral vector.

13.2 Supported zero acts in an explicitly computed distribution space

Let $\widetilde{D} = G(D) = D \sqcup \{\tau_D\}$ with its standard split-semimodule scalar action. For supported $n \in \mathbb{Z}$, $n \cdot d = nd$ is supported and $n \cdot \tau_D = \tau_D$. For the absent scalar τ , every product is τ_D . Let

$$\mathcal{T} = \mathbb{C}^{(D)} \oplus \mathbb{C} \delta_{\tau_D}, \quad \mathcal{T}' = \mathbb{C}^D \oplus \mathbb{C}, \quad \langle f, h \rangle = \sum_{d \in D} f(d) h(d) + c_f c_h. \quad (\text{IA6})$$

The first factor of $\mathcal{T}$ consists of finite-support functions, so the displayed pairing is defined for every $h \in \mathcal{T}'$. It is the full algebraic dual pairing. No square-summability condition is imposed on a distribution in $\mathcal{T}'$.

Pullback by the scalar maps defines operators R_a on all functions $\mathcal{T}'$ satisfying $R_a R_b = R_{ab}$ and $R_1 = I$. For nonzero supported n , R_n restricts to (IA1) on D and fixes the absent coordinate. For the two distinct zeros it gives, exactly,

$$R_e(f, c) = (f(0_D)\mathbf{1}_D, c), \quad R_\tau(f, c) = (c\mathbf{1}_D, c). \quad (\text{IA7})$$

In particular

$$R_e\delta_{0_D} = (\mathbf{1}_D, 0), \quad R_\tau\delta_{0_D} = 0, \quad R_e\delta_{\tau_D} = \delta_{\tau_D}, \quad R_\tau\delta_{\tau_D} = (\mathbf{1}_D, 1). \quad (\text{IA8})$$

Every equality follows by evaluating at a supported d and at τ_D . They prove, rather than identify away, the difference between the action of the integer zero and the absent scalar. Composition of the original scalar maps proves the representation law also when a or b is either zero. This is a representation of the *multiplicative scalar monoid*; no false additive identity $R_{a+b} = R_a + R_b$ is asserted for pullbacks of functions.

There is an exact reason to keep the distribution space (IA6). The nonzero function $\mathbf{1}_D$ has infinite counting-measure norm. Thus R_e has no bounded extension on $\ell^2(D)$ agreeing with (IA8). Its full action nonetheless exists on $\mathcal{T}'$ and its restriction $\mathcal{T} \rightarrow \mathcal{T}'$ is given by (IA7). Its Fourier transform on the arithmetic component is

$$f(0_D)\mathbf{1}_D \longleftrightarrow \left(\int_{\widehat{\mathbb{Z}}} \mathcal{F}f(x) dx \right) \delta_0. \quad (\text{IA9})$$

To prove this without a convergence claim, test against a character polynomial $F = \sum_d a_d \chi_d$. The distribution with all Fourier coefficients one takes the value $\sum_d a_d = F(0)$, which is exactly δ_0 . Orthogonality (PD5) gives $f(0_D) = \int \mathcal{F}f$. This proves both sides of (IA9) on every test. The algebraic transpose of R_e on distributions is also represented on finite-support tests: it sends (f, c) to $((\sum_d f(d))\delta_{0_D}, c)$. Direct substitution in (IA6) verifies the transpose identity. Thus the mass and supported-zero evaluation terms are explicit adjoint boundary maps.

The integer 0 has therefore remained in every scalar map. Its action lands at a boundary distribution that the Hilbert completion alone does not contain. The maps (IA7)–(IA9) specify this enlargement and its relation to the positive Hilbert operators (IA1)–(IA4). Any purity or duality argument using that Hilbert realization must retain this computed zero-scalar boundary and its transpose, rather than quietly delete the integer zero from the programme.

14 The whole finite support spectrum and its relative cohomology

The coefficient source is the original globalization note and the full lattice extension [1, 3]. The complete construction and proofs follow.

14.1 The original semiring and its exact topology

Let L be a finite nontrivial bounded distributive lattice. Its bottom and top are 0_L and 1_L . For a nonzero ring R , retain the semiring

$$\mathbf{G}_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{a} = (a, 1_L) : a \in R\} \subset R \times L. \quad (\text{FL1})$$

The overlap is $e = z_{1_L} = \widehat{0}$, and the additive identity is $\tau = z_{0_L} \neq e$. Addition is coefficient addition and lattice join; multiplication is coefficient multiplication and lattice meet. In particular

$$z_\lambda + z_\mu = z_{\lambda \vee \mu}, \quad z_\lambda z_\mu = z_{\lambda \wedge \mu}, \quad \widehat{a} + z_\lambda = \widehat{a}, \quad \widehat{a} z_\lambda = z_\lambda. \quad (\text{FL2})$$

Every coefficient and every support label remains present.

Proposition 14.1 (All strata and all opens). *Put $S = \text{Spec}_{\text{lat}} L$ and $X = \text{Spec } R$. The prime spectrum of $\mathbb{G}_L(R)$ is the disjoint union $Y = S \sqcup X$ with primes*

$$P_{\mathfrak{a}} = \{z_\lambda : \lambda \in \mathfrak{a}\}, \quad Q_{\mathfrak{p}} = \{z_\lambda : \lambda \in L\} \cup \{\widehat{a} : a \in \mathfrak{p}\}. \quad (\text{FL3})$$

Here $\mathfrak{a}$ is a prime lattice ideal and $\mathfrak{p}$ a ring prime. The order in either stratum is its original prime-inclusion order, and $P_{\mathfrak{a}} \subsetneq Q_{\mathfrak{p}}$ for every pair. The opens of Y are exactly the opens of S and the sets $S \cup U$ with nonempty open $U \subset X$. In particular $D(e) = S$ and $V(e) = X$.

Proof. An ideal containing a supported element $\widehat{a}$ contains e because $\widehat{a}e = e$, and then contains every z_λ because $ez_\lambda = z_\lambda$. Its supported coefficients form a ring ideal: closure under negatives is multiplication by $\widehat{-1}$. Its primality is exactly primality of that ring ideal. An ideal with no supported element consists of zero-coefficient labels from a proper lattice ideal. Addition gives join closure and multiplication by z_μ for $\mu \leq \lambda$ gives downward closure. Its primality is exactly the meet-prime condition. Conversely these constructions give ideals and verify the corresponding primality conditions, proving (FL3). A pure support prime omits e whereas every arithmetic prime contains the whole zero-coefficient lattice. This proves every stated inclusion. Finally the basic opens are

$$D(z_\lambda) = \{P_{\mathfrak{a}} : \lambda \notin \mathfrak{a}\}, \quad D(\widehat{a}) = S \cup D_R(a). \quad (\text{FL4})$$

Unions of these give exactly the asserted topology and both stratum identifications. $\square$

For completeness, the support spectrum here is nonempty and T_0 . Choose a maximal proper lattice ideal I , possible because L is finite and nontrivial. If $a, b \notin I$ but $a \wedge b \in I$, maximality gives $u \vee a = 1_L$ and $v \vee b = 1_L$ for some $u, v \in I$. Distributivity expands $(u \vee a) \wedge (v \vee b)$ as a join of four elements of I , including $a \wedge b$, a contradiction. Thus I is prime. Distinct prime ideals differ on some label λ , and the basic open $D(z_\lambda)$ separates them, proving T_0 .

More generally, take any nonempty finite T_0 space S and any nonempty irreducible space X and give $Y = S \triangleright X$ the same topology. Write $j : S \hookrightarrow Y$ for the open inclusion and $i : X \hookrightarrow Y$ for the closed inclusion. The applications below use $X = \text{Spec } \mathbb{Z}$. No semiring additive-module action is implicitly imposed on the abelian sheaves: the receiving category throughout is the category of abelian sheaves on this whole topology.

We use the specialization convention that $s \leq t$ means that s generalizes t . Thus opens in a finite T_0 space are lower sets, the smallest neighbourhood of s is $\downarrow s$, and a sheaf has a restriction map from its stalk at t to its stalk at s when $s \leq t$. This fixes all cochain orientations below.

14.2 Sheaves, universal coupling, and derived support

Let $c_X B$ denote the constant sheaf on the irreducible space X : its value on every nonempty open is B and every such restriction is the identity. Let $c_S B$ denote the usual constant sheaf on S ; its value on an open $V \subset S$ is one copy of B for each connected component of V . For an abelian sheaf $\mathcal{A}$ on X define

$$g\mathcal{A} = \varinjlim_{\emptyset \neq U \subset X} \mathcal{A}(U). \quad (\text{FL5})$$

The nonempty opens form a filtered restriction system because X is irreducible. The functor g is exact. Indeed a sheaf epimorphism lifts a section on some nonempty member of an open cover, which suffices for its generic germ, and a generic germ with zero image becomes zero on a smaller nonempty open, where the kernel sheaf gives exactness. There are adjunctions

$$g \dashv c_X, \quad c_S \dashv \Gamma(S, -). \quad (\text{FL6})$$

The first expresses that compatible maps $\mathcal{A}(U) \rightarrow B$ are exactly maps from their colimit; the second follows because a map from constant sections is determined by their global images and

then sheafification. Both constant-sheaf functors are exact, as is seen on stalks. In particular c_X preserves injectives, and applying it to an injective resolution of B proves

$$R\Gamma(X, c_X B) = B[0]. \quad (\text{FL7})$$

Theorem 14.2 (Full gluing over a finite support space). *An abelian sheaf on Y is exactly a triple*

$$\mathcal{F} = (\mathcal{A}, \mathcal{B}, \phi), \quad \phi : \mathcal{A} \longrightarrow c_X \Gamma(S, \mathcal{B}), \quad (\text{FL8})$$

with $\mathcal{A}$ a sheaf on X and $\mathcal{B}$ a sheaf on S . Its sections on $S \cup U$ are $\mathcal{A}(U)$ and its sections on an open $V \subset S$ are $\mathcal{B}(V)$. Restriction from $S \cup U$ to V is ϕ_V followed by restriction from S to V . A morphism (a, b) satisfies $\phi'a = c_X \Gamma(S, b)\phi$. Exactness is detected on the two components. The elementary functors are

$$\begin{aligned} i^* \mathcal{F} &= \mathcal{A}, & i_* \mathcal{A} &= (\mathcal{A}, 0, 0), & i^! \mathcal{F} &= \ker \phi, \\ j^* \mathcal{F} &= \mathcal{B}, & j_! \mathcal{B} &= (0, \mathcal{B}, 0), & j_* \mathcal{B} &= (c_X \Gamma(S, \mathcal{B}), \mathcal{B}, \text{id}). \end{aligned} \quad (\text{FL9})$$

Moreover the exact left adjoint of i^* is

$$U_S \mathcal{A} = (\mathcal{A}, c_S g \mathcal{A}, \epsilon_S), \quad (\text{FL10})$$

where ϵ_S sends a section to the constant support section with its original generic germ. This is the universal coupling to the entire support space.

Proof. Starting with a sheaf on Y , set $\mathcal{A}(U) = \mathcal{F}(S \cup U)$ for nonempty U and $\mathcal{B} = j^* \mathcal{F}$. Any pair of nonempty arithmetic opens intersects. Consequently the sheaf gluing condition for the opens $S \cup U_a$ is exactly that for the corresponding opens U_a of X . These values therefore form a sheaf $\mathcal{A}$. Restriction to all of S gives ϕ . The mixed open $S \cup U$ is the smallest open of Y containing U , so the resulting sheaf is precisely $i^* \mathcal{F}$.

Conversely a triple gives the displayed restrictions. A cover of a mixed open has mixed members covering its arithmetic part. Sections on those members glue in $\mathcal{A}$; compatibility with any member lying wholly in S is exactly its prescribed restriction from $\Gamma(S, \mathcal{B})$. Covers lying wholly in S use the sheaf axiom for $\mathcal{B}$. This proves the sheaf axiom and the inverse classification. The morphism equation follows from restriction to S . Kernels and cokernels are computed on the two strata: they are the inverse-image sheaves on X and S , and stalkwise exactness on these disjoint strata is exactness on Y . The induced cokernel arrow is obtained by composing to $\Gamma(S, \text{coker } b)$; no exactness of $\Gamma(S, -)$ is being assumed.

The formulas for $i^*, i_*, i^!, j^*, j_!, j_*$ follow by the Hom adjunctions: a map from $i_* \mathcal{C}$ lands in $\ker \phi$, and a map to $j_* \mathcal{D}$ has its arithmetic part forced by its support part. The putative $j_!$ has zero sections on every mixed open. This is a sheaf because a cover of a mixed open has a mixed member whose intersection with S is all of S , forcing any compatible support section to vanish. It has the required left adjunction to j^* .

For the final adjunction, a map $U_S \mathcal{C} \rightarrow \mathcal{F}$ with arithmetic part a has its support map uniquely determined by the composite $\mathcal{C} \xrightarrow{a} \mathcal{A} \xrightarrow{\phi} c_X \Gamma(S, \mathcal{B})$: apply successively the two adjunctions in (FL6). Conversely that map satisfies the gluing equation. Thus $U_S \dashv i^*$. Exactness of U_S follows from exactness of g, c_S , and the componentwise exactness already proved. All assertions follow without dropping a support point. $\square$

The distinction from a single open point is that $\Gamma(S, -)$, and hence j_* , can have higher derived functors. They must be retained. We use cohomological degrees, so $B[-1]$ has B in degree one.

Theorem 14.3 (Derived gluing including support higher cohomology). *For $\mathcal{F} = (\mathcal{A}, \mathcal{B}, \phi)$ there are natural formulas*

$$Ri^! \mathcal{F} \simeq \text{Cone}(\mathcal{A} \longrightarrow c_X R\Gamma(S, \mathcal{B}))[-1], \quad (\text{FL11})$$

$$R\Gamma(Y, \mathcal{F}) = R\Gamma(X, \mathcal{A}), \quad R\Gamma_X(Y, \mathcal{F}) \simeq \text{Cone}(R\Gamma(X, \mathcal{A}) \longrightarrow R\Gamma(S, \mathcal{B}))[-1]. \quad (\text{FL12})$$

The maps are induced by the actual restriction to S in (FL8). The localization triangle is

$$i_* Ri^! \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow Rj_* \mathcal{B} \longrightarrow i_* Ri^! \mathcal{F}[1]. \quad (\text{FL13})$$

Here $R\Gamma_X(Y, -)$ denotes derived sections supported in the closed arithmetic stratum. The formula does not replace $R\Gamma(S, \mathcal{B})$ by its zeroth cohomology.

Proof. The functors i^* and j^* are exact. Their right adjoints i_* and j_* preserve injectives. Also i^* preserves injectives because its left adjoint U_S is exact, and j^* preserves injectives because $j_!$ is exact. If $\mathcal{I}$ is injective on Y , write $\mathcal{B}_I = j^* \mathcal{I}$. Extend the identity support map $j_! \mathcal{B}_I \rightarrow \mathcal{I}$ along the monomorphism $j_! \mathcal{B}_I \rightarrow j_* \mathcal{B}_I$. Injectivity supplies a map $j_* \mathcal{B}_I \rightarrow \mathcal{I}$ right inverse to the restriction $\mathcal{I} \rightarrow j_* \mathcal{B}_I$: on S its composite is the identity, and on X the gluing equation forces the identity on $c_X \Gamma(S, \mathcal{B}_I)$. Thus

$$0 \rightarrow i_* i^! \mathcal{I} \rightarrow \mathcal{I} \rightarrow j_* j^* \mathcal{I} \rightarrow 0$$

is split exact. Apply this to an injective resolution of $\mathcal{F}$. Its support restriction is an injective resolution of $\mathcal{B}$; the corresponding j_* complex computes $Rj_* \mathcal{B}$. The kernel complex computes $i_* Ri^! \mathcal{F}$, proving (FL13). Restriction to X turns the j_* complex into $c_X \Gamma(S, j^* \mathcal{I}^\bullet)$, which is exactly $c_X R\Gamma(S, \mathcal{B})$. This proves (FL11). Global sections on Y are the sections of $\mathcal{A}$ on X , and i^* preserves injectives, proving the first formula in (FL12). Applying global sections to (FL11) and using (FL7) proves the second. At degree zero the kernel consists precisely of sections restricting to zero on the open stratum, which verifies the asserted support interpretation. $\square$

14.3 Finite cochains with their complete orientation

For the next calculation, identify a finite T_0 space with its poset under the convention above. A sheaf $\mathcal{B}$ is equivalently the diagram of stalks $B_s = \mathcal{B}(\downarrow s)$ and restriction maps $r_{t,s} : B_t \rightarrow B_s$ for $s \leq t$. Indeed a section on a lower open is a compatible family of these germs, since the smallest neighbourhoods cover it, and sheaf gluing recovers exactly such families.

Theorem 14.4 (The full finite cochain resolution). *For a sheaf of rational vector spaces on a finite T_0 space S , $R\Gamma(S, \mathcal{B})$ is represented by*

$$C^n(S, \mathcal{B}) = \prod_{s_0 < \dots < s_n} B_{s_0}, \quad (\text{FL14})$$

with differential

$$(dc)_{s_0 < \dots < s_{n+1}} = r_{s_1, s_0} c_{s_1 < \dots < s_{n+1}} + \sum_{a=1}^{n+1} (-1)^a c_{s_0 < \dots < \widehat{s_a} < \dots < s_{n+1}}. \quad (\text{FL15})$$

For constant coefficients $\mathbb{Q}$ every restriction is the identity, so the degree-zero differential is $(db)_{s < t} = b_t - b_s$. No chain with a repeated point is included.

Proof. Computing with rational sheaves gives the same derived sections as computing with their underlying abelian sheaves. Indeed the forgetful functor is right adjoint to sheafwise rationalization $\mathbb{Q} \otimes_{\mathbb{Z}} -$; rationalization is exact because it is localization of abelian groups. Thus an injective rational sheaf remains injective after forgetting its rational scalar action. For $t \in S$ and a rational vector space M , let $I_t(M)$ be the sheaf with value M on an open containing t and zero otherwise. This is direct image from the one-point subspace. Its stalk at q is M if $t \leq q$ and zero otherwise. It is right adjoint to the exact stalk functor at t . Since every rational vector space is injective, $I_t(M)$ is an injective sheaf.

Define the sheaf

$$\mathcal{C}^n = \prod_{s_0 < \dots < s_n} I_{s_n}(B_{s_0}).$$

Its value on a lower open V is the product over all chains contained in V , since having $s_n \in V$ forces every preceding point into V . The formula (FL15) therefore defines a sheaf differential. Expanding d^2 cancels every twice-deleted face with the opposite order of deletion; the two terms using a restriction cancel because $r_{s_2, s_0} = r_{s_1, s_0} r_{s_2, s_1}$. The augmentation $\mathcal{B} \rightarrow \mathcal{C}^0$ is the family of germs.

At a stalk q , the augmented complex is that on $\downarrow q$, which has maximal point q . Its contracting map from degree zero to the augmentation is evaluation at q . For $n \geq 1$, its contracting map from degree n to degree $n - 1$ is

$$(hc)_{s_0 < \dots < s_{n-1}} = \begin{cases} (-1)^n c_{s_0 < \dots < s_{n-1} < q}, & s_{n-1} < q, \\ 0, & s_{n-1} = q. \end{cases}$$

The identity $dh + hd = 1$ follows by pairing the faces omitting one of $s_0, \dots, s_{n-1}$, which cancel, leaving only the face omitting q . In degree zero the additional term is the restriction of the value at q , so that identity includes the augmentation. This proves stalkwise exactness of the augmented complex. The sheaves $\mathcal{C}^n$ are finite products of injectives and vanish above the maximum strict-chain length. They give an injective resolution, whose global sections are precisely (FL14). This proves the claimed derived computation with all its signs. $\square$

For an order-preserving map $f : S' \rightarrow S$, the induced pullback on constant-coefficient cochains is

$$(f^*c)_{t_0 < \dots < t_n} = \begin{cases} c_{f(t_0) < \dots < f(t_n)}, & f(t_0), \dots, f(t_n) \text{ are distinct,} \\ 0, & \text{otherwise.} \end{cases} \quad (\text{FL16})$$

At degree zero it is the usual coordinate pullback. This commutes with d . If the image chain has no repetitions it is immediate from (FL15). If it has a repetition, terms deleting either member of an adjacent equal pair have the same value and opposite signs; all other terms still have a repetition and are zero. This proves the identity in that case as well. The construction is functorial and gives the map induced by restriction on the injective cochain resolutions and their augmentations. More explicitly, at a point $t \in S'$ it pulls cochains on $\downarrow f(t)$ back to cochains on $\downarrow t$, by the displayed formula. These maps commute with restriction to smaller minimal opens, so define a sheaf cochain map from the inverse-image resolution to the resolution on S' . Its augmentation is the identity on constant coefficients. This identifies its global cochain map with the actual derived pullback. Equivalently it is the map on the cochains of strict chains after removing repeated chains; the cancellation just proved verifies that removal directly.

14.4 The universal arithmetic sheaf for every finite support

Take $X = \text{Spec } \mathbb{Z}$, so its generic stalk is $\mathbb{Q}$. Define the actual sheaf on the whole ordinal-sum space

$$\mathcal{U}_S = U_S \mathcal{O}_X = (\mathcal{O}_X, \mathbb{Q}_S, \text{generic constant restriction}). \quad (\text{FL17})$$

The morphism sends each original regular integer-valued rational section to its same rational generic value at every support point. In particular it is diagonal on disconnected support components.

We will use $R\Gamma(X, \mathcal{O}_X) = \mathbb{Z}[0]$, including its vanishing in higher degrees. Here is a direct verification in this case. On a nonempty open $U = D(N)$, regular functions are the rational numbers whose denominator has prime factors only among those dividing N . The quotient sheaf $\mathcal{P}_X = \mathbb{Q}_X / \mathcal{O}_X$ has

$$\mathcal{P}_X(U) = \bigoplus_{p \in U \text{ closed}} \mathbb{Q} / \mathbb{Z}_{(p)}. \quad (\text{FL18})$$

The rational section map onto this direct sum is surjective: represent each of finitely many prescribed p classes by a_p/p^{n_p} and sum them; each other summand is integral at that prime. Its kernel is precisely $\mathcal{O}_X(U)$. Formula (FL18) is a sheaf, since opens of this Noetherian space are quasi-compact and compatible finite-support sections glue with finite support. Restrictions discard the removed primes and are surjective, so $\mathcal{P}_X$ is flasque. The rational constant sheaf is also flasque. Their exact sequence is therefore an acyclic resolution of $\mathcal{O}_X$. The surjectivity just proved at $U = X$ makes its global-sections complex have only $\mathbb{Z}$ in degree zero, proving the claimed calculation. The acyclicity of flasque sheaves follows from the usual resolution argument: local lifts modulo a flasque kernel extend and glue by successive extension over open sets, so a flasque resolution has an exact global-sections complex in positive degrees. This argument applies to both displayed sheaves.

Theorem 14.5 (A complete relative complex for the full support lattice). *Put $C^\bullet = C^\bullet(S, \mathbb{Q})$ with the exact differential (FL15). The relative complex of $\mathcal{U}_S$ is represented by*

$$K_S^0 = \mathbb{Z}, \quad K_S^n = C^{n-1} \ (n \geq 1), \quad d^0(z) = (z)_{s \in S}, \quad d^n = d_C^{n-1} \ (n \geq 1). \quad (\text{FL19})$$

These signs identify it with the mapping fibre in (FL12). Explicitly use fibre coordinates $(a, c) \in A^n \oplus C^{n-1}$ with differential $(a, c) \mapsto (d_A a, f(a) - d_C c)$; this is a model of $\text{Cone}(f)[-1]$. In support degree $n - 1$ multiply this coordinate by $(-1)^{n-1}$ to obtain the displayed positive support differential. Let c be the number of connected components of S . Then

$$H_X^0(Y, \mathcal{U}_S) = 0, \quad H_X^1(Y, \mathcal{U}_S) = \mathbb{Q}^c / \Delta \mathbb{Z}, \quad H_X^n(Y, \mathcal{U}_S) = H^{n-1}(S, \mathbb{Q}) \ (n \geq 2). \quad (\text{FL20})$$

On the arithmetic stratum, the entire derived object is

$$Ri^! \mathcal{U}_S \simeq [\mathcal{O}_X \longrightarrow c_X C^0 \longrightarrow c_X C^1 \longrightarrow \cdots], \quad (\text{FL21})$$

with $\mathcal{O}_X$ in degree zero, the displayed generic constant map in degree zero, and the same support differential thereafter. In particular

$$R^1 i^! \mathcal{U}_S = (\mathbb{Q}^c)_X / \Delta \mathcal{O}_X, \quad R^n i^! \mathcal{U}_S = c_X H^{n-1}(S, \mathbb{Q}) \ (n \geq 2). \quad (\text{FL22})$$

Proof. Theorem 14.4 computes the actual support derived sections, and (FL17) sends a rational number to its constant zero-cochain. Substitute these maps and $R\Gamma(X, \mathcal{O}_X) = \mathbb{Z}[0]$ into Theorem 14.3. The indicated sign change gives exactly (FL19) and (FL21); it does not change their maps on degree-zero constants. The kernel of d_C^0 consists of functions equal along every comparable pair, hence constant on each connected component of the comparability graph. Those graph components are exactly the connected components of the finite space: a union of graph components is open and closed, whereas a partition into two open-and-closed sets cannot separate a comparable pair. Thus that kernel is $\mathbb{Q}^c$. The constant integer map is injective since S is nonempty. Taking cohomology now gives (FL20). Stalkwise the original local domain embeds into its generic field, and the same calculation of the kernel and higher differentials gives (FL22). This proves the derived formulas and all connecting maps from the actual sheaf data. $\square$

There is a canonical diagonal inclusion of the one-point relative complex

$$[\mathbb{Z} \hookrightarrow \mathbb{Q}] \longrightarrow K_S^\bullet, \quad (\text{FL23})$$

which is the identity on $\mathbb{Z}$, the all-points constant map on $\mathbb{Q}$, and zero in higher degrees. At the sheaf level this is $[\mathcal{O}_X \hookrightarrow \mathbb{Q}_X] \rightarrow Ri^! \mathcal{U}_S$. Its quotient is the shifted reduced support complex

$$\tilde{C}^0 = C^0 / \Delta \mathbb{Q}, \quad \tilde{C}^n = C^n \ (n \geq 1), \quad \text{placed starting in relative degree one.} \quad (\text{FL24})$$

Thus the original principal parts and the complete remaining support complex are linked by an explicit short exact sequence of complexes. For any specified support point s , evaluation of C^0 at s , zero in positive support degrees, and the identity on $\mathbb{Z}$ give a retraction of (FL23). It is a cochain map because the target has no positive support terms. This splits the complexes once that point is chosen. The diagonal inclusion itself requires no chosen point.

14.5 Every Boolean branch and every deletion map

Let I be a nonempty finite set of cardinality r and take $L = \mathcal{P}(I) = \mathbb{B}^r$. Its prime lattice ideals are $\mathfrak{a}_b = \{A \subset I : b \notin A\}$ for $b \in I$. Indeed a meet-prime ideal must omit some atom; it cannot omit two distinct atoms because their intersection is empty, and any ideal omitting the atom $\{b\}$ must contain every other atom. Hence these are all primes. They are incomparable, so S_I is discrete.

Theorem 14.6 (The complete Boolean relative object). *For $Y_I = S_I \triangleright \text{Spec } \mathbb{Z}$ the universal arithmetic coupling has*

$$Ri^! \mathcal{U}_I = [\mathcal{O}_X \xrightarrow{\Delta} \mathbb{Q}_X^I], \quad H_X^1(Y_I, \mathcal{U}_I) = \mathbb{Q}^I / \Delta \mathbb{Z}, \quad (\text{FL25})$$

and all other relative cohomology groups vanish. Its exact canonical sequence is

$$0 \longrightarrow \mathbb{Q}/\mathbb{Z} \xrightarrow{d} \mathbb{Q}^I / \Delta \mathbb{Z} \xrightarrow{q} \mathbb{Q}^I / \Delta \mathbb{Q} \longrightarrow 0, \quad d(a + \mathbb{Z}) = [(a)_{b \in I}], \quad q([v]) = v + \Delta \mathbb{Q}. \quad (\text{FL26})$$

For a chosen branch $b \in I$, the branch map is $\beta_b([v]) = v_b + \mathbb{Z}$ and a splitting of the sequence is

$$v + \Delta \mathbb{Q} \longmapsto [v - v_b \mathbf{1}_I]. \quad (\text{FL27})$$

In particular the explicit branch coordinates are

$$\mathbb{Q}^I / \Delta \mathbb{Z} \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z} \oplus \mathbb{Q}^{I \setminus \{b\}}, \quad [v] \longmapsto (v_b + \mathbb{Z}, (v_a - v_b)_{a \neq b}). \quad (\text{FL28})$$

Proof. For discrete S_I , the support complex is $\mathbb{Q}^I$ in degree zero and zero elsewhere. Theorem 14.5 gives (FL25). In (FL26), the first map is injective because a diagonal vector is an integer diagonal exactly when its coefficient is an integer. The kernel of the second consists exactly of rational diagonal vectors modulo integer diagonals; the second map is surjective. This proves the sequence. Changing a representative v by a rational diagonal changes v_b by the same rational, so $v - v_b \mathbf{1}_I$ is unchanged. Its class projects to $v + \Delta \mathbb{Q}$, proving the splitting. Coordinate evaluation is well defined modulo integers and is a left inverse of d . Finally an inverse to (FL28) sends $(a + \mathbb{Z}, (u_c)_{c \neq b})$ to the class of the vector with entry a at b and entry $a + u_c$ at c ; changing a by an integer changes the vector by an integer diagonal. These checks prove every formula. $\square$

Proposition 14.7 (Actual branch, deletion, and duplication morphisms). *For a map of nonempty finite sets $f : J \rightarrow I$, inverse image of subsets is a bounded lattice homomorphism $\mathcal{P}(I) \rightarrow \mathcal{P}(J)$. It induces the original semiring map $\mathbf{G}_{\mathcal{P}(I)}(\mathbb{Z}) \rightarrow \mathbf{G}_{\mathcal{P}(J)}(\mathbb{Z})$, with unchanged supported coefficients, and the corresponding map of full spectra is*

$$f_Y : Y_J \rightarrow Y_I, \quad f_Y|_{S_J} = f, \quad f_Y|_X = \text{id}_X. \quad (\text{FL29})$$

The pullback of universal arithmetic sheaves is the universal sheaf on Y_J . Its induced relative map is exactly

$$\mathbb{Q}^I / \Delta \mathbb{Z} \longrightarrow \mathbb{Q}^J / \Delta \mathbb{Z}, \quad [v] \longmapsto [(v_{f(b)})_{b \in J}]. \quad (\text{FL30})$$

For a nonempty subset $J \subset I$ this is the deletion projection, with the exact sequence

$$0 \longrightarrow \mathbb{Q}^{I \setminus J} \longrightarrow \mathbb{Q}^I / \Delta \mathbb{Z} \longrightarrow \mathbb{Q}^J / \Delta \mathbb{Z} \longrightarrow 0. \quad (\text{FL31})$$

The first map inserts zeros on J . For arbitrary f , the kernel is $\mathbb{Q}^{I \setminus f(J)}$ by the same insertion, and the cokernel is $\mathbb{Q}^J / f^* \mathbb{Q}^I$; it consists of the differences between duplicated coordinates in each fibre of f .

Proof. The preimage operation preserves bottom, top, intersections, and unions, so (FL2) verifies the semiring map. The inverse image of the lattice prime at $b \in J$ is the lattice prime at $f(b)$; arithmetic primes contract to themselves. This proves (FL29) on the entire spectrum, not only on one stratum. A mixed open pulls back to the mixed open with the same arithmetic part; the constant rational support data pulls back by the same coordinate map. Thus it identifies the universal sheaves and gives the map of relative complexes that is the identity on $\mathbb{Z}$ and f^* on degree one. This proves (FL30).

For inclusion of J , a kernel element restricts to one common integer on J . Subtract its integer diagonal to give zero on J , uniquely because J is nonempty; all remaining coordinates are free rationals. Surjectivity is extension of a vector by zeros outside J . This proves (FL31). The same reasoning uses $f(J)$ in the general case. Since $\Delta\mathbb{Z} \subset f^*\mathbb{Q}^I$, taking the quotient of the target gives the stated cokernel. More explicitly it is $\bigoplus_{a \in f(J)} \mathbb{Q}^{f^{-1}(a)} / \Delta\mathbb{Q}$. $\square$

Deleting all support points is a different endpoint: the remaining space is X , its relative complex is $\mathbb{Z}[0]$, and the map of complexes from (FL25) is the identity in degree zero and zero in degree one. Its relative degree-one target is zero. The kernel formula (FL31) was stated only for nonempty J and is not extended incorrectly to this endpoint.

14.6 A finite lattice whose support has a cycle

Let P have six elements

$$v_1, v_2, v_3, e_{12}, e_{23}, e_{31},$$

and exactly the strict relations

$$v_1 < e_{12}, v_2 < e_{12}, v_2 < e_{23}, v_3 < e_{23}, v_3 < e_{31}, v_1 < e_{31}. \quad (\text{FL32})$$

These are the vertices and edges of the boundary of a triangle ordered by face inclusion; there is no filled triangular face. Set $L = \text{Down}(P)$, with union as join and intersection as meet.

Proposition 14.8 (The eighteen-element lattice and its exact spectrum). *The lattice L has exactly eighteen elements. For any finite poset P , the map*

$$p \mapsto \mathfrak{a}_p = \{D \in \text{Down}(P) : p \notin D\} \quad (\text{FL33})$$

is a homeomorphism $P \rightarrow \text{Spec}_{\text{lat}} \text{Down}(P)$ when P has its lower-set topology. It preserves the stated specialization order: $p \leq q$ if and only if $\mathfrak{a}_p \subseteq \mathfrak{a}_q$.

Proof. A downset in the triangle poset first chooses a subset V of its three vertices and then an arbitrary subset of those edges whose two endpoints lie in V . For $|V| = 0, 1, 2, 3$, the numbers of choices are respectively 1, 3, $3 \cdot 2$, 2^3 . Their sum is $1 + 3 + 6 + 8 = 18$.

For a general finite P , each displayed $\mathfrak{a}_p$ is a proper lattice ideal and is prime: membership in a union is downward-ideal closure, and if p belongs to both downsets it belongs to their intersection. To show exhaustion, let $h : \text{Down}(P) \rightarrow \{0, 1\}$ be the indicator of the complement of a prime ideal. It preserves union and intersection by the ideal and prime conditions. Every nonempty downset is the union of the principal downsets of its members. Choose a minimal p for which $h(\downarrow p) = 1$; one exists because $h(P) = 1$. If $h(\downarrow q) = 1$, then $h(\downarrow p \cap \downarrow q) = 1$. Decomposing that intersection into principal downsets yields some $r \leq p, q$ with $h(\downarrow r) = 1$. Minimality forces $r = p$, so $p \leq q$. It follows that for every downset D , $h(D) = 1$ exactly when $p \in D$. This proves exhaustion and uniqueness. The basic spectral open at a downset D is exactly $\{\mathfrak{a}_p : p \in D\}$, which proves the homeomorphism. Finally if $p \leq q$, absence of p from a downset implies absence of q , proving the ideal inclusion. Conversely if $p \not\leq q$, the downset $\downarrow q$ belongs to $\mathfrak{a}_p$ but not $\mathfrak{a}_q$, disproving inclusion. $\square$

Theorem 14.9 (Every triangle support cochain and every relative group). *For $S = \text{Spec}_{\text{lat}} L$ with the lattice of Proposition 14.8, the full constant-rational support complex is*

$$C^0 = \mathbb{Q}^6 \xrightarrow{D} C^1 = \mathbb{Q}^6, \quad C^n = 0 \quad (n \geq 2), \quad (\text{FL34})$$

where the degree-zero coordinates are $(a_1, a_2, a_3, b_{12}, b_{23}, b_{31})$, the degree-one coordinates are in exactly the order of (FL32), and

$$D(a, b) = (b_{12} - a_1, b_{12} - a_2, b_{23} - a_2, b_{23} - a_3, b_{31} - a_3, b_{31} - a_1). \quad (\text{FL35})$$

Its cohomology is $H^0(S, \mathbb{Q}) = \mathbb{Q}$ by constant sections and $H^1(S, \mathbb{Q}) = \mathbb{Q}$ by the quotient map

$$\omega(c) = c_1 - c_2 + c_3 - c_4 + c_5 - c_6. \quad (\text{FL36})$$

The actual relative complex over $\text{Spec } \mathbb{Z}$ is

$$\mathbb{Z} \xrightarrow{z \mapsto (z, z, z, z, z, z)} \mathbb{Q}^6 \xrightarrow{D} \mathbb{Q}^6 \quad \text{in degrees } 0, 1, 2. \quad (\text{FL37})$$

It has precisely

$$H_X^1(Y, \mathcal{U}_S) = \mathbb{Q}/\mathbb{Z}, \quad H_X^2(Y, \mathcal{U}_S) = \mathbb{Q}, \quad H_X^n(Y, \mathcal{U}_S) = 0 \quad (n \neq 1, 2). \quad (\text{FL38})$$

At the sheaf level $R^1 i^! \mathcal{U}_S = \mathcal{P}_X$ and $R^2 i^! \mathcal{U}_S = \mathbb{Q}_X$, with no other cohomology sheaves.

Proof. The six points and the six relations in (FL32) are all strict chains of lengths zero and one. There is no strict chain of length two. Theorem 14.4 therefore gives exactly (FL34) and (FL35). Its kernel forces equal values at every incident vertex and edge, hence forces all six values equal. Substitution gives $\omega D = 0$.

Conversely let $c \in \mathbb{Q}^6$ satisfy $\omega(c) = 0$. Put

$$\begin{aligned} a_1 &= 0, & b_{12} &= c_1, & a_2 &= c_1 - c_2, \\ b_{23} &= c_1 - c_2 + c_3, & a_3 &= c_1 - c_2 + c_3 - c_4, \\ b_{31} &= c_1 - c_2 + c_3 - c_4 + c_5. \end{aligned} \quad (\text{FL39})$$

The first five equations of $D(a, b) = c$ hold by direct substitution; the sixth holds because $\omega(c) = 0$. Thus $\text{im } D$ is exactly $\ker \omega$. The functional ω is surjective, as is seen on $(1, 0, 0, 0, 0, 0)$, proving the support cohomology and fixing its generator and orientation.

Insert this complete support complex into Theorem 14.5. The resulting relative differentials are precisely (FL37). Its first cohomology consists of a rational constant six-vector modulo an integer constant six-vector, identified by $q \mapsto q + \mathbb{Z}$. Its second cohomology is the quotient by D , identified by (FL36). Its degree-zero kernel is zero, and there are no further terms. The sheaf-level calculation has the same support matrix, with the initial diagonal $\mathcal{O}_X \hookrightarrow \mathbb{Q}_X$; its first quotient is $\mathbb{Q}_X/\mathcal{O}_X = \mathcal{P}_X$, and its second is $\mathbb{Q}_X$. $\square$

14.7 Maps retaining or deleting the actual cycle

Proposition 14.10 (Support maps and their relative chain maps). *For an order-preserving map $f : S' \rightarrow S$, the map $f_Y : S' \triangleright X \rightarrow S \triangleright X$ that is f on support and the identity on X is continuous. Its inverse image takes $\mathcal{U}_S$ to $\mathcal{U}_{S'}$. On relative complexes its map is the identity on $\mathbb{Z}$ and the cochain pullback (FL16) in all support degrees. If the support spaces are spectra of downset lattices, this is the actual spectral map induced by the lattice homomorphism $D \mapsto f^{-1}(D)$ and the coefficient-preserving map of the original semirings.*

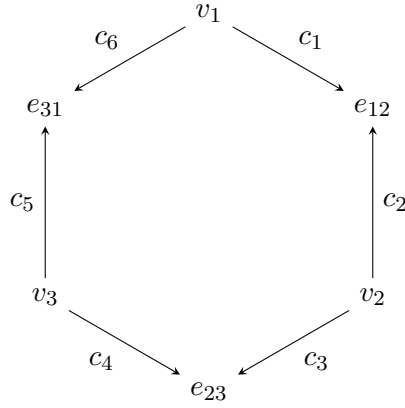


Figure 8: The complete support specialization diagram for the eighteen-element lattice. Arrows run from a generalizing vertex prime to a specializing edge prime, so opens are lower sets. The six arrow coordinates are exactly those of (FL35). The oriented cycle functional is $c_1 - c_2 + c_3 - c_4 + c_5 - c_6$; there is no two-simplex filling it. Theorem 14.9 proves that this contributes the additional relative group $\mathbb{Q}$ in degree two. The drawing is an order diagram and asserts no metric on the prime space.

Proof. The preimage of a support open is a support open because f is order preserving. The preimage of $S \cup U$ is $S' \cup U$, proving continuity on every open. The arithmetic sheaf is unchanged and the inverse image of a constant rational sheaf on support is the constant rational sheaf. The generic-value map stays constant with the same value, proving the sheaf identification. Its maps on the relative resolutions are exactly those displayed, by the natural augmentations, restriction maps, and the cochain computation in Theorem 14.4. The cochain identity was checked including collapsed chains after (FL16). Finally f^{-1} preserves unions and intersections and both bounds. The preimage of the prime at s' , applied to a downset D , tests $s' \notin f^{-1}D$, which is exactly $f(s') \notin D$. Thus its spectral map is f , while supported coefficients are unchanged. This proves the last assertion on both strata. $\square$

For the triangle, delete the maximal element e_{31} to obtain the lower open subposet

$$T = \{v_1, v_2, v_3, e_{12}, e_{23}\}.$$

Its support cochains are $\mathbb{Q}^5 \rightarrow \mathbb{Q}^4$, with the first four coordinates of (FL35). The explicit preimage formulas for these four coordinates are the first five assigned quantities in (FL39); hence that differential is surjective and its kernel is the rational constants. The relative groups for T are therefore $\mathbb{Q}/\mathbb{Z}$ in degree one and zero in degree two. The actual inclusion map $T \triangleright X \rightarrow S \triangleright X$ pulls back the relative complex by deleting b_{31} from degree one and c_5, c_6 from degree two. It induces the identity on the principal-parts $\mathbb{Q}/\mathbb{Z}$ and the zero map from the original cycle $\mathbb{Q}$. There is no claim that $T \triangleright X$ is an open subset of the whole space: it is the indicated continuous subspace embedding, and the maps are the proved pullbacks of pairs.

The map $S \rightarrow \{*\}$ gives a map from the triangle full space to the one-point full space. Its relative pullback is the canonical diagonal inclusion (FL23). Choosing any point $* \rightarrow S$ gives the evaluation retraction. Thus the ordinary principal-parts contribution is retained with exact section and retraction maps, while the new degree-two support cycle is kept in the kernel of each branch evaluation. These statements specify the full morphisms, not merely a comparison of dimensions.

14.8 Scope and proof provenance

The original coefficient/support definition and its complete prime classification were read in the existing programme sources `01_spectra.tex`, the theorem `thm:rebuilt-full-support`, and

08_full_stalks.tex, formulas (TL1)–(TL5). They are reproved in Proposition 14.1 to fix the topology and prevent a change of support object. The earlier independent `split_prime_recollement_pairing.tex`, formulas (RP3)–(RP12) and (RP21)–(RP25), treated a single extra point and its principal parts. The present proofs retain the entire finite support and the higher derived support sections, and do not infer them from that one-point calculation.

The extra groups in (FL20) and (FL38) are actual derived invariants of the original full-support prime topology and the specified universal arithmetic coupling. Their receiving category is the abelian-sheaf category of that topology. They have not been given an incompatible additive-group semiring-module action. The proofs construct no Frobenius endomorphism and assign no weight to the new cycle; such an endomorphism and its trace would have to be computed from actual maps on the displayed complexes. No Riemann-hypothesis implication is asserted here.

15 Full support degree and the original arithmetic quotient

The question addressed here is Deligne’s purity condition, with its actual receiving maps. The supplied English mathematical edition of Weil II, (3.3.4)–(3.3.6), identifies the image of compactly supported cohomology in ordinary cohomology as the pure object, after proving opposite weight bounds and using duality [9]. We do not assume an analogous assertion over the split base. We calculate first what the full support changes in the relative arithmetic group, then construct an exact map from its prime-degree operator to the original arithmetic companion action. This includes the attained metric of that map and its entire distributional extension criterion.

The prime point $\{\tau, e\}$ is retained from the original globalization paper [1]; it is not a discovery of this continuation. The scalar recollement and prime operators are proved in the preceding chapters, (RP21)–(RP35), (PD1)–(PD21), and (IA1)–(IA9). The full finite-support recollement is proved in the accompanying chapter. The arithmetic matrix is exactly the retained programme matrix in CLM13–CLM23 of the complete reconstruction source supplied with this edition. No canonical metric or sheaf-theoretic trace is inferred from a formal resemblance of operators.

15.1 The canonical diagonal and all additional support classes

For a nonempty finite support space S , let c be its number of connected components. The full recollement calculation gives

$$D_S = H_X^1(Y, U\mathcal{O}_X) = \mathbb{Q}^c / \Delta\mathbb{Z}, \quad X = \operatorname{Spec} \mathbb{Z}, \quad (\text{FD1})$$

where $\Delta t = (t, \dots, t)$ has every component present. The remaining positive relative degrees are $H_X^{j+1}(Y, U\mathcal{O}_X) = H^j(S, \mathbb{Q})$ for $j \geq 1$. We shall use the complete group in (FD1), without selecting a branch. There is a canonical exact sequence

$$0 \longrightarrow D = \mathbb{Q}/\mathbb{Z} \xrightarrow{\Delta} D_S \xrightarrow{\pi} \mathbb{Q}^c / \Delta\mathbb{Q} \longrightarrow 0. \quad (\text{FD2})$$

The first map is injective because $\Delta q \in \Delta\mathbb{Z}$ exactly when $q \in \mathbb{Z}$. Its image is the kernel of the quotient map π . That map is onto by the definition of its codomain. This proves exactness, including the diagonal lattice, rather than merely an abstract dimension count. Moreover the torsion subgroup of D_S is exactly ΔD : if $n[x] = 0$, then $nx \in \Delta\mathbb{Z}$, so $x \in \Delta\mathbb{Q}$; conversely each diagonal rational class has finite order. This characterization makes the first term canonical.

For a nonzero integer n , multiplication $[x] \mapsto [nx]$ is onto, since $[x/n]$ is a preimage, and

$$\ker(n : D_S \rightarrow D_S) = \Delta(n^{-1}\mathbb{Z}) / \Delta\mathbb{Z}, \quad |\ker n| = |n|. \quad (\text{FD3})$$

Every fibre is a translate of this displayed kernel. On $D_S / \Delta D$ the same multiplication is an automorphism, because that quotient is a rational vector space. Multiplication by n on each $H^j(S, \mathbb{Q})$, $j \geq 1$, is also an automorphism. Thus disconnected support adds actual classes but

does not replace the arithmetic degree $|n|$ by $|n|^c$. A cycle in the support adds higher relative cohomology, whose multiplication-by- n degree is one.

For Boolean support with $c = r$ labelled points, forgetting any set of coordinates, with at least one coordinate retained, gives the actual map $D_r \rightarrow D_t$ by coordinate projection. Its kernel is isomorphic to $\mathbb{Q}^{r-t}$: subtract the common integer from the retained coordinates of a representative whose image vanishes, leaving those coordinates zero. The deleted coordinates are then uniquely determined rationals. These maps commute with (FD3), and are the identity on the diagonal torsion subgroup. This proves their relation to the scalar branch, including the entire kernel of every deletion.

15.2 The Hilbert receiver of the entire relative group

Use counting measure on the countable group D_S , and set

$$(A_{n,S}f)(d) = f(nd), \quad A_{n,S}^*g(u) = \sum_{nd=u} g(d), \quad A_{n,S}^*A_{n,S} = |n|I. \quad (\text{FD4})$$

The formulas first hold on finitely supported functions. Collecting the sum by fibres in (FD3) proves both the adjoint formula and the exact norm identity $\|A_{n,S}f\|^2 = |n|\|f\|^2$; hence the bounded extensions and their adjoints obey (FD4). They retain all support components and every diagonal torsion class.

Extension by zero along $\Delta : D \rightarrow D_S$ gives an isometric map

$$J_S : \ell^2(D) \hookrightarrow \ell^2(D_S), \quad A_{n,S}J_S = J_SA_n, \quad A_{n,S}^*J_S = J_SA_n^*. \quad (\text{FD5})$$

Indeed nd is torsion if and only if d is torsion, since the quotient in (FD2) is uniquely divisible and torsion free. The fibre above a torsion element is consequently entirely in ΔD ; the fibre above a nontorsion element is entirely outside it. These observations prove both identities point by point. The canonical torsion subspace and its orthogonal complement are therefore both invariant under the operator and its adjoint. The original vector δ_0 belongs to this torsion subspace. The original scalar Weil receiver (PD14)–(PD18) is recovered exactly there; the additional classes have not been assigned invented coefficients in that scalar receiver.

For a fixed prime p , put $V_{p,S} = p^{-1/2}A_{p,S}$. It is an isometry and is pure in the precise operator-theoretic sense

$$\bigcap_{m \geq 0} V_{p,S}^m \ell^2(D_S) = \{0\}. \quad (\text{FD6})$$

If f belongs to this intersection, then for every m it is constant on each coset of $\ker p^m$, because it is in the range of pullback by p^m . Thus it is constant on every coset of the infinite subgroup $\Delta(\mathbb{Z}[1/p]/\mathbb{Z}) = \bigcup_m \ker p^m$. Square summability forces each such constant to vanish. This proves (FD6). Here “pure isometry” is its stated intersection property; it is not being identified with Deligne’s purity theorem.

For completeness an explicit minimal unitary extension follows without assuming a dilation theorem. Put $W_S = \ker V_{p,S}^*$. The projections $V^m V^{*m} - V^{m+1} V^{*(m+1)}$ have mutually orthogonal ranges $V^m W_S$. Their finite sums telescope to $I - V^{M+1} V^{*(M+1)}$. Decreasing orthogonal projections converge strongly to the projection onto their intersection: the squared norm differences telescope, so the projected vectors are Cauchy, and their limit lies in every range. By (FD6) that limit is zero. Hence $\ell^2(D_S) = \bigoplus_{m \geq 0} V^m W_S$. On $\tilde{H}_S = \bigoplus_{m \in \mathbb{Z}} W_S$ take the bilateral shift $\tilde{V}$ and embed the nonnegative summands as just identified. The operator

$$\tilde{A}_{p,S} = \sqrt{p} \tilde{V}, \quad \tilde{A}_{p,S}^* \tilde{A}_{p,S} = \tilde{A}_{p,S} \tilde{A}_{p,S}^* = pI \quad (\text{FD7})$$

extends $A_{p,S}$. The space W_S is nonzero: otherwise V is a surjective isometry on a nonzero space, contradicting (FD6). The bilateral shift has spectrum the unit circle and no eigenvectors. To

verify these claims directly, a unit $w \in W_S$ gives orthonormal translates w_m ; the unit vectors $(2M+1)^{-1/2} \sum_{m=-M}^M \lambda^{-m} w_m$ have residual squared norm $2/(2M+1)$ for $|\lambda| = 1$. Neumann series invert the shift minus λ for $|\lambda| \neq 1$. An eigenvector on the circle would have equal component norms at every integer index and is thus zero. Scaling proves the full circle $|z| = \sqrt{p}$ for (FD7). This extends the proved local weight-one norm identity to every class of the full relative group (FD1).

The Hilbert construction has a precise relation to additive complexification, which is a different receiver. On finite-support functions define

$$\kappa_S : \mathbb{C}^{(D_S)} \longrightarrow D_S \otimes_{\mathbb{Z}} \mathbb{C} \simeq \mathbb{C}^c / \Delta \mathbb{C}, \quad \kappa_S(\delta_d) = d \otimes 1. \quad (\text{FD8})$$

The quotient of the free complex vector space by the relations $\delta_0 = 0$ and $\delta_{x+y} - \delta_x - \delta_y = 0$ has exactly the universal property of the displayed tensor product: a complex-linear map out of it is an additive map from D_S to a complex vector space, and extends uniquely by $d \otimes z \mapsto zf(d)$. This proves both the map and its complete kernel presentation. The tensor product kills every torsion element, because its order is invertible in $\mathbb{C}$. Using (FD2), or its explicit coordinates after one branch is specified, then proves the last isomorphism. In particular $\kappa_S \delta_0 = 0$ and the entire diagonal torsion subspace is killed.

For every positive integer n , this exact receiver satisfies

$$\kappa_S A_{n,S} = \kappa_S, \quad \kappa_S A_{n,S}^* = n \kappa_S. \quad (\text{FD9})$$

Indeed $A_{n,S} \delta_u = \sum_{nd=u} \delta_d$. Each of its n preimages has tensor value $(u \otimes 1)/n$, since the kernel in (FD3) is torsion. Their sum is $u \otimes 1$, proving the first identity. The adjoint sends δ_d to δ_{nd} , which proves the second. Thus the prime pullback on counting functions, the degree- p operation on the relative group, and its additive complexification are linked by specified maps with their exact factors. They are not silently identified. Retaining the supported zero and torsion is essential for the original integral vector and the Weil receiver: they disappear under (FD8), whose kernel has now been calculated rather than treated as absent data.

The supported integer zero is still present. On the full algebraic dual of $\mathbb{C}^{(D_S)} \oplus \mathbb{C} \delta_\tau$, its pullback is $(f, c) \mapsto (f(0) \mathbf{1}_{D_S}, c)$, whereas absent τ gives $(f, c) \mapsto (c \mathbf{1}_{D_S}, c)$. These distinct maps are proved by evaluating the two scalar actions, precisely as in (IA7). The first does not preserve $\ell^2(D_S)$, since D_S is infinite. Its exact target remains the stated algebraic dual; no integer has been deleted.

15.3 An explicit quotient onto the original arithmetic action

Keep the original polynomial, its coordinate, and its arithmetic prime:

$$\begin{aligned} q &= (k+1)^2, \quad k \geq 17, \quad k \equiv 1 \pmod{4}, \quad \delta, \gamma > 0, \quad p \text{ prime}, \\ Q_k(y) &= \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/(Q_k), \quad M[f] = [yf], \\ F_p &= p^{k/2} \exp(i(\log p)M), \quad B_p = \exp(i(\log p)M). \end{aligned} \quad (\text{AQ1})$$

The letter B_p denotes this complete centred matrix, not a changed prime action: $F_p = p^{k/2} B_p$ is retained in every intertwining formula. At each original cutoff $N \in \{q-1, q, 2q-1, 2q\}$, let G_N be the original positive definite metric on E in its original monomial coordinates. No metric is chosen in its place.

The k -fold local relative operator (PD20) acts on $L^2(\mathbb{Q}_p^k)$ as $p^k f(x/p)$; its associated unitary operator is $\mathcal{U}^{(k)} f(x) = p^{k/2} f(x/p)$. Set

$$A = \mathbb{Z}_p^\times \times \mathbb{Z}_p^{k-1}, \quad h = \mathbf{1}_A, \quad h_m = (\mathcal{U}^{(k)})^m h, \quad w_p = \|h_m\|^2 = 1 - p^{-1}. \quad (\text{AQ2})$$

The supports of the h_m are disjoint because the first coordinate has valuation m . Thus their inner products are $w_p \mathbf{1}_{m=n}$. The Haar factor w_p is retained, and the unchanged k -fold degree is p^k . Let $\Phi_{p,N}$ be the finite sums $\sum_m h_m \otimes v_m$ in $L^2(\mathbb{Q}_p^k) \otimes (E, G_N)$, with the inherited norm. Define

$$\Psi_p \left(\sum_m h_m \otimes v_m \right) = \sum_m B_p^m v_m \in E. \quad (\text{AQ3})$$

Uniqueness of the orthogonal coefficients makes this map well defined; the sum is finite. It is onto since $h_0 \otimes v$ maps to v . For the actual degree operator $T_p = (T_p^{\text{rel}})^{\otimes k} \otimes I_E$,

$$\Psi_p T_p = F_p \Psi_p, \quad T_p(h_m \otimes v) = p^{k/2} h_{m+1} \otimes v. \quad (\text{AQ4})$$

Both sides of the first identity, on a basis summand, equal $p^{k/2} B_p^{m+1} v$. This is an exact typed, surjective morphism from a concrete subspace of the new relative-prime receiver to the original arithmetic matrix, with its uncentred factor. It does not assert that the morphism is bounded in the Hilbert norm or is a geometric compact-to-ordinary cohomology map. Those are mathematical properties to calculate, rather than inferred identifications.

Its full algebraic kernel is also explicit. Write a finite sequence as the Laurent polynomial $v(t) = \sum_m t^m v_m$. Then

$$\ker \Psi_p = (tI_E - B_p)E[t, t^{-1}]. \quad (\text{AQ5})$$

The right side is in the kernel by evaluation. Conversely subtract $v(B_p)$ from $v(t)$ term by term. For $m > 0$, $t^m I - B_p^m = (tI - B_p) \sum_{j=0}^{m-1} t^{m-1-j} B_p^j$. For $m = -r < 0$ use $t^{-r} I - B_p^{-r} = -t^{-r} B_p^{-r} (t^r I - B_p^r)$. These identities hold since t commutes with the matrix and B_p is invertible. Their sum proves (AQ5), including all negative powers. The support-preserving lift of (AQ3)–(AQ5) on synchronized full- L carriers is $(v, \lambda) \mapsto (\Psi_p v, \lambda)$; its layerwise kernel is $(\ker \Psi_p)_L^\uparrow$. Direct evaluation of addition, scalar multiplication and (AQ4) proves these assertions, with the zero map $(v, \lambda) \mapsto (0, \lambda)$ retained. This is the retraction category and its supported zero of CR15–CR23; it does not identify that map with the constant- τ map on a free semimodule.

15.4 The attained quotient metric, including every cross term

Restrict (AQ3) to $-J \leq m \leq J$, for an integer $J \geq 0$. Its source metric is $w_p \text{diag}(G_N, \dots, G_N)$. The minimum squared norm among all preimages of $v \in E$ is

$$\min_{\sum_{m=-J}^J B_p^m v_m = v} w_p \sum_{m=-J}^J v_m^* G_N v_m = v^* H_{p,J;N} v, \quad H_{p,J;N} = w_p \left(\sum_{m=-J}^J B_p^m G_N^{-1} (B_p^*)^m \right)^{-1}. \quad (\text{AQ6})$$

To prove the minimum, put $C = \sum_m B_p^m G_N^{-1} (B_p^*)^m$. It is positive definite because the $m = 0$ term is. The candidate $v_m^0 = G_N^{-1} (B_p^*)^m C^{-1} v$ has the required sum. For any other preimage write $v_m = v_m^0 + u_m$, so $\sum_m B_p^m u_m = 0$. The cross term in its squared norm is $2w_p \text{Re}(v^* C^{-1} \sum_m B_p^m u_m) = 0$. The remaining extra term is nonnegative and vanishes only for $u_m = 0$. The candidate's squared norm is $w_p v^* C^{-1} v$, proving (AQ6). This is a new, explicitly identified attained quotient metric; it has not been substituted for any older native conductor metric.

All original roots are distinct for $\delta, \gamma > 0$. Let $V_{(a,b),j} = y_{ab}^j$ be their original evaluation matrix, $0 \leq j < q$, and set

$$y_{ab} = (2b - k)\gamma - i(2a - k)\delta, \quad \text{quad} \alpha_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}, \quad D_N = V G_N^{-1} V^*. \quad (\text{AQ7})$$

Polynomial interpolation proves V invertible and $V B_p V^{-1} = \text{diag}(\alpha_{ab})$. Consequently the complete covariance in (AQ6) satisfies

$$(V C V^*)_{ij} = (D_N)_{ij} \mathcal{D}_J(\alpha_i \overline{\alpha_j}), \quad \mathcal{D}_J(z) = \sum_{m=-J}^J z^m = \begin{cases} z^{-J} (1 - z^{2J+1}) / (1 - z), & z \neq 1, \\ 2J + 1, & z = 1. \end{cases} \quad (\text{AQ8})$$

Matrix multiplication gives each summand, and the finite geometric sum gives the last expression. It retains every pair of roots, every phase resonance, and every original metric entry. In particular the variance of the i th original root functional is exactly $w_p^{-1}(D_N)_{ii}\mathcal{D}_J(|\alpha_i|^2)$.

For a rigorous size comparison, define the actual constants $d_{-,N} = \lambda_{\min}(D_N) > 0$ and $d_{+,N} = \lambda_{\max}(D_N)$. Multiplying $d_-I \preceq D_N \preceq d_+I$ by each diagonal power, summing, and inverting positive matrices gives

$$\frac{w_p}{d_{+,N}}\mathbf{V}^* \text{diag}(\mathcal{D}_J(|\alpha_i|^2)^{-1})\mathbf{V} \preceq H_{p,J;N} \preceq \frac{w_p}{d_{-,N}}\mathbf{V}^* \text{diag}(\mathcal{D}_J(|\alpha_i|^2)^{-1})\mathbf{V}. \quad (\text{AQ9})$$

The upper bound is at most $w_p d_{-,N}^{-1} p^{-2J\delta} \mathbf{V}^* \mathbf{V}$, because k is odd and $|2a - k| \geq 1$: choose the endpoint $m = J$ or $m = -J$ in the positive sum to get $\mathcal{D}_J(|\alpha_{ab}|^2) \geq p^{2J|2a-k|\delta}$. This is a proved exponential collapse of this quotient metric for the displayed off-line family, with all comparison constants specified. It is not a conclusion about the existence of actual zeta zeros.

For the four original cutoff signs $\varepsilon_{q-1} = \varepsilon_q = 1$ and $\varepsilon_{2q-1} = \varepsilon_{2q} = -1$, the exact return of this new metric is

$$\sum_N \varepsilon_N \log \det H_{p,J;N} = - \sum_N \varepsilon_N \log \det [(D_N)_{ij} \mathcal{D}_J(\alpha_i \bar{\alpha}_j)]_{ij}. \quad (\text{AQ10})$$

Indeed the exact determinant is

$$\det H = \frac{w_p^q |\det \mathbf{V}|^2}{\det(\mathbf{V} \mathbf{C} \mathbf{V}^*)}.$$

Both displayed common factors cancel since $\sum_N \varepsilon_N = 0$. The D_N factors do not cancel by this argument and have all been retained. Formula (AQ10) is an exact receiving identity, not an asserted asymptotic evaluation.

15.5 The precise distribution space selected by the quotient

The original action now determines which sequences it can receive. For any invertible matrix B on a finite-dimensional metric space, consider the already defined finite-sequence map $\Psi_B((v_m)) = \sum_m B^m v_m$. Define the sequence space

$$\mathcal{S}(\mathbb{Z}; E) = \{(v_m) : \sup_m (1 + |m|)^r \|v_m\|_{G_N} < \infty \text{ for every integer } r \geq 0\}. \quad (\text{AQ11})$$

Its topology is given by exactly these seminorms. We have the complete criterion

$$\Psi_B \text{ has a continuous extension to } \mathcal{S}(\mathbb{Z}; E) \iff |\lambda| = 1 \text{ for every eigenvalue } \lambda \text{ of } B. \quad (\text{AQ12})$$

Here is a proof including nonsemisimple matrices. In a Jordan basis, a block is $\lambda I + K$, $K^d = 0$. For every integer m ,

$$(\lambda I + K)^m = \sum_{j=0}^{d-1} \binom{m}{j} \lambda^{m-j} K^j. \quad (\text{AQ13})$$

For nonnegative m this is the binomial formula. For negative m it follows by multiplying the finite formal inverse series modulo K^d , or inducting downwards from $m = 0$ with that inverse. The coefficients $\binom{m}{j}$ are polynomials in m of degree j . If all eigenvalues have modulus one, the finite Jordan change of basis gives an explicit constant C_B with $\|B^m\|_{G_N} \leq C_B(1 + |m|)^{d_{\max}-1}$. Thus the series defining Ψ_B converges absolutely on (AQ11), bounded by $C_B \sum_m (1 + |m|)^{-2}$ times its seminorm of order $d_{\max} + 1$. It is continuous and extends the finite sum.

Conversely continuity of a linear map from (AQ11) to finite-dimensional E bounds it by $C \sup_m (1 + |m|)^r \|v_m\|$ for some finite C, r : continuity at zero supplies finitely many seminorms, and the largest dominates all of them. Test a sequence supported at the single index m with

coefficient an eigenvector v . It gives $|\lambda|^m \|v\| \leq C(1+|m|)^r \|v\|$ for all integers m . Letting m tend to positive and negative infinity proves $|\lambda| = 1$. This also proves uniqueness, since truncation converges in every Schwartz seminorm (use a seminorm of one higher order for the tail).

For (AQ1), the actual eigenvalues in (AQ7) have moduli $p^{(2a-k)\delta}$. Thus (AQ12) fails at every $\delta > 0$ in the specified family. The exact root functional at $i = (a, b)$ has norm on the finite word space

$$\|\mathrm{ev}_{y_i} \Psi_p\|^2 = \frac{(D_N)_{ii}}{w_p} \sum_{m=-J}^J p^{2m(2a-k)\delta}. \quad (\text{AQ14})$$

This is both the diagonal case of (AQ8) and a direct finite dual norm calculation. It gives the complete rate of its failure to have a polynomial sequence bound; no projected sign or total determinant has been used to assign that rate.

At the literal collision $\delta = 0$, keep the same polynomial rather than replacing it by its square-free part: $Q_k(y) = \prod_{b=0}^k (y - (2b-k)\gamma)^{k+1}$. The Chinese remainder map gives $k+1$ blocks $\mathbb{C}[u]/u^{k+1}$ with $M = (2b-k)\gamma + u$ on the b th block. On that block the original centred powers are

$$B_p^m = e^{im(2b-k)\gamma \log p} \sum_{j=0}^k \frac{(im \log p)^j}{j!} u^j. \quad (\text{AQ15})$$

This follows from the terminating exponential in the same quotient algebra. Its growth is at most polynomial of degree k , so (AQ12) holds without deleting any repeated root or nilpotent jet. The exact formula (AQ6) also remains valid at this collision; only the distinct-root coordinate formula (AQ8) is replaced by (AQ15).

For every parameter, including the collision, the finite-sequence map has a receiving space selected by its full matrix:

$$\mathcal{A}_B = \{(v_m) : \sum_m \|B^m v_m\|_{G_N} < \infty\}, \quad \|(v_m)\|_{\mathcal{A}_B} = \sum_m \|B^m v_m\|_{G_N}. \quad (\text{AQ16})$$

The coordinatewise map $(v_m) \mapsto (B^m v_m)$ is an isometric bijection to $\ell^1(\mathbb{Z}; E)$, so this is a complete normed space. The sum map extends (AQ3) continuously and surjectively to it. The sequence shift and its inverse are bounded there, with norms at most $\|B\|$ and $\|B^{-1}\|$, respectively, and the exact intertwining relation (AQ4) persists. This constructs the object defined by the failed Schwartz extension, rather than discarding the arithmetic quotient. The connecting inclusion of finite sequences into both (AQ11) and (AQ16), and their common map (AQ3), are explicit. No off-circle eigenvalue has been removed by declaration.

15.6 The compact-to-ordinary map on the computed affine base

For the universal coupling on $Y = S \triangleright \mathrm{Spec} \mathbb{Z}$, the full recollement proof gives $R\Gamma(Y, U\mathcal{O}_X) = R\Gamma(X, \mathcal{O}_X)$ and

$$R\Gamma_X(Y, U\mathcal{O}_X) \longrightarrow R\Gamma(X, \mathcal{O}_X) \longrightarrow R\Gamma(S, \mathbb{Q}). \quad (\text{AQ17})$$

The middle complex is $\mathbb{Z}$ in degree zero and zero in positive degrees. This can be proved directly here: the exact principal-parts sequence $0 \rightarrow \mathcal{O}_X \rightarrow \mathbb{Q}_X \rightarrow \bigoplus_p i_{p*}(\mathbb{Q}_p/\mathbb{Z}_p) \rightarrow 0$ has flasque last two terms. Constants on nonempty opens of the irreducible X give surjective restrictions in $\mathbb{Q}_X$; restriction in the direct sum of point sheaves is deletion of coordinates. The global principal-parts map is onto by writing a finite collection of p -power fractions and adding them. Its kernel is $\mathbb{Z}$. This computes the asserted cohomology, including all primes.

Consequently the natural relative-to-ordinary map in (AQ17) has zero image in every positive degree, while its boundary groups are exactly (FD1) and the higher support groups. In degree zero its relative source is zero, since $\mathbb{Z} \rightarrow \mathbb{Q}^c$ is injective. This calculates the particular image; it does not identify a Zariski support functor with the compact support functor of an arithmetic

curve containing its Archimedean data. The exact relation is the triangle (AQ17). The relative boundary is retained as its cone, with the operators (FD3)–(FD7), and is the source of the arithmetic quotient (AQ3).

These calculations locate the next purity issue precisely. The full relative spectrum supplies an actual degree operator and its pure circle norm identity. It also supplies a distributional quotient onto the original arithmetic action, whose metric and extension space have now been evaluated. The Zariski relative-to-ordinary image on the affine universal coupling is zero; it cannot be substituted for the unconstructed global image involving the Archimedean term and both pole maps. Establishing that global map’s regularity and dual pairing requires further mathematics. The statements (AQ12)–(AQ16) specify the exact objects and growth distinction that such a map must address; they are complete calculations, not an assumed purity theorem or an RH conclusion.

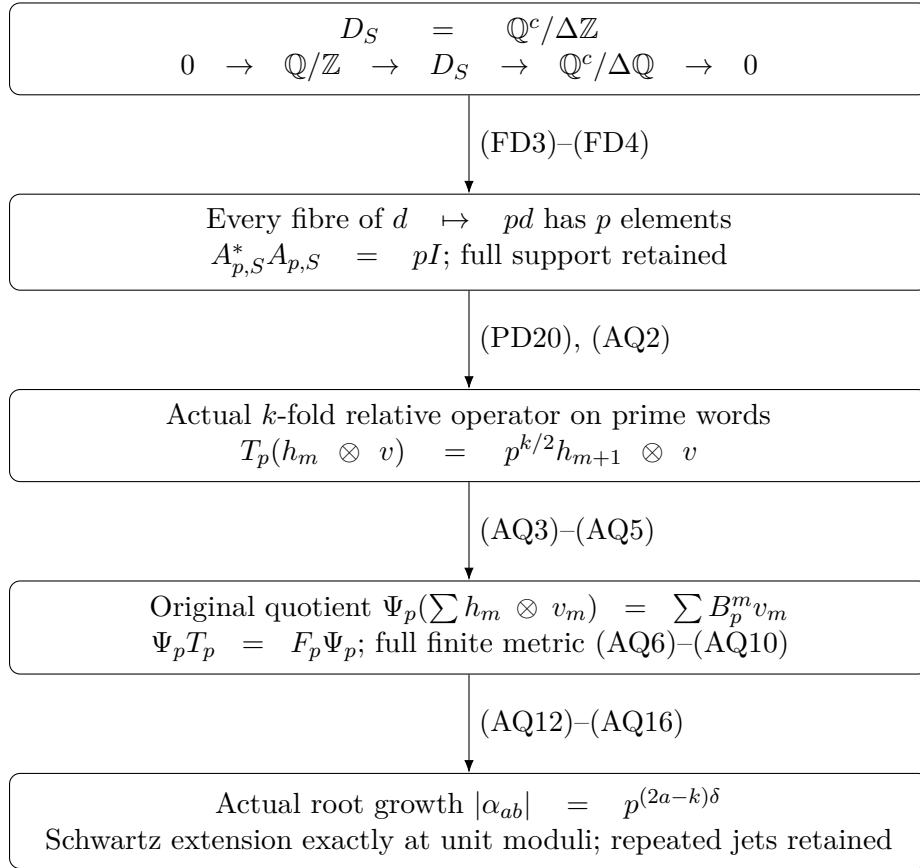


Figure 9: The exact link from the added prime and full support to the original arithmetic action. The prime degree fixes its square-root factor. The arithmetic quotient keeps every original eigenvalue, coordinate and metric; its growth is calculated rather than assigned. The diagram does not identify this quotient with Deligne’s geometric image. The latter comparison is (AQ17) and the scope following it; Deligne’s human-source theorem is Weil II, (3.3.6) [9].

16 The completed prime-word cohomology and its exact image

The quotient (AQ3) now permits an actual image calculation, rather than a proposed regularity assumption. The source is the prime-word subspace of the relative operator (AQ2), with its physical valuation index $m \in \mathbb{Z}$. We compute its finite-support and rapid-decay cohomology, the map between them, and the full relative object recording the classes that this map kills. This is

a local comparison built from the split relative-prime operator. Its construction is not identified with global arithmetic compactly supported cohomology.

16.1 Three exact coefficient complexes

Keep E, G_N, k, p, F_p, B_p from (AQ1), with their original factors. For an arbitrary invertible complex matrix B on E , let

$$\begin{aligned} \Phi &= E^{(\mathbb{Z})}, & \mathcal{S} &= \mathcal{S}(\mathbb{Z}; E), & \Pi &= E^{\mathbb{Z}}, \\ (sv)_m &= v_{m-1}, & d_B &= sI - B. \end{aligned} \quad (\text{PC1})$$

The first is finite-support sequences, the second is exactly (AQ11), and the third is all sequences, with no decay imposed. The differential is $(d_B v)_m = v_{m-1} - Bv_m$ in each case. Put each complex in degrees zero and one. Its degree action is $p^{k/2}s$ on both terms; this commutes with d_B , so it induces an action on cohomology. For $B = B_p$ this is the actual tensor relative-prime degree in (AQ4).

The finite complex has the exact sequence

$$0 \longrightarrow \Phi \xrightarrow{d_B} \Phi \xrightarrow{\Psi_B} E \longrightarrow 0, \quad \Psi_B(v) = \sum_m B^m v_m. \quad (\text{PC2})$$

Surjectivity and the kernel equality are (AQ3) and (AQ5). Injectivity of d_B follows directly: for a nonzero finite sequence, take its largest nonzero index m ; then $(d_B v)_{m+1} = v_m \neq 0$. Consequently its cohomology is E in degree one and zero in degree zero. The degree action on that cohomology is $p^{k/2}B$ by (AQ4).

At the other endpoint the all-sequence complex has

$$H^0(\Pi \xrightarrow{d_B} \Pi) = \{(B^{-m}v)_{m \in \mathbb{Z}} : v \in E\} \simeq E, \quad H^1(\Pi \xrightarrow{d_B} \Pi) = 0. \quad (\text{PC3})$$

The homogeneous recurrence is exactly $v_m = B^{-1}v_{m-1}$, proving the kernel formula in both directions. For a prescribed sequence w , choose $v_0 = 0$, solve $v_m = B^{-1}(v_{m-1} - w_m)$ successively for $m \geq 1$, and solve $v_{m-1} = w_m + Bv_m$ successively for $m \leq 0$. This produces a two-sided sequence with $d_B v = w$ and proves surjectivity. The degree action on its kernel again corresponds to $p^{k/2}B$, since $(sv)_m = B^{-(m-1)}v = B^{-m}Bv$. Thus the full algebraic receiver retains every original eigenvalue, in its stated cohomological degree.

16.2 Rapid sequences and smooth division on the original circle

The Fourier series map is

$$\mathcal{F} : \mathcal{S} \xrightarrow{\sim} C^\infty(\mathbb{T}; E), \quad (\mathcal{F}v)(z) = \sum_{m \in \mathbb{Z}} z^m v_m, \quad |z| = 1. \quad (\text{PC4})$$

For a rapid sequence each differentiated series converges absolutely and uniformly, since multiplication of its m th coefficient by any power of m still gives an absolutely summable sequence. Conversely the coefficient formula $v_m = (2\pi)^{-1} \int_0^{2\pi} f(e^{i\theta}) e^{-im\theta} d\theta$ and integration by parts any number of times give every seminorm in (AQ11). The Fourier coefficients determine the smooth function: for example convolution with the finite Fejer kernels, whose coefficients are $1 - |m|/(M+1)$ for $|m| \leq M$, tends uniformly to a continuous periodic function. This follows directly from their nonnegativity, integral one, and the formula $(M+1)^{-1} |\sum_{j=0}^M e^{ij\theta}|^2$, which tends uniformly to zero outside any neighbourhood of zero. Thus zero coefficients force the function to be zero, and the two displayed operations are inverse. These estimates also prove continuity in both smooth and sequence seminorms. In these exact coordinates, d_B is multiplication by $zI - B$.

Let E_λ be the generalized eigenspace of B at λ , P_λ its spectral projection, and $N_\lambda = (B - \lambda I)|_{E_\lambda}$ with nilpotence index d_λ . The decomposition and its projections are algebraic: Bezout's identity for the pairwise coprime powers $(t - \lambda)^{d_\lambda}$ gives their Chinese remainder idempotents, evaluated at B . This proves the direct-sum decomposition and does not require semisimplicity or distinct eigenvalues. Put

$$E_o = \bigoplus_{|\lambda|=1} E_\lambda, \quad E_{\text{off}} = \bigoplus_{|\lambda| \neq 1} E_\lambda, \quad P_o = \sum_{|\lambda|=1} P_\lambda. \quad (\text{PC5})$$

For $z = e^{i\theta}$, write $\partial_z = (iz)^{-1} \partial_\theta$. This is a specified complex differential operator on smooth functions on the circle; it is not an assumption of a holomorphic extension. Define the complete jet receiver

$$\mathcal{J}_B(f) = \sum_{|\lambda|=1} \sum_{j=0}^{d_\lambda-1} \frac{N_\lambda^j}{j!} (\partial_z^j P_\lambda f)(\lambda) \in E_o. \quad (\text{PC6})$$

The sum of finitely many derivative evaluations is continuous. For a constant function $f = v \in E_o$, it has value v , so it is surjective.

Theorem 16.1 (The entire rapid-decay cohomology). *There is an exact sequence of complex vector spaces and continuous maps*

$$0 \longrightarrow C^\infty(\mathbb{T}; E) \xrightarrow{zI-B} C^\infty(\mathbb{T}; E) \xrightarrow{\mathcal{J}_B} E_o \longrightarrow 0. \quad (\text{PC7})$$

The first map has closed range. Consequently the complex on $\mathcal{S}$ in (PC1) has $H^0 = 0$, $H^1 = E_o$; no off-circle eigenspace or unit-circle Jordan block is omitted from this assertion.

Proof. Away from the finitely many unit-circle eigenvalues, $zI - B$ is invertible. A smooth function in its kernel vanishes on that dense open set and hence everywhere, proving injectivity.

On a primary component with $|\lambda| \neq 1$, the inverse $\sum_{j=0}^{d_\lambda-1} (z - \lambda)^{-j-1} N_\lambda^j$ is smooth on the entire circle. There is therefore no cokernel contribution from that component.

For $|\lambda| = 1$, a smooth component f_λ has a local finite expansion

$$f_\lambda(z) = \sum_{r=0}^{d_\lambda-1} a_r (z - \lambda)^r + (z - \lambda)^{d_\lambda} g(z), \quad a_r = \frac{\partial_z^r f_\lambda(\lambda)}{r!}, \quad (\text{PC8})$$

with g smooth. To justify this for smooth, possibly nonholomorphic data, use a real coordinate $t = \theta - \theta_0$. The quotient $(z(t) - \lambda)/t$ is smooth and nonzero at zero. The fundamental theorem of calculus gives a smooth quotient by t for every smooth function vanishing at zero. Repeating this division gives the finite expansion; applying ∂_z successively identifies its coefficients as the displayed derivatives.

Multiplication of (PC8) by the finite inverse above has, at the negative power $(z - \lambda)^{-l}$, exactly the coefficient

$$\sum_{r=0}^{d_\lambda-l} N_\lambda^{r+l-1} a_r = N_\lambda^{l-1} \sum_{r=0}^{d_\lambda-1} N_\lambda^r a_r, \quad 1 \leq l \leq d_\lambda. \quad (\text{PC9})$$

Terms added on the right have powers at least d_λ and are zero. The coefficient for $l = 1$ is precisely the λ component of (PC6); all other negative coefficients are its displayed multiples. Thus if (PC6) vanishes, the inverse applied to f_λ has a smooth extension at λ . It is smooth elsewhere already and solves $(zI - B)g = f$ on the whole circle. Conversely, if $f = (zI - B)g$ with smooth g , its inverse away from the point is that smooth function. A nonzero highest negative Laurent power in (PC9) is unbounded in some linear coordinate as $t \rightarrow 0$; hence all negative coefficients must vanish, including $l = 1$. This proves that the kernel of (PC6) is exactly the range. Surjectivity was proved after (PC6), completing exactness. The range is closed as the kernel of a continuous map to a finite-dimensional space. $\square$

16.3 The comparison image and every lost class

For a finite Laurent polynomial $f(z) = \sum z^m v_m$, the terminating Taylor formula on each primary block, including negative powers, gives

$$\mathcal{J}_B(f) = P_\circ \sum_m B^m v_m = P_\circ \Psi_B(v). \quad (\text{PC10})$$

For negative powers this is exactly the inverse-block binomial calculation (AQ13). Hence the inclusion of finite sequences into rapid sequences induces the fully evaluated cohomology map

$$\boxed{H^1(\Phi \xrightarrow{d_B} \Phi) = E \xrightarrow{P_\circ} H^1(\mathcal{S} \xrightarrow{d_B} \mathcal{S}) = E_\circ.} \quad (\text{PC11})$$

Its image is the entire right side, its kernel is E_{off} , and its cokernel is zero. Its degree action is the actual $p^{k/2}B|_{E_\circ}$. Every eigenvalue on the image has absolute value $p^{k/2}$. This proves a local purity-circle image statement for these specified complexes. It does not assert the arithmetic integrality of these eigenvalues or Deligne's purity at every embedding of an algebraic coefficient field.

The relative object of this completion is also completely determined. Let $C_\Phi, C_\mathcal{S}$ denote the two complexes in (PC1). The homotopy fibre of their actual inclusion has

$$H^1(\text{Cone}(C_\Phi \rightarrow C_\mathcal{S})[-1]) = E_{\text{off}}, \quad H^j = 0 \quad (j \neq 1). \quad (\text{PC12})$$

Indeed the long exact sequence of the explicit cone uses the cohomology already proved, and its only nonzero portion is $0 \rightarrow H^1 \rightarrow E \xrightarrow{P_\circ} E_\circ \rightarrow H^2 \rightarrow 0$. The last map is surjective, so the assertion follows with its exact kernel. This retains the removed classes as a derived object with their unchanged original degree action $p^{k/2}B|_{E_{\text{off}}}$. The finite, rapid, and all-sequence constructions (PC2), (PC7), and (PC3) therefore specify both which receiver retains each class and its cohomological degree. No class is declared nonexistent merely because one completion does not receive it.

For the actual arithmetic family (AQ1) with $\delta > 0$, odd k implies $2a - k \neq 0$ for every original root. Thus (AQ7) gives

$$E_\circ = 0, \quad E_{\text{off}} = E, \quad H^1(C_\mathcal{S}) = 0, \quad H^1(\text{Cone}(C_\Phi \rightarrow C_\mathcal{S})[-1]) = E. \quad (\text{PC13})$$

This is a calculation for the displayed hypothetical off-line family. The finite source and the all-sequence kernel still contain all q dimensions and their original phases. It is not a contradiction unless another proved geometric map requires a nonzero such class to survive precisely the comparison (PC11).

At $\delta = 0$, keep every repeated root and nilpotent coefficient of (AQ15). All eigenvalues of B_p have modulus one, including any coincident phases at the selected prime. Hence $E_\circ = E$, and (PC11) is the identity on the original q -dimensional coefficient algebra. The receiver (PC6) retains derivatives through order k on the size- $k + 1$ blocks of (AQ15). When different blocks have the same exponential eigenvalue, their direct sum is the primary space used in (PC6); no block is merged into a single vector or discarded. The full relative object (PC12) is then acyclic.

16.4 What has been supplied for the global purity problem

The construction uses the additional arithmetic prime through its relative principal parts, the actual p -to-one multiplication, and the resulting k -fold degree action. Its comparison is now a proved map with a computed image, kernel, and full collision specialization. The point at which Deligne's method would require additional global information is explicit: a geometric compact-to-ordinary map must be related to this local completion by actual chain maps, retaining the two pole classes, the Archimedean contribution, and the original observation and conductor

maps. The affine comparison (AQ17), whose image was calculated as zero, is not substituted for those maps. Equations (PC11)–(PC13) prevent a false shortcut: choosing a completion that keeps only a purity circle proves the stated image calculation, but does not by itself prove that a class from the original global trace has to survive it. The exact relative object where such a class can reside is (PC12), not an unrecorded error term.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \Phi & \xrightarrow{sI-B} & \Phi & \xrightarrow{\Psi_B} & E & \longrightarrow & 0 \\
& & \downarrow & & \downarrow & & \downarrow P_o & & \\
0 & \longrightarrow & \mathcal{S} & \xrightarrow{sI-B} & \mathcal{S} & \xrightarrow{\mathcal{J}_B \mathcal{F}} & E_o & \longrightarrow & 0
\end{array}$$

Figure 10: The exact local cohomology comparison (PC2), (PC7), and (PC10). The upper terms are finite prime-word sequences in the relative-degree receiver; the lower terms retain every rapid sequence. The right-hand kernel is exactly E_{off} , retained by (PC12). Both rows carry the unchanged degree action $p^{k/2}s$, inducing $p^{k/2}B$ on their images. This is a computed local comparison, with the global Deligne comparison and its human-source theorem specified in the text.

17 The exact spectrum of support operations and transfers

17.1 The full support object and the endomorphism to be computed

Let L be a finite nontrivial bounded distributive lattice, $S = \text{SpecLat } L$, $X = \text{Spec } \mathbb{Z}$, and $Y = S \triangleright X$. The opens of Y are the opens of S and the sets $S \cup U$ with nonempty open $U \subset X$. This is the entire spectrum of the original semiring

$$\mathbf{G}_L(\mathbb{Z}) = \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in \mathbb{Z}\}. \quad (\text{SE1})$$

The supported zero $e = (0, 1_L)$ and the additive identity $\tau = (0, 0_L)$ remain distinct; $D(e) = S$ and $V(e) = X$. We use the original specialization order: $s \leq t$ means that s generalizes t , so finite opens are lower sets.

For completeness, the passage from the full lattice to this finite space loses no lattice element. An element $j \neq 0$ is join irreducible if $j = a \vee b$ forces $j = a$ or $j = b$. Distributivity makes it join prime: if $j \leq a \vee b$, then $j = (j \wedge a) \vee (j \wedge b)$, so $j \leq a$ or $j \leq b$. Every lattice element is the join of the join irreducibles below it, by induction in the finite order. Each j defines the prime ideal $I_j = \{\lambda : j \not\leq \lambda\}$. Conversely, the complement of a prime ideal is a nonempty finite filter; its meet j generates it, and primeness makes j join prime and nonzero. Thus these are exactly all prime ideals. Consequently

$$L \longrightarrow \text{Down}(S), \quad \lambda \longmapsto D(\lambda) = \{I_j : j \leq \lambda\}$$

is an isomorphism: joins and meets give union and intersection, injectivity follows from the join decomposition, and a downset of join irreducibles is obtained from the join of its elements. For the last assertion, a join prime below that finite join lies below one member and hence lies in the downset. An order map on S therefore gives an actual lattice map on this original L by inverse image of downsets. No Boolean replacement of L is made.

The receiving category in this note is abelian sheaves on the whole space Y . A sheaf is given by

$$(\mathcal{A}, \mathcal{B}, \phi), \quad \phi : \mathcal{A} \longrightarrow c_X \Gamma(S, \mathcal{B}), \quad (\text{SE2})$$

where c_X is the constant sheaf on the irreducible arithmetic space. Its mixed-open sections are $\mathcal{A}(U)$ and its support-open sections are $\mathcal{B}(V)$; the map ϕ gives their restrictions. This receiving

category is not silently replaced by additive-group modules over the split semiring sheaf. That latter action kills its additively idempotent support stalks, as proved in (RP17)–(RP20) of `split_prime_recollement_pairing.tex`.

Let Λ be a finite free abelian group and set $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{Q}$. The actual universal coefficient sheaf is

$$\mathcal{U}_{S,\Lambda} = (\mathcal{O}_X \otimes_{\mathbb{Z}} \Lambda, V_S, \text{constant generic restriction}). \quad (\text{SE3})$$

Every point of the support receives the same original generic value. For $\Lambda = \mathbb{Z}$ this is the previously constructed universal arithmetic coupling, with no branch selected.

For constant rational coefficients define the full strict-chain complex

$$C^q(S, \mathbb{Q}) = \prod_{s_0 < \dots < s_q} \mathbb{Q}, \quad (dc)_{s_0 < \dots < s_{q+1}} = \sum_{a=0}^{q+1} (-1)^a c_{s_0 < \dots < \widehat{s_a} < \dots < s_{q+1}}. \quad (\text{SE4})$$

Its exact relative arithmetic complex is

$$K_{S,\Lambda}^0 = \Lambda, \quad K_{S,\Lambda}^{q+1} = C^q(S, \mathbb{Q}) \otimes_{\mathbb{Q}} V, \quad d^0(\lambda) = (\lambda)_{s \in S}, \quad d^{q+1} = d_C \otimes 1. \quad (\text{SE5})$$

The proof that this is the actual closed-support derived complex is given in (FL8)–(FL22) of `finite_support_recollement.tex`: full sheaf gluing gives the mapping fibre $\text{Cone}(R\Gamma(X, \mathcal{O}_X \otimes \Lambda) \rightarrow R\Gamma(S, V_S))[-1]$; the first term is $\Lambda[0]$. The support complex is the global-sections complex of an injective resolution: its degree- q sheaf is the product over strict chains of the direct image of the first stalk from the chain's last point. At a point s , append the maximal point s to chains in $\downarrow s$ with sign $(-1)^q$; this contracts its augmented stalk complex. Thus no support degree has been omitted in (SE5). The signs shown are obtained from the usual shifted cone by multiplying its support coordinate in degree $q+1$ by $(-1)^q$.

17.2 The exact cone endomorphism and its gluing constraint

An order-preserving map $f : S \rightarrow S$ induces an actual map $f_Y : Y \rightarrow Y$, equal to the identity on X and to f on S . For support spectra of downset lattices it is induced by $D \mapsto f^{-1}(D)$ on the lattice and the unchanged integer coefficient on (SE1). The pullback cochain map is

$$(f^*c)_{s_0 < \dots < s_q} = \begin{cases} c_{f(s_0) < \dots < f(s_q)}, & \text{the image points are distinct,} \\ 0, & \text{otherwise.} \end{cases} \quad (\text{SE6})$$

It commutes with d : on a chain with an adjacent pair of equal image points, the two terms deleting that pair cancel, and all other terms retain a repetition; on a chain without such a repetition it is the usual alternating-face identity. The maps are functorial under composition, including noninjective maps.

Theorem 17.1 (An endomorphism of the actual sheaf and its cone). *Let $A \in \text{End}_{\mathbb{Z}}(\Lambda)$ and let A_V be its rational extension. The maps A on the arithmetic coefficients and A_V on the pulled-back constant support sheaf define a morphism $f_Y^{-1}\mathcal{U}_{S,\Lambda} \rightarrow \mathcal{U}_{S,\Lambda}$. On the exact relative complex its endomorphism is*

$$T_{A,f}^0 = A, \quad T_{A,f}^{q+1} = f^* \otimes A_V. \quad (\text{SE7})$$

For a general compatible cochain diagram $u : A^\bullet \rightarrow B^\bullet$, with endomorphisms a, b satisfying $bu = ua$, the corresponding endomorphism of $\text{Cone}(u)[-1]$ is exactly (a, b) on its two entries. There is no extra scalar in this construction.

Proof. For a regular coefficient section, taking its generic value and then applying A_V gives the generic value of its original image under A . Its support restriction is constant, and pullback by

any f preserves that constant section. This is precisely the commutative gluing square in (SE2), so it constructs the sheaf morphism. The degree-zero map of (SE5) therefore commutes with the constant generic inclusion. In higher degrees A_V commutes with restrictions and f^* commutes with d , proving (SE7) is its chain endomorphism.

For the general statement use the shifted-cone convention $R^n = A^n \oplus B^{n-1}$ with $d_R(x, y) = (d_A x, u(x) - d_B y)$. Applying (a, b) before or after this differential gives the same result exactly when $bu = ua$. There is no multiplication other than a and b in either formula. Changing to the positive support-differential convention of (SE5) conjugates both endomorphisms by the stated sign change and introduces no scalar. $\square$

The absolute and relative receivers have an exact comparison as well. Let $i : X \hookrightarrow Y$ be the arithmetic inclusion. Its restriction functor has exact left adjoint

$$U_S(\mathcal{A}) = (\mathcal{A}, c_S \mathcal{A}_\eta, \text{generic constant restriction}),$$

where η is the original arithmetic generic point, not a point of S . To check the adjunction, a map on the arithmetic component uniquely forces the support map: under $\text{Hom}(c_S \mathcal{A}_\eta, \mathcal{B}) = \text{Hom}(\mathcal{A}_\eta, \Gamma(S, \mathcal{B}))$ it is its generic restriction followed by the target gluing map. Both the generic-stalk functor and the constant-sheaf functor are exact, which proves exactness of U_S . Thus i^* preserves injectives. Since $\Gamma(Y, \mathcal{F}) = \Gamma(X, i^* \mathcal{F})$ by the triple description, injective resolutions give the exact equivariant identity

$$(R\Gamma(Y, \mathcal{U}_{S, \Lambda}), T_{A, f}) \simeq (\Lambda[0], A). \quad (\text{SE7a})$$

Here the arithmetic acyclicity was proved in (FL18)–(FL20) by the rational-sections and principal-parts resolution. Consequently the absolute rational receiver has exactly the coefficient eigenvalues. The new relative receiver is the actual cone (SE5); its additional degrees, possible zero blocks, and coefficient/support factors are calculated below. This gives both sides of the comparison, rather than substituting an absolute cohomology identity for the relative calculation.

Proposition 17.2 (When rescaling is a proved operation). *Over a coefficient field, replacing only the support endomorphism b by λb while retaining a defines an endomorphism of the same cone exactly when*

$$(\lambda - 1)ua = 0. \quad (\text{SE8})$$

In particular for the universal scalar arithmetic action $a = n \neq 0$ and its injective generic map, it forces $\lambda = 1$. Multiplying both a and b by a scalar is instead a genuine new coefficient operation whenever that scalar preserves their coefficient objects. On the original scalar lattice $\mathbb{Z}$, such a scalar is an integer; an invertible scalar preserving this lattice in both directions is 1 or -1 .

Proof. The new compatibility condition is $\lambda bu = ua$. Subtract the original equality $bu = ua$ to obtain exactly (SE8). The injectivity and nonzero scalar case gives the stated conclusion. Simultaneous scalar multiplication commutes with both differentials and preserves the equality. For a scalar preserving $\mathbb{Z}$, its value on 1 must be an integer; if its inverse preserves $\mathbb{Z}$ too, both it and its reciprocal are integers, so it is ± 1 . $\square$

Thus a one-dimensional coefficient representation with an assigned endomorphism λ would multiply the eigenvalues by λ because its tensor product does so. That is an additional specified operation. Neither the new prime topology nor the cone shift supplies that representation or an unproved Tate twist.

17.3 All eigenvalues and the nilpotent part of a finite support map

Theorem 17.3 (The complete eventual-periodicity decomposition). *For every self-map f of the finite support poset there are integers $N \geq 0$ and $m \geq 1$ such that $f^{N+m} = f^N$. Consequently, on every support cochain group and every rational support cohomology group,*

$$F^N(F^m - I) = 0, \quad F = f^*. \quad (\text{SE9})$$

Choose $k \geq N$ divisible by m and set $S_\infty = f^N(S)$. Then $r = f^k : S \rightarrow S_\infty$ is a retraction of the inclusion $a : S_\infty \hookrightarrow S$. The restriction f_∞ is an order automorphism of finite order dividing m . In each cohomological degree,

$$H^q(S, \mathbb{Q}) = \underbrace{r^* H^q(S_\infty, \mathbb{Q})}_{\text{periodic part}} \oplus \underbrace{\ker(a^* : H^q(S, \mathbb{Q}) \rightarrow H^q(S_\infty, \mathbb{Q}))}_{\text{nilpotent part}}. \quad (\text{SE10})$$

On the first part f^* is intertwined with f_∞^* by r^* ; on the second $(f^*)^N = 0$. The nonzero complex eigenvalues are roots of unity, with no nontrivial Jordan blocks. Every remaining Jordan block has eigenvalue zero and size at most N .

Proof. Every orbit of a finite function eventually reaches a cycle. Take N at least the maximum number of steps before a cycle and let m be a common multiple of all cycle lengths. This proves the equality of functions. Their pullbacks are functorial by (SE6), so the same polynomial identity holds already on cochains and hence on cohomology. The stable image consists of precisely the periodic points. There f is a permutation with inverse f^{m-1} . Both are order preserving, so the restricted map is an order automorphism. The map $r = f^k$ fixes every stable point and takes every point into the stable image, giving $ra = \text{id}$.

On cochains and cohomology, $E = r^* a^* = F^k$ is idempotent. The identities $a^* r^* = I$ and $E^2 = E$ give exactly the direct sum in (SE10). Its image is the finite-order part, since $F^m = I$ there. To verify the nilpotence bound, note that every $F^N v$ lies in this periodic image: the polynomial identity makes it fixed by F^m , and E acts as the identity on such a vector. If $Ev = 0$, then $F^N v = EF^N v = F^N Ev = 0$. Over $\mathbb{C}$, the polynomial $x^m - 1$ has distinct roots, since its derivative mx^{m-1} has no common root with it. It therefore diagonalizes the periodic part with roots of unity as eigenvalues. The other part is annihilated by x^N , which gives the stated zero eigenvalues and Jordan bound. $\square$

This decomposition retains both the stable image and its exact complement, with explicit projection and connecting maps. It is not an instruction to replace the full support by its periodic points.

It also computes all multiplicities, rather than only bounding the possible eigenvalues. In any fixed support degree, put $H = H^q(S, \mathbb{C})$, $E = F^k$ as above, and $N_0 = F|_{\ker E}$. For every m th root of unity ζ , define the exact projector

$$\Pi_\zeta = \frac{1}{m} \sum_{j=0}^{m-1} \zeta^{-j} F^j E. \quad (\text{SE10a})$$

On a periodic eigenvector of eigenvalue η , the coefficient is $m^{-1} \sum_j (\eta/\zeta)^j$, which is one for $\eta = \zeta$ and zero otherwise; on $\ker E$ it vanishes. Thus the multiplicity of ζ is exactly $\text{rank } \Pi_\zeta$, computable from the finite cochain kernels and quotients in (SE4). Put $d_a = \dim \ker N_0^a$, so $d_0 = 0$ and $d_a = \dim \ker E$ for $a \geq N$. The number of nilpotent Jordan blocks of size exactly $a \geq 1$ is

$$b_a = 2d_a - d_{a-1} - d_{a+1}. \quad (\text{SE10b})$$

Indeed a nilpotent block of size b contributes $\min(a, b)$ to d_a . Its first difference is one exactly for $a \leq b$, and its second difference in the displayed order is one exactly for $a = b$. Summing over

all blocks proves the formula. Therefore the finite cochain matrices, the recurrent retraction, and these formulas specify every eigenvalue and every Jordan block over $\mathbb{C}$, starting from the original rational cohomology action.

Nonzero nilpotent blocks really occur. Let S be the disjoint union of three copies of the six-point triangle face poset. Map the first copy isomorphically to the second, the second to the third, and every point of the third to one of its vertices. This is order preserving. Each triangle has rational $H^1 = \mathbb{Q}$, with the six-incidence alternating cycle coordinate from (FL35)–(FL39). The induced action is

$$F(a_1, a_2, a_3) = (a_2, a_3, 0) \quad \text{on } H^1(S, \mathbb{Q}) = \mathbb{Q}^3. \quad (\text{SE11})$$

It is a single zero-eigenvalue Jordan block of size three, because $F^3 = 0$ and $F^2 \neq 0$. The constant map on the third triangle kills its one-cochains under (SE6), which verifies the last coordinate without a homotopy assumption.

Theorem 17.4 (Combining support and coefficient spectra). *Let A be any endomorphism of a finite-dimensional complex coefficient space V , and let $F = f^*$ act on a support cohomology group H . Write $V = V_0 \oplus V_\times$ for the generalized zero eigenspace and the sum of its nonzero generalized eigenspaces. Write $H = H_{\text{nil}} \oplus H_{\text{per}}$ as in (SE10). Then the exact zero-eigenvalue part of $A \otimes F$ is*

$$(V_0 \otimes H_{\text{per}}) \oplus (V \otimes H_{\text{nil}}), \quad (\text{SE12})$$

and its complementary invertible part is $V_\times \otimes H_{\text{per}}$. On that part the eigenvalues are exactly $\lambda\zeta$, with multiplicity, where λ is a nonzero coefficient eigenvalue and ζ a support root of unity. The coefficient Jordan-block sizes are retained on each support eigenspace. In particular

$$|\lambda\zeta| = |\lambda|. \quad (\text{SE13})$$

Proof. Both displayed decompositions are invariant. On H_{per} choose the eigenspace decomposition provided by Theorem 17.3. Each summand with eigenvalue ζ gives the operator ζA on V . This has precisely the coefficient Jordan blocks with their eigenvalues multiplied by ζ . On $V \otimes H_{\text{nil}}$, the N th power is zero because $(A \otimes F)^N = A^N \otimes F^N$. On $V_0 \otimes H_{\text{per}}$, a power annihilating $A|_{V_0}$ annihilates the tensor. On the remaining summand both factors are invertible. This proves the exact generalized-zero space and every assertion about the complementary spectrum. $\square$

For completeness, the sizes of the combined zero blocks are also fixed by the original matrices. Set $A_0 = A|_{V_0}$ and $N_0 = F|_{H_{\text{nil}}}$. The rank of the a th power on the zero-eigenvalue space (SE12) is exactly

$$\text{rank}(A_0^a) \dim H_{\text{per}} + \text{rank}(A^a) \text{rank}(N_0^a), \quad a \geq 0. \quad (\text{SE12a})$$

This follows from the two invariant direct summands in that formula, invertibility of F^a on its periodic part, and $\text{rank}(B \otimes C) = \text{rank}(B) \text{rank}(C)$. To see the last equality, choose bases in which the image and kernel of each map have coordinate bases; their tensor map has exactly the products of the nonzero image basis vectors as an image basis. Subtract (SE12a) from the dimension of (SE12) and apply (SE10b) to obtain every combined zero block.

Consequently the statement that every eigenvalue has modulus one can change upon passing to the full support object: it acquires a failure if a nonzero nilpotent support part is retained. Its nonzero periodic eigenvalues do not acquire a new modulus. They retain the actual coefficient moduli by (SE13). A coefficient eigenvalue off the unit circle remains off it on every nonzero periodic support channel. If that channel is annihilated by the support map, the exact annihilated space is (SE12), rather than a corrected unit-modulus mode. If a supplied coefficient action is $p^{w/2}B$, its nonzero combined eigenvalues are $p^{w/2}\beta\zeta$; removing that scalar would be the specified coefficient operation $p^{-w/2}$, not a consequence of cohomological degree.

17.4 Relative degrees and their actual Euler determinant factors

For a bounded finite-dimensional complex $C^\bullet$ with chain endomorphism T , fix the convention

$$\mathcal{E}(C, T; t) = \prod_q \det(1 - tH^q(T))^{(-1)^{q+1}}. \quad (\text{SE14})$$

This rational function retains its stated convention throughout.

Proposition 17.5 (Cone determinant and degree change). *For a compatible map $(A^\bullet, a) \rightarrow (B^\bullet, b)$ and its actual relative cone (R, T) from Theorem 17.1,*

$$\mathcal{E}(R, T; t) = \mathcal{E}(A, a; t) / \mathcal{E}(B, b; t). \quad (\text{SE15})$$

A shift $C[-1]$ inverts $\mathcal{E}(C, T; t)$ and retains the same endomorphisms on the shifted cohomology spaces. More specifically, the actual relative cohomology has equivariant exact sequences

$$0 \rightarrow \text{coker}(H^{q-1}A \rightarrow H^{q-1}B) \rightarrow H^q R \rightarrow \ker(H^q A \rightarrow H^q B) \rightarrow 0. \quad (\text{SE16})$$

Thus its characteristic polynomial is the product of those of the two indicated actual subquotients. The degree shift changes the Euler exponent; kernels, cokernels, and possible extensions determine which eigenvalues and Jordan data the cone retains.

Proof. For an invariant short exact sequence of finite-dimensional vector spaces, a basis adapted to its subspace makes the endomorphism upper triangular by blocks, so its determinant $\det(1 - tT)$ is the product of those on the subspace and quotient. Apply this first to boundaries inside cycles and then to cycles inside cochains. All boundary factors cancel in the alternating product, proving that (SE14) can also be computed on cochains. The cochains of R are $A^q \oplus B^{q-1}$ with the diagonal endomorphism (a, b) , so their alternating product is exactly (SE15). The shift assertion follows by replacing q by $q + 1$. The long exact cohomology sequence of the cone, with the maps induced by u , gives (SE16). Its maps commute with T by $bu = ua$, and the same block determinant argument proves the characteristic-polynomial assertion. $\square$

For the actual lattice complex (SE5), its first relative group is

$$D_{S, \Lambda} = V^c / \Delta \Lambda, \quad c = |\pi_0(S)|, \quad (\text{SE17})$$

and its higher groups are $H^q(S, \mathbb{Q}) \otimes V$ in relative degree $q + 1$ for $q \geq 1$. On (SE17),

$$[v] \mapsto [(A_V v_{\bar{f}(b)})_{b \in \pi_0(S)}], \quad (\text{SE18})$$

where $\bar{f}$ is the actual map of connected components. Its canonical torsion subgroup is $\Delta(V/\Lambda)$, acted on by A ; the rational quotient is $V^c / \Delta V$, acted on by $\bar{f}^* \otimes A_V$. Indeed torsion in (SE17) means that some integer multiple of the vector is an integer-lattice diagonal, forcing it to be a rational diagonal. This proves the assertion and retains that torsion.

After rationalizing the complex, and only in that receiver, the torsion is killed. Its Euler determinant is then

$$\mathcal{E}(K_{S, \Lambda} \otimes \mathbb{C}, T_{A, f}; t) = \prod_{q \geq 0} \det \left(1 - t(A_{\mathbb{C}} \otimes \tilde{f}_q^*) \right)^{(-1)^q}, \quad (\text{SE19})$$

where degree zero is reduced support cohomology $\tilde{H}^0(S, \mathbb{C}) = H^0(S, \mathbb{C}) / \Delta \mathbb{C}$, and in positive degrees $\tilde{H}^q = H^q$. This is exactly the relative shift in (SE5). Formula (SE19) does not encode the discarded V/Λ ; its group and Hilbert receivers are treated in the next section.

For Boolean support, let the eventual cycles of the component map have lengths $\ell_1, \dots, \ell_b$. The transient coordinates give zero eigenvalues, and the periodic coordinates give these cycles. Thus the exact determinant in (SE19) is

$$\frac{\prod_{a=1}^b \det(1 - t^{\ell_a} A_{\mathbb{C}}^{\ell_a})}{\det(1 - tA_{\mathbb{C}})}. \quad (\text{SE20})$$

For a cycle of length ℓ , its eigenvalues are the ℓ th roots of unity. Taking their product against each coefficient eigenvalue gives $\prod_{\zeta^\ell=1} (1 - t\lambda\zeta) = 1 - (t\lambda)^\ell$, which proves the numerator. The diagonal coefficient space is invariant with action $A_{\mathbb{C}}$, proving the denominator. In particular with r fixed Boolean branches and scalar coefficient a it is $(1 - at)^{r-1}$.

For triangle support, its rational H^1 is one-dimensional, with the exact alternating incidence generator (FL36). The identity map acts there by 1; reflection of the triangle reverses the generator and acts by -1 ; the constant map acts by 0 by (SE6). With scalar coefficient a , the respective Euler factors are

$$(1 - at)^{-1}, \quad (1 + at)^{-1}, \quad 1. \quad (\text{SE21})$$

The last value does not say that the relative degree-two space has vanished: it is still $\mathbb{Q}$, with zero endomorphism. These examples compute the actual spectral effect of the topology and its map.

17.5 Exact scalar-branch maps and the degree action (FD1)–(FD9)

Write K_* for the one-support-point version of (SE5). Its diagonal inclusion in K_S sends the generic coefficient to the constant zero-cochain and is the identity on the arithmetic lattice. Evaluation at $s \in S$, zero on positive support degrees, gives the branch map $\beta_s : K_S \rightarrow K_*$. The actual intertwining identities are

$$T_{A,f}\Delta = \Delta T_{A,*}, \quad \beta_s T_{A,f} = T_{A,*} \beta_{f(s)}. \quad (\text{SE22})$$

They follow by substituting a constant cochain and by evaluation in (SE6). If s has period m , iteration gives $\beta_s T_{A,f}^m = T_{A,*}^m \beta_s$. There is also an equivariant retraction without choosing one branch: average evaluation over the finite stable set S_∞ in degree zero of support, use zero in positive support degrees, and use the identity on Λ . This is a chain map: its degree-one component composed with d^0 is the original inclusion $\Lambda \rightarrow V$, and every differential out of a positive degree of the target vanishes. It retracts Δ because the average of a constant cochain is its original value. The stable map permutes this finite set, so the retraction commutes with the endomorphisms. Its rational average is taken in V and sends every diagonal lattice element back to that same lattice element. Thus it splits off the actual scalar relative complex equivariantly while retaining the entire remaining support complex.

Now specialize to the scalar lattice $\Lambda = \mathbb{Z}$ and the positive integer coefficient n . Set

$$D_S = \mathbb{Q}^c / \Delta\mathbb{Z}, \quad T = nF, \quad (Fv)_b = v_{\bar{f}(b)}, \quad J = \bar{f}(\pi_0(S)), \quad r = |J|. \quad (\text{SE23})$$

Then the exact group kernel and cokernel are

$$\ker T \cong (n^{-1}\mathbb{Z}/\mathbb{Z}) \oplus \mathbb{Q}^{c-r}, \quad \text{coker } T \cong \mathbb{Q}^c / F\mathbb{Q}^c. \quad (\text{SE24})$$

For the kernel, the coordinates of a representative indexed by J must all equal a common k/n , with $k \in \mathbb{Z}$. Subtract that rational diagonal and retain the unrestricted coordinates off J . This gives the displayed isomorphism, independent of changing the representative by an integer diagonal. The image is $F\mathbb{Q}^c / \Delta\mathbb{Z}$, because $n\mathbb{Q} = \mathbb{Q}$ and the diagonal is included, which proves the cokernel. On the canonical diagonal torsion subgroup $\mathbb{Q}/\mathbb{Z}$ the action is exactly multiplication by n for every support map.

Theorem 17.6 (When the full counting pullback exists). *On finitely supported functions on D_S , consider the actual pullback $P_T h(d) = h(Td)$. If $\bar{f}$ is not a permutation, it does not map the vector δ_0 to $\ell^2(D_S)$: its support is the infinite kernel in (SE24). Its exact target is then the space of all functions on D_S , not a bounded counting operator.*

If $\bar{f}$ is a permutation, T is surjective with exactly n points in every fibre, and

$$P_T^* h(u) = \sum_{Td=u} h(d), \quad P_T^* P_T = nI. \quad (\text{SE25})$$

The diagonal torsion subspace is reducing, and extension by zero $J_S : \ell^2(\mathbb{Q}/\mathbb{Z}) \rightarrow \ell^2(D_S)$ satisfies

$$P_T J_S = J_S P_n, \quad P_T^* J_S = J_S P_n^*. \quad (\text{SE26})$$

For the additive receiver $\kappa_S(\delta_d) = d \otimes 1 \in \mathbb{C}^c / \Delta \mathbb{C}$,

$$\kappa_S P_T = \bar{F}^{-1} \kappa_S, \quad \kappa_S P_T^* = n \bar{F} \kappa_S. \quad (\text{SE27})$$

When $f = \text{id}$ these are exactly (FD3)–(FD5) and (FD8)–(FD9) of `16_full_support_degree_and_arithmetic_quotient.tex`.

Proof. The failure in the first case follows from $P_T \delta_0 = \mathbf{1}_{\ker T}$ and the nonzero rational summand in (SE24). If $\bar{f}$ is a permutation, F is a group automorphism and commutes with multiplication by n . The latter is surjective with kernel $\Delta(n^{-1}\mathbb{Z})/\Delta\mathbb{Z}$, of order n . Group the counting sum by these fibres to obtain the adjoint and norm formulas in (SE25).

The quotient of D_S by its torsion is $\mathbb{Q}^c/\Delta\mathbb{Q}$; on it $n\bar{F}$ is invertible. Thus Td is torsion exactly when d is torsion. Every fibre above torsion stays in that subgroup, and every other fibre stays outside it. This proves both reducing and the two pointwise identities (SE26).

The pullback of a basis vector is the sum of its n preimages. After additive complexification every such preimage has the same value $n^{-1}\bar{F}^{-1}(u \otimes 1)$, because the differences are torsion. Their sum proves the first identity in (SE27). The adjoint sends δ_d to δ_{Td} , proving the second. The complete kernel of κ_S consists of the linear span of δ_0 and $\delta_{a+b} - \delta_a - \delta_b$: this is the universal presentation of the additive tensor product. It in particular kills the entire diagonal torsion sector. This proves the receiver statements without discarding that sector beforehand. $\square$

For a nonpermutational component map there is still a precise map on finitely supported functions: the pushforward $T_* \delta_d = \delta_{Td}$. It obeys

$$\kappa_S T_* = (n\bar{F}) \kappa_S. \quad (\text{SE28})$$

Its algebraic dual is pullback on all functions, so the noninvertible support map is retained in exact domains and codomains. For the zero coefficient $n = 0$, the kernel is all of the infinite group D_S and the same counting-pullback obstruction holds. The abelian coefficient cone then has zero endomorphism; this statement does not identify the original semiring elements e and τ .

For a prime p and a permutational component map, write $P_T = U_F P_p$, with $U_F h(d) = h(Fd)$. The two factors commute and U_F is a finite-order unitary. Its normalized pullback is still a pure isometry in the exact sense of (FD6): the m th range consists of functions constant on cosets of the kernel of multiplication by p^m , since F^m is an automorphism. A vector in every such range is constant on cosets of the infinite group $\Delta(\mathbb{Z}[1/p]/\mathbb{Z})$ and square summability forces zero. The telescoping range projections therefore give the same bilateral shift extension as in (FD7), with coefficient $\sqrt{p}$ and with the support permutation retained. The norm factor is produced by the proved p -element fibre, not by the relative cohomological shift.

17.6 A proved covering operation that does produce a new multiplier

The finite-self-map restriction in Theorem 17.3 matters. Actual maps between different finite support lattices can have nonunit degree on their cohomology. The following construction gives all maps explicitly.

For $m \geq 1$, let P_m be the face poset of a cycle with $3m$ vertices v_i and $3m$ edges e_i , indexed modulo $3m$, with exactly $v_i < e_i$ and $v_{i+1} < e_i$. It has no two-dimensional face. Put $L_m = \text{Down}(P_m)$ and define

$$q_m : P_m \rightarrow P_1, \quad v_i \mapsto v_{i \bmod 3}, \quad e_i \mapsto e_{i \bmod 3}. \quad (\text{SE29})$$

It is an order map and an m -sheeted covering of the finite spaces: the preimage of a minimal vertex open is m disjoint vertex opens, and that of $\{v_i, v_{i+1}, e_i\}$ is m disjoint copies of that three-point minimal edge open. Inverse image of downsets gives a bounded lattice homomorphism $L_1 \rightarrow L_m$ and the corresponding coefficient-preserving semiring and full spectral maps.

For the complete cochain complex of P_m , use zero-coordinates a_i, b_i and incidence coordinates

$$(Da)_{i,0} = b_i - a_i, \quad (Da)_{i,1} = b_i - a_{i+1}.$$

Its cycle functional is

$$\omega_m(c) = \sum_{i \bmod 3m} (c_{i,0} - c_{i,1}). \quad (\text{SE30})$$

Its kernel is the image of D : set $a_0 = 0$, define successively $b_i = c_{i,0} + a_i$ and $a_{i+1} = b_i - c_{i,1}$, and the closing equation $a_{3m} = a_0$ is exactly $\omega_m(c) = 0$. The kernel of D itself is the constant functions on all vertices and edges. Since ω_m is surjective on the coordinate vector $c_{0,0} = 1$, these formulas prove $H^0(P_m, \mathbb{Q}) = \mathbb{Q}$ and $H^1(P_m, \mathbb{Q}) = \mathbb{Q}$ with the exact stated orientation.

Theorem 17.7 (Cover pullback, transfer, and their full gluing). *The actual cochain pullback q_m^* repeats every coordinate m times. The transfer Tr_{q_m} sums the coordinates over all lifts of each vertex, edge, and incidence chain. They are cochain maps and satisfy*

$$\text{Tr}_{q_m} q_m^* = mI. \quad (\text{SE31})$$

On support cohomology their exact maps are

	H^0	H^1
q_m^*	1	m
Tr_{q_m}	m	1.

(SE32)

The pullback of full universal arithmetic sheaves has arithmetic map the identity. Extending the support transfer across the arithmetic stratum forces its arithmetic map to be multiplication by m . With that map it gives an actual sheaf transfer, and the composite transfer after pullback is multiplication by m on the entire universal sheaf and its relative complex. On relative cohomology the individual maps are

	$H_X^1 = \mathbb{Q}/\mathbb{Z}$	$H_X^2 = \mathbb{Q}$
q_m^*	1	m
Tr_{q_m}	m	1.

(SE33)

Proof. Every base vertex, edge, and incidence has exactly m lifts. Each lift of a face is the face of exactly one lifted incidence of the fixed base type. Thus taking a differential and summing lifts commute, proving the transfer is a cochain map; pullback is already (SE6). Summing the m identical pulled-back coordinates proves (SE31) on every cochain, not only on cohomology. Constants pull back unchanged and transfer to m times themselves. Formula (SE30) gives $\omega_m(q_m^* c) = m\omega_1(c)$ and $\omega_1(\text{Tr}_{q_m} d) = \omega_m(d)$, proving (SE32).

The full map $Y_m \rightarrow Y_1$ is the identity on X and q_m on support. Pushforward on the support is exact because locally it is a finite product over the covering sheets; on arithmetic sheaves it is the identity. The full pushforward therefore is exact by the two-component sheaf description. The support trace is the sheaf map that sums these local sheets. A constant generic rational value a restricts on them to m copies of a , whose trace is ma . The gluing equation in (SE2) consequently requires the arithmetic image of a to have generic value ma . The injective map $\mathcal{O}_X \rightarrow \mathbb{Q}_X$ forces arithmetic multiplication by m . This map satisfies the equation and therefore defines the full sheaf transfer.

On relative complexes the pullback is the identity on the initial $\mathbb{Z}$ and coordinate repetition thereafter. The transfer is m on that $\mathbb{Z}$ and the just-proved cochain trace thereafter. Their composite is m in every degree. A constant rational modulo the original integer diagonal is sent to itself under pullback and multiplied by m under transfer, while the cycle coordinates are those already computed. This proves (SE33). $\square$

$$\begin{array}{ccccc}
\mathbb{Z} & \xrightarrow{\Delta} & C^0(P_1, \mathbb{Q}) & \xrightarrow{D} & C^1(P_1, \mathbb{Q}) \\
\parallel & & \downarrow q_m^* & & \downarrow q_m^* \\
\mathbb{Z} & \xrightarrow{\Delta} & C^0(P_m, \mathbb{Q}) & \xrightarrow{D} & C^1(P_m, \mathbb{Q}) \\
\downarrow m & & \downarrow \text{Tr}_{q_m} & & \downarrow \text{Tr}_{q_m} \\
\mathbb{Z} & \xrightarrow{\Delta} & C^0(P_1, \mathbb{Q}) & \xrightarrow{D} & C^1(P_1, \mathbb{Q})
\end{array}$$

Figure 11: The actual relative complexes, in degrees 0, 1, 2 from left to right. The support covering fixes the arithmetic coefficient under pullback. Summing its m support sheets forces multiplication by m on that coefficient under transfer. Every square commutes, and each composite vertical map is mI , by Theorem 17.7 and (SE31)–(SE33). The diagram records actual maps and assigns no metric or weight to the prime topology.

In particular, the multiplier m on the new relative degree-two class has a proved source: the actual covering map and its oriented cycle coordinate. It is a map between the displayed support spaces; it is not the action of a self-map of one fixed finite space with eventual periodicity. The transfer–pullback composite is an actual endomorphism, and its multiplier is the proved degree. Its extension over the arithmetic stratum obeys, rather than bypasses, the scalar constraint (SE8). The induced determinant factor on the rational relative degree-two line is $(1 - mt)^{-1}$ under the convention (SE14). This distinguishes an actual degree operation from a scalar invented from a degree label.

17.7 Proof scope and source locators

The existing reading ledger and the complete finite-support proof `finite_support_recollement.tex`, especially (FL8)–(FL22) and (FL32)–(FL39), fix the topology, derived complex, and triangle orientation used here. The actual root source `16_full_support_degree_and_arithmetic_quotient.tex`, (FD1)–(FD9), was read in full through its degree-action receiver. The identities (SE25)–(SE27) prove the comparison including noninvertible support maps and the complete domains in which their pullbacks exist.

No assertion in this note is labelled as Deligne purity. The exact finite-support operation gives roots of unity and a computed nilpotent part; the coefficient operation supplies its actual scalar moduli; the cone supplies computed subquotients and Euler exponents; and the covering/transfer construction supplies its explicitly proved degree. No Tate twist, scalar multiplier, or positive

metric is inferred from the topology without the corresponding operation. The Connes–Consani source is not needed for these finite-support derivations and is not claimed as read here.

18 The class-field functor and the actual adelic coefficient sheaf over the split base

We use Alain Connes and Caterina Consani’s original author source *Knots, primes and class field theory*, arXiv:2501.06560v1 [14]. The inputs are their associated covers in `sec2`, the class-field functor `main`, and, in Section 4, `localv`, `structure`, and `structure1`. The source archive and its author TeX are retained without alteration. The following constructions apply the full spectrum (FL3) to those objects. In particular they do not replace the support spectrum by one selected branch. All sheaf cohomology below is cohomology of the underlying complex vector sheaf; a multiplication is asserted only where its compatibility is proved.

18.1 Extending the cover functor, with the inertia retained

Put $S = \operatorname{Spec}_{\text{lat}} L$, where L is a finite nontrivial bounded distributive lattice. For a topological space Z define

$$T_S Z = S \sqcup Z, \quad \operatorname{Open}(T_S Z) = \operatorname{Open}(S) \cup \{S \cup U : \emptyset \neq U \in \operatorname{Open}(Z)\}. \quad (\text{CCF1})$$

This is a topology: a union involving an old open contains all of S , and an intersection of two such opens either has the displayed form or equals S . It is functorial in continuous maps on both strata. The inverse image of $S \cup U$ under $f \sqcup h$ is $S' \cup h^{-1}U$, interpreted as the open S' when $h^{-1}U$ is empty. The foundational prime classification gives the actual homeomorphism $\operatorname{Spec} G_L(R) = T_S \operatorname{Spec} R$ for every nonzero ring R .

Use the original spaces and group

$$Y_{\mathbb{Q}} = \mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}, \quad K = \widehat{\mathbb{Z}}^{\times}, \quad X_{\mathbb{Q}} = Y_{\mathbb{Q}}/K, \quad \chi : K \rightarrow G, \quad (\text{CCF2})$$

where G is finite abelian and χ is continuous. Connes–Consani’s cover is $X_{\mathbb{Q}}^{\chi} = (Y_{\mathbb{Q}} \times G)/K$, with $w \cdot (y, g) = (yw, \chi(w)g)$. Extend the K action trivially on S . There are canonical, G -equivariant homeomorphisms

$$(T_S Y_{\mathbb{Q}})/K \simeq T_S X_{\mathbb{Q}}, \quad ((T_S Y_{\mathbb{Q}}) \times G)/K \simeq T_S X_{\mathbb{Q}}^{\chi}. \quad (\text{CCF3})$$

For the first statement, a saturated old open is the inverse image of an old quotient open, and a support open is unchanged. For the second, write an open in $(T_S Y_{\mathbb{Q}}) \times G$ by its G slices. Invariance under K and surjectivity of χ imply that either all slices have no old point, with the same support open, or every slice contains all of S . In the latter case their old parts form an open invariant subset of $Y_{\mathbb{Q}} \times G$. This proves the quotient topology in (CCF3), not only a bijection of points.

The map in (CCF3) over an old point has the original fibre and stabilizer. In particular, for a representative a with $a_{\infty} \neq 0$, put $Z(a) = \{v : a_v = 0\}$. Its fibre and inertia are

$$G/\chi(H_{Z(a)}), \quad \chi(H_{Z(a)}), \quad H_{Z(a)} = \prod_{v \in Z(a)} \mathbb{Z}_v^{\times} \subset K. \quad (\text{CCF4})$$

Indeed $aw = qa$ forces $q = 1$ at the real coordinate; at a nonzero finite coordinate it forces $w_v = 1$, and at a zero coordinate it imposes no condition. This proves the stabilizer formula directly. For a support point $s \in S$ the fibre is one coarse point and its G stabilizer is all of G . Thus the extension changes the inertia geometry along every support point.

The group action retained at that point is explicit. The action groupoid of K on $\{s\} \times G$ is equivalent to the one-object groupoid $B \ker \chi$: the orbit is transitive and the automorphisms of

$(s, 1)$ are exactly $\ker \chi$. Its map to the base groupoid BK is induced by $\ker \chi \hookrightarrow K$. Consequently the finite covering data at the new point are

$$B \ker \chi \longrightarrow BK, \quad K / \ker \chi \simeq G. \quad (\text{CCF5})$$

This also computes the exact transfer on constant complex coefficients. The induced coefficient space is $\mathbb{C}[G]$; the unit sends z to the constant vector $(z)_{g \in G}$, and the counit sums its entries. Their composite is $|G|z$. Coarse fibre cardinality one therefore does not compute this transfer: the stabilizer data in (CCF5) give its degree $|G|$.

For a finite abelian extension $E/\mathbb{Q}$, take the source's reciprocity map $\chi_E : K \rightarrow G_E$. Given an embedding $\sigma : E_1 \rightarrow E_2$, write it as inclusion followed by $h \in G_{E_1}$, with restriction $r : G_{E_2} \rightarrow G_{E_1}$. The source's map is

$$\rho_\sigma(g) = r(g)h, \quad [y, g] \longmapsto [y, \rho_\sigma(g)]. \quad (\text{CCF6})$$

It descends since $r\chi_{E_2} = \chi_{E_1}$ and both groups are abelian. Extend it by the identity on S . The slice proof above shows continuity, and composition agrees with composition of the ρ_σ . Hence (CCF3) defines an actual contravariant extension of the supplied class-field functor. The reciprocity theorem used to supply χ_E is the human-source input; no new proof of class field theory is asserted here.

There is a compatible algebraic action on the original base. Every $g \in G_E$ acts on $G_L(\mathcal{O}_E)$ by $(a, \lambda) \mapsto (g(a), \lambda)$. On prime spectra it fixes each support prime and acts on the arithmetic primes by the usual Galois action. Its map to $\text{Spec } G_L(\mathbb{Z})$ is the identity on S and the original arithmetic prime map on the other stratum. This follows by taking inverse images in (FL3). In particular the fixed support points in (CCF3) have the same coefficient-action origin as the fixed support primes of the actual semiring.

The original periodic orbit C_p and its mapping torus remain old subspaces, with their original topology. In the source's orientation their period and monodromy are

$$\text{length}(C_p) = \log p, \quad g \longmapsto \chi_E(p)g \quad \text{when } \chi_E(\mathbb{Z}_p^\times) = 1. \quad (\text{CCF7})$$

This is also obtained directly from $(h, \lambda) \sim (ph, p\lambda)$ in the source mapping torus. The new support stratum is fixed by the scaling action. The unramified monodromy in (CCF7) is a permutation on $\mathbb{C}[G_E]$: its eigenvalues are roots of unity. The new stabilizers, transfers, and the cohomology calculated next must all be retained in applying this observation to the split base.

18.2 A complete resolution of the original semilocal sheaves

Let $X = \text{Spec } \mathbb{Z}$, with generic point η . In this subsection F denotes a finite set of finite primes, not the support space S . Retain the actual Bruhat–Schwartz functions and group algebras

$$V = \mathcal{S}(\mathbb{A}_{\mathbb{Q}}), \quad B = V \rtimes \mathbb{Q}^\times, \quad \mathcal{O}(X \setminus F) = \mathcal{S}(\mathbb{A}_{F \cup \{\infty\}}), \quad \mathcal{A}(X \setminus F) = \mathcal{O}(X \setminus F) \rtimes \mathbb{Z}[F^{-1}]^\times. \quad (\text{CCF8})$$

Every inclusion into the generic space inserts $1_{\mathbb{Z}_p}$ at the missing finite primes. These are exactly the sheaves **structure** and **structure1** of Connes–Consani, including their specified transition maps.

Write $V^{(p)}$ for the restricted tensor product over all places other than p , including the real factor, and put

$$V_p = \mathbb{C}1_{\mathbb{Z}_p} \otimes V^{(p)}, \quad B_p = \bigoplus_{r \in \mathbb{Q}^\times : v_p(r)=0} V_p U_r. \quad (\text{CCF9})$$

They are the actual stalks of $\mathcal{O}$ and $\mathcal{A}$ at the closed point p . To see this, the neighbourhoods of p may exclude any finite set not containing p . Their union allows arbitrary finite coefficients away from p but retains precisely the displayed unramified factor and the condition $v_p(r) = 0$. Define the marked local defect spaces

$$D_V = \bigoplus_p V/V_p, \quad D_B = \bigoplus_p B/B_p, \quad \delta_V(f) = ([f]_p)_p, \quad \delta_B(b) = ([b]_p)_p. \quad (\text{CCF10})$$

The maps land in direct sums: a Bruhat–Schwartz function involves only finitely many non-standard finite factors, and a rational number has nonzero valuation at only finitely many primes.

There are exact flasque resolutions

$$\begin{aligned} 0 \longrightarrow \mathcal{O} \longrightarrow c_X V \longrightarrow \bigoplus_p i_{p*}(V/V_p) \longrightarrow 0, \\ 0 \longrightarrow \mathcal{A} \longrightarrow c_X B \longrightarrow \bigoplus_p i_{p*}(B/B_p) \longrightarrow 0. \end{aligned} \tag{CCF11}$$

The displayed arrows are the quotient maps at every included closed point. Their kernel on $X \setminus F$ is exactly the space in (CCF8): a coefficient is in every retained V_p exactly when it has the unramified factor at every $p \notin F$, and a group label must be a unit at those same primes. The source’s local characterization `localv`, or its factorization $f(x, y) = 1_{\mathbb{Z}_p}(x)f(0, y)$, proves this coefficient claim. At p the cokernel arrow is onto by definition, and at η its target stalk is zero. This proves sheaf exactness.

For clarity, the direct sums in (CCF11) have finite-support sections on each nonempty open. A section has zero generic germ, so vanishes on a smaller cofinite open. The remaining support is finite. Conversely any finite collection of stalk values defines a section of the corresponding sum of skyscraper sheaves. Its restriction map forgets the excluded components and is onto. These sheaves are therefore flasque. The constant sheaves are flasque because every nonempty open of X is connected and their nonempty restrictions are identities. It follows that

$$R\Gamma(X, \mathcal{O}) = [V \xrightarrow{\delta_V} D_V], \quad R\Gamma(X, \mathcal{A}) = [B \xrightarrow{\delta_B} D_B], \tag{CCF12}$$

in degrees zero and one, with all higher groups zero. In particular their degree-zero groups are exactly $\mathcal{S}(\mathbb{R})$ and $\mathcal{S}(\mathbb{R}) \rtimes \{\pm 1\}$, as in the original source. This calculation also determines the previously unspecified higher sheaf group.

Here is its explicit mixed-place content. Define $E_p = \ker(f \mapsto f(0) : \mathcal{S}(\mathbb{Q}_p) \rightarrow \mathbb{C})$. Since $1_{\mathbb{Z}_p}(0) = 1$, the original space splits as $\mathbb{C}1_{\mathbb{Z}_p} \oplus E_p$. Expanding each finite tensor therefore gives, with all factors retained,

$$V = \bigoplus_{T \text{ finite}} V_T, \quad V_T = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_{p \in T} E_p. \tag{CCF13}$$

Factors outside T are $1_{\mathbb{Z}_p}$. For $r \in \mathbb{Q}^\times$ put $D(r) = \{p : v_p(r) \neq 0\}$ and $T_r = T \cup D(r)$. Then (CCF12) is the direct sum of the actual diagonal complexes

$$[V_T \longrightarrow V_T^T] \quad \text{or} \quad [V_T U_r \longrightarrow (V_T U_r)^{T_r}], \tag{CCF14}$$

respectively. An empty target is zero. Indeed $V_T U_r$ survives in B/B_p precisely when $p \in T_r$; the quotient sends its coefficient unchanged to that marked component. Thus

$$\begin{aligned} H^1(X, \mathcal{O}) &= \bigoplus_{|T| \geq 2} V_T \otimes (\mathbb{C}^T / \Delta \mathbb{C}), \\ H^1(X, \mathcal{A}) &= \bigoplus_{r, T: |T_r| \geq 2} V_T U_r \otimes (\mathbb{C}^{T_r} / \Delta \mathbb{C}). \end{aligned} \tag{CCF15}$$

Equations (CCF11)–(CCF15) retain every mixed finite-place coefficient and every original rational label. The decomposition (CCF13) is not claimed to be invariant under nonunit finite scalings; their exact defect is computed below.

18.3 Relative cohomology after adjoining the whole support spectrum

For either $(\mathcal{H}, W, D) = (\mathcal{O}, V, D_V)$ or $(\mathcal{A}, B, D_B)$ take the universal extension $U_S \mathcal{H}$ of (FL12). Its restriction to S is the constant sheaf $c_S W$, and its arithmetic-to-support map is the actual generic inclusion, followed by the constant diagonal. If S has one point, (CCF12) and the relative cone give the particularly simple complete answer

$$R\Gamma_X(T_S X, U_S \mathcal{H}) \simeq D[-1]. \quad (\text{CCF16})$$

Here $D[-1]$ means D in cohomological degree one. To prove the identity with signs specified, use the complex $W \rightarrow D \oplus W$, $w \mapsto (\delta w, w)$. The map $(d, w) \mapsto d - \delta w$ kills this differential and induces an isomorphism on its cokernel. Its kernel consists exactly of $(\delta w, w)$; its degree-zero kernel is zero. This is a full quasi-isomorphism, not a subtraction of dimensions.

For arbitrary S , let $C^\bullet(S, W)$ be the complete finite cochain complex (FL18), with its stated specialization orientation. An explicit relative complex is

$$\begin{aligned} K^0 &= W, & K^1 &= D \oplus C^0(S, W), & K^n &= C^{n-1}(S, W) \quad (n \geq 2), \\ d^0 w &= (\delta w, \Delta w), & d^1(d, c) &= d_S c, & d^n &= d_S \quad (n \geq 2). \end{aligned} \quad (\text{CCF17})$$

Changing the support coordinate signs in the mapping fibre gives precisely this convention. Since $d_S \Delta = 0$, it is a complex. Let $\tilde{C}^\bullet(S, W)$ denote $C^\bullet(S, W)$ modulo the constant diagonal in degree zero. Projection in (CCF17) gives an exact sequence of complexes whose kernel is $[W \rightarrow D \oplus \Delta W]$. The preceding explicit contraction identifies that kernel with $D[-1]$. Therefore

$$\begin{aligned} H_X^0 &= 0, & 0 &\longrightarrow D \longrightarrow H_X^1 \longrightarrow W \otimes \tilde{H}^0(S, \mathbb{C}) \longrightarrow 0, \\ H_X^{j+1} &= W \otimes H^j(S, \mathbb{C}) \quad (j \geq 1). \end{aligned} \quad (\text{CCF18})$$

No support component or support cycle has been suppressed. Choosing $s_0 \in S$ splits the degree-one sequence by the retraction $[(d, c)] \mapsto d - \delta(c_{s_0})$. It is well defined on closed representatives modulo $(\delta w, \Delta w)$. The choice is stated explicitly; the injection of D is canonical.

The original arithmetic map is also explicit. Forgetting the support part in (CCF17) sends $[(d, c)]$ to $[d]$ in $D/\delta W = H^1(X, \mathcal{H})$. On the canonical D summand it is the quotient map

$$D \longrightarrow D/\delta W. \quad (\text{CCF19})$$

It is onto. Thus the image in the original semilocal sheaf cohomology is exactly (CCF15). In particular the vanishing image computed for the different coefficient sheaf $\mathcal{O}_X$ in (AQ17) cannot be substituted here: (CCF19) computes the required comparison for these actual Connes–Consani coefficients.

18.4 Both pole characters and Fourier reflection on the new groups

Use additive Haar measure with $\text{vol}(\mathbb{Z}_p) = 1$ at finite places and dx at the real place. Define on V

$$\epsilon_0(f) = f(0), \quad \epsilon_1(f) = \int_{\mathbb{A}_{\mathbb{Q}}} f(x) dx, \quad (\vartheta_\lambda f)(x_\infty, x_f) = f(\lambda^{-1} x_\infty, x_f), \quad \lambda > 0. \quad (\text{CCF20})$$

Insertion of $1_{\mathbb{Z}_p}$ preserves both functionals, so they define a sheaf morphism $\mathcal{O} \rightarrow c_X \mathbb{C}^2$. It is onto on every nonempty open. For example $e^{-\pi x^2}$ has moments $(1, 1)$ and $x^2 e^{-\pi x^2}$ has moments $(0, 1/(2\pi))$, and both have standard factors at every finite place. The usual Gaussian integral gives these values; it also follows by differentiating $\int e^{-a\pi x^2} dx = a^{-1/2}$ at $a = 1$. The exact pole action is

$$(\epsilon_0, \epsilon_1)\vartheta_\lambda = \text{diag}(1, \lambda)(\epsilon_0, \epsilon_1). \quad (\text{CCF21})$$

The first equality is evaluation and the second is change of variables; no factor has been inferred from the height of a prime.

Put $\mathcal{O}^0 = \ker(\epsilon_0, \epsilon_1)$ and $V^0 = \ker((\epsilon_0, \epsilon_1) : V \rightarrow \mathbb{C}^2)$. At every finite prime the canonical map

$$V^0/(V^0 \cap V_p) \xrightarrow{\sim} V/V_p \quad (\text{CCF22})$$

is an isomorphism. Injectivity is its displayed kernel. For surjectivity, subtract from a given f an element of V_p with the same two moments, using the two real test functions above. This calculation retains the two pole conditions rather than dropping them without a map. For one support point, $\mathcal{O}^0$ and $\mathcal{O}$ therefore have the same relative degree-one group D_V . For general S , their difference is exactly the relative complex of the constant pole sheaf:

$$R\Gamma_X(U_{ScX}\mathbb{C}^2) = \tilde{C}^\bullet(S, \mathbb{C}^2)[-1], \quad (\text{CCF23})$$

with the degree convention of (CCF17). This follows either by the same contraction with $D = 0$, or by the short exact sequence of the three coefficient sheaves. Its two scaling characters are still 1 and λ on every support term.

Take the original additive Fourier transform, with character $e^{-2\pi ixy}$ at infinity and the compatible conductor $\mathbb{Z}_p$ characters at finite places. Character orthogonality gives $\widehat{1_{\mathbb{Z}_p}} = 1_{\mathbb{Z}_p}$: the integral over $\mathbb{Z}_p$ is one when the character is trivial there and zero otherwise, by translating it by an element with nontrivial character value. The tensor Fourier transform $\mathcal{F}$ thus preserves every V_p , every semilocal transition, and V^0 . It exchanges the two moments and obeys the exact identities

$$\mathcal{F}^2 f(x) = f(-x), \quad \mathcal{F}\vartheta_\lambda = \lambda\vartheta_{\lambda^{-1}}\mathcal{F}. \quad (\text{CCF24})$$

The second follows by $x_\infty = \lambda t$ in the defining integral. The first is Fourier inversion on the real Schwartz space and on finite local test functions; the tensor identity then holds on every original adelic test. Consequently (CCF24) holds on D_V , all complexes (CCF17), and their arithmetic images (CCF19). It is an actual weight-one reflection on the reconstructed objects, including their support cycles.

There is a precise multiplicative target for the crossed product as well. Fourier transforms pointwise multiplication to additive convolution. If $\alpha_r f(x) = f(r^{-1}x)$, then the product formula gives $|r|_{\mathbb{A}} = 1$, and $\mathcal{F}\alpha_r = \alpha_{r^{-1}}\mathcal{F}$. Hence

$$\sum f_r U_r \mapsto \sum \widehat{f}_r U_{r^{-1}} \quad (\text{CCF25})$$

is an isomorphism from the pointwise crossed-product sheaf to the convolution crossed-product sheaf. In fact it sends $f_r \alpha_r(g_s) U_{rs}$ to $\widehat{f}_r * \alpha_{r^{-1}}(\widehat{g}_s) U_{r^{-1}s^{-1}}$, which is exactly the target product. At a semilocal stage the same product formula holds for its unit group, and Fourier fixes the inserted standard factors. Thus (CCF25) respects every original restriction and induces the corresponding relative-cohomology isomorphism. Its scaling relation is still (CCF24), with the transported convolution action $\lambda\vartheta_{\lambda^{-1}}$ written in full.

18.5 The exact positive receiver and the nonunit-prime defect

The additive Haar metric gives a concrete positive receiver on (CCF16). It is not introduced by assigning a weight to a topological degree. In the p factor, the orthogonal projection onto the original unramified line is

$$(P_p f)(x_p, x^{(p)}) = 1_{\mathbb{Z}_p}(x_p) \int_{\mathbb{Z}_p} f(t, x^{(p)}) dt. \quad (\text{CCF26})$$

It preserves Bruhat–Schwartz functions and has image V_p . Consequently $[f]_p \mapsto (1 - P_p)f$ is a well-defined isomorphism $V/V_p \rightarrow (1 - P_p)V$. Give this actual quotient the positive norm

$\|(1 - P_p)f\|_{L^2(\mathbb{A})}$ and give D_V the orthogonal direct-sum norm. It vanishes only on the zero quotient class, since $(1 - P_p)f = 0$ implies $f \in V_p$.

On D_B , apply the same quotient projection to a coefficient f_r when $v_p(r) = 0$, and leave it unchanged when $v_p(r) \neq 0$, then sum squared norms over (p, r) . This is exactly the quotient by the subspace B_p of (CCF9), with counting measure on rational group labels. All sums are finite on the algebraic spaces under discussion. Both forms satisfy

$$\langle \vartheta_\lambda d, \vartheta_\lambda d' \rangle = \lambda \langle d, d' \rangle, \quad (\text{CCF27})$$

because P_p acts only on a finite-place coordinate and real Haar measure changes by λ . Fourier is an isometry for these forms and commutes with P_p ; (CCF25) also permutes the rational labels without changing their unit condition. For the full support space, any chosen s_0 in (CCF18), together with a fixed positive form on the finite-dimensional support cohomology, extends (CCF27) to every relative group. The choice of the support form and splitting is part of that receiver and is not asserted to be canonical.

The receiving map (CCF19) has its actual kernel δW . Its attained positive quotient is calculated using the closure of this kernel. Let $\overline{D}_V$ be the Hilbert completion of the above orthogonal sum. If F contains all nonstandard finite factors of a test function f , the commuting projections give the exact orthogonal decomposition

$$f = \sum_{T \subset F} \left(\prod_{p \in T} (1 - P_p) \prod_{p \in F \setminus T} P_p \right) f. \quad (\text{CCF28})$$

This follows by expanding $\prod_{p \in F} (P_p + (1 - P_p)) = I$; different choices are orthogonal in at least one factor. For two different primes and any test f , set $R_{pq} = (1 - P_p)(1 - P_q)f$. It is the same mixed function whether extracted from the p or q component of $\delta_V f$. Thus the bounded maps on $\overline{D}_V$

$$d \mapsto (1 - P_q)d_p - (1 - P_p)d_q \quad (\text{CCF29})$$

vanish on $\delta_V V$ and on its closure. They may be nonzero: take $d_p = h_p \otimes h_q \otimes h_\infty$, $d_q = 0$, with $h_p \perp 1_{\mathbb{Z}_p}$, $h_q \perp 1_{\mathbb{Z}_q}$ and nonzero Schwartz factors. This also proves directly that the arithmetic mixed image in (CCF19) is not lost under completion.

An exact orthogonal decomposition computes the whole quotient. Put $E_p^\perp = \ker(\int_{\mathbb{Z}_p})$ in the local Bruhat–Schwartz space. Replace E_p in (CCF13) by this orthogonal complement, and call the resulting summand $V_T^\perp$. All unramified factors have norm one. Then

$$\overline{D}_V = \widehat{\bigoplus_{T \neq \emptyset} V_T^\perp} \otimes \mathbb{C}^T, \quad \overline{\delta_V V} = \widehat{\bigoplus_{T \neq \emptyset} V_T^\perp} \otimes \Delta \mathbb{C}, \quad (\text{CCF30})$$

and the attained Hilbert quotient is

$$\overline{D}_V / \overline{\delta_V V} \simeq \widehat{\bigoplus_{|T| \geq 2} V_T^\perp} \otimes (\Delta \mathbb{C})^\perp. \quad (\text{CCF31})$$

To prove these identities, expand any finite tensor in the mutually orthogonal choices $\mathbb{C}1_{\mathbb{Z}_p}$ and $E_p^\perp$. Its T component appears in D_V exactly at the marked primes $p \in T$, and δ_V is the diagonal on that component. Such finite tensors are dense in the restricted adelic L^2 tensor product, and finite collections of their diagonal images are obtained from one finite sum in V . Taking closures therefore gives exactly (CCF30). Orthogonal projection onto the diagonal in each finite $\mathbb{C}^T$ gives (CCF31). The projection is the literal arithmetic mean $|T|^{-1} \sum_{p \in T}$, so all multiplicities are retained. The natural algebraic map from (CCF15) into (CCF31) is injective: a finite component has zero projection precisely when its marked values are diagonal, already the algebraic relation. This is the complete quotient calculation, not a supposition that the image of a dense subspace is closed.

Exactly the same proof for B replaces T by $T_r = T \cup D(r)$ and adds its original label r . It yields

$$\overline{D}_B / \overline{\delta_B B} \simeq \bigoplus_{r, T: |T_r| \geq 2} \widehat{\bigoplus} \overline{V}_T^\perp U_r \otimes H_{T_r}, \quad H_{T_r} = \{(c_p)_{p \in T_r} : \sum_{p \in T_r} c_p = 0\}. \quad (\text{CCF32})$$

The finite factor uses its inherited Euclidean metric. Equations (CCF27) and (CCF24) pass to both attained quotients because the diagonal relation and its closure are invariant.

Finally the operation of a finite prime must be traced through these maps, rather than inferred from (CCF27). Let $(\alpha_p f)(x) = f(p^{-1}x)$ act at all adelic places. The product formula makes it an isometry on $L^2(\mathbb{A})$, but it does not preserve V_p : its local standard factor becomes $1_{p\mathbb{Z}_p}$. The exact nonunit defect is the map

$$\partial_p : V_p \longrightarrow V/V_p, \quad f \longmapsto [\alpha_p f]_p. \quad (\text{CCF33})$$

For $f = 1_{\mathbb{Z}_p} \otimes g$ it is represented by $1_{p\mathbb{Z}_p} \otimes \alpha_p^{(p)} g$; subtracting its orthogonal unramified projection gives the exact vector

$$(1_{p\mathbb{Z}_p} - p^{-1}1_{\mathbb{Z}_p}) \otimes \alpha_p^{(p)} g. \quad (\text{CCF34})$$

Its squared norm is $(p^{-1} - p^{-2}) \|\alpha_p^{(p)} g\|^2 = (1 - p^{-1}) \|g\|^2$, because the product of the other local moduli of the rational p is p . Thus the defect is nonzero and has a calculated norm on the original coordinates. It is an actual element of the supported relative group, not a discarded failure of an endomorphism.

At a prime $\ell \neq p$, the same rational p is a local unit, fixes $1_{\mathbb{Z}_\ell}$, and preserves V_ℓ . Thus (CCF33) is localized at the original prime p . On the crossed product its coefficientwise form is the defect of $\text{Ad } U_p$ on B_p , with group labels unchanged. The real scaling ϑ_p is an endomorphism of every sheaf and every quotient above, with the exact factor p in (CCF27); the rational-prime action has the additional boundary (CCF34). These two actions are connected by their explicit adelic components, not identified by dropping that boundary. The calculation supplies the full support and mixed-place objects on which a global trace comparison must now be performed.

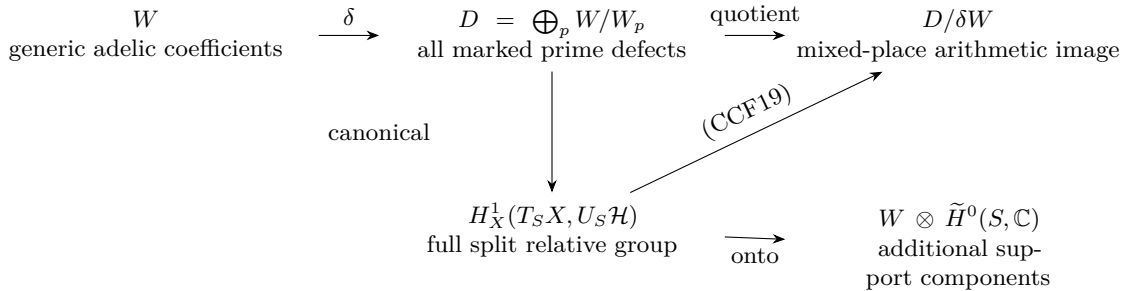


Figure 12: The actual Connes–Consani coefficient sheaf after the split-base extension. The bottom row is completed on the left by $0 \rightarrow D$ in (CCF18); its kernel is exactly the marked defect space. Every higher support cycle contributes $W \otimes H^j(S, \mathbb{C})$ in degree $j + 1$. Equations (CCF11)–(CCF19) prove the maps, and (CCF30)–(CCF32) compute the attained positive arithmetic quotient. Human coefficient source: Connes–Consani [14], Section 4.

19 The complete marked metric and its attained relative quotient

19.1 Source, coefficients, and exact scope

The human source is Alain Connes and Caterina Consani, *Knots, primes and class field theory*, arXiv:2501.06560v1. The original author file `arxiv-pisa.tex`, lines 880–1035, was read,

including Section 4 and the original labels `localv`, `structure`, and `structure1`. These supply the actual semilocal transition maps, closed stalks, generic Bruhat–Schwartz coefficients, and rational crossed-product labels used below. No replacement of the coefficient sheaf is made. The independent calculation verifies (CCF8)–(CCF19) and (CCF26)–(CCF34) in the local programme source `CONNES_SUPPORT_FUNCTOR_AND_COHOMOLOGY.tex`. The formulas below prove the cohomology, closure, and positive receiver directly. Their assertions concern underlying complex vector sheaves and the explicitly defined Hilbert spaces.

Put $X = \operatorname{Spec} \mathbb{Z}$, $V = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$, and $B = \bigoplus_{r \in \mathbb{Q}^\times} VU_r$, retaining its original crossed-product multiplication when an algebra is required. For each finite prime p , let $\xi_p = 1_{\mathbb{Z}_p}$ and let $V^{(p)}$ contain all factors other than p , including the real factor. The original stalks are

$$V_p = \mathbb{C}\xi_p \otimes V^{(p)}, \quad B_p = \bigoplus_{r: v_p(r)=0} V_p U_r. \quad (\text{MQ1})$$

They follow directly by taking the union of all semilocal stages whose excluded finite-prime set does not contain p . The other finite factors and the rational label are unrestricted away from p , while the displayed conditions persist at p .

There are two versions of the calculation. In the coefficient version let $W = V$ and $W_p = V_p$. In the labelled version let $W = B$ and $W_p = B_p$. Write $\mathcal{H}$ for the corresponding original sheaf and set

$$D_W = \bigoplus_p W/W_p, \quad \delta w = ([w]_p)_p. \quad (\text{MQ2})$$

The sum is finite for every w : each adelic test has only finitely many nonstandard finite factors and each rational label has only finitely many nonzero valuations. In particular this construction retains every prime mark rather than identifying equal coefficients at different marks.

19.2 The sheaf cohomology and its full rational labels

Theorem 19.1 (The actual arithmetic resolution). *There is an exact sequence of sheaves*

$$0 \longrightarrow \mathcal{H} \longrightarrow c_X W \xrightarrow{\delta} \bigoplus_p i_{p*}(W/W_p) \longrightarrow 0, \quad (\text{MQ3})$$

whose two right-hand sheaves are flasque. Consequently

$$R\Gamma(X, \mathcal{H}) = [W \xrightarrow{\delta} D_W], \quad H^1(X, \mathcal{H}) = D_W / \delta W, \quad H^n(X, \mathcal{H}) = 0 \quad (n \geq 2). \quad (\text{MQ4})$$

In degree zero the groups are respectively $\mathcal{S}(\mathbb{R})$ and $\mathcal{S}(\mathbb{R}) \rtimes \{\pm 1\}$, as in the human source.

Proof. On $U = X \setminus F$, membership in the kernel means that the coefficient has the factor ξ_p at every $p \notin F$ and, in the labelled case, that each surviving label is a unit at those primes. The source’s local factorization lemma proves the first condition: support in $\mathbb{Z}_p$ together with invariance under its additive translations is equivalent to $f(x_p, y) = \xi_p(x_p)f(0, y)$. Applying this to every retained prime gives exactly the original semilocal sections. Independence of the group-algebra labels proves the assertion coefficient by coefficient. At a closed point the quotient map is onto by definition; at the generic point the target stalk is zero. This proves sheaf exactness.

Sections of the direct sum on a nonempty open have finite support. Indeed their generic germ is zero, so the section vanishes on a smaller cofinite open. Its remaining support is finite. Conversely any finite collection of marked values gives a section, by the skyscraper descriptions. Restrictions delete excluded marks and are onto. This proves flasqueness. Every nonempty open of X is irreducible, so the constant sheaf has value W there and identity restrictions, hence is flasque too. Applying global sections to this flasque resolution proves (MQ4). Its kernel consists of tests standard at every finite prime; a label standard at every prime is precisely 1 or -1 . This proves the last claim. $\square$

Every local algebraic splitting $\mathcal{S}(\mathbb{Q}_p) = \mathbb{C}\xi_p \oplus E_p$ expands the original restricted tensor product as the algebraic direct sum

$$V = \bigoplus_{T \text{ finite}} V_T, \quad V_T = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_{p \in T} E_p, \quad (\text{MQ5})$$

with standard factors outside T . This follows by expanding each finite tensor. Uniqueness follows by applying the two complementary local projections within any finite set containing the factors of a proposed finite relation.

For uniform notation, in the unlabelled version a channel a is T and its marked set is $J_a = T$. In the labelled version it is $a = (r, T)$ and

$$D(r) = \{p : v_p(r) \neq 0\}, \quad J_a = T \cup D(r), \quad W_a = V_T U_r. \quad (\text{MQ6})$$

In the unlabelled version set $W_a = V_T$. A channel survives in W/W_p exactly when $p \in J_a$. Thus the entire arithmetic complex, with no rational labels removed, is

$$[W \longrightarrow D_W] = \bigoplus_a [W_a \xrightarrow{\Delta} W_a^{J_a}]. \quad (\text{MQ7})$$

An empty marked set gives zero target. Taking this finite diagonal quotient in every channel proves

$$H^1(X, \mathcal{H}) = \bigoplus_{a: |J_a| \geq 2} W_a \otimes (\mathbb{C}^{J_a} / \Delta \mathbb{C}). \quad (\text{MQ8})$$

The evaluation complement $E_p = \ker(f \mapsto f(0))$ gives exactly the original (CCF13)–(CCF15). Nothing in this computation requires it to be preserved by Fourier transform or by nonunit scaling.

19.3 The full support extension and the arithmetic image

Let $S = \text{Spec}_{\text{lat}} L$ for the original finite nontrivial bounded distributive lattice, and let $Y = S \triangleright X$. Its opens are the opens of S and $S \cup U$ for nonempty arithmetic opens U . Use the universal sheaf $U_S \mathcal{H}$, whose arithmetic component is $\mathcal{H}$, whose support component is $c_S W$, and whose gluing map is the original generic coefficient, constant on S . Write $C^\bullet(S, W)$ for the strict-chain finite cochains, with the original lower-open specialization convention. Its differential is alternating deletion of chain entries. The complete finite cochain resolution and recollement are proved in `finite_support_recollement.tex`, (FL8)–(FL22).

The actual relative complex is

$$\begin{aligned} K^0 &= W, & K^1 &= D_W \oplus C^0(S, W), & K^n &= C^{n-1}(S, W) \quad (n \geq 2), \\ d^0 w &= (\delta w, \Delta w), & d^1(d, c) &= d_S c, & d^n &= d_S \quad (n \geq 2). \end{aligned} \quad (\text{MQ9})$$

Indeed (MQ3) gives the arithmetic complex and the generic morphism is the identity on its degree-zero W and zero on its degree-one D_W . The mapping fibre of the map to $C^\bullet(S, W)$ has precisely these terms and, after multiplying successive support coordinates by alternating signs, these differentials. This fixes both its coefficient map and its sign.

Proposition 19.2 (The image in the original arithmetic group). *The full relative cohomology has the exact sequence*

$$0 \longrightarrow D_W \xrightarrow{d \mapsto [(d, 0)]} H_X^1(Y, U_S \mathcal{H}) \longrightarrow W \otimes \tilde{H}^0(S, \mathbb{C}) \longrightarrow 0, \quad (\text{MQ10})$$

and $H_X^{j+1} = W \otimes H^j(S, \mathbb{C})$ for $j \geq 1$. The actual map from relative to arithmetic cohomology is

$$[(d, c)] \longmapsto [d] \in D_W / \delta W. \quad (\text{MQ11})$$

Its restriction to the canonical D_W is the quotient map and is surjective. Hence (CCF19) has exactly the claimed arithmetic image (MQ8).

Proof. A closed degree-one representative has $d_{\mathcal{S}}c = 0$, so c is constant on each support component. Modulo the global diagonal its class gives the right-hand map of (MQ10). It is onto. If $c = \Delta w$, subtract the boundary $(\delta w, \Delta w)$ to obtain $(d - \delta w, 0)$. This proves exactness. The indicated injection has zero kernel because $(d, 0) = (\delta w, \Delta w)$ implies $w = 0$. All higher groups follow directly from the remaining support cochains. Projection of (MQ9) onto $[W \rightarrow D_W]$ is a chain map: it is the identity on W , sends (d, c) to d , and is zero thereafter. This proves (MQ11), including its well-definedness and surjectivity on the canonical summand. $\square$

Choosing a point s_0 gives the algebraic retraction $[(d, c)] \mapsto d - \delta c_{s_0}$. It is well defined because adding $(\delta w, \Delta w)$ adds equal terms. It is the identity on the canonical D_W . The choice is explicit; the injection and the arithmetic quotient in (MQ10)–(MQ11) are canonical.

19.4 Orthogonal unramified projections and the attained quotient

Use additive Haar measure with $\text{vol}(\mathbb{Z}_p) = 1$ and the real measure dx . Set $H = L^2(\mathbb{A}_{\mathbb{Q}})$ and

$$P_p f = \xi_p \otimes \int_{\mathbb{Z}_p} f(t, \cdot) dt, \quad Q_p = I - P_p. \quad (\text{MQ12})$$

This is the orthogonal projection onto $\mathbb{C}\xi_p \widehat{\otimes} H^{(p)}$, since $\|\xi_p\| = 1$. It preserves adelic tests and its image on them is exactly V_p . The projections for different primes commute because they act on different tensor factors. Use from now on

$$E_p^\perp = \{f \in \mathcal{S}(\mathbb{Q}_p) : \int_{\mathbb{Z}_p} f = 0\}, \quad V_T^\perp = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_{p \in T} E_p^\perp. \quad (\text{MQ13})$$

The same proof as (MQ5) gives an algebraic direct sum of mutually orthogonal spaces. Let H_T be the Hilbert closure of $V_T^\perp$ in H . Their orthogonal sum is all of H : finite adelic tensors are dense, and their finite orthogonal expansions lie in these summands. This also proves that each local orthogonal test space is dense in its local Hilbert complement, by applying $I - P_p$ to a dense family of local tests.

In the labelled case use $H_B = \widehat{\bigoplus_r} H U_r$, with counting measure on the original rational labels. Put

$$Q_{p,r} = \begin{cases} I - P_p, & v_p(r) = 0, \\ I, & v_p(r) \neq 0. \end{cases} \quad (\text{MQ14})$$

It realizes the quotient by B_p label by label. On $H_T U_r$ it is the identity exactly for $p \in T \cup D(r)$ and zero otherwise. In the unlabelled case use Q_p and omit r . Consequently the Hilbert completion of the marked quotient is

$$\overline{D}_W = \widehat{\bigoplus_{a: k_a > 0}} H_a \otimes \mathbb{C}^{J_a}, \quad k_a = |J_a|, \quad (\text{MQ15})$$

where $H_a = H_T$ or $H_T U_r$. All finite-factor and rational-label marks are included in J_a exactly as in (MQ6).

Theorem 19.3 (The closure and the injective positive arithmetic receiver). *The closure of the actual generic diagonal is*

$$\overline{\delta W} = \widehat{\bigoplus_{a: k_a > 0}} H_a \otimes \Delta \mathbb{C}. \quad (\text{MQ16})$$

The attained Hilbert quotient is therefore

$$\mathcal{M}_W = \overline{D}_W / \overline{\delta W} \simeq \widehat{\bigoplus_{a: k_a \geq 2}} H_a \otimes (\Delta \mathbb{C})^\perp \subset \widehat{\bigoplus_a} H_a \otimes \mathbb{C}^{J_a}, \quad (\text{MQ17})$$

where the displayed last containment is through zero-sum marked vectors. The natural map $D_W/\delta W \rightarrow \mathcal{M}_W$ is injective and has dense image. For $d_a = (d_{a,p})_{p \in J_a}$ put $\bar{d}_a = k_a^{-1} \sum_{p \in J_a} d_{a,p}$. Its exact norm is

$$\| [d] \|^2 = \sum_{a: k_a > 0} \sum_{p \in J_a} \|d_{a,p} - \bar{d}_a\|^2 = \sum_{a: k_a > 0} \frac{1}{k_a} \sum_{\{p,q\} \subset J_a} \|d_{a,p} - d_{a,q}\|^2. \quad (\text{MQ18})$$

The unordered-pair sum counts every distinct pair once. Thus (CCF31)–(CCF32) give the claimed injective positive realizations, including every rational-label mixed component.

Proof. On an algebraic channel W_a , the map is exactly its diagonal into its k_a marks. Each W_a is dense in H_a , and finite collections of their diagonal vectors are images of one finite sum in W . Their closure is therefore the indicated Hilbert sum of diagonals. Conversely that sum is closed: it is the range of the bounded orthogonal projection taking the arithmetic mean in every finite marked factor. This proves equality in (MQ16), including its closedness.

The complement of a diagonal vector space consists of vectors with zero sum. Subtracting $\bar{d}_a$ gives the corresponding orthogonal projection, hence proves (MQ17) and the first norm formula. Expand the squared differences in the second formula: each squared norm occurs $k_a - 1$ times and each distinct inner product occurs with coefficient $-2 \operatorname{Re}$. Dividing by k_a gives $\sum_p \|d_{a,p}\|^2 - k_a \|\bar{d}_a\|^2$, which is exactly the first expression.

An algebraic d has finitely many channels and each coordinate lies in its corresponding algebraic W_a . If its projection is zero, all its marked coordinates in a channel equal one common $w_a \in W_a$. The finite sum $w = \sum_a w_a$ then satisfies $d = \delta w$. Thus $D_W \cap \bar{\delta W} = \delta W$, proving injectivity. Finite algebraic zero-sum vectors are dense in the displayed orthogonal Hilbert sum, proving density. This proof also shows why changing from evaluation complements to orthogonal complements does not change injectivity: both decompose the same original algebraic D_W and the same original diagonal δW . $\square$

For an explicit nonzero class, choose $p \neq q$ and a nonzero $h \in V_{\{p,q\}}^\perp$. Give it the p mark and give the q mark zero. Its projected marked vector is $(h/2, -h/2)$ and its squared quotient norm is $\|h\|^2/2$. There is also a mixed class involving no nonstandard finite coefficient at all: take the original label $r = pq$, the channel $T = \emptyset$, and $h \in \mathcal{S}(\mathbb{R}) \otimes \bigotimes_\ell \xi_\ell$. Its marked set is again exactly $\{p, q\}$ and the same norm calculation applies to hU_{pq} . This verifies the rational-label part independently of coefficient ramification.

Proposition 19.4 (All simultaneous compatibility equations). *For $d = (d_{p,r}) \in \bar{D}_B$, regard $d_{p,r}$ as a vector of $Q_{p,r}H$. Then*

$$\bar{\delta B} = \bigcap_{r \in \mathbb{Q}^\times} \bigcap_{p \neq q} \ker(d \mapsto Q_{q,r}d_{p,r} - Q_{p,r}d_{q,r}). \quad (\text{MQ19})$$

The unlabelled assertion is obtained by omitting r and using Q_p . Every displayed map is bounded with norm at most $\sqrt{2}$.

Proof. Coordinate extraction and each Q are contractions. The triangle and Cauchy–Schwarz inequalities give the stated bound. Commutation of the Q projections proves that these maps kill every generic diagonal and hence its closure. Conversely, decompose into the orthogonal channels. In channel $a = (r, T)$, its p coordinate is zero outside J_a . For $p, q \in J_a$ the displayed equation reads $d_{a,p} = d_{a,q}$, because both projections act as the identity. Thus every channel is diagonal. Theorem 19.3 gives exactly (MQ19). $\square$

19.5 The generic diagonal as an actual closed unbounded operator

The preceding positivity does not require the generic diagonal to be a bounded operator in the original generic L^2 norm. In fact it is unbounded. The precise receiving operator is as follows.

Theorem 19.5 (Domain, adjoint, and closed range). *Let $\mathcal{H}_W = \widehat{\bigoplus_a H_a}$ denote H or H_B . The closure of the algebraic operator $\delta : W \rightarrow \overline{D}_W$ is*

$$\text{Dom } \delta_{\max} = \left\{ h = (h_a) : \sum_a k_a \|h_a\|^2 < \infty \right\}, \quad (\delta_{\max} h)_{a,p} = h_a \quad (p \in J_a). \quad (\text{MQ20})$$

Here h is also required to belong to $\mathcal{H}_W$, so the unmarked $k_a = 0$ part has its usual square summability. Its range is the closed subspace (MQ16). Its adjoint is

$$(\delta_{\max}^* d)_a = \sum_{p \in J_a} d_{a,p}, \quad \text{Dom } \delta_{\max}^* = \left\{ d : \sum_a \left\| \sum_{p \in J_a} d_{a,p} \right\|^2 < \infty \right\}. \quad (\text{MQ21})$$

On an unmarked channel the adjoint value is zero. The positive operator $\delta_{\max}^* \delta_{\max}$ is the multiplication operator $N : h_a \mapsto k_a h_a$, with domain $\sum_a k_a^2 \|h_a\|^2 < \infty$ in $\mathcal{H}_W$.

Proof. The squared norm of an algebraic diagonal is exactly $\sum_a k_a \|h_a\|^2$. Convergence of a vector and its diagonal implies convergence in each channel; the limiting marked coordinates are therefore the same coefficient in each mark. This proves closedness of the operator in (MQ20). Conversely truncate a vector to finitely many channels and approximate each remaining h_a by its dense algebraic tests. The graph norm on that channel is $(1 + k_a)^{1/2} \|h_a\|$, so these approximations converge in graph norm. Thus this closed operator is exactly the closure of the original operator, not an unrelated extension.

A vector in (MQ16) has coefficients z_a repeated k_a times with $\sum k_a \|z_a\|^2 < \infty$. Setting the unmarked coefficients to zero makes (z_a) a vector of $\mathcal{H}_W$, since every nonzero k_a is at least one. It belongs to the displayed domain and maps to that vector. This proves the closed-range assertion. Pairing against arbitrary finite-channel tests shows that the adjoint is the marked sum; this functional is bounded on $\mathcal{H}_W$ exactly under the square-summability condition (MQ21). Applying it to a diagonal multiplies by k_a . The domains then give precisely the asserted domain of N . $\square$

The restriction to the orthogonal complement of its kernel has a bounded inverse on the range. There is also a bounded generalized inverse on all of $\overline{D}_W$: send a marked vector to $\bar{d}_a$ for $k_a > 0$ and to zero for $k_a = 0$. Indeed

$$\sum_a k_a \|\bar{d}_a\|^2 \leq \sum_{a,p} \|d_{a,p}\|^2. \quad (\text{MQ22})$$

Thus this mean lies in the actual closed domain and its diagonal is the orthogonal projection onto (MQ16).

For a concrete failure of boundedness, choose finite sets T_n with $|T_n| = n$, and a test $h_n \in V_{T_n}^\perp$ with $\|h_n\| = 1$. Such tests exist: at p the nonzero test $1_{p\mathbb{Z}_p} - p^{-1}\xi_p$ is orthogonal to ξ_p , and its normalization, together with a normalized real Schwartz function, gives the required tensor. Then $\|\delta h_n\| = \sqrt{n}$. Also δW itself need not be closed: choose distinct primes p_n and unit tests in the corresponding singleton channels. The convergent marked sum $\sum_n 2^{-n} \delta h_n$ lies in (MQ16), but cannot lie in δW since it has infinitely many nonzero channels. These facts verify the need for the closure used in (CCF30)–(CCF32).

19.6 The relative completion and the positive arithmetic image

The algebraic splitting in (MQ10) gives the positive relative receiver chosen in (CCF27): give its D_W the marked norm and its remaining support groups chosen positive finite-dimensional forms tensored with the generic L^2 norm. All these forms are strictly positive on the original algebraic

groups. This is a valid specified receiver. It is useful to compute its exact relation to completing the terms of (MQ9) in their original coefficient norms.

Give support cochains the norm that sums squared coefficient norms over strict chains. Put $M = |S| > 0$. On closed degree-one support cochains, c_s is constant on each connected component. Write

$$\bar{c} = M^{-1} \sum_{s \in S} c_s, \quad c'_s = c_s - \bar{c}, \quad \sum_s c'_s = 0. \quad (\text{MQ23})$$

The number of points in each support component is thus retained in the sum; it is not replaced by a component count. The closed degree-zero boundary is the graph $w \mapsto (\delta_{\max} w, \Delta w)$ on $\text{Dom } \delta_{\max}$, and it has closed range because its second coordinate has norm $\sqrt{M} \|w\|$.

On an algebraic closed representative define

$$q = d - \delta \bar{c}, \quad c' = c - \Delta \bar{c}. \quad (\text{MQ24})$$

These are invariant under its actual boundaries. In a channel with $k_a > 0$, write $q_a = q_a^\circ + \Delta \bar{q}_a$, where $\sum_{p \in J_a} q_{a,p}^\circ = 0$. Its exact quotient norm in this closed-graph Hilbert complex is

$$\|q_a^\circ\|^2 + \frac{k_a M}{k_a + M} \|\bar{q}_a\|^2 + \sum_{s \in S} \|c'_{a,s}\|^2. \quad (\text{MQ25})$$

For $k_a = 0$ only the last summand remains. To prove the formula, all representatives in that channel have the form $(q_a + \Delta h, \Delta h + c'_a)$. Orthogonality of the two mean-zero parts leaves the minimization

$$\inf_h \left\{ \sum_{p \in J_a} \|q_{a,p} + h\|^2 + M \|h\|^2 \right\}.$$

Completing the square gives the minimizing value $h = -(k_a + M)^{-1} \sum_p q_{a,p}$ and the value $\|q_a\|^2 - \|\sum_p q_{a,p}\|^2 / (k_a + M)$, which is exactly (MQ25). Finite algebraic closed representatives are dense in the full closed degree-one space: truncate its channels and approximate the coefficient of each support component by algebraic tests, retaining equality within that component. Thus taking the orthogonal sum over all channels proves the formula for the completed receiver. The coordinate q after this completion belongs to the weighted space specified by (MQ25); its diagonal part need not belong to the original unweighted $\bar{D}_W$. This is why no unjustified application of $\delta_{\max}$ to an arbitrary completed generic coefficient is required.

The diagonal q coefficient has norm factor $k_a M / (k_a + M)$ in this receiver. On the subspace $c' = 0$, the algebraically split D_W coordinate for every point choice is exactly q , and its norm has factor k_a . Since k_a is unbounded, the identity on the algebraic relative groups does not identify these two completions by a bounded inverse. This is an exact distinction between two specified relative norms, not a failure of the arithmetic quotient. Indeed the relative-to-arithmetic map on either receiver is

$$(q, c') \mapsto q^\circ \in \mathcal{M}_W. \quad (\text{MQ26})$$

It is bounded and onto in the closed-graph receiver, by (MQ25); set $q = q^\circ$ and $c' = 0$ for an isometric right inverse. On algebraic classes it is exactly (MQ11), because $\delta \bar{c}$ is an actual diagonal. Hence both completions retain exactly the same attained positive mixed-place arithmetic quotient, with all labels and marked multiplicities. The additional support components and all higher support cycles remain present in their respective relative groups.

19.7 Scaling, Fourier, and the nonunit-prime defect

The real action $\vartheta_\lambda f(x_\infty, x_f) = f(\lambda^{-1} x_\infty, x_f)$ commutes with every P_p and has squared-norm factor λ . It preserves every orthogonal channel and its marked set. It therefore gives on the actual arithmetic quotient

$$\langle \vartheta_\lambda[d], \vartheta_\lambda[d'] \rangle = \lambda \langle [d], [d'] \rangle. \quad (\text{MQ27})$$

This holds for all rational labels as well. It also preserves the closed diagonal domain, since the weights k_a are unaffected. The finite-place Fourier transform fixes ξ_p under the specified conductor- $\mathbb{Z}_p$ characters and Haar measures. Its unitarity therefore makes it commute with P_p : both the line $\mathbb{C}\xi_p$ and its orthogonal complement are invariant. Full Fourier is consequently unitary on each marked quotient and preserves the generic diagonal and its closure. In the crossed product the map additionally sends r to r^{-1} ; the equality $D(r^{-1}) = D(r)$ preserves every marked set. These observations prove that the positive arithmetic receiver in (MQ17) carries the claimed Fourier and real-scaling maps. Their exact relation is $\mathcal{F}\vartheta_\lambda = \lambda\vartheta_{\lambda^{-1}}\mathcal{F}$, obtained by the real change of variable $x_\infty = \lambda t$.

For comparison, the original rational-prime action $\alpha_p f(x) = f(p^{-1}x)$ scales all places. The product formula makes it an isometry of the global L^2 space. On an original unramified coefficient $f = \xi_p \otimes g$, however,

$$Q_p \alpha_p f = (1_{p\mathbb{Z}_p} - p^{-1}\xi_p) \otimes \alpha_p^{(p)} g, \quad (\text{MQ28})$$

since $P_p 1_{p\mathbb{Z}_p} = p^{-1}\xi_p$. Its squared norm is

$$(p^{-1} - p^{-2}) \|\alpha_p^{(p)} g\|^2 = (1 - p^{-1}) \|g\|^2. \quad (\text{MQ29})$$

The other local moduli have product p , proving the second equality. Equations (CCF33) and (CCF34) are thereby verified with their original normalizations. At $\ell \neq p$ the rational p is a local unit and preserves the unramified line there. In a crossed-product stalk B_p all its original labels are units at p ; conjugation by U_p leaves those labels unchanged and gives exactly this coefficient defect. Thus no endomorphism of the p -marked quotient is inferred from a map that fails to preserve its kernel. The actual defect map and its positive norm remain specified.

$$\begin{array}{ccc} D_W & \longrightarrow & D_W/\delta W \\ \downarrow & & \downarrow \\ \overline{D}_W & \longrightarrow & \mathcal{M}_W \end{array}$$

Figure 13: The exact positive arithmetic receiver. The upper map is the canonical relative arithmetic image in Proposition 19.2; the lower map subtracts the arithmetic mean in every marked channel. Its kernel is the closed diagonal of Theorem 19.3. The right-hand map is injective because the algebraic marked space meets that closed diagonal exactly in δW . Formula (MQ18) retains every marked-prime multiplicity and every rational label.

The calculations affirm the claimed arithmetic image and attained positive realization. The additional analytic distinction is the proved unboundedness of the generic diagonal and the resulting difference between specified relative completion norms. Its exact closed domain, range, adjoint, norm comparison, and common arithmetic quotient have all been constructed above. No purity or zeta-zero claim is inferred from these positive receiving spaces alone.

20 Prime boundary classes in the actual adelic theta receiver

The preceding coefficient calculation can be carried through the original rational-scaling relation and actual theta map. The source here is Connes [4], Section III, equations (6)–(19): its two moment conditions, weighted periodization norm, and the factor $|x|^{1/2}$ are all retained. We compute an exact kernel which would be lost by taking coinvariants before imposing the two conditions. It lies in the marked relative group (CCF16). This calculation uses original author TeX, not a PDF extraction.

Connes–Consani [15], `h0coinv`, `propmapE`, and `propmapE1`, already prove the unconditioned spherical coinvariant and trace comparison. The calculations below reprove its maps and compute

the extra moment-first kernel and its receiving relative classes. Their `quotbruhat` supplies the real Laplacian context, which is applied with its exact range at the end.

20.1 The entire spherical coefficient module

Write $K = \widehat{\mathbb{Z}}^\times$, $\Gamma = \mathbb{Q}^\times$, $\Gamma_+ = \mathbb{Q}_{>0}^\times$, and $R = \mathbb{C}[\Gamma_+]$. A group-algebra basis element is denoted t_a for $a \in \Gamma_+$. Let $I = \ker(\text{aug} : R \rightarrow \mathbb{C})$. For $a > 0$ rational set

$$\eta_a = \bigotimes_p 1_{p^{v_p(a)}\mathbb{Z}_p}, \quad (\alpha_a f)(x) = f(a^{-1}x), \quad s(h) = \eta_1 \otimes h. \quad (\text{PM1})$$

The local action and all rational coefficients are original ones. Every $\mathbb{Z}_p^\times$ -invariant local Schwartz function is a unique finite linear combination of $1_{p^n\mathbb{Z}_p}$, $n \in \mathbb{Z}$. In fact its values on valuation shells are zero below a finite bound and constant above a finite bound; successive differences give the coefficients of these nested balls. Their independence follows by evaluating on successive shells. Averaging a finite tensor over the compact unit groups is still a finite tensor sum; only finitely many nonstandard places occur. The global invariant space is therefore

$$V^K = \bigoplus_{a \in \Gamma_+} \eta_a \otimes \mathcal{S}(\mathbb{R}), \quad \mathcal{J} : R \otimes \mathcal{S}(\mathbb{R}) \xrightarrow{\sim} V^K, \quad t_a \otimes h \mapsto \alpha_a s(h). \quad (\text{PM2})$$

Indeed $\alpha_a s(h) = \eta_a \otimes h(x/a)$, so the change between the two displayed coordinates is invertible on every summand. Rational positive scaling acts on R by left multiplication and leaves the second coordinate unchanged. The element $-1 \in \Gamma$ acts by $h(x) \mapsto h(-x)$.

Both moments of $\alpha_a s(h)$ equal those of h . For the integral this is the exact product $a \prod_p p^{-v_p(a)} = 1$, and evaluation at zero is unchanged. Consequently the moment map on (PM2) is $\text{aug} \otimes m$, where

$$m(h) = \left(h(0), \int_{\mathbb{R}} h(x) dx \right), \quad H = \mathcal{S}(\mathbb{R})^{\text{even}}, \quad H_0 = \ker(m|_H), \quad M_0 = (V^K)^0. \quad (\text{PM3})$$

The superscript zero means both original moment conditions. The odd functions have both moments zero and vanish under $\{\pm 1\}$ coinvariants: $h - (-h) = 2h$. Averaging by $\frac{1}{2}(1 + \alpha_{-1})$ is an exact projection, so all coinvariant calculations reduce to the even summand without losing a remaining odd class.

20.2 Taking the moments first leaves two classes per prime

Let $\Lambda = \Gamma_+ \otimes_{\mathbb{Z}} \mathbb{C} = \bigoplus_p \mathbb{C}\ell_p$, where ℓ_p records the original rational prime generator. There is a canonical split exact sequence of complex vector spaces

$$0 \longrightarrow \Lambda \otimes \mathbb{C}^2 \xrightarrow{b} (M_0)_\Gamma \xrightarrow{\Phi} H_0 \longrightarrow 0. \quad (\text{PM4})$$

The section is the actual unramified source $h \mapsto [s(h)]$. In the original finite-place coordinates, the quotient map is

$$\Phi \left[\sum_a \eta_a \otimes h_a \right] (x) = \frac{1}{2} \sum_a (h_a(ax) + h_a(-ax)). \quad (\text{PM5})$$

The sum is finite. For $u \in \mathbb{C}^2$, choose any even Schwartz function h with $m(h) = u$. Then

$$b(\ell_p \otimes u) = [(\alpha_p - 1)s(h)]. \quad (\text{PM6})$$

The vector in brackets is in M_0 although $s(h)$ itself need not satisfy the moments. Formula (PM6) is independent of the lift: a difference of lifts belongs to H_0 , so its difference under $\alpha_p - 1$ is one of the relations defining $(M_0)_\Gamma$.

Here is a full proof of exactness, including its injectivity. The two even functions

$$h_0(x) = (1 - 2\pi x^2)e^{-\pi x^2}, \quad h_1(x) = 2\pi x^2 e^{-\pi x^2} \quad (\text{PM7})$$

have moments $(1, 0)$ and $(0, 1)$. Thus $H = H_0 \oplus \text{span}\{h_0, h_1\}$. In the even part of (PM2), the moment kernel is exactly

$$R \otimes H_0 \oplus I \otimes \mathbb{C}^2. \quad (\text{PM8})$$

Positive-rational coinvariants quotient a module by I times that module. They therefore turn (PM8) into $H_0 \oplus (I/I^2) \otimes \mathbb{C}^2$. The canonical map $I/I^2 \rightarrow \Lambda$ sends $t_a - 1$ to $\sum_p v_p(a)\ell_p$. Products satisfy $t_{ab} - 1 = (t_a - 1) + (t_b - 1) + (t_a - 1)(t_b - 1)$, and $t_{p-1} - 1 \equiv -(t_p - 1) \pmod{I^2}$. Hence the classes $t_p - 1$ span. Their independence follows by the derivations $D_p(t_a) = v_p(a)$: each D_p vanishes on I^2 and takes value δ_{pq} on $t_q - 1$. This proves $I/I^2 \simeq \Lambda$. It proves (PM4) and every map (PM5)–(PM6). Although (PM7) was used to verify exactness, the resulting injection is independent of those lifts and the section is already canonical.

The same calculation specifies the higher algebraic group homology:

$$H_n(\Gamma, M_0) \simeq \bigwedge^{n+1} \Lambda \otimes \mathbb{C}^2 \quad (n \geq 1). \quad (\text{PM9})$$

For completeness, on any finitely generated subgroup of Γ_+ use the tensor product of the exact two-term resolutions with differential $t_p - 1$ for its prime generators. The one-variable resolution is exact because multiplication by $t_p - 1$ is injective on its Laurent polynomial ring and its quotient is evaluation at one. Tensoring over $\mathbb{C}$ preserves exactness, giving the finite Koszul resolution. Tensoring this resolution with a trivial module annihilates every differential, yielding the exterior powers of its generator space. Increasing the finite set of generators and taking the exact filtered union gives $H_j(\Gamma_+, \mathbb{C}^2) = \bigwedge^j \Lambda \otimes \mathbb{C}^2$. The even V^K in (PM2) is a free R module and hence has zero higher group homology. The order-two factor has an exact averaging projection over $\mathbb{C}$, so adds no higher group. The long exact sequence of $0 \rightarrow M_0 \rightarrow V^K \rightarrow \mathbb{C}^2 \rightarrow 0$ now proves (PM9), and its degree-zero portion is exactly (PM4). These are algebraic homology statements; no Hilbert completion is hidden in them.

20.3 The complete map to Connes's periodization

On the trivial compact-character sector, represent an idele class by real $x > 0$ and finite components one. The original periodization is

$$(\mathcal{E}f)(x) = x^{1/2} \sum_{q \in \mathbb{Q}^\times} f(qx). \quad (\text{PM10})$$

It is invariant under α_q , by reindexing the sum. On an unramified even source the finite conditions force $q \in \mathbb{Z} \setminus \{0\}$, and therefore

$$\mathcal{E}f(x) = x^{1/2} \Theta \Phi[f](x), \quad \Theta h(x) = 2 \sum_{n \geq 1} h(nx). \quad (\text{PM11})$$

This holds first on each summand $\alpha_a s(h)$ of (PM2) and then by linearity. The factor two is the sum over positive and negative integers; the factor $x^{1/2}$ is the source's specified measure comparison, not a new convention.

The map Θ is injective on H . Indeed, for fixed $x > 0$, rapid decay implies absolute convergence of $\sum_{m,n \geq 1} |h(mnx)|$: for $A > 2$ it is bounded by a constant times $x^{-A} \zeta(A)^2$. With the integer Möbius function μ , the exact divisor identity $\sum_{d|r} \mu(d) = 1$ for $r = 1$ and zero otherwise gives

$$h(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta h(nx). \quad (\text{PM12})$$

The divisor identity follows by expanding $\prod_{p|r}(1-1)$ for $r > 1$ and its empty product for $r = 1$. This supplies a direct proof of injectivity with no assumption about zeta zeros. Continuity gives $h(0)$ as well, and evenness determines negative arguments.

For $h \in H_0$, both $h(0)$ and $\widehat{h}(0) = \int h$ vanish. Poisson summation gives

$$\Theta h(x) = x^{-1} \widehat{\Theta h}(1/x). \quad (\text{PM13})$$

One proof in these exact conventions is to periodize $t \mapsto h(xt)$ on $\mathbb{R}/\mathbb{Z}$. Its uniformly convergent Fourier series has coefficients $x^{-1} \widehat{h}(n/x)$; evaluation at zero gives the equality of the sums over all integers. Both zero-index terms vanish, which gives (PM13). All derivatives are controlled by the same Schwartz bounds. Thus Θh decays to arbitrary order at infinity and, by (PM13), at zero after multiplication by any fixed power. In particular the original norm

$$\|f\|_\delta^2 = \int_0^\infty |\Theta \Phi[f](x)|^2 (1 + \log^2 x)^{\delta/2} x \frac{dx}{x} \quad (\delta > 1) \quad (\text{PM14})$$

is finite. It is exactly Connes's norm in this sector, with its original weight. Equation (PM12) proves the exact kernel after algebraic coinvariants:

$$\ker(\mathcal{E} : (M_0)_\Gamma \rightarrow C^\infty(\mathbb{R}_{>0})) = b(\Lambda \otimes \mathbb{C}^2). \quad (\text{PM15})$$

The same kernel is killed by (PM14). Consequently the moment-first quotient contains every prime boundary class, while the original periodization seminorm explicitly removes them. This is a calculated kernel with its unchanged representatives, rather than an identification of those classes with absent τ .

20.4 Their actual support map, scaling characters, and Fourier action

Use the invariant marked relative space $D^K = D_V^K$ from (CCF10). Unit averaging commutes with its projections, and finite averaging on every local test proves exactness of taking these invariants. Let

$$R_0 = \text{span}\{(\alpha_q - 1)f : q \in \Gamma, f \in M_0\}, \quad D_{\Gamma,0} = D^K / \delta_V R_0. \quad (\text{PM16})$$

This is the stated quotient of the original relative group; no action on D^K is presumed when a rational prime fails to preserve a local unramified line. The map δ_V nevertheless induces an exact injection

$$\Lambda \otimes \mathbb{C}^2 \hookrightarrow D_{\Gamma,0}, \quad \ell_p \otimes u \longmapsto [\delta_V((\alpha_p - 1)s(h))]. \quad (\text{PM17})$$

The only nonzero marked component of this representative is at p , where it is $[\alpha_p s(h)]_p$, the actual defect (CCF33). Other primes see a unit and retain the standard factor, so their quotient class is zero. This connects the new classes to the original prime, not to an auxiliary index.

To prove injectivity of (PM17), suppose a representative $v \in M_0$ of a class in the kernel of Φ satisfies $\delta_V v \in \delta_V R_0$. Then $v - r \in \ker \delta_V$ for some $r \in R_0$. By (CCF11) its value is $s(h)$ with $h \in \mathcal{S}(\mathbb{R})$; since v, r have both moments zero, h has both moments zero. In coinvariants its class is $[s(h^{\text{even}})]$, whose image under Φ is h^{even} . The class under consideration had image zero, so $h^{\text{even}} = 0$. Its odd part is already an order-two relation. Therefore $[v] = 0$ in $(M_0)_\Gamma$, which proves injectivity. The image of the full induced map $(M_0)_\Gamma \rightarrow D_{\Gamma,0}$ is precisely (PM17), since the unramified section of H_0 has $\delta_V s(h) = 0$.

Archimedean scaling commutes with every rational action and with δ_V . It therefore acts on all these groups. Its exact action on the new prime classes is

$$\vartheta_\lambda b(\ell_p \otimes (u_0, u_1)) = b(\ell_p \otimes (u_0, \lambda u_1)). \quad (\text{PM18})$$

Apply (CCF21) to the chosen lift h ; (PM6) is independent of the remaining zero-moment difference. This proves (PM18) without choosing homogeneous Schwartz lifts. The section $H_0 \rightarrow (M_0)_\Gamma$ also commutes with ϑ_λ , so (PM4) is equivariantly split for this action.

Fourier inversion on the original adelic space satisfies $\mathcal{F}\alpha_p = \alpha_{p^{-1}}\mathcal{F}$ and $\mathcal{F}s(h) = s(\widehat{h})$. It induces a map on $(M_0)_\Gamma$. In I/I^2 inversion sends $t_p - 1$ to its negative, while Fourier exchanges the two moments. Hence

$$\mathcal{F}b(\ell_p \otimes (u_0, u_1)) = -b(\ell_p \otimes (u_1, u_0)). \quad (\text{PM19})$$

One may verify the sign directly by $t_p^{-1} - 1 = -t_p^{-1}(t_p - 1)$ and reducing modulo I^2 . Moreover $\Phi\mathcal{F} = \widehat{\Phi}$, by (PM2) and the same inversion of rational labels. Together (PM13), (PM18), and (PM19) give the exact reflection and both pole characters before and after periodization. For example $\mathcal{E}(\mathcal{F}f)(x) = \mathcal{E}f(1/x)$ for $f \in M_0$, including the retained factor $x^{1/2}$.

Finally the map to the original arithmetic image (CCF19) sends the representatives (PM17) to zero because they are in $\delta_V V$. Thus the complete receiving diagram has three specified operations: passage to the split relative group, rational-scaling relations imposed after the moment conditions, and the arithmetic or theta receiver. The new prime classes survive the first two operations as (PM17) and have the exactly computed kernel in the third. Their survival in the relative space and their removal by the periodization norm are both part of the reconstructed mathematics.

20.5 The Laplacian range and the full missing moment-first module

Retain the human source's differential operator $H = x\partial_x$ and $\Delta = H(1 + H)$, acting at the real coordinate. The map $\Delta : H \rightarrow H_0$ is an isomorphism [15, `quotbruhat schwartz`]; in this sentence the first H denotes the space of even Schwartz functions from (PM3), and in the formulas below we write that space explicitly to avoid confusion with the operator. Here is a direct inverse in the original coordinate. For $f \in H_0$ and $x > 0$ set

$$u(x) = -\int_x^\infty \frac{f(t)}{t} dt, \quad g(x) = -\frac{1}{x} \int_x^\infty u(t) dt. \quad (\text{PM20})$$

Extend both evenly. Since f is even and $f(0) = 0$, $f(t)/t$ is smooth odd and vanishes to first order at zero. Its primitive u is smooth even and rapidly decreasing at infinity. It obeys $Hu = f$. Integration by parts gives $\int_0^\infty u = -\int_0^\infty f = 0$. Thus near zero the second expression equals $x^{-1} \int_0^x u(t)dt$, which is smooth even; its expression as a tail proves rapid decay. It satisfies $(1 + H)g = u$, hence $\Delta g = f$. The kernel of H on Schwartz functions is zero, and the kernel of $1 + H$ there is zero as well, since its solutions have xg constant away from zero. This proves injectivity and the full inverse claim.

In the even spherical adelic coordinates (PM2), Δ acts as $1 \otimes \Delta$; it commutes with every rational scaling. It follows that its entire range is $R \otimes H_0$. The exact extra quotient before taking coinvariants is

$$M_0^{\text{even}}/\Delta(V^{K,\text{even}}) \xrightarrow{\sim} I \otimes \mathbb{C}^2, \quad \sum_a t_a \otimes h_a \mapsto \sum_a t_a \otimes m(h_a). \quad (\text{PM21})$$

The moment condition says exactly that the target has augmentation zero. The kernel is $R \otimes H_0$, and surjectivity follows using (PM7) coefficientwise. After coinvariants the exact remaining quotient is $(I/I^2) \otimes \mathbb{C}^2$, namely (PM4)'s prime classes. Thus the real Laplacian theorem and the full adelic moment-first space have an explicit comparison with a calculated cokernel. The original theta image is unchanged by this cokernel, because (PM15) is exactly its kernel.

On the positive arithmetic Hilbert quotient (CCF31)–(CCF32), the same original operator has a fully determined realization. On finite Schwartz channels integration by parts gives

$$H^* = -1 - H, \quad \Delta = -H^*H = (H + \tfrac{1}{2})^2 - \tfrac{1}{4}, \quad \langle f, \Delta f \rangle = -\|(H + \tfrac{1}{2})f\|^2 - \tfrac{1}{4}\|f\|^2. \quad (\text{PM22})$$

The boundary terms $x\bar{f}g$ vanish at zero and infinity. Each finite-place projection and marked mean commutes with H , so the same calculation holds on all zero-sum marked channels. This proves the exact sign and the retained 1/4 on the actual attained receiver.

Its closed realization and spectrum can also be calculated. On the positive and negative half-lines, respectively, the maps $f \mapsto e^{t/2}f(e^t)$ and $f \mapsto e^{t/2}f(-e^t)$ are unitary from additive L^2 to $L^2(\mathbb{R}, dt)$. They send $H + 1/2$ to ∂_t . The logarithmic Fourier transform here is explicitly $(2\pi)^{-1/2} \int e^{-it\xi} h(t) dt$; it therefore sends Δ to multiplication by $-\xi^2 - 1/4$, on its exact domain

$$\{h \in L^2(\mathbb{R}) : (\xi^2 + 1/4)h(\xi) \in L^2(\mathbb{R})\}. \quad (\text{PM23})$$

Real multiplication on this domain is selfadjoint: its resolvents at nonreal parameters are the bounded multipliers $(-\xi^2 - 1/4 - z)^{-1}$, and the adjoint condition obtained by testing compactly supported functions gives exactly the same domain. Compact smooth functions away from $x = 0$, in each half-line and each finite channel, give a core by truncation and smooth approximation in the multiplication graph norm. Consequently its spectrum on any nonzero such arithmetic channel is $(-\infty, -1/4]$; approximate functions supported near a frequency show every value in that interval occurs in the spectrum, while outside it the reciprocal is bounded. Each level set has measure zero, so there is no nonzero eigenvector at a single spectral value. For an orthogonal sum, its domain additionally requires the sum of the integrals $\int (\xi^2 + 1/4)^2 |h_a(\xi)|^2 d\xi$ over all channels to be finite. This follows directly by applying the multiplication operator to the whole Hilbert direct sum. Its spectrum is the same interval when at least one channel is nonzero. This is a complete spectral calculation for the specified adelic Haar receiver. The source's full zeta-jet receiver and this Hilbert receiver remain connected by the explicit source maps and kernels (PM4)–(PM21); no identification of their spectral quotients is assumed.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Lambda \otimes \mathbb{C}^2 & \xrightarrow{b} & (M_0)_\Gamma & \xrightarrow{\Phi} & H_0 \longrightarrow 0 \\ & & \downarrow & & \downarrow \delta_V & & \downarrow x^{1/2}\Theta \\ & & D_{\Gamma,0} & & D_{\Gamma,0} & & C^\infty(\mathbb{R}_{>0}) \end{array}$$

Figure 14: The entire moment-first kernel of the original theta receiver. The top row is (PM4), with canonical unramified section. The first two vertical maps agree on b by (PM17); the right vertical map is injective by (PM12). The composite along the top and right is the actual periodization (PM10). Each prime contributes both pole characters, with Fourier sign and exchange as in (PM19). Source of the periodization and weight: Connes [4], Section III (6)–(19).

21 All moment-first classes, higher cycles, and their actual completions

21.1 Original source and coefficient coordinates

The source is Alain Connes, *Trace formula in noncommutative Geometry and the zeros of the Riemann zeta function*, arXiv:math/9811068. The supplied original author file `math_9811068.tex` was read at lines 646–825: Section III, equations (6)–(19), including the two moments, the weighted norm, its Haar normalization, the periodization factor, and its exact scaling identity. This is an independent derivation of (PM1)–(PM19) of the programme source `PRIME_MEMORY_AND_ACTUAL_THETA_RECEIVER.tex`. The new maps below also identify precisely what happens in the marked Hilbert receiver, rather than identifying algebraic coinvariants with a completed quotient.

Put $V = \mathcal{S}(\mathbb{A}_\mathbb{Q})$, $K = \widehat{\mathbb{Z}}^\times$, $\Gamma = \mathbb{Q}^\times$, $\Gamma_+ = \mathbb{Q}_{>0}^\times$, and $R = \mathbb{C}[\Gamma_+]$. Its basis is t_a for positive rational a , and $I = \ker \text{aug}$ is its augmentation ideal. Use the actual action and the actual unramified section

$$\alpha_a f(x) = f(a^{-1}x), \quad s(h) = h \otimes \bigotimes_p 1_{\mathbb{Z}_p}, \quad \eta_a = \bigotimes_p 1_{p^{v_p(a)}\mathbb{Z}_p}. \quad (\text{MC1})$$

Every $\mathbb{Z}_p^\times$ -invariant local test is a unique finite linear combination of $1_{p^n \mathbb{Z}_p}$, $n \in \mathbb{Z}$. Its shell values vanish for sufficiently negative valuations and become constant for sufficiently positive valuations; successive differences give exactly the claimed coefficients. Independence follows by testing successive shells. Expanding finite adelic tensors therefore gives the exact isomorphism

$$J : R \otimes \mathcal{S}(\mathbb{R}) \xrightarrow{\sim} V^K, \quad J(t_a \otimes h) = \alpha_a s(h) = \eta_a \otimes h(x/a). \quad (\text{MC2})$$

Compact-unit averaging is valid here without an integration completion: a local test has an open stabilizer in its compact unit group, so its orbit is finite. Every relevant averaging is a finite average and remains an original adelic test. This also proves exactness of invariants on the modules and marked quotients used below. Positive rationals act on the first factor of (MC2) by left multiplication. The element -1 acts on the second factor by real reflection.

Let

$$m(h) = \left(h(0), \int_{\mathbb{R}} h(x) dx \right), \quad (\text{MC3})$$

$$H = \mathcal{S}(\mathbb{R})^{\text{even}}, \quad H_0 = \ker(m|_H), \quad M_0 = \ker(m_{\mathbb{A}} : V^K \rightarrow \mathbb{C}^2).$$

Both adelic moments of $\alpha_a s(h)$ equal $m(h)$: evaluation is fixed, and the integral scales by $a \prod_p p^{-v_p(a)} = 1$. Hence the moment map in (MC2) is exactly $\text{aug} \otimes m$. Odd functions already have both moments zero. Their coinvariant classes vanish because reflection acts by -1 and the relation is twice the vector. Averaging by $\frac{1}{2}(1 + \alpha_{-1})$ is an exact projection, so the even reduction loses no surviving coinvariant class.

21.2 All moment-first coinvariants and their explicit inverse coordinates

Write

$$\Lambda = \bigoplus_p \mathbb{C} \ell_p, \quad \ell(a) = \sum_p v_p(a) \ell_p. \quad (\text{MC4})$$

The sums are finite, with the original prime exponents and signs. Let R_0 be the span of $(\alpha_q - 1)f$ with $q \in \Gamma$ and $f \in M_0$. Then $(M_0)_{\Gamma} = M_0/R_0$.

Theorem 21.1 (The exact canonical split sequence). *There is a canonical split exact sequence*

$$0 \longrightarrow \Lambda \otimes \mathbb{C}^2 \xrightarrow{b} (M_0)_{\Gamma} \xrightarrow{\Phi} H_0 \longrightarrow 0. \quad (\text{MC5})$$

Its section is $h \mapsto [s(h)]$, and its injection is

$$b(\ell_p \otimes u) = [(\alpha_p - 1)s(h)] \quad \text{for any } h \in H \text{ with } m(h) = u. \quad (\text{MC6})$$

For a representative $f = \sum_a \alpha_a s(h_a) \in M_0$, its complete inverse coordinates are

$$\Phi[f] = \sum_a h_a^{\text{even}}, \quad \beta[f] = \sum_a \ell(a) \otimes m(h_a), \quad [f] = [s(\Phi[f])] + b(\beta[f]). \quad (\text{MC7})$$

In the original coordinates $f = \sum_a \eta_a \otimes g_a$, the first formula is $\Phi[f](x) = \frac{1}{2} \sum_a (g_a(ax) + g_a(-ax))$.

Proof. The even functions

$$u(x) = (1 - 2\pi x^2)e^{-\pi x^2}, \quad v(x) = 2\pi x^2 e^{-\pi x^2} \quad (\text{MC8})$$

have moments $(1, 0)$ and $(0, 1)$, because $\int e^{-\pi x^2} = 1$ and $\int x^2 e^{-\pi x^2} = 1/(2\pi)$. Thus $H = H_0 \oplus \mathbb{C}u \oplus \mathbb{C}v$. The even moment kernel is exactly the R -module

$$R \otimes H_0 \oplus I \otimes \mathbb{C}^2. \quad (\text{MC9})$$

Taking positive-rational coinvariants quotients by I times this module and gives $H_0 \oplus (I/I^2) \otimes \mathbb{C}^2$. The map $t_a - 1 \mapsto \ell(a)$ identifies I/I^2 with Λ . Indeed $t_{ab} - 1 = (t_a - 1) + (t_b - 1) + (t_a - 1)(t_b - 1)$ and $t_{p^{-1}} - 1 \equiv -(t_p - 1) \pmod{I^2}$ show spanning by the prime classes. The linear derivation $D_p(t_a) = v_p(a)$ vanishes on I^2 and takes value δ_{pq} on $t_q - 1$, proving independence. This proves the exact sequence.

Changing the lift in (MC6) changes it by $(\alpha_p - 1)s(h_0)$ with $h_0 \in H_0$, an actual relation in R_0 . Thus b is independent of the moment lifts. The first formula of (MC7) is augmentation followed by even projection. Its moments are zero because $f \in M_0$. For the second, applying $(\alpha_c - 1)$ to a representative changes the indicated sum by $\ell(c) \otimes \sum_a m(h_a) = 0$. Reflection also leaves it unchanged, since both moments of an odd function vanish. Thus β is defined on the stated coinvariants. It is zero on the section $s(H_0)$ and takes $b(\ell_p \otimes u)$ to $\ell_p \otimes u$. The just-proved direct sum therefore proves the final equality and all canonicity claims. $\square$

For comparison, quotienting V^K by rational relations first gives H ; imposing its two moments then gives only H_0 . The precise defect is the kernel in (MC5). More explicitly, if $I_\Gamma V^K$ denotes the span of all $(\alpha_q - 1)f$ with $f \in V^K$, then its moments vanish and

$$I_\Gamma V^K / I_\Gamma M_0 \simeq \Lambda \otimes \mathbb{C}^2, \quad I_\Gamma M_0 = R_0. \quad (\text{MC10})$$

This follows because augmentation is the complete quotient $V^K \rightarrow H$ and its restriction to M_0 is Φ .

21.3 Higher pole classes with explicit cycles and actions

Theorem 21.2 (All higher algebraic group homology). *For $n \geq 1$ there is the canonical connecting isomorphism*

$$H_n(\Gamma, M_0) \simeq \bigwedge^{n+1} \Lambda \otimes \mathbb{C}^2. \quad (\text{MC11})$$

For distinct ordered primes $p_0, \dots, p_n$ and a lift h of $u \in \mathbb{C}^2$, the class corresponding to $\ell_{p_0} \wedge \dots \wedge \ell_{p_n} \otimes u$ is represented in the finite-prime Koszul complex by the exact cycle

$$\sum_{i=0}^n (-1)^i (\alpha_{p_i} - 1) s(h) \otimes e_{p_0} \wedge \dots \widehat{e_{p_i}} \dots \wedge e_{p_n}. \quad (\text{MC12})$$

The same formula for $n = 0$ is (MC6).

Proof. For a finite set of prime generators, tensor the two-term Laurent polynomial resolutions with differential $t_p - 1$. Each is exact because this multiplication is injective and its cokernel is evaluation at one. Tensor products over the coefficient field preserve exactness. On a trivial module every differential becomes zero, so the result in degree j is its tensor with the j th exterior power of the prime-generator space. The subgroups generated by finite prime sets are cofinal among the finite subsets of Γ_+ , and every group-homology chain uses finitely many group elements. Their exact filtered union therefore gives $H_j(\Gamma_+, \mathbb{C}^2) = \bigwedge^j \Lambda \otimes \mathbb{C}^2$.

The module in (MC2) is free over R , and remains free when restricted to any such subgroup. Hence its higher positive-rational homology is zero. The order-two group is exact under averaging over $\mathbb{C}$, so adds no positive-degree homology. The long exact sequence of $0 \rightarrow M_0 \rightarrow V^K \rightarrow \mathbb{C}^2 \rightarrow 0$ gives (MC11). The Koszul differential of $s(h) \otimes e_{p_0} \wedge \dots \wedge e_{p_n}$ is exactly (MC12). Each coefficient has both moments zero, and its next differential is zero because the Koszul differential squares to zero. Changing h by a zero-moment lift changes the cycle by a boundary in the M_0 complex itself. This verifies the stated cycle and its independence without an implicit lift choice. $\square$

Let ϑ_λ scale the real coordinate by $f(x_\infty, x_f) \mapsto f(\lambda^{-1}x_\infty, x_f)$. It commutes with rational scaling and acts on the two moments by $\text{diag}(1, \lambda)$. Thus it acts on each right-hand side of (MC5) and (MC11) by the identity on the exterior prime factor and that exact two-character action on $\mathbb{C}^2$. The adelic Fourier transform satisfies $\mathcal{F}\alpha_a = \alpha_{a^{-1}}\mathcal{F}$ and $\mathcal{F}s(h) = s(\widehat{h})$. It exchanges the two moments. Inversion on Γ_+ negates each generator of Λ . Consequently its exact action on the higher prime module is

$$\mathcal{F}(\omega \otimes (u_0, u_1)) = (-1)^{n+1}\omega \otimes (u_1, u_0) \quad \text{in degree } n \geq 0. \quad (\text{MC13})$$

One can also check the sign on chains: inversion sends the Koszul basis e_p to $-t_p^{-1}e_p$, since $t_p^{-1} - 1 = -t_p^{-1}(t_p - 1)$. On trivial coefficients this is exactly the stated exterior sign. Degree zero gives (PM19).

21.4 The original periodization and its complete kernel

In the compact-unit invariant sector an idele class has representative with real coordinate $x > 0$ and finite coordinates one. Connes's actual map is

$$(\mathcal{E}f)(x) = x^{1/2} \sum_{q \in \mathbb{Q}^\times} f(qx). \quad (\text{MC14})$$

For $\alpha_a s(h)$ the finite conditions force $q/a \in \mathbb{Z}$, so the nonzero rational sum is precisely the sum of $h(nx)$ over nonzero integers n . In particular it is absolutely convergent for each $x > 0$. By the finite expansion (MC2), all of (MC14) is therefore

$$\mathcal{E}f(x) = x^{1/2}\Theta\Phi[f](x), \quad \Theta h(x) = 2 \sum_{n \geq 1} h(nx). \quad (\text{MC15})$$

Reindexing its absolutely convergent rational sum proves invariance under rational scaling. The two retained constants, 2 and $x^{1/2}$, have their original sources: sign summation and comparison of the multiplicative and weighted measures.

The theta map on H is injective. For fixed $x > 0$ and $A > 2$, Schwartz decay bounds the double absolute sum $\sum_{m, n \geq 1} |h(mnx)|$ by $C_A x^{-A} \zeta(A)^2$. Thus the integer Möbius identity can be rearranged to give

$$h(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta h(nx). \quad (\text{MC16})$$

Indeed the coefficient of $h(rx)$ is $\sum_{n|r} \mu(n)$, equal to one for $r = 1$ and zero otherwise by expanding the product $\prod_{p|r} (1 - 1)$. Continuity determines $h(0)$ and evenness determines its negative arguments. This proves injectivity.

For $h \in H_0$, periodizing the Schwartz function $t \mapsto h(xt)$ on $\mathbb{R}/\mathbb{Z}$ gives a uniformly convergent Fourier series with coefficients $x^{-1}\widehat{h}(n/x)$. Evaluation at zero therefore yields Poisson summation. Both zero-index terms vanish because both moments vanish, and hence

$$\Theta h(x) = x^{-1}\Theta\widehat{h}(1/x). \quad (\text{MC17})$$

Schwartz bounds at infinity and this exact reflection imply decay to every positive order at both endpoints, even after multiplying by a fixed power. Thus the original Section III(7) norm in this sector is finite and is exactly

$$\|f\|_\delta^2 = \int_0^\infty |\Theta\Phi[f](x)|^2 (1 + \log^2 x)^{\delta/2} x \frac{dx}{x}, \quad \delta > 1. \quad (\text{MC18})$$

The source's Section III(14) Haar normalization gives compact-unit volume one, so there is no extra compact-volume multiplier here. The formula is a seminorm until its exact kernel is removed. Equations (MC5), (MC15), and (MC16) prove

$$\ker(\mathcal{E} : (M_0)_\Gamma \rightarrow C^\infty(\mathbb{R}_{>0})) = b(\Lambda \otimes \mathbb{C}^2). \quad (\text{MC19})$$

The same is the exact kernel of (MC18), because a continuous theta function of zero weighted integral is zero. Equivalently the kernel on M_0 before coinvariants is $I_\Gamma V^K$, while the coinvariant relations imposed after the moments are only $I_\Gamma M_0$. This verifies (PM15) with the two original moments imposed at the stated stage.

21.5 The full marked extension beyond the injected prime classes

Use $D = D_V = \bigoplus_p V/V_p$ and $\delta : V \rightarrow D$ from `connes_marked_quotient_verification.tex`, (MQ1)–(MQ4). The compact-unit action preserves every V_p , so acts on D and commutes with δ . Exact finite averaging identifies $(D/\delta V)^K = D^K/\delta V^K$. Define the exact algebraic receiver

$$D_{\Gamma,0} = D^K/\delta R_0. \quad (\text{MC20})$$

No rational action on D is assumed in this definition. A rational prime does not preserve its local unramified line; the formula uses only the actual images under δ of existing relations in M_0 .

Theorem 21.3 (The complete marked extension and all of its maps). *There is a canonical exact sequence*

$$0 \longrightarrow \Lambda \otimes \mathbb{C}^2 \xrightarrow{\delta b} D_{\Gamma,0} \longrightarrow (D/\delta V)^K \longrightarrow 0. \quad (\text{MC21})$$

The first map sends $\ell_p \otimes u$ to the class of $\delta((\alpha_p - 1)s(h))$ for any lift $m(h) = u$. Its only nonzero marked component is the original prime p . The right-hand map is the original arithmetic quotient. The induced map from $(M_0)_\Gamma$ to $D_{\Gamma,0}$ has kernel exactly $s(H_0)$ and image exactly the displayed prime kernel.

Proof. Independence of the moment lift follows because the change belongs to δR_0 . At a prime $\ell \neq p$, rational p is a local unit and keeps the factor $1_{\mathbb{Z}_\ell}$, so the marked class is zero; at p the original standard term is itself zero in V/V_p , leaving $[\alpha_p s(h)]_p$.

Suppose $f \in M_0$ maps to zero in $D_{\Gamma,0}$. Then $\delta f = \delta r$ for some $r \in R_0$. The kernel of δ is the original unramified $s(\mathcal{S}(\mathbb{R}))$, by the arithmetic resolution, so $f - r = s(h)$. Both moments of h vanish because those of f and r do. Its odd part is a reflection relation; its even part lies in H_0 . This proves that the induced kernel is exactly $s(H_0)$, whose injection is already proved by $\Phi s = 1$. Restricting to $\ker \Phi$ therefore proves injectivity of δb and identifies the whole induced image with the prime module.

Finally $\delta M_0 = \delta V^K$. Indeed subtract from any $f \in V^K$ an unramified $s(h)$ with the same two moments, using (MC8). The difference belongs to M_0 and its marked image equals δf . Therefore the kernel of the right-hand quotient in (MC21) is exactly $\delta M_0/\delta R_0$, which is the prime image just computed. Surjectivity and the remaining maps follow from the displayed successive quotients. $\square$

Thus the injected prime classes are the entire extra kernel over the original mixed-place arithmetic receiver. All that receiver's finite-place components, and its real odd coefficients as well, remain in the right-hand term. Taking K invariants does not impose real evenness there: the rational relations in this theorem were imposed on generic coefficient images, not on an assumed rational action on the marked space.

21.6 Their exact fate in the inherited marked Hilbert quotient

Give D the original norm from the orthogonal unramified projections $P_p f = 1_{\mathbb{Z}_p} \otimes \int_{\mathbb{Z}_p} f(t, \cdot) dt$. Let $\overline{D}^K$ be the Hilbert completion of D^K . It equals the invariant subspace of $\overline{D}$: averaging is a bounded orthogonal projection and preserves the dense algebraic tests. The marked diagonal closure and its faithful arithmetic quotient were computed in (MQ15)–(MQ19).

Theorem 21.4 (The prime kernel has zero inherited quotient norm). *In the original marked Hilbert norm,*

$$\overline{\delta R_0} = \overline{\delta V^K} \quad \text{inside } \overline{D}^K. \quad (\text{MC22})$$

Consequently the separated Hilbert completion of $D^K/\delta R_0$ in its inherited quotient seminorm is exactly

$$\overline{D}^K/\overline{\delta V^K}, \quad (\text{MC23})$$

the positive arithmetic mixed-place receiver. On its original algebraic quotient, the radical is exactly $\delta b(\Lambda \otimes \mathbb{C}^2)$.

Proof. There is an explicit moment-preserving approximation. For the functions u, v in (MC8), put

$$u_n(x) = u(nx), \quad v_n(x) = n^{-1}v(x/n), \quad h_n = h - h(0)u_n - \left(\int h\right)v_n. \quad (\text{MC24})$$

For every even Schwartz h , the moments of u_n and v_n remain $(1, 0)$ and $(0, 1)$, so $h_n \in H_0$. Moreover $\|u_n\|_2 = n^{-1/2}\|u\|_2$ and $\|v_n\|_2 = n^{-1/2}\|v\|_2$, hence $h_n \rightarrow h$ in L^2 . All approximants remain original Schwartz functions.

For a fixed prime, the exact defect norm (MQ28)–(MQ29) is

$$\|\delta((\alpha_p - 1)s(h))\|^2 = (1 - p^{-1})\|h\|_2^2. \quad (\text{MC25})$$

Thus the same marked vectors with h_n converge to the vector with h , while each approximating vector belongs to δR_0 because $s(h_n) \in M_0$. Every prime generator of (MC21) therefore lies in $\overline{\delta R_0}$. The exact sequence and $\delta M_0 = \delta V^K$ imply that every algebraic element of δV^K is a sum of an element of δR_0 and finitely many such prime vectors. This proves one inclusion in (MC22); the other is immediate.

To identify the radical exactly, the faithful arithmetic quotient calculation (MQ16)–(MQ18) gives $D \cap \overline{\delta V} = \delta V$. Averaging shows $D^K \cap \overline{\delta V^K} = \delta V^K$: an invariant algebraic vector in the full closure has a lift $f \in V$, and averaging that lift gives one in V^K . Its class modulo δR_0 is precisely the kernel in (MC21). This proves both the exact radical and (MC23). $\square$

The approximation in (MC24) does not converge in the Schwartz topology in general: its first correction concentrates at zero and its second spreads to large scale. It is therefore not a relation erasing the original algebraic classes. It proves their fate under the specific inherited L^2 quotient operation. This is distinct from the exact theta kernel, though both receivers kill the same prime module by the maps just proved.

21.7 The positive weight constraint and a retained indefinite form

On the prime module the actual operators are

$$T_\lambda(\ell_p \otimes (u_0, u_1)) = \ell_p \otimes (u_0, \lambda u_1), \quad F(\ell_p \otimes (u_0, u_1)) = -\ell_p \otimes (u_1, u_0). \quad (\text{MC26})$$

The second sign is the original Fourier inversion of the prime generator in (MC13), not a chosen orientation.

Proposition 21.5 (The complete positive obstruction and its receiver). *Every positive semidefinite Hermitian form on $D_{\Gamma,0}$ satisfying the original marked scaling identity*

$$Q(T_\lambda z, T_\lambda z) = \lambda Q(z, z) \quad (\lambda > 0) \quad (\text{MC27})$$

annihilates the entire prime module in (MC21). Hence it factors through that actual arithmetic quotient. The inherited marked quotient has exactly this radical, by Theorem 21.4. There is nevertheless an explicit nondegenerate Hermitian form on the retained algebraic prime module:

$$J(z, w) = \sum_p (z_{p,0} \overline{w_{p,1}} + z_{p,1} \overline{w_{p,0}}). \quad (\text{MC28})$$

It satisfies $J(T_\lambda z, T_\lambda w) = \lambda J(z, w)$ and $J(Fz, Fw) = J(z, w)$. It has signature $(1, 1)$ at every prime, retains both original characters, and is not a positive realization.

Proof. For a prime vector z with coordinates $(u_0, 0)$, its action is fixed. Equation (MC27) gives $Q(z, z) = \lambda Q(z, z)$, so $Q(z, z) = 0$ by taking $\lambda \neq 1$. For the other coordinate it gives $\lambda^2 Q(z, z) = \lambda Q(z, z)$ and the same conclusion. The Cauchy–Schwarz inequality for a positive semidefinite Hermitian form then makes these null vectors orthogonal to every vector. Finite sums give the full prime module in the radical, proving factorization through the actual quotient. No completion assumption is needed for this argument.

In (MC28), every vector has finite prime support, so the sum is defined. Its local matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is invertible and has eigenvalues 1 and -1 . Substituting (MC26) gives the two asserted identities term by term; the two Fourier minus signs cancel. For clarity, the form is an explicit additional construction from the original moment coordinates. Its constant has not been inferred from a cohomological degree. The whole family on one prime compatible with (MC27) has matrix $\begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}$: the same scaling equation forces both diagonal entries to vanish. Fourier invariance forces c real, and nondegeneracy is precisely $c \neq 0$. Formula (MC28) specifies the value $c = 1$ in the displayed original coordinates. $\square$

Completing this last explicitly specified form with the positive coordinate norm $\sum_p (|z_{p,0}|^2 + |z_{p,1}|^2)$ gives $\ell^2(\{\text{primes}\}) \otimes \mathbb{C}^2$, with a bounded invertible fundamental symmetry given by the same exchange matrix. This is a concrete indefinite Hilbert-space form retaining the prime classes. Its scaling has the two actual characters, and its Fourier operator is the stated negative exchange. It is not the inherited marked quotient completion (MC23); the exact maps and kernels relating that positive completion to the algebraic objects were proved above.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Lambda \otimes \mathbb{C}^2 & \xrightarrow{\delta b} & D_{\Gamma,0} & \longrightarrow & (D/\delta V)^K \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & \longrightarrow & \overline{D}^K / \overline{\delta V}^K & \xlongequal{\quad} & \overline{D}^K / \overline{\delta V}^K \end{array}$$

Figure 15: The exact marked extension and its inherited positive completion. The top row is Theorem 21.3. The explicit Schwartz moment corrections (MC24) prove the zero map on the left and the closure equality in Theorem 21.4. The arithmetic right-hand map is injective by the orthogonal marked-mean quotient calculation. The separate indefinite form (MC28) retains the prime classes with their two original scaling characters and Fourier sign.

All moment-first coinvariant, higher homology, and theta-kernel claims checked here are algebraically valid. Their exact positive completion has the additional kernel proved above. No spectral purity or zeta-zero conclusion is inferred from these computations.

22 The exact Euler domains and their receiving maps

This independent addendum checks (PM20)–(PM23) in `PRIME_MEMORY_AND_ACTUAL_THETA_RECEIVER.tex`. It uses the complete module and map calculations in `moment_first_coinvariants_verification.tex`, (MC5)–(MC7), (MC21)–(MC28), and the marked Hilbert receiver

constructed in `connes_marked_quotient_verification.tex`, (MQ15)–(MQ18). It specifies the logarithmic Fourier convention and the full orthogonal-sum domain explicitly. The differential operator, original real coordinate, moments, and retained constant 1/4 are unchanged.

22.1 The exact real inverse and the full adelic quotient

Write $\mathcal{S} = \mathcal{S}(\mathbb{R})^{\text{even}}$, $m(h) = (h(0), \int h)$, $\mathcal{S}_0 = \ker m$, $E = x\partial_x$, and $\Delta = E(1 + E)$. The letter E in this addendum denotes the differential operator, not periodization. First $\Delta\mathcal{S} \subset \mathcal{S}_0$: its evaluation at zero is zero and integration by parts gives $\int \Delta g = -\int (1 + E)g = 0$.

Theorem 22.1 (The complete inverse). *For $f \in \mathcal{S}_0$, put for $x > 0$*

$$u(x) = -\int_x^\infty f(t)t^{-1} dt, \quad g(x) = -x^{-1} \int_x^\infty u(t) dt. \quad (\text{LD1})$$

Their even extensions are Schwartz functions, and $\Delta g = f$. This defines the inverse of the continuous linear isomorphism $\Delta : \mathcal{S} \rightarrow \mathcal{S}_0$.

Proof. Because f is even and $f(0) = 0$, its quotient $f(t)/t$ extends smoothly and oddly across zero and vanishes there. Differentiating the first integral gives $u' = f/x$, hence $Eu = f$. Its even extension is smooth. Its tail and each derivative decay rapidly at infinity. Fubini is valid since the corresponding double absolute integral equals $\int_0^\infty |f(t)|dt$, and gives $\int_0^\infty u = -\int_0^\infty f = 0$. Therefore near zero

$$g(x) = x^{-1} \int_0^x u(t) dt = \int_0^1 u(tx) dt.$$

This proves smooth even extension of g , including all derivatives. The tail expression in (LD1) proves rapid decay of it and its derivatives. Differentiating xg gives $(1 + E)g = u$ and hence $\Delta g = f$.

The kernel of E in Schwartz space is zero: a solution is constant on each half-line and rapid decay forces each constant to vanish. The kernel of $1 + E$ is zero too, because xg is constant on each half-line. Their product therefore has zero kernel on Schwartz space. This proves the inverse assertion. For continuity of the inverse, on $0 \leq t \leq 1$ use $|f(t)| \leq t^2 \sup_{[0,1]} |f''|/2$. The integral defining $u(0)$ and the derivatives of u are then bounded by finitely many Schwartz seminorms of f . The displayed average for g does the same near zero. On $x \geq 1$ the tail formulas and their derivatives give any requested weighted bound from finitely many weighted bounds for f and its derivatives. Thus every Schwartz seminorm of g is controlled by finitely many of those of f . The direct differential operator is continuous, proving the claimed topological isomorphism as well as its explicit inverse. $\square$

In the original spherical module $V^{K,\text{even}} = R \otimes \mathcal{S}$, rational scaling acts on R and E acts on the second factor: Euler differentiation commutes with $x \mapsto x/a$. The entire range is therefore $R \otimes \mathcal{S}_0$. In the original moment-first module the exact quotient is

$$M_0^{\text{even}}/\Delta(V^{K,\text{even}}) \xrightarrow{\sim} I \otimes \mathbb{C}^2, \quad \sum_a t_a \otimes h_a \mapsto \sum_a t_a \otimes m(h_a). \quad (\text{LD2})$$

The zero total moment is exactly the augmentation-zero condition. Its kernel and surjectivity follow coefficient by coefficient from (LD1) and the two explicit moment lifts in (MC8). This verifies (PM21) on the full original coefficient module.

The domain of the induced operator on moment-first coinvariants is different from the domain in (LD2). Put $P = \Lambda \otimes \mathbb{C}^2$. The canonical splitting (MC5)–(MC7) gives the exact operator

$$(M_0)_\Gamma = \mathcal{S}_0 \oplus P, \quad \Delta = (\Delta|_{\mathcal{S}_0}) \oplus 0. \quad (\text{LD3})$$

Indeed the unramified section commutes with the operator. On a prime class, applying Δ gives $[(\alpha_p - 1)s(\Delta h)] = 0$, because $\Delta h \in \mathcal{S}_0$ and this is an actual relation imposed after the moments.

Equivalently E acts by 0 and -1 on the two pole coordinates, so $E(1 + E)$ kills both. The same argument kills every higher pole homology group (MC11).

Consequently its exact kernel and cokernel are

$$\ker \Delta = P, \quad \text{coker } \Delta \simeq \mathbb{C}^2 \oplus P. \quad (\text{LD4})$$

For the second equality the actual quotient map on the first factor is $f \mapsto m(\Delta^{-1}f)$ from $\mathcal{S}_0$ to $\mathbb{C}^2$. It is onto by applying Δ to arbitrary moment lifts, and its kernel is exactly $\Delta\mathcal{S}_0$. This proves every map in (LD4).

22.2 The Haar realization, constants, and full domain

On a Schwartz coefficient in additive real L^2 , integration by parts gives the exact formulas

$$E^* = -1 - E, \quad \Delta = -E^*E = (E + \tfrac{1}{2})^2 - \tfrac{1}{4}, \quad \langle f, \Delta f \rangle = -\|(E + \tfrac{1}{2})f\|^2 - \tfrac{1}{4}\|f\|^2. \quad (\text{LD5})$$

The boundary term is $x\bar{f}g$, which vanishes at zero and infinity. The middle identity uses the skew adjointness of $E + 1/2$ on this core. Every orthogonal finite-place projection and every marked arithmetic mean acts on another factor and commutes with these operations. Thus the identities hold on the actual zero-sum marked arithmetic channels, including rational group labels.

On the positive and negative half-lines define respectively

$$(U_+f)(t) = e^{t/2}f(e^t), \quad (U_-f)(t) = e^{t/2}f(-e^t). \quad (\text{LD6})$$

Changing variables gives unitary maps to $L^2(\mathbb{R}, dt)$. Direct differentiation sends $E + 1/2$ to ∂_t . For the multiplier $-\xi^2 - 1/4$ the required logarithmic Fourier convention is explicitly

$$(\mathcal{F}_{\text{ang}}g)(\xi) = (2\pi)^{-1/2} \int_{\mathbb{R}} e^{-it\xi} g(t) dt. \quad (\text{LD7})$$

It sends ∂_t to $i\xi$. If instead the earlier real Fourier convention $\int e^{-2\pi it\xi} g(t) dt$ is retained, the multiplier is $-4\pi^2\xi^2 - 1/4$. These are the exact coordinate change $\xi_{\text{ang}} = 2\pi\xi$, not different operators or a change of the retained constant $1/4$.

Let $\mathcal{H}_a$ include all finite-place and zero-sum marked coefficient factors in an actual arithmetic channel. The full Hilbert sum, after (LD6)–(LD7), has the selfadjoint realization

$$(\Delta h)_{a,\pm}(\xi) = (-\xi^2 - \tfrac{1}{4})h_{a,\pm}(\xi), \quad \text{Dom } \Delta = \left\{ h : \sum_{a,\pm} \int_{\mathbb{R}} (\xi^2 + \tfrac{1}{4})^2 \|h_{a,\pm}(\xi)\|_{\mathcal{H}_a}^2 d\xi < \infty \right\}. \quad (\text{LD8})$$

Here h also belongs to the Hilbert sum itself. For an even real sector the unitary map from the entire even real L^2 space to one copy is $f \mapsto \sqrt{2}e^{t/2}f(e^t)$; the same multiplier and weighted domain hold. The summability in (LD8) is essential: membership in each individual channel's domain does not suffice for an infinite orthogonal sum.

To prove selfadjointness, test the adjoint against compactly supported frequency functions with finitely many coefficient channels. The test determines its value to be the same real multiplier and imposes exactly (LD8). Conversely every vector in that domain satisfies the adjoint identity by Cauchy–Schwarz. Its nonreal resolvents are bounded multipliers $(-\xi^2 - 1/4 - z)^{-1}$, which verifies the entire resolvent assertion as well. Finite channel truncation and frequency cutoff approximate every domain vector in graph norm. Smooth compact frequency approximations then give Schwartz functions in t . Multiplying those by smooth cutoffs tending to one approximates them with their first two derivatives in $L^2(dt)$. Their inverse images under (LD6) are smooth and compactly supported away from $x = 0$. Thus the original finite Schwartz-channel operator is a core for this realization.

On every nonzero channel, frequencies in arbitrarily small neighbourhoods of a prescribed ξ_0 give approximate eigenvectors of value $-\xi_0^2 - 1/4$. Outside $(-\infty, -1/4]$ the reciprocal multiplier is bounded. Every level set of the multiplier has measure zero, so no nonzero Hilbert eigenvector exists. This proves the full spectrum $(-\infty, -1/4]$ and all domain claims in the stated angular convention. In particular zero is in its resolvent set.

22.3 The exact comparison with the retained prime kernel

The operator identities above do not identify their different receiving modules. Their actual maps, already proved in (MC21) and (MC22), give the diagram

$$(M_0)_\Gamma = \mathcal{S}_0 \oplus P \xrightarrow{\delta} D_{\Gamma,0} \longrightarrow \overline{D}^K / \overline{\delta V^K}. \quad (\text{LD9})$$

$\searrow \quad \quad \quad \nearrow$
 0

The first map has kernel exactly $\mathcal{S}_0$ and image exactly P . The second map kills exactly P on the algebraic marked quotient, and its image is the faithfully embedded dense arithmetic mixed-place quotient. Therefore the displayed composite is exactly zero. All arrows commute with E and Δ on their stated algebraic domains. On the prime module $\Delta = 0$; on the attained positive Haar quotient its closed realization has the strictly negative spectral gap just proved. This agreement is accounted for by the exact kernel in (LD9). It supplies no identification of the theta receiver with the Haar arithmetic receiver and no transfer of a zeta-zero assertion between them.

23 Full jet duality and all prime-compatible positive forms

23.1 The original coefficients and the complete local pairing

Keep

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \iota\rho = 1 - \bar{\rho}. \quad (\text{FJ1})$$

Let Z be a nonempty finite set of distinct nontrivial zeros, invariant under ι . Every multiplicity m_ρ is the actual multiplicity. Write $t_\rho = s - \rho$ and

$$g(\rho + t) = u_\rho(t)t^{m_\rho}, \quad u_\rho(t) = 2\lambda_\rho e^{h_\rho(t)}, \quad \lambda_\rho = \xi^{(m_\rho)}(\rho)/m_\rho!, \quad h_\rho(0) = 0. \quad (\text{FJ2})$$

All equalities below use this coordinate and this unit. The finite algebra is $A = A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho/(g)$; its coordinate identification with $\prod_\rho \mathbb{C}[t_\rho]/(t_\rho^{m_\rho})$ follows because u_ρ is invertible, and does not replace g in any residue. These are the finite jets in the supplied calculation [16], Sections 2–4, and the original tau coefficient model [17], Sections 5–6. Connes and Consani's original author source [15], Proposition `laplacian`, describes the finite spectral quotient by jets and the triangular multiplication action; the arguments below give the pairing and support maps directly.

For $f \in A$, put $f_\rho^\dagger(t) = \overline{f_{\iota\rho}(-\bar{t})}$. The functional equation and reality give $g^\dagger = g$ and $m_{\iota\rho} = m_\rho$, hence this is a well-defined antilinear algebra involution. Define

$$R(f, h) = \sum_{\rho \in Z} \text{Res}_{t=0} \frac{f_\rho^\dagger(t)h_\rho(t)}{u_\rho(t)t^{m_\rho}} dt, \quad (\Pi_p f)_\rho(t) = p^\rho e^{t \log p} f_\rho(t) \pmod{t^{m_\rho}}. \quad (\text{FJ3})$$

Here p is any positive prime. Multiplication by p^s is a linear automorphism of A , not a unital algebra homomorphism $A \rightarrow A$.

Theorem 23.1 (Exact coefficient matrix, flags, and duality). *Write $u_\rho(t)^{-1} = \sum_{r \geq 0} v_{\rho,r} t^r$ and $f_\rho = \sum_{j < m_\rho} f_{\rho,j} t^j$. Then*

$$R(f, h) = \sum_{\rho \in Z} \sum_{j+k \leq m_\rho-1} (-1)^j v_{\rho, m_\rho-1-j-k} \overline{f_{\iota\rho,j}} h_{\rho,k}. \quad (\text{FJ4})$$

It is perfect and skew-Hermitian. The linear isomorphism

$$D_R : \bar{A} \longrightarrow A^*, \quad D_R(\bar{f})(h) = R(f, h) \quad (\text{FJ5})$$

obeys, without a change of the prime factor,

$$R(\Pi_p f, \Pi_p h) = pR(f, h), \quad D_R \bar{\Pi}_p = p(\Pi_p^{-1})^* D_R. \quad (\text{FJ6})$$

For $F_\rho^a = t_\rho^a A_\rho$, $0 \leq a \leq m_\rho$, the annihilator of F_ρ^b in its reflected partner is $F_{\iota\rho}^{m_\rho-b}$. The resulting perfect pairing of complementary associated quotients is

$$(F_{\iota\rho}^a / F_{\iota\rho}^{a+1}) \times (F_\rho^{m_\rho-1-a} / F_\rho^{m_\rho-a}) \longrightarrow \mathbb{C}, \quad (ct^a, dt^{m_\rho-1-a}) \longmapsto \frac{(-1)^a \bar{c}d}{2\lambda_\rho}. \quad (\text{FJ7})$$

On every quotient F_ρ^a / F_ρ^{a+1} , the prime eigenvalue is exactly p^ρ , at every jet order a .

Proof. Taking the coefficient of t^{-1} proves (FJ4). Changing a representative by a multiple of g adds a holomorphic integrand, so the formula is well defined. Its antidiagonal coefficient is $(-1)^j v_{\rho,0} = (-1)^j / (2\lambda_\rho) \neq 0$. Reversing the order of the columns makes this block triangular with nonzero diagonal. Thus every local block pairs perfectly with its reflected block and their direct sum is perfect. For a meromorphic germ g , reflection and conjugation send the coefficient of t^{-1} to minus its conjugate. Applying this to $g = f^\dagger h / g$ and summing the reflected points proves $R(h, f) = -\overline{R(f, h)}$. Furthermore $(p^s)^\dagger = p^{1-s}$, so multiplication of the two factors inside the residue gives exactly p . Replacing h by $\Pi_p^{-1} h$ gives the second identity in (FJ6). Products of elements of orders a, b have order at least $a + b$. Consequently F^{m-b} annihilates F^b . If f has lowest nonzero order $a < m - b$, choose $h = u_\rho(t) t^{m-1-a}$ times a scalar at the partner point. It belongs to F^b and its residue against f is nonzero. This proves the reverse containment. At orders adding to $m - 1$ only $v_{\rho,0}$ occurs, giving (FJ7). Finally $e^{t \log p} = 1 + t \log p + \dots$ proves the action on each quotient. $\square$

The equality $\overline{p^\rho} p^\rho = p$ is the eigenvalue identity in (FJ7). It pairs two reflected points; it does not force either point individually to be fixed by ι . All higher jets remain in (FJ4), with the same eigenvalue and their nonzero Jordan entries $p^\rho (\log p)^j / j!$.

Theorem 23.2 (The Jacobian defines its exact quotient and image). *Let $N = \prod_\rho (t_\rho)$, $E = A/N = \mathbb{C}^Z$, and $\epsilon : A \rightarrow E$ be evaluation. Set $J_g = M_{g'}$ and*

$$j_g : E \longrightarrow A, \quad (a_\rho) \longmapsto (2m_\rho \lambda_\rho a_\rho t_\rho^{m_\rho-1})_\rho. \quad (\text{FJ8})$$

Then $J_g = j_g \epsilon$, $\ker J_g = N$, and $\text{im } J_g = \prod_\rho \mathbb{C} t_\rho^{m_\rho-1}$, including the whole factor when $m_\rho = 1$. All maps are prime equivariant. The Hermitian form

$$T(f, h) = R(f, J_g h) = \sum_\rho m_\rho \overline{f(\iota\rho)} h(\rho) = \text{Tr}(M_{f^\dagger h} \mid A) \quad (\text{FJ9})$$

has radical exactly N , and $T(\Pi_p f, \Pi_p h) = pT(f, h)$.

Proof. Differentiating the unaltered local equation gives $g' = mu(t)t^{m-1} + u'(t)t^m \equiv 2m\lambda t^{m-1} \pmod{t^m}$. This proves (FJ8), its kernel, and its image. Multiplication operators commute, and the prime acts by p^ρ both on values and on the displayed image, proving equivariance. Also $g'/g = m/t + u'/u$; the second term is holomorphic. Its residue against $f^\dagger h$ is zero, proving the sum in (FJ9) with every multiplicity. Multiplication by a germ on the m -dimensional local factor has its value on all m diagonal entries, proving the trace identity. The involution permutes equal multiplicities, proving Hermitian symmetry. Test against the value vector supported at the reflected point to detect any nonzero value of f ; this proves the radical assertion. The first identity of (FJ6) and commutation with J_g prove the final assertion. $\square$

$$\begin{array}{ccccc}
A & \xrightarrow{\epsilon} & E & \xrightarrow{j_g} & \text{im } J_g \subset A \\
D_R^r \downarrow & & & & \downarrow D_R^r \\
(\bar{A})^* & & & & (\bar{A})^*
\end{array}$$

Figure 16: The upper composite is J_g , with kernel N , and its pullback through the residue duality gives T . In this diagram $D_R^r(h)$ means the equivalent map $\bar{f} \mapsto R(f, h)$. The full coefficient matrix is FJ4; the actual factor $2m_\rho \lambda_\rho$ is FJ8. No Boolean observation occurs.

23.2 The actual tau chart and every retained support label

Use the original spaces $V, \mathcal{B}$ of [17], Section 2: V consists of even Schwartz functions ϕ on $\mathbb{R}$ with $\phi(0) = \widehat{\phi}(0) = 0$, with Fourier kernel $e^{-2\pi i x \xi}$; $\mathcal{B}$ consists of smooth functions on $\mathbb{R}_{>0}$ for which every $x^b(-x\partial_x)^k F$ is bounded, $b \in \mathbb{Z}$, $k \geq 0$. Set $\Theta\phi(x) = \sum_{n \neq 0} \phi(nx)$, $\mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x$, and $JF(x) = x^{-1}F(1/x)$. The four-point chart has $+ < \eta < \sigma$ and $- < \eta < \sigma$ with upper-set topology. The sheaf has $V, V, \mathcal{B}, 0$ at these points and maps $\Theta, J\Theta = \Theta\mathcal{F}$ to η . Its cohomology complex and actual prime action are

$$C_\tau = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(\phi, \psi) = \Theta(\phi - \widehat{\psi}), \quad U_p^0(\phi, \psi) = (U_p\phi, pU_{1/p}\psi), \quad U_p^1 F(x) = F(x/p). \quad (\text{FJ10})$$

For each nonempty $B \subseteq \{+, -\}$ retain exactly its source legs, the same $\mathcal{B}$ at η , and their original signs. The empty label has zero complex. Transports insert zero legs and are the identity at η . This is the complete original label lattice $L = \mathcal{P}(\{+, -\})$, not a chosen branch.

Theorem 23.3 (Surjection and dual injection over the whole tau lattice). *Every nonempty degree-one fibre is $Q = \mathcal{B}/\Theta V$. There is a surjective morphism of support diagrams*

$$q_Z : \{Q_B\}_{B \in L} \longrightarrow \{A_B\}_{B \in L}, \quad q_Z[F]_\rho = \sum_{j < m_\rho} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} t_\rho^j, \quad (\text{FJ11})$$

where $A_B = A$ for $B \neq \emptyset$, $A_\emptyset = 0$, and every nonempty transport is the identity. It intertwines (FJ10) with Π_p at every label. On opposite diagrams, precomposition and (FJ5) give an injection

$$q_Z^* D_R : \{\bar{A}_B\}_{L^{\text{op}}} \hookrightarrow \{\text{Hom}(Q_B, \mathbb{C}(1))\}_{L^{\text{op}}}, \quad \bar{f} \mapsto ([F] \mapsto R(f, q_Z[F])). \quad (\text{FJ12})$$

Here $\mathbb{C}(1)$ is the actual moment line with prime action p , and the dual action is $\ell \mapsto p\ell U_p^{-1}$. For the chart's coefficient diagram $\mathcal{T}$ and right adjoint $K_\tau(\mathbb{C}(1))$, the target fibre is exactly

$$H^{-1} R \text{Hom}(\mathcal{T}, K_\tau(\mathbb{C}(1))) = \text{Hom}(Q, \mathbb{C}(1)).$$

The contracted map $\bar{f} \mapsto ([F] \mapsto R(f, J_g q_Z[F]))$ has kernel $\bar{N}$ in every nonempty source fibre. Labels of these zero functionals remain distinct.

Proof. Fourier inversion on even Schwartz functions makes $\mathcal{F} : V \rightarrow V$ an automorphism, so every nonempty leg has image ΘV . The change of variable $x = py$ gives $\mathcal{M}U_p = p^s \mathcal{M}$. The original Fourier convention gives $\widehat{U_p \phi} = pU_{1/p} \widehat{\phi}$, proving $dU_p^0 = U_p^1 d$. Every one of these operators is invertible on its retained leg. For $\Re s > 1$, absolute summation gives $\mathcal{M}\Theta\phi(s) = 2\zeta(s) \int_0^\infty \phi(x)x^{s-1} dx$. The integral is holomorphic for $\Re s > -2$ because evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$ at zero. Near any nontrivial zero the other factors of g are nonzero and holomorphic. Analytic continuation therefore shows that $\mathcal{M}\Theta\phi$ is divisible there by the actual g . Thus (FJ11) kills the relations.

To prove surjectivity already on compact smooth tests, use $y = \log x$. Any functional annihilating all their jets would give a finite sum $\sum_\rho e^{\rho y} P_\rho(y) = 0$ on an arbitrary open interval,

where $\deg P_\rho < m_\rho$. To see this, the integral of that sum against every compact smooth test on the interval is zero; continuity forces the sum itself to vanish. Analyticity extends the equality to all y . For a fixed ρ , apply $\prod_{\eta \neq \rho} (\partial_y - \eta)^{m_\eta}$. All other terms vanish, while its operator on P_ρ is $\prod_{\eta \neq \rho} (\partial_y + \rho - \eta)^{m_\eta}$. It is invertible on polynomials of degree $< m_\rho$, being triangular with nonzero constant diagonal. Hence $P_\rho = 0$ for every ρ . The jet map has zero annihilator in a finite-dimensional target, so it is onto. Differentiating $p^s \mathcal{M}F$ proves its equivariance.

All nonempty support transports are identities on these quotients, which proves the diagram assertion. Precomposition reverses the arrows. Surjectivity and the perfect pairing prove the injection; FJ6 proves its equivariance. For completeness, the projective resolution of the constant chart diagram is $0 \rightarrow P_\eta \rightarrow P_+ \oplus P_- \rightarrow \mathbb{C} \rightarrow 0$, with maps $v \mapsto (v, -v)$ and $(u, v) \mapsto u + v$. It is exact at each of the four points. Applying Hom gives (FJ10). Its dual injective complex is $[I_\eta \mathbb{C}(1) \rightarrow I_+ \mathbb{C}(1) \oplus I_- \mathbb{C}(1)]$ in degrees $-1, 0$, with maps $(+\text{res}, -\text{res})$. Hom into this complex is the dual of (FJ10); its degree -1 kernel is precisely the functionals annihilating $\text{im } d$, namely $\text{Hom}(Q, \mathbb{C}(1))$. Finally FJ8–9 identify the contracted kernel. $\square$

Reconstruction gives $\mathcal{Q} = \bigsqcup_{B \in L} Q_B$ and $\mathcal{A} = \bigsqcup_{B \in L} A_B$, retaining each label even at zero. The actual full base

$$G_L(\mathbb{C}) = \{(r, 1_L) : r \in \mathbb{C}\} \cup \{z_B = (0, B) : B \in L\} \quad (\text{FJ13})$$

acts by $(r, 1_L)(B, v) = (B, rv)$ and $z_C(B, v) = (B \cap C, 0)$. Addition transports to the union label and adds there. The maps q_Z, J_g, ϵ, j_g preserve these actions: each is linear within a fibre and carries its zero to the zero with the same label. Thus they are maps over the entire lattice base. Dual reconstruction uses L^{op} and precomposition. Its least element is the original full-support label and has a nonzero dual fibre; replacing it by zero would destroy FJ12.

23.3 All finite support cochains and their exact dual receiver

Let S be any nonempty finite T_0 support space, ordered so that opens are lower sets, as in [18], formulas (FL14)–(FL16). Its full constant-coefficient complex is

$$C^n(S, A) = \prod_{s_0 < \dots < s_n} A, \quad (dc)_{s_0 < \dots < s_{n+1}} = \sum_{a=0}^{n+1} (-1)^a c_{s_0 < \dots < \widehat{s_a} < \dots < s_{n+1}}. \quad (\text{FJ14})$$

There is no degree truncation. Empty products in degrees with no chains mean the zero vector space. Define $C_n(S, A)$ by the same finite coordinate sets and the boundary matrix which is the transpose of this displayed differential. The scalar incidence coefficients are the original integers $0, 1, -1$.

Theorem 23.4 (Perfect support cohomology–homology duality). *The prime acts on every coefficient in FJ14 by Π_p and commutes with every differential and boundary. The pairing*

$$\mathcal{R}_n(a, c) = \sum_{s_0 < \dots < s_n} R(a_{s_0 < \dots < s_n}, c_{s_0 < \dots < s_n}) \quad (\text{FJ15})$$

descends to a perfect pairing $H_n(S, A) \times H^n(S, A) \rightarrow \mathbb{C}$ and obeys the exact prime similitude $\mathcal{R}_n(\Pi_p a, \Pi_p c) = p \mathcal{R}_n(a, c)$. There are canonical isomorphisms

$$H^n(S, A) = H^n(S, \mathbb{C}) \otimes_{\mathbb{C}} A, \quad H_n(S, A) = H_n(S, \mathbb{C}) \otimes_{\mathbb{C}} A. \quad (\text{FJ16})$$

The Jacobian acts as $1 \otimes J_g$. Its kernel and image on either side are, respectively, $H^n(S, \mathbb{C}) \otimes N$ and $H^n(S, \mathbb{C}) \otimes \text{im } J_g$, with H_n in the homological formula. Consequently the contraction of FJ15 by J_g has exactly these radicals and retains the factor m_ρ on its value quotient.

For an order-preserving map $f : S' \rightarrow S$, use

$$(f^*c)_{t_0 < \dots < t_n} = \begin{cases} c_{f(t_0) < \dots < f(t_n)}, & \text{all image points distinct,} \\ 0, & \text{otherwise,} \end{cases} \quad (\text{FJ17})$$

and its transpose chain map f_* . They commute with all prime and Jacobian maps, and $\mathcal{R}_n(f_*a, c) = \mathcal{R}_n(a, f^*c)$ on chains and homology.

Proof. Expanding d^2 pairs every two deleted vertices in its two possible orders with opposite signs; thus it is zero. The chain assertion is its transpose. Scalar incidence matrices commute with coefficient multiplication. Transposition and the real signs give $\mathcal{R}_n(a, db) = \mathcal{R}_{n-1}(\partial a, b)$. It follows that cycles pair trivially with boundaries, on either side. Each coordinate pairing is perfect by Theorem 23.1. In a finite-dimensional perfect pairing, the annihilator of a kernel is the image of the adjoint matrix: containment follows by evaluation, and equality follows because both dimensions equal the rank of that matrix. Apply this once to each of the two adjacent differentials. The induced pairing on the two homology quotients is perfect.

Every chain module in FJ14 is its scalar chain module tensored with A . Choosing any vector-space complements to its kernels and images proves that tensoring with A preserves both kernels and images. The resulting isomorphisms FJ16 are canonical because their maps send $[c] \otimes a$ to $[c \otimes a]$ independently of those complements. Apply the same exactness to FJ8 and use perfectness of the homology pairing; this proves all kernel, image, and radical assertions. Coefficientwise FJ6 gives the prime factor p .

For FJ17, if there is an adjacent repetition in the image chain, the two terms deleting its equal members cancel and every other term still has a repetition. Without a repetition the ordinary face formula gives $df^* = f^*d$. Transposition gives the chain identity. The defining finite sums give the adjointness, and coefficientwise operators commute with these scalar matrices. All identities therefore descend to the stated groups. $\square$

Corollary 23.5 (The exact augmented support quotient). Put $K^0 = A$, $K^n = C^{n-1}(S, A)$ for $n \geq 1$, with $d^0a = (a)_{s \in S}$ and $d^n = d_C^{n-1}$. If S has c connected components, then

$$H^0(K) = 0, \quad H^1(K) = A^c / \Delta A, \quad H^n(K) = H^{n-1}(S, \mathbb{C}) \otimes A \quad (n \geq 2). \quad (\text{FJ18})$$

The first nonzero group is paired perfectly with $\{(a_1, \dots, a_c) \in A^c : \sum_i a_i = 0\}$ by $\sum_i R(a_i, b_i)$. This pairing, its Jacobian contraction, every support pullback, and the prime similitude follow from FJ15 without discarding a support component.

Proof. The diagonal is injective because $S \neq \emptyset$. Degree-zero cocycles of the support complex are exactly functions constant on each connected component, proving FJ18. Changing a representative b by Δh changes the pairing by $R(\sum_i a_i, h) = 0$. Conversely this annihilator of ΔA is exactly the stated kernel by perfectness of R . The quotient pairing is therefore perfect. The remaining assertions are the cochain identities already proved, including the augmentation. $\square$

For any declared support-trajectory lattice of the incidence matrices, restrict FJ14 to its upward-closed coordinate trajectories. Extension by zero gives the original cochain maps: closure ensures that no nonzero matrix entry leaves the next support. Tensoring each such complex with A , as in FJ16, preserves its entire cohomology diagram. Its dual diagram reverses arrows by matrix transpose. In particular an inclusion on cochains need not be injective on cohomology; its dual map is the transpose of the actual induced map, with the full kernel preserved. This is the precise receiver for the full support construction of [18], not a replacement by Boolean stalks.

23.4 The spaces defined by the positivity and descent defects

Theorem 23.6 (Every positive semidefinite prime-similitude form). *The Hermitian positive semidefinite forms B on the entire A_Z satisfying $B(\Pi_p f, \Pi_p h) = pB(f, h)$ for both $p = 2, 3$ are exactly*

$$B(f, h) = \sum_{\substack{\rho \in Z \\ \Re \rho = 1/2}} b_\rho \overline{f(\rho)} h(\rho), \quad b_\rho \geq 0. \quad (\text{FJ19})$$

Thus their common forced radical is the prime-invariant ideal

$$\mathfrak{D}_Z = \left(\prod_{\Re \rho \neq 1/2} A_\rho \right) \oplus \left(\prod_{\Re \rho = 1/2} (t_\rho) \right), \quad A_Z / \mathfrak{D}_Z = \mathbb{C}^{Z \cap (1/2 + i\mathbb{R})}. \quad (\text{FJ20})$$

The quotient map is a prime-equivariant algebra homomorphism and is the universal quotient through which all forms FJ19 factor. A positive definite prime-similitude form on A_Z exists exactly when every point of Z lies on the critical line and every $m_\rho = 1$.

Proof. Let N_B be the radical of a positive semidefinite Hermitian form. Cauchy–Schwarz follows by evaluating $B(v + zw, v + zw) \geq 0$ for complex z , and proves that $B(v, v) = 0$ is equivalent to $v \in N_B$. The two prime similitudes and invertibility of Π_p make N_B invariant under Π_p and its inverse. On A/N_B each $p^{-1/2}\Pi_p$ is unitary. A finite-dimensional unitary operator is diagonalizable: an eigenvector exists over $\mathbb{C}$, its orthogonal complement is invariant under the operator and its inverse, and induction produces an orthonormal eigenbasis. Its eigenvalues have modulus one by its norm identity.

The local image of A_ρ in this quotient is annihilated by $(\Pi_2 - 2^\rho)^{m_\rho}$. If $\Re \rho \neq 1/2$, the eigenvalue 2^ρ has the wrong modulus, so the image is zero. On a critical factor, diagonalizability forces $\Pi_2 - 2^\rho$ to annihilate the image. But $\Pi_2 - 2^\rho = 2^\rho t_\rho a(t_\rho)$, with $a(0) = \log 2 \neq 0$, so its image ideal is exactly (t_ρ) . Thus every nonconstant jet in a critical factor also maps to zero.

For remaining value vectors at ρ, η , equivariance gives $(\overline{p^\rho} p^\eta - p)B(e_\rho, e_\eta) = 0$ for $p = 2, 3$. At critical points a nonzero cross term requires $p^\rho = p^\eta$ for both primes. Then $(\rho - \eta) \log 2 = 2\pi i k$ and $(\rho - \eta) \log 3 = 2\pi i l$. A zero integer forces $\rho = \eta$; otherwise division gives a rational relation between $\log 2$ and $\log 3$, contradicting unique factorization after exponentiation. Therefore distinct critical values are orthogonal. Their diagonal weights are nonnegative, proving FJ19. Conversely that formula gives a positive semidefinite form and its prime similitudes directly. Choosing every weight strictly positive proves that the intersection of all their radicals is exactly FJ20. The final claim follows because FJ20 is zero exactly in the stated case. $\square$

Corollary 23.7 (The full support multiplicity spaces). For a finite-dimensional complex vector space W , all positive semidefinite Hermitian forms on $W \otimes A$ compatible with $1 \otimes \Pi_2$ and $1 \otimes \Pi_3$ are exactly

$$B(v, w) = \sum_{\Re \rho = 1/2} B_\rho(v(\rho), w(\rho)),$$

where each B_ρ is an arbitrary positive semidefinite Hermitian form on W . Their common forced radical is $W \otimes \mathfrak{D}_Z$. This applies to each of $W = H^n(S, \mathbb{C})$, $W = H_n(S, \mathbb{C})$ and $W = \mathbb{C}^c / \Delta \mathbb{C}$ in FJ16–18.

Proof. The quotient-by-radical argument of Theorem 23.6 annihilates each off-line factor and each $W \otimes (t_\rho)$ in a critical factor. Distinct critical eigenvalue pairs are orthogonal by the same two-prime calculation. Within a fixed eigenvalue pair the prime action is scalar on W , so it imposes no further condition on the positive semidefinite form on that space. Conversely any displayed family gives both prime similitudes. Choosing a positive definite form on every nonzero W proves the common-radical assertion. $\square$

For the original T , each critical point contributes the weight m_ρ . A reflected nonfixed pair contributes $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ on its values and has a negative vector $(1, -1)$. Thus the exact similitude FJ6 and FJ9 exists before any positivity. FJ19 specifies all possible positive replacements and their precise lost data; it does not establish that the actual T belongs to that set for all zeros. In particular a finite positive restriction of the original trace proves no universal positivity assertion.

There is also an exact obstruction to obtaining a scalar-valued pairing from support labels alone. For a diagram M_B on a finite join lattice with top 1, let $\mathcal{M} = \bigsqcup_B M_B$ with addition by transport to joins. Its universal map to an additive group is

$$\varepsilon_{\text{top}} : \mathcal{M} \rightarrow M_1, \quad (B, v) \mapsto h_{B1}v. \quad (\text{FJ21})$$

Indeed every label zero $(B, 0)$ is additively idempotent and must map to zero in a group, while $(B, v) + (1, 0) = (1, h_{B1}v)$. This proves both factorization and its uniqueness. Consequently any complex-valued pairing additive in each entry factors through FJ21 in both entries. The exact object killed in fibre B is $\ker h_{B1}$; retaining its diagram is necessary to preserve all support cohomology. The opposite dual diagram used in FJ12 and FJ15 does preserve it and supplies the proved connecting maps. At the still coarser coefficient-support observation of [19], vectors $(1, 1)$ and $(1, -1)$ on a reflected pair have identical support but opposite values $2m, -2m$ under T . This computes a failure of factorization, while FJ11–18 give the full coefficient and cohomological maps which do exist.

23.5 The prime action on every declared jet support

Fix an actual factor A_ρ of multiplicity m and its unaltered coordinate basis $1, t_\rho, \dots, t_\rho^{m-1}$. A declared support is a subset $B \subseteq \{0, \dots, m-1\}$. Its carrier element (B, x) has $x \in \mathbb{C}^B$, and zero coordinates in B remain supported zeros. Outside B there is external absence. Put $\uparrow B = \{k : \text{some } j \in B \text{ has } j \leq k\}$.

Theorem 23.8 (Full prime support lift and its cancellation cells). *The actual prime matrix has entries*

$$(P_{p,\rho})_{kj} = \begin{cases} p^\rho (\log p)^{k-j} / (k-j)!, & k \geq j, \\ 0, & k < j. \end{cases} \quad (\text{FJ22})$$

Every entry in its lower triangle is nonzero. Its declared-support lift to $G(\mathbb{C})^m$ is the exact semimodule map

$$\tilde{\Pi}_{p,\rho}(B, x) = (\uparrow B, (P_{p,\rho}x)|_{\uparrow B}). \quad (\text{FJ23})$$

For positive integers $a, b > 1$ these lifts satisfy $\tilde{\Pi}_a \tilde{\Pi}_b = \tilde{\Pi}_{ab}$. For the action with parameter $1/p$, the composite with FJ23 is $(B, x) \mapsto (\uparrow B, x^{\text{ext}})$, where x^{ext} adds supported zeros. It is the identity precisely on the subdiagram of upward-closed supports, whose coefficient fibres are the full ideals $F_\rho^a = t_\rho^a A_\rho$.

There is an exact linear isomorphism to the source's derivative stalk

$$s_\rho : A_\rho \longrightarrow E_\rho = t_\rho \mathbb{C}[t_\rho] / (t_\rho^{m+1}), \quad f \mapsto t_\rho f, \quad (\text{FJ24})$$

intertwining multiplication by $p^\rho e^{t_\rho \log p}$ on both spaces. Its composition with the original pointed map σ_ρ records the actual nonzero coefficients, with index shifted by one. For $m \geq 2$ no map on these numerical support values alone intertwines this prime action for all $f \in A_\rho$.

The exact replacement is the finite family of coefficient spaces

$$\mathcal{C}_{p,B,C} = \left\{ x \in \mathbb{C}^m : \begin{array}{l} x_j \neq 0 \iff j \in B, \\ p^\rho \sum_{j=0}^k \frac{(\log p)^{k-j}}{(k-j)!} x_j \neq 0 \iff k \in C \end{array} \right\}. \quad (\text{FJ25})$$

They are disjoint locally closed algebraic subsets, some possibly empty, which partition A_ρ . Multiplication by $P_{p,\rho}$ gives the explicit isomorphism $\mathcal{C}_{p;B,C} \rightarrow \mathcal{C}_{1/p;C,B}$ with inverse $P_{1/p,\rho}$. Thus the failed support descent determines a specified family of coefficient spaces and exact maps.

Proof. Expanding the original exponential gives FJ22. Matrix multiplication in $G(\mathbb{C})$ gives support $\uparrow B$ even if an amplitude cancels; this proves FJ23 including all its supported zeros. The exponential identity gives $P_a P_b = P_{ab}$, with every prime factor and logarithm unchanged. Since $\uparrow\uparrow B = \uparrow B$, the supports of the two lifted sides also agree for $a, b > 1$. The inverse-parameter matrix is lower triangular with every lower entry nonzero as well. Its ring composite is the identity matrix, but its declared composite retains every lower triangular path. Thus its value on (B, x) has exactly the support and amplitudes stated after FJ23. Equality with (B, x) is precisely $B = \uparrow B$. Such subsets are the terminal segments $\{a, \dots, m-1\}$ and the empty set, with coefficient spaces exactly F_ρ^a . Their restriction maps are inclusions; FJ7 is the full reflected duality of this flag diagram.

The map s_ρ is well defined and its inverse divides its polynomial representative by t_ρ , modulo t_ρ^m . Commutation with coefficient multiplication proves its equivariance. For failure of the numerical support action, put $L = \log p$ and compare $f = 1 + Lt_\rho$ and $h = 1 - Lt_\rho$, both having the same nonzero coefficient indices $\{0, 1\}$. The coefficient of t_ρ in their prime images is respectively $2p^\rho L \neq 0$ and zero. This remains valid for every $m \geq 2$, regardless of later coefficients. Therefore no function of the input numerical support can give the output numerical support for both inputs. Finally FJ25 consists of linear equations together with inequalities that finitely many linear polynomials are nonzero, proving local closedness. Every x has one input and one output support, proving the partition. Invertibility of $P_{p,\rho}$ and its explicit inverse give both the asserted bijection and its regular inverse. $\square$

The closure $B \mapsto \uparrow B$ preserves joins and bottom; it does not preserve every meet. For $m \geq 2$, $B = \{0\}$ and $C = \{1\}$ have empty intersection, whereas their upper closures intersect. The exact general comparison is $\uparrow(B \cap C) \subseteq \uparrow B \cap \uparrow C$. Consequently the unrestricted support lift is a map of the original $G(\mathbb{C})$ semimodules with this support transport; it is not silently a scalar-base automorphism of $G_{\mathcal{P}(\{0, \dots, m-1\})}(\mathbb{C})$. On the complete terminal-flag diagram the support map is the identity, the entire prime action is invertible, and every full jet is still present in its largest coefficient fibre. The products of these constructions over the actual $\rho \in Z$ retain every multiplicity, reflection partner, and support coordinate.

24 The finite projective receiver of the full arithmetic jets

The new full-support reconstruction supplied with this calculation uses finite projective modules to receive ordinary matrices. We prove that construction here and apply it to the complete jet pairing above. The scalar object is the original $S = G_L(\mathbb{C})$ of version 11 [3]; L is any nontrivial bounded distributive lattice. No choice of Boolean branch or replacement of an amplitude zero by absence occurs in these maps.

For positive integers n let P_n have entries $1^\bullet$ on the diagonal and e off the diagonal. The free identity has τ off the diagonal. Multiplication in the original semiring gives

$$P_n^2 = P_n, \quad \mathcal{V}_n := P_n S^n = \{(a, x) : a = 1_L, x \in \mathbb{C}^n\} \sqcup \{(a, 0) : a < 1_L\}. \quad (\text{JC1})$$

The displayed coordinates identify $(a, 0)$ with the constant vector z_a , and $(1_L, x)$ with its fully supported coordinate vector. To verify the image exactly, if one input coordinate has top support, each row acquires top support through multiplication by $1^\bullet$ or e , and its amplitude is the original coordinate amplitude. If all supports are lower, every entry in the output has the common join of those supports and zero amplitude; the join is allowed to be top. Conversely each displayed vector is fixed by P_n . This proves both the image and idempotence and realizes $\mathcal{V}_n$ as a retract of the free module, hence as finite projective.

For a matrix $A \in M_{m,n}(\mathbb{C})$ let $\widehat{A}$ have all entries supported, including every zero. Then

$$P_m \widehat{A} P_n = \widehat{A}, \quad \widehat{A} \widehat{B} = \widehat{AB}, \quad \widehat{I}_n = P_n, \quad \widehat{0} = e P_n. \quad (\text{JC2})$$

Each matrix entry is a nonempty sum of supported products, so its amplitude is the ordinary matrix product and its support is top even when the sum cancels. This proves every formula in (JC2). Consequently all invertible ordinary matrices become automorphisms of these actual projective modules, with inverse $\widehat{A}^{-1}$.

More precisely, for arbitrary $X \in M_{m,n}(S)$, the entry supports of $P_m X P_n$ are all the join of the supports of every entry of X ; its amplitude matrix is the original amplitude matrix. Therefore

$$P_m M_{m,n}(S) P_n = \{\widehat{A} : A \in M_{m,n}(\mathbb{C})\} \sqcup \{z_a \mathbf{1}_{m,n} : a < 1_L\}. \quad (\text{JC3})$$

Every matrix in the displayed set occurs. In the square case this is the exact semiring $G_L(M_n(\mathbb{C}))$, with its possibly noncommutative amplitude multiplication and identity P_n . The support-preserving maps in (JC3) are precisely the fully supported ones: they act by $(1, x) \mapsto (1, Ax)$ and fix every lower support. A lower constant matrix instead meets all input supports with a and cannot preserve top. Composition is exactly ordinary matrix composition. The relative zero map is $\widehat{0}$. The relative zero object is $eS \simeq L$, with zero-dimensional arithmetic fibre; the unique support-preserving arrows to and from it justify that description. This includes dimension zero without identifying the object with the absent semimodule.

With lexicographic ordering of tensor coordinates, $P_n \otimes P_m = P_{nm}$: a diagonal entry is $1^\bullet$ and every other entry has one or two factors e , whose product is e . The usual explicit tensor basis maps of finite free semimodules identify $\mathcal{V}_n \otimes_S \mathcal{V}_m$ with the image of this projector. Thus the projective realization also retains every ordinary tensor operation.

For original positive Hermitian matrices G, H , give $\mathcal{V}_n$ the pairing which equals $(x^* G y)^\bullet$ on the top fibre and $z_{a \wedge b}$ whenever one input is in lower support. Sesquilinearity follows by checking whether a join reaches top: lower inputs have zero amplitude, and their joins and meets follow distributivity in L ; on the top fibre it is ordinary sesquilinearity. Hence the actual adjoint identity is

$$\widehat{A}^{\dagger G, H} = \widehat{G^{-1} A^* H}. \quad (\text{JC4})$$

Testing both top inputs proves the amplitudes, and a lower input on either side gives the same meet of supports. No new metric is substituted for the source or target metric.

The trace on a finite projective retract is computed by inclusion, the endomorphism, and the retraction in its free ambient module. It is independent of the retract presentation: if maps U, V give inverse projective identifications, the equality $\text{tr}(XY) = \text{tr}(YX)$ follows by writing both finite double sums over the commutative scalar semiring and interchanging their indices. Applying that identity to the retract maps proves invariance. For (JC2) it gives

$$\text{tr}_S(\widehat{A}) = (\text{Tr}_{\mathbb{C}} A)^\bullet, \quad p \text{tr}_S(\widehat{A}) = \text{Tr}_{\mathbb{C}} A, \quad \chi \text{tr}_S(\widehat{A}) = 1_L. \quad (\text{JC5})$$

The last equality holds even if the amplitude trace is zero. After scalar extension by $p : S \rightarrow \mathbb{C}$, the projector is I_n . After extension by localization $\chi : S \rightarrow L = S[e^{-1}]$, it is the all-one matrix, whose image is the common-coordinate copy of L ; every $\widehat{A}$ becomes the identity on this image. These follow by tensoring the two retract arrows, so identify the actual base changes, including their trace maps.

Now take the full packet A_Z of (FJ1)–(FJ3), of dimension $d = \sum_\rho m_\rho$, with every local unit retained. Choose the ordered monomial jet coordinates already used there. Its perfect residue pairing has an invertible matrix D_R . Equation (JC2) therefore transports its full dual isomorphism to the projective corner, with inverse $\widehat{D_R^{-1}}$. The transported pairing has the same full amplitude R and value $z_{a \wedge b}$ on lower supports. The prime matrix and the exact duality give

$$\widehat{D_R} \widehat{\Pi_p} = p(\widehat{\Pi_p^{-1}})^* \widehat{D_R}. \quad (\text{JC6})$$

This is the lifted equality of the original matrices in (FJ4)–(FJ7), proved by (JC2), and retains every higher jet and the original factor p . In particular the inverse prime matrix is an actual projective automorphism without assuming it is a free split automorphism.

The Jacobian map J_g from (FJ8)–(FJ10) has ordinary kernel the nilradical and image the socle. Its supported-zero kernel on the top fibre is precisely that same kernel, and every lower support is retained with zero amplitude. Thus the exact support-preserving kernel and quotient are

$$\mathcal{V}(\text{nil } A_Z) \hookrightarrow \mathcal{V}(A_Z) \longrightarrow \mathcal{V}(A_Z/\text{nil } A_Z), \quad (a, x) \longmapsto (a, [x]). \quad (\text{JC7})$$

Here $\mathcal{V}(W)$ is (JC1) with top fibre W ; for $W = 0$ it is L . The last map has exactly the displayed fibrewise kernel, is onto on every fibre, and preserves all support transitions. The full residue pairing is perfect before this quotient; the ordinary arithmetic trace pairing factors through it by (FJ10). After localization every object and map in (JC7) is L with the identity, including its fibrewise zero object. This describes the entire degeneration rather than assigning that identity a complex trace of one. The amplitude and support trace receivers are exactly (JC5), with the actual connecting scalar maps stated there.

25 The full-addition base product and its surviving support geometry

The supplied arithmetic-specificity calculation computes the self-product of the scalar globalization. Here we retain the entire finite lattice of the original version 11 definition [3]. The result specifies which coordinates agree in the product, which remain independent, and the new relative cohomology they support. The blueprint conventions are those of Lorscheid [12], original author TeX `blueprints-v2.tex`, axioms (A1)–(A9), `lemma_proper`, and Section 1.12’s tensor construction. We use the term properization for its quotient by equality relations between individual elements. Semiring completion also adds the values of formal sums; the two operations below have different explicit images.

25.1 The integer coordinate in every common target

Put $S = G(\mathbb{Z})$ and denote its supported integers by $j(n)$, with $j(0) = e$ and its additive identity τ . For maps f, g from the full-addition blueprint of S to a common proper blueprint, set $u = f(j(-1))$ and $v = g(j(-1))$. Both satisfy the original relations

$$1 + u + 1 \equiv 1, \quad u + 1 + u \equiv u, \quad 1 + v + 1 \equiv 1, \quad v + 1 + v \equiv v. \quad (\text{BP1})$$

In the commutative monoid of formal sums modulo preaddition, these regular inverses of 1 are unique. Indeed

$$u \equiv u + 1 + v + 1 + u \equiv u + 1 + v, \quad v \equiv v + 1 + u + 1 + v \equiv u + 1 + v. \quad (\text{BP2})$$

The first equality in each line replaces 1 by $1 + v + 1$ or $1 + u + 1$ inside the corresponding regular-inverse identity; the second uses $u + 1 + u \equiv u$ or $v + 1 + v \equiv v$. Properness implies $u = v$. Then $f(e) \equiv 1 + u \equiv g(e)$, so these also agree. Repeated addition determines every positive integer from 1 and every negative integer from u ; both maps send τ to the unique zero. Thus $f = g$ on the entire scalar S . This proof uses the regular inverse, not an additive inverse of 1 with respect to τ .

Consequently, for any specified blueprint base $B \rightarrow S$,

$$(B_S \otimes_B B_S)^{\text{prop}} \simeq B_S, \quad S \otimes_{\mathbb{N}} S \simeq S. \quad (\text{BP3})$$

For either assertion, pairs of agreeing maps into the relevant common target are exactly single maps, by (BP2). The inverse arrows are the inclusion $a \mapsto a \otimes 1$ and multiplication $a \otimes b \mapsto ab$.

Their composites are identities by that universal property. This proves the scalar calculation in the arrival. In particular $e \otimes 1 = 1 \otimes e$; the actual prime (e) remains present and does not become two different arithmetic coordinates.

25.2 The full lattice product in commutative semirings

Let P, Q be nonempty finite posets, with their topology of lower sets. Put $L = \text{Down}(P)$, $M = \text{Down}(Q)$, and $T = \text{Down}(P \times Q)$. The exact coproduct is

$$G_L(\mathbb{Z}) \otimes_{\mathbb{N}} G_M(\mathbb{Z}) \simeq G_T(\mathbb{Z}). \quad (\text{BP4})$$

The structure maps send every supported integer to the same integer and send the zero-amplitude supports by

$$A \longmapsto A \times Q, \quad B \longmapsto P \times B. \quad (\text{BP5})$$

Their product on zero-amplitude elements has support $A \times B$. All formulas preserve both τ and supported e in their exact roles.

Here is a proof of the whole universal property. Given semiring maps $f : G_L(\mathbb{Z}) \rightarrow C$ and $g : G_M(\mathbb{Z}) \rightarrow C$, (BP2) makes their supported integer maps agree. Write E for their common value of e , and $a_A = f(0, A)$, $b_B = g(0, B)$. Each is additively and multiplicatively idempotent, is absorbed by E under addition, and has E as a multiplicative identity. The finite sums of their products form a bounded distributive lattice with join given by addition, meet by multiplication, bottom 0_C and top E . For detail, sums of commuting multiplicative idempotents which are also additively idempotent remain idempotent under either operation. Cross absorption holds since $b_B + E = E$ implies $a_A b_B + a_A E = a_A E = a_A$. Distributivity and this identity prove both absorption laws on the generated set. The two original lattice maps therefore take values in this common bounded lattice.

The universal lattice map from T has the explicit value

$$F(U) = \bigvee_{(p,q) \in U} a_{\downarrow p} b_{\downarrow q}. \quad (\text{BP6})$$

It preserves unions immediately. For intersections, distribute the two joins and use $\downarrow p \cap \downarrow p' = \bigcup_{r \leq p, p'} \downarrow r$, and the identical formula in Q . The resulting terms are exactly those indexed by $U \cap U'$. It preserves bottom and top. The same expansion proves $F(A \times Q) = a_A$ and $F(P \times B) = b_B$. Every lower set is the union of its principal rectangles, so (BP6) is the unique such map.

Extend F to $G_T(\mathbb{Z})$ by the common supported integer map. A supported integer absorbs each generator under addition and acts identically on it by multiplication; these identities hold first in its original factor and then for products and joins. They also hold for a negative supported integer, because both factors have the identical map $j(-1)$ proved in (BP2). Thus the extension preserves the mixed operations. It agrees at e , so is well-defined. It is the required unique semiring map. This proves (BP4).

For arbitrary finite nontrivial bounded distributive lattices this description includes every case, rather than a chosen subfamily. To see this directly, write P for the nonzero join-irreducible elements of L , with inherited order. Every element is the join of the join-irreducibles below it, by induction in the finite order. Distributivity implies a join-irreducible below a finite join lies below one summand: for two summands intersect it with the join and use its defining irreducibility, then induct. Hence $a \mapsto \{p : p \leq a\}$ is a lattice isomorphism $L \simeq \text{Down}(P)$, with inverse the join. The same proof applies to M .

25.3 Proper blueprint product, semiring completion, and their exact map

Use the base blueprint with zero $\mathbb{F}_1^{\text{mon}} = \{0, 1\}$, with only its zero relation, and let $\mathcal{B}$ be the raw blueprint tensor of the two full-addition factors. In the description of (BP4), its properization is

the subblueprint with underlying set

$$\mathcal{A}_{P,Q} = \{(n, P \times Q) : n \in \mathbb{Z}\} \cup \{(0, A \times B) : A \in L, B \in M\}, \quad (\text{BP7})$$

and preaddition the equality of its finite sums in $G_T(\mathbb{Z})$. In particular the exact connecting maps are

$$\mathcal{B} \longrightarrow \mathcal{B}^{\text{prop}} \simeq \mathcal{A}_{P,Q} \hookrightarrow B_{G_T(\mathbb{Z})}, \quad (\mathcal{B}^{\text{prop}})_{\mathbb{N}} = G_T(\mathbb{Z}). \quad (\text{BP8})$$

Indeed the raw underlying elements are pure tensors; the finite sums modulo their preaddition form its universal semiring completion. Maps from that completion to C are the two original semiring maps, so (BP4) identifies this completion exactly. A pure tensor has the value $(nm, P \times Q)$ when both amplitudes are nonzero, and otherwise has amplitude zero and rectangular support as in (BP7). All elements of (BP7) occur. Equality of images of two raw elements in the semiring completion is exactly their preaddition equivalence, by its definition as the quotient of formal sums. Properization consequently has precisely (BP7)'s elements and induced preaddition. Adding all their finite sums gives all lower sets by principal rectangles, proving the last assertion of (BP8). Multiplication of rectangles closes (BP7), so its indicated subblueprint is defined.

This identifies a material extra operation. If $P = Q$ consists of two incomparable points, the union $\{(p_1, p_1), (p_2, p_2)\}$ is a lower set but is not a rectangle. Its zero-amplitude value is a sum in the semiring completion, while it is absent from the underlying set (BP7). This is an explicit example and exact embedding, with no identification of properization with adding all sums. For scalar one-point P, Q , every lower set is a rectangle and (BP8) reduces to (BP3).

25.4 The changed prime spectrum and the diagonal

The full spectrum theorem already proved in this manuscript gives

$$\text{Spec } G_T(\mathbb{Z}) = (P \times Q) \blacktriangleright \text{Spec } \mathbb{Z}. \quad (\text{BP9})$$

For completeness the lattice prime associated to (p, q) is $\{U : (p, q) \notin U\}$: a union avoids the point exactly when both sets do, and an intersection avoids it exactly when one does. Every finite lattice prime has this form, since its complement is a prime filter, whose smallest element is a principal lower set; the join-irreducibility proof following (BP6) identifies its generator. The general support/arithmetic classification then proves (BP9). The two projection maps on spectra are

$$(p, q) \mapsto p, \quad (p, q) \mapsto q; \quad \mathfrak{p} \mapsto \mathfrak{p} \quad (\mathfrak{p} \in \text{Spec } \mathbb{Z}). \quad (\text{BP10})$$

Testing inverse images against (BP5) proves these assertions point by point. There are no support/arithmetic mixed points: the two pulled-back supported zeros are the same element e , so any prime contains both or neither. Every integer and its original prime vanishing set is retained on the single arithmetic stratum.

For $P = Q$, multiplication of the two equal factors induces the surjective map

$$G_{\text{Down}(P \times P)}(\mathbb{Z}) \longrightarrow G_{\text{Down}(P)}(\mathbb{Z}), \quad (0, U) \mapsto (0, \{p : (p, p) \in U\}), \quad j(n) \mapsto j(n). \quad (\text{BP11})$$

It preserves joins and meets because it is inverse image under the diagonal, so is a semiring map; it is surjective using $A \times P$. Its spectrum map is the diagonal on the full support and the identity on the arithmetic stratum. Its exact congruence identifies two null supports precisely when their diagonal intersections agree, and identifies no distinct nonzero amplitudes. This gives the complete defect object and comparison map of the self-product calculation.

25.5 An explicit product with surviving relative cohomology

Take P to be the six-element face poset of the boundary of an oriented triangle: three vertices and its three edges, ordered by incidence. This is the earlier (FL) support example. The poset $P \times P$ is the face poset of the product cell structure on the product of the two boundary triangles. Its order complex is the barycentric subdivision of that product. An explicit realization sends a pair of faces to the product of their barycentres and extends affinely on each chain. On each product face, the chains ending in that face form the cone from its barycentre over the subdivided boundary. Induction over dimensions 0, 1, 2 shows these cones tile each convex product face with disjoint interiors and agree on common boundaries. Thus this realization is a homeomorphism, including every one of the 36 original support points.

Here is also a direct integral calculation. The triangle boundary has cellular boundary $\partial e_i = v_{i+1} - v_i$ (indices modulo three). Choose e_0, e_1 as a spanning tree. Their boundaries and v_0 form an integral basis of the vertex group; the edge basis $e_0, e_1, z = e_0 + e_1 + e_2$ splits the chain complex into two identity complexes and a copy of $\mathbb{Z}$ in each degree 0, 1. The product cells have boundary $\partial(a \times b) = \partial a \times b + (-1)^{\deg a} a \times \partial b$, as follows by orienting each rectangular face by its two edge coordinates. Tensoring the displayed splitting leaves its two identity factors contractible and gives

$$H^j(P \times P, \mathbb{Q}) = \begin{cases} \mathbb{Q} & j = 0, 2, \\ \mathbb{Q}^2 & j = 1, \\ 0 & j > 2. \end{cases} \quad (\text{BP12})$$

Write α_1, α_2 for the edge-circuit cohomology classes pulled back from the factors, each evaluating to one on z . Their product evaluates to one on the oriented product $z \times z$. The diagonal pulls both to α and their degree-two product to zero, since the one-dimensional boundary has no degree-two cochains.

Apply the actual relative complex (FL14), $[\mathbb{Z} \rightarrow C^\bullet(P \times P, \mathbb{Q})]$, with the integer term in degree zero and the support terms shifted by one. The constants map is injective, and (BP12) gives

$$H_X^1 = \mathbb{Q}/\mathbb{Z}, \quad H_X^2 = \mathbb{Q}\alpha_1 \oplus \mathbb{Q}\alpha_2, \quad H_X^3 = \mathbb{Q}(\alpha_1\alpha_2), \quad H_X^{j>3} = 0. \quad (\text{BP13})$$

The morphism (BP11) pulls degree one identically, degree two by $(a, b) \mapsto a + b$, and degree three to zero. Thus the support product has two independent degree-two relative classes and one degree-three class, and the diagonal's exact lost classes are $\mathbb{Q}(\alpha_1 - \alpha_2)$ and $\mathbb{Q}(\alpha_1\alpha_2)$. They are produced by the complete original support product. Their existence establishes these groups and maps; a Frobenius action or a positive Weil pairing is not assigned to them without a construction.

$$\begin{array}{ccc} (P \times P) \blacktriangleright \text{Spec } \mathbb{Z} & \xrightleftharpoons[\text{pr}_2]{\text{pr}_1} & P \blacktriangleright \text{Spec } \mathbb{Z} \\ & \searrow \text{diag} & \\ & \mathbb{Q}\alpha_1 \oplus \mathbb{Q}\alpha_2 & \xrightarrow{(a,b) \mapsto a+b} \mathbb{Q}\alpha \end{array}$$

Figure 17: The full-support product and its diagonal, (BP9)–(BP13). The top row displays maps of spectra; the bottom displays the pullback of relative degree-two cohomology along the diagonal. Both arithmetic maps are the identity. For the six-point triangle-boundary support, the product retains all 36 support points and the additional relative degree-three class. Sources of the coefficient object and blueprint operations: version 11 [3] and Lorscheid [12].

26 Primitive representatives of the complete arithmetic jet packets

26.1 The two endpoint maps and their exact theta splitting

Keep the original objects

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \quad D = -x\partial_x, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} |x^b D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V. \end{aligned} \tag{PL1}$$

Mellin transforms use dx/x : $M_+\phi(s) = \int_0^\infty \phi(x)x^s dx/x$ and $\mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x$. The Fourier transform on $\mathbb{R}$ has kernel $e^{-2\pi i x \xi}$. Poisson summation, with both moments in V zero, shows that $\Theta V \subset \mathcal{B}$: the sum decays with all Euler derivatives at infinity, while $\Theta\phi(x) = x^{-1}\Theta\widehat{\phi}(1/x)$ gives the same property at zero. The two source conventions and their exact scale action are retained from [17, 23]; Meyer's original treatment [20], Theorem `the:Lap_range`, provides the human source context for the full Mellin quotient. Here no closure is added to the algebraic relation space ΘV .

Use the actual completion and seed:

$$\begin{aligned} a(s) &= s(s-1), \quad g(s) = 2\xi(s) = a(s)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g(0) = g(1) = 1, \\ \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad A_*(s) = M_+\phi_*(s) = \frac{1}{2}a(s)\pi^{-s/2}\Gamma(s/2). \end{aligned} \tag{PL2}$$

Gaussian integration gives $\phi_*(0) = 0$ and $\int_{\mathbb{R}} \phi_* = 4\pi^2(3/(4\pi^2)) - 6\pi(1/(2\pi)) = 0$. The substitution $t = \pi x^2$ gives the Mellin identity in PL2: its two terms are $2\pi^{-s/2}\Gamma(s/2+2)$ and $-3\pi^{-s/2}\Gamma(s/2+1)$. The gamma recurrence gives exactly the stated factor. No seed coefficient is changed. For $\Re s > 1$, absolute summation and substitution $t = nx$ give

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)M_+\phi(s), \quad \mathcal{M}\Theta\phi_* = g, \quad \mathcal{M}DF = s\mathcal{M}F. \tag{PL3}$$

The first identity continues across the critical strip and at its endpoints: $M_+\phi$ is holomorphic for $\Re s > -2$, by evenness and $\phi(0) = 0$, and $M_+\phi(1) = 0$. Using $\zeta(0) = -1/2$ and the residue one at $s = 1$ gives

$$\ell_0(\phi) := \mathcal{M}\Theta\phi(0) = -M_+\phi(0), \quad \ell_1(\phi) := \mathcal{M}\Theta\phi(1) = 2M'_+\phi(1). \tag{PL4}$$

In particular both endpoint values of $\Theta\phi_*$ are one. The operator D preserves V : it preserves evenness and the Schwartz conditions, its value at zero is zero, and integration by parts gives $\int D\phi = \int \phi = 0$. Thus every polynomial in D preserves V , and commutes with Θ .

Theorem 26.1 (A primitive representative for every original class). *Set*

$$\alpha_0 = (1-D)\phi_*, \quad \alpha_1 = D\phi_*, \quad E_i = \Theta\alpha_i, \quad \mathcal{B}_{\text{pr}} = \ker(\mathcal{M}(\cdot)(0), \mathcal{M}(\cdot)(1)). \tag{PL5}$$

The exact continuous projections

$$PF = F - \mathcal{M}F(0)E_0 - \mathcal{M}F(1)E_1, \quad P_V\phi = \phi - \ell_0(\phi)\alpha_0 - \ell_1(\phi)\alpha_1 \tag{PL6}$$

have images $\mathcal{B}_{\text{pr}}$ and

$$V_{\text{pr}} = \{\phi \in V : M_+\phi(0) = 0, M'_+\phi(1) = 0\}, \tag{PL7}$$

respectively. They satisfy $P\Theta = \Theta P_V$ and give the exact decompositions and quotient isomorphism

$$\mathcal{B} = \mathcal{B}_{\text{pr}} \oplus \mathbb{C}E_0 \oplus \mathbb{C}E_1, \quad \Theta V = \Theta V_{\text{pr}} \oplus \mathbb{C}E_0 \oplus \mathbb{C}E_1, \quad Q \cong \mathcal{B}_{\text{pr}}/\Theta V_{\text{pr}}. \tag{PL8}$$

The last isomorphism is induced by inclusion, with inverse $[F] \mapsto [PF]$.

Proof. PL3 gives $\mathcal{M}E_0 = (1-s)g(s)$ and $\mathcal{M}E_1 = sg(s)$. Their endpoint pairs are $(1,0)$ and $(0,1)$, since $g(0) = g(1) = 1$. This proves idempotence, the asserted images and direct sums. PL4 gives $\ell_i(\alpha_j) = \delta_{ij}$, hence the same calculation on V proves its projection. Direct substitution proves $\Theta = \Theta_P$. Furthermore $\Theta V \cap \mathcal{B}_{\text{pr}} = \Theta V_{\text{pr}}$, by PL4. Subtracting the two theta terms in PL6 preserves every original cohomology class, proving surjectivity and the explicit inverse in PL8. Its kernel is exactly the stated intersection. All endpoint functionals are continuous: split their defining integrals at one and bound by two suitable seminorms from PL1. All other operators used in PL6 are fixed continuous linear maps. $\square$

The source conditions $\phi(0) = \int \phi = 0$ and the logarithmic conditions in PL7 have now been joined by actual maps, without identifying them. Also Θ is injective: in $\Re s > 1$ the Euler product is nonzero, so $\Theta\phi = 0$ forces $M_+\phi = 0$. Fourier uniqueness in $\log x$ on any line $\Re s = c > 1$ then forces $\phi = 0$.

For $p > 1$, write $U_p F(x) = F(x/p)$, also on V . It preserves the two primitive spaces, because its endpoint action is $(b_0, b_1) \mapsto (b_0, pb_1)$. The full projection defect is

$$(PU_p - U_p P)F = \Theta[\mathcal{M}F(0)(U_p \alpha_0 - \alpha_0) + \mathcal{M}F(1)(U_p \alpha_1 - p\alpha_1)]. \quad (\text{PL9})$$

Both bracketed differences belong to V_{pr} : their two endpoint values vanish by the same scaling calculation. Thus the projection is prime-equivariant on Q , with its precise representative defect retained.

26.2 Every multiplicity and every exact packet relation

Let Z be a finite nonempty set of actual nontrivial zeros. Keep every full multiplicity m_ρ , and fix the exact polynomial

$$h(s) = c_h \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad c_h \neq 0, \quad d = \deg h, \quad U_h(s) = g(s)/h(s), \quad A_h = \mathbb{C}[s]/(h). \quad (\text{PL10})$$

The chosen c_h is retained throughout. At ρ , U_h is a holomorphic unit. Write $u_h = [U_h] \in A_h^\times$. For $u \in A_h$, define the unique polynomial of degree less than d

$$P_u = \text{rem}_h(u_h^{-1}u), \quad R_u = \text{rem}_h([a]^{-1}P_u). \quad (\text{PL11})$$

The inverse $[a]^{-1}$ exists because no selected zero is 0 or 1. Both inverses are full inverses in the nonreduced algebra.

Lemma 26.2 (The exact inverse Euler operators). *If $H \in \mathcal{B}$ and $\mathcal{M}H(\rho) = 0$, then*

$$S_\rho H(x) = x^{-\rho} \int_x^\infty H(t)t^{\rho-1} dt \quad (\text{PL12})$$

belongs to $\mathcal{B}$, satisfies $(D - \rho)S_\rho H = H$, and has Mellin transform $\mathcal{M}H(s)/(s - \rho)$. Consequently

$$S_h = c_h^{-1} \prod_{\rho \in Z} S_\rho^{m_\rho} : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B}, \quad h(D)S_h = 1 \quad (\text{PL13})$$

is well defined, independent of the order of the factors, and continuous. Here j_h denotes all Taylor coefficients modulo h .

Proof. Differentiation of the tail gives the displayed differential equation. Put $x = e^y$, $\sigma = \Re \rho$, and $H(e^y) = b(y)$. For $y \geq 0$, $|b(y)| \leq C_N e^{-Ny}$ gives $|S_\rho H(e^y)| \leq C_N e^{-Ny}/(N - \sigma)$ whenever $N > \sigma$. The zero moment changes the same tail at $y \leq 0$ into minus the integral from $-\infty$ to y . The bound $|b(y)| \leq C'_N e^{Ny}$ then gives $|S_\rho H(e^y)| \leq C'_N e^{Ny}/(N + \sigma)$ for $N + \sigma > 0$. The differential equation bounds every derivative in the same way. These are exactly the seminorm

estimates in PL1, and give continuity. Integration by parts gives the Mellin quotient identity, with its removable value at ρ . After one factor is removed, the remaining zero orders of the transform allow the next factor to be removed; thus PL13 is defined with the entire multiplicity retained. Each $D - \rho$ is injective on $\mathcal{B}$, since a homogeneous solution $Cx^{-\rho}$ fails rapid decrease at at least one end unless $C = 0$. Their product is injective. Every order of PL13 solves the same equation, so uniqueness proves order independence. $\square$

Define the original section and its primitive lift by actual elements of $\mathcal{B}$:

$$s_h u = S_h \Theta(P_u(D)\phi_*), \quad \text{Prim}_h(u) = a(D)S_h \Theta(R_u(D)\phi_*). \quad (\text{PL14})$$

Their Mellin transforms are $U_h P_u$ and $a U_h R_u$. The domain conditions of PL13 hold because the theta transforms are $g P_u$ and $g R_u$, both divisible by the unaltered h .

Theorem 26.3 (Full-jet same-class lift and its exact affine relation). *For every $u \in A_h$,*

$$j_h \mathcal{M} s_h u = j_h \mathcal{M} \text{Prim}_h(u) = u, \quad \text{Prim}_h = \mathcal{P} s_h, \quad [\text{Prim}_h(u)] = [s_h u] \text{ in } Q. \quad (\text{PL15})$$

More explicitly, the exact division identity and its quotient are

$$a(s)R_u(s) - P_u(s) = h(s)T_u(s), \quad T_u(s) = -(1-s)\frac{P_u(0)}{h(0)} - s\frac{P_u(1)}{h(1)}, \quad (\text{PL16})$$

and

$$\text{Prim}_h(u) - s_h u = \Theta(T_u(D)\phi_*). \quad (\text{PL17})$$

For every polynomial P , the whole polynomial packet family has the exact relation calculation

$$[S_h \Theta(P(D)\phi_*)] = 0 \text{ in } Q \iff h \mid P. \quad (\text{PL18})$$

For the primitive family $a(D)S_h \Theta(P(D)\phi_)$, the same condition is necessary and sufficient; if $P = hT$, its theta primitive is exactly $a(D)T(D)\phi_* \in V_{\text{pr}}$.*

Proof. The full unit inverse in PL11 proves the first jet identity. The second follows because $aR_u - P_u$ is divisible by h . That difference has degree at most $d+1$; hence T_u has degree at most one. At $s = 0, 1$, it has the two values $-P_u(0)/h(0)$ and $-P_u(1)/h(1)$; these uniquely determine the affine polynomial displayed in PL16, including c_h . Multiplication of PL16 by U_h gives gT_u , whose inverse Mellin transform is $\Theta(T_u(D)\phi_*)$ by PL3. Mellin injectivity proves PL17. The endpoint values of $s_h u$ are $P_u(0)/h(0)$ and $P_u(1)/h(1)$, since $g(0) = g(1) = 1$. PL6 and PL16 therefore show that its primitive lift is exactly $\mathcal{P} s_h u$. This proves all of PL15.

If the element in PL18 is $\Theta\phi$, its full jets at the selected zeros vanish, by PL3. Its jets are $u_h[P]$; the unit u_h is invertible, so $[P] = 0$ in A_h , which is exactly $h \mid P$. Conversely, for $P = hT$, PL13 and injectivity give $S_h \Theta(P(D)\phi_*) = \Theta(T(D)\phi_*)$. Inserting $a(D)$ proves sufficiency in the primitive case, and necessity follows because a is another unit modulo h . Its primitive source belongs to V_{pr} because its theta transform is $a(s)g(s)T(s)$, with both endpoint values zero. $\square$

In particular the class of $S_h \Theta\phi_*$ has annihilator exactly (h) under $\mathbb{C}[D]$, and its cyclic subspace is $\mathbb{C}[s]/(h)$ with its entire nilpotent structure. Neither PL15 nor PL17 asserts an isometry of a previously chosen finite source metric or preservation of its polynomial cutoff. The primitive transform numerator has degree at most $d+1$, while P_u, R_u have degree at most $d-1$.

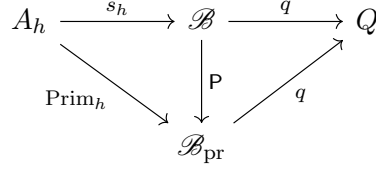


Figure 18: The two routes to Q agree exactly by PL16–17. Their representative difference is the original theta relation with the displayed affine polynomial T_u . The full local unit u_h^{-1} occurs in both routes through PL11.

26.3 The exact primitive theta relation for every prime action

Let $M_s : A_h \rightarrow A_h$ be multiplication by s , and $\Pi_p = M_{p^s}$, with its full finite Taylor expansion at every zero. For $v \in A_h$, define the scalar

$$\kappa_h(v) = \frac{sR_v(s) - R_{M_s v}(s)}{h(s)}. \quad (\text{PL19})$$

This quotient is constant: its numerator is divisible by h and has degree at most d . It retains the factor c_h^{-1} in polynomial division. Both remainder operations in PL11 commute with multiplication modulo h , so

$$D \text{Prim}_h(v) - \text{Prim}_h(M_s v) = \Theta(\kappa_h(v) a(D)\phi_*). \quad (\text{PL20})$$

This follows by taking Mellin transforms: the difference is $aU_h(sR_v - R_{M_s v}) = ag\kappa_h(v)$.

Theorem 26.4 (The full prime defect is a primitive source relation). *With $L_p = \log p$, define the actual Schwartz source*

$$\psi_{p,u} = \int_0^{L_p} \kappa_h(e^{tM_s} u) U_{e^{L_p-t}}(a(D)\phi_*) dt. \quad (\text{PL21})$$

Then $\psi_{p,u} \in V_{\text{pr}}$ and

$$U_p \text{Prim}_h(u) - \text{Prim}_h(\Pi_p u) = \Theta\psi_{p,u}. \quad (\text{PL22})$$

Thus the lifted packet action is exactly prime-equivariant in the primitive quotient of PL8, with a specified relation before taking that quotient. The defect is zero at all actual full zero jets.

Proof. The integrand is a smooth family in V on a compact interval, so its integral is in V , with every Schwartz seminorm finite. The seed $a(D)\phi_*$ has theta transform ag ; its endpoint values vanish. Every positive dilation preserves those zero endpoint values, proving that the integral lies in V_{pr} . Differentiate $U_{e^{L_p-t}} \text{Prim}_h(e^{tM_s} u)$ with respect to t . By PL20 its derivative is $-\Theta[\kappa_h(e^{tM_s} u) U_{e^{L_p-t}} a(D)\phi_*]$. Integration between 0 and L_p proves PL22 with the displayed sign. The relation space has zero full jets at every actual zero, by PL3; this proves the last assertion. $\square$

Apply every map in PL6–22 at each nonempty original tau support label, with zero map on the empty label. The transports at degree one are identities, so all these maps commute with them. Their kernels remain labelled internal zeros. Thus PL8, PL15 and PL22 also hold in the full degree-one support diagram and its reconstruction over $G_L(\mathbb{C})$. No numerical coefficient-support observation is used in this passage.

26.4 The original two-leg complex and its endpoint quotient

Here is the exact chain calculation, including its degree-zero term. Write $\mathcal{F}$ for the original Fourier transform, $\ell = (\ell_0, \ell_1)$, $b(F) = (\mathcal{M}F(0), \mathcal{M}F(1))$, $S(b_0, b_1) = (b_1, b_0)$, and $A(b_0, b_1) = b_0\alpha_0 + b_1\alpha_1$.

The functional equation $g(1-s) = g(s)$ and Poisson identity give $\Theta \mathcal{F} \phi_* = J \Theta \phi_*$, with Mellin transform $g(1-s) = g(s)$. Injectivity of theta gives $\mathcal{F} \phi_* = \phi_*$. Fourier differentiation gives $\mathcal{F} D = (1-D)\mathcal{F}$; therefore

$$\mathcal{F} A = A S, \quad \ell \mathcal{F} = S \ell, \quad \mathcal{F} P_V = P_V \mathcal{F}, \quad \mathcal{F} V_{\text{pr}} = V_{\text{pr}}. \quad (\text{PL45})$$

The identity for ℓ also follows directly from $\mathcal{M} J F(s) = \mathcal{M} F(1-s)$. These formulas prove the asserted commutation by substitution into $P_V = 1 - A \ell$.

For each original mask $I \subseteq \{+, -\}$, let

$$K_I = [V^I \xrightarrow{d_I} \mathcal{B}], \quad d_I(\phi, \psi) = \Theta(\phi - \mathcal{F} \psi) \quad (\text{PL46})$$

when $I \neq \emptyset$, with a missing summand interpreted as the zero coefficient at that retained mask. Let $K_\emptyset = 0$. Define $K_{I,\text{pr}}$ by replacing the present source copies by V_{pr} and the target by $\mathcal{B}_{\text{pr}}$. PL45 proves that this is a subcomplex and that the original projection is the chain map (P_V^I, P) . Its full quotient complex is

$$\begin{aligned} E_I &= [(\mathbb{C}^2)^I \longrightarrow \mathbb{C}^2], \quad (b_+, b_-) \longmapsto b_+ - S b_-, \quad E_\emptyset = 0, \\ 0 &\longrightarrow K_{I,\text{pr}} \longrightarrow K_I \xrightarrow{(\ell^I, b)} E_I \longrightarrow 0. \end{aligned} \quad (\text{PL47})$$

The identity $b d_I = \ell \phi - S \ell \psi$ proves that the displayed surjection is a chain map. Its kernels are exactly the indicated primitive spaces. A chain section is $(A^I, \Theta A)$: its compatibility follows from $\mathcal{F} A = A S$, and its composite with the quotient is the identity by PL4-6. Thus the sequence splits as complexes. The quotient, section, and projection all commute with the mask inclusions, which are coordinate inclusions in degree zero and identities in degree one.

For a singleton mask the differential of E_I is either 1 or $-S$, hence is invertible. For the joint mask its kernel is $\{(b, S b) : b \in \mathbb{C}^2\}$, and its differential is onto. Consequently

$$H^1(E_I) = 0, \quad H^0(E_I) = \begin{cases} \mathbb{C}^2, & I = \{+, -\}, \\ 0, & \text{otherwise,} \end{cases} \quad H^1(K_{I,\text{pr}}) \xrightarrow{\cong} H^1(K_I). \quad (\text{PL48})$$

The last map is induced by inclusion. Its inverse is the original primitive projection on degree-one classes. This proves exactly which degree-zero directions are removed; it does not identify the entire two complexes.

Finally retain the source action $(U_p, p U_{1/p})$ and target action U_p from the original tau complex. Put $D_p = \text{diag}(1, p)$. On E_I the respective actions are

$$\begin{aligned} (D_p, S D_p S) &\text{ on the present source legs,} \quad D_p \text{ on the target,} \\ D_p(b_+ - S b_-) &= D_p b_+ - S(S D_p S) b_-. \end{aligned} \quad (\text{PL49})$$

Thus the short exact sequence PL47 and the isomorphism in PL48 are prime-equivariant. On its joint-mask $H^0 = \mathbb{C}^2$, use $b \mapsto (b, S b)$; the induced action is exactly D_p . The displayed section is not asserted to be prime-equivariant. Its target defect is $U_p \Theta A - \Theta A D_p$, and the two source defects are $U_p A - A D_p$ and $p U_{1/p} A - A S D_p S$, respectively. Each lands in its primitive space because its two endpoint values are zero. These are the chain components of the exact representative discrepancy PL9, with the Fourier swap and both prime factors retained.

26.5 The raw pairing and the whole Weil form of this lift

Assume now that Z is invariant under $\iota \rho = 1 - \bar{\rho}$. For the actual local coordinates $t_\rho = s - \rho$, retain

$$g(\rho + t_\rho) = u_\rho(t_\rho) t_\rho^{m_\rho}, \quad u_\rho(0) = 2\lambda_\rho, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}. \quad (\text{PL23})$$

On $A_h \cong \prod_{\rho \in Z} \mathcal{O}_\rho / (g)$, use

$$R_h(u, v) = \sum_{\rho \in Z} \text{Res}_\rho \frac{u^\dagger(s)v(s)}{g(s)} ds, \quad J_g = M_{g'}. \quad (\text{PL24})$$

The germs in this equation may be any representatives. Divisibility by g makes a change of representative holomorphic. The local coefficient matrix of the residue is $(-1)^j [t^{m_\rho-1-j-k}] u_\rho(t)^{-1}$; its antidiagonal entries are $(-1)^j / (2\lambda_\rho) \neq 0$. It is therefore perfect, without replacing the unit. Since $g'/g = m_\rho/t_\rho + u'_\rho/u_\rho$,

$$R_h(u, J_g v) = T_h(u, v) = \sum_{\rho \in Z} m_\rho \overline{u(\iota \rho)} v(\rho). \quad (\text{PL25})$$

The full kernel of J_g , and radical of T_h , is $\mathfrak{n}_h = \prod_{\rho \in Z} (t_\rho)$. Indeed locally $J_g v = 2m_\rho \lambda_\rho v(\rho) t_\rho^{m_\rho-1}$. The raw pairing detects the complementary higher jets and stays perfect on those directions.

For $F, H \in \mathcal{B}$, define the full reflected zero form

$$\mathcal{W}_g(F, H) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{\mathcal{M}F(1-\bar{\rho})} \mathcal{M}H(\rho). \quad (\text{PL26})$$

The sum is absolutely convergent. Repeated integration by parts in $\log x$ gives arbitrary inverse powers of $|\Im \rho|$, uniformly on $0 \leq \Re \rho \leq 1$. The required zero-count bound also follows directly from the retained seed. For $x \geq 1$, its Gaussian series satisfies $|\Theta \phi_*(x)| \leq Cx^4 e^{-\pi x^2}$: factor the first exponential out of each summand and bound the remaining series by $\sum_{n \geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. The symmetry $J\Theta \phi_* = \Theta \phi_*$, proved in PL45, gives

$$\begin{aligned} g(s) &= \int_1^\infty \Theta \phi_*(x) (x^s + x^{1-s}) \frac{dx}{x}, \\ \max_{|s| \leq R} |g(s)| &\leq C\pi^{-(R+5)/2} \Gamma((R+5)/2) \\ &\leq \exp(C'(R+1) \log(R+2)) \quad (R \geq 1). \end{aligned} \quad (\text{PL50})$$

The gamma bound follows by splitting its integral at one and bounding its tail by $n!$, with $n = \lceil (R+5)/2 \rceil$; $n! \leq n^n$. To count zeros, choose a radius r between $2R$ and $3R$ whose circle has no zeros. Factor the finitely many zeros inside that circle out of g . The logarithm of the modulus of the remaining nonvanishing factor is harmonic. Its circle average equals its central value; the average of $\log |re^{i\theta} - \rho|$ is $\log r$ for $|\rho| < r$. Since $g(0) = 1$, subtraction yields $\sum_{|\rho| < r} m_\rho \log(r/|\rho|) = (2\pi)^{-1} \int_0^{2\pi} \log |g(re^{i\theta})| d\theta$. Each zero with $|\rho| \leq R$ contributes at least $\log 2$. PL50 therefore bounds their total multiplicity by $C''(R+1) \log(R+2)$. Dyadic summation now proves $\sum_\rho m_\rho (1 + |\Im \rho|)^{-2N} < \infty$ for $N \geq 2$, because the zeros lie in the original critical strip. This proves the absolute convergence asserted in PL26. The human explicit formula with this transform is recalled below from [21].

Theorem 26.5 (Complete packet pullback, with its whole radical). *For all full jets $u, v \in A_h$,*

$$\mathcal{W}_g(\text{Prim}_h(u), \text{Prim}_h(v)) = T_h(u, v) = R_h(u, J_g v). \quad (\text{PL27})$$

The same full Weil form is obtained using s_h in either entry. No zero outside Z is omitted: its summand is exactly zero. The primitive family $\text{Prim}_h(A_h)$ has trace radical $\text{Prim}_h(\mathfrak{n}_h)$, whose elements are nonzero cohomology classes whenever their jets are nonzero. The raw residue form transported through the inverse jet map on this family is perfect.

Proof. At selected zeros, both representative maps have precisely the full jets u, v , by PL15. At every other actual zero, the transforms $U_h P_u$ and $aU_h R_u$ have the full zero order of g , because

h does not vanish there. Thus PL26 reduces exactly to PL25. This proves all equalities with both representatives. PL15 makes each representative map injective, also after passage to Q , since theta relations have zero actual jets. The radical is therefore the exact inverse image of $\mathfrak{n}_h$. Perfectness of the transported raw form follows from PL24. $\square$

A fixed reflected zero contributes $m_\rho|u(\rho)|^2$. A nonfixed reflection pair contributes the matrix $m_\rho \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ on its values. In particular the exact primitive lift transports both signs of this matrix when such an actual pair is present. It does not create positivity. The prime identity

$$\mathcal{W}_g(U_p F, U_p H) = p \mathcal{W}_g(F, H) \quad (\text{PL28})$$

follows term by term from $\overline{p^{1-\bar{\rho}}} p^\rho = p$. PL22 shows that the same identity is induced by Π_p on the complete primitive packet classes.

26.6 The same primitive functions for the actual and control zeros

Keep the supplied exact control constants $\delta_0 = 1/1000$, $\gamma_0 = 3$. Let γ_1, γ_2 be the actual first two positive critical ordinates, and put

$$\begin{aligned} Q_1(s) &= ((s - \tfrac{1}{2})^2 + \gamma_1^2)((s - \tfrac{1}{2})^2 + \gamma_2^2), & U(s) &= g(s)/Q_1(s), \\ Q_0(s) &= ((s - \tfrac{1}{2})^2 - (\delta_0 + i\gamma_0)^2)((s - \tfrac{1}{2})^2 - (\delta_0 - i\gamma_0)^2), & \tilde{g}(s) &= U(s)Q_0(s). \end{aligned} \quad (\text{PL29})$$

The selected actual zeros are simple. The low-height verified theorem and exhaustive sign-change method in the original author source [22], Theorem **bank** and Section 2, supply these input facts: the zero count counts multiplicities, and matching it by distinct sign changes forces each counted zero to be simple. The same low-height theorem excludes the off-critical points of Q_0 from the actual zero set. Thus U is a unit at every root of $Q_0 Q_1$. No rounded ordinate is used.

For $P \in \mathbb{C}[y]_{\leq 3}$, put $y(s) = (s - \frac{1}{2})/i$ and retain the same entire transform in both forms:

$$G_P(s) = a(s)P(y(s))U(s), \quad F_P = \mathcal{M}^{-1}G_P. \quad (\text{PL30})$$

Its element $F_P \in \mathcal{B}_{\text{pr}}$ is constructed by PL12–14, using Q_1 and polynomial $P(y(s))$; no unproved inverse-transform membership is required. Let Z_- and Z_+ be the roots of Q_1, Q_0 , respectively. On coefficient vectors $p = (p_0, p_1, p_2, p_3)$, the two evaluation maps are

$$(E_{\pm} p)_z = w_z \sum_{k=0}^3 p_k y(z)^k, \quad w_z = a(z)U(z), \quad z \in Z_{\pm}. \quad (\text{PL31})$$

Both maps are invertible: their diagonal factors w_z are nonzero and their Vandermonde nodes are distinct. At $z = \tau \in Z_-$, the factor is exactly $w_\tau = \tau(\tau - 1)g'(\tau)/Q_1'(\tau)$.

Theorem 26.6 (Actual/control full-Weil contrast and its paired defect). *For the same functions F_P, F_Q ,*

$$\mathcal{W}_g(F_P, F_Q) = p^* E_-^* E_- q, \quad \mathcal{W}_{\tilde{g}}(F_P, F_Q) = p^* E_+^* R_{\text{ref}} E_+ q, \quad (\text{PL32})$$

where R_{ref} exchanges the two members of each of the two reflected pairs in Z_+ . The forms have inertia $(4, 0)$ and $(2, 2)$, respectively. These are the entire zero sums. The exact connecting map and signed defect space are

$$C = E_+ E_-^{-1} : \mathbb{C}^{Z_-} \longrightarrow \mathbb{C}^{Z_+}, \quad \Gamma_C = \{(u, Cu) : u \in \mathbb{C}^{Z_-}\}, \quad (\text{PL33})$$

$$\mathfrak{d}((u, Cu), (v, Cv)) = T_{\tilde{g}, Z_+}(Cu, Cv) - T_{g, Z_-}(u, v). \quad (\text{PL34})$$

This is the restriction of the two actual Jacobian pairings with their indicated signs; it is not declared to be an isometry.

Proof. At every actual zero outside Z_- , the transform G_P vanishes, so the actual sum is exactly over Z_- . Those four zeros are critical and simple, giving the first matrix. For the control, the only remaining nonzero values occur at Z_+ : the retained zeros of U still annihilate G_P . The four inserted zeros are simple and form two nonfixed reflected pairs, giving the second matrix. A Vandermonde matrix is invertible because a cubic polynomial vanishing at four distinct nodes is zero. Congruence by each $E_\pm$ therefore preserves the inertia. The two reflection blocks have eigenvectors $(1, 1), (1, -1)$ with eigenvalues $1, -1$, proving the stated counts. Eliminating the common polynomial coordinates gives exactly C ; subtraction of the two forms gives PL34. $\square$

More explicitly, order one reflected pair as $\rho_1, \rho_2 = \iota\rho_1$ and put $\omega_j = y(\rho_j)$ for all four inserted roots. For the full Lagrange polynomials $L_j(y) = \prod_{k \neq j} (y - \omega_k)/(\omega_j - \omega_k)$, take

$$P(y) = \frac{L_1(y)}{a(\rho_1)g(\rho_1)/Q_1(\rho_1)} - \frac{L_2(y)}{a(\rho_2)g(\rho_2)/Q_1(\rho_2)}. \quad (\text{PL35})$$

Its inserted values are exactly $(1, -1, 0, 0)$, so $\mathcal{W}_{\tilde{g}}(F_P, F_P) = -2$. It is a nonzero polynomial; the first form in PL32 is positive definite, so $\mathcal{W}_g(F_P, F_P) > 0$. Every actual unit value and phase remains in PL35.

The graph also has an exact prime-action defect. Retain the actual diagonal actions $(\Pi_{p,-}u)_\tau = p^\tau u_\tau$ and $(\Pi_{p,+}v)_z = p^z v_z$, and define

$$\Delta_{p,n} = C\Pi_{p,-}^n - \Pi_{p,+}^n C, \quad \mathfrak{D}_p(u) = (\Delta_{p,1}u, \dots, \Delta_{p,4}u) \in (\mathbb{C}^{Z_+})^4. \quad (\text{PL51})$$

For every $p > 1$, this stacked defect map is injective. To prove it, put $q_p(X) = \prod_{\tau \in Z_-} (X - p^\tau)$. Then $q_p(\Pi_{p,-}) = 0$. If $\mathfrak{D}_p(u) = 0$, the four power identities together with the automatic zeroth identity give $q_p(\Pi_{p,+})Cu = Cq_p(\Pi_{p,-})u = 0$. Every p^τ has modulus $p^{1/2}$, whereas the inserted eigenvalues have moduli $p^{1/2+\delta_0}$ or $p^{1/2-\delta_0}$. These are different, so $q_p(\Pi_{p,+})$ is invertible. Since C is invertible, $u = 0$, proving injectivity. The image of $\mathfrak{D}_p$, with its displayed inclusion in $(\mathbb{C}^{Z_+})^4$, is the exact four-step defect space. In particular Γ_C contains no nonzero subspace invariant under the ambient action $(\Pi_{p,-}, \Pi_{p,+})$: iteration on such a subspace would force all four defects to vanish. This does not erase the connecting map C or its paired defect.

There is a corresponding exact formula for that pairing. Set $\Delta_p = \Delta_{p,1}$, $T_- = T_{g,Z_-}$, and $T_+ = T_{\tilde{g},Z_+}$. The individual prime similitudes on T_- and T_+ give

$$\begin{aligned} & \mathfrak{d}((\Pi_{p,-}u, C\Pi_{p,-}u), (\Pi_{p,-}v, C\Pi_{p,-}v)) - p\mathfrak{d}((u, Cu), (v, Cv)) \\ &= T_+(\Pi_{p,+}Cu, \Delta_p v) + T_+(\Delta_p u, \Pi_{p,+}Cv) + T_+(\Delta_p u, \Delta_p v). \end{aligned} \quad (\text{PL52})$$

Indeed substitute $C\Pi_{p,-} = \Pi_{p,+}C + \Delta_p$ into the two slots of PL34 and expand its sesquilinear form. The two terms without a Δ_p scale by p ; their subtraction leaves precisely the three terms displayed. All phases and unit factors in $C = E_+E_-^{-1}$ are retained.

For the full-multiplicity extension, take Q_- to contain the full actual multiplicities of a reflection-stable packet and let Q_+ have a disjoint reflection-stable set of nontrivial-strip roots, none an actual zero. Use equal total degree and the same chosen leading coefficient in Q_+ and Q_- . Reflection then multiplies both polynomials by the same factor, so $(Q_+/Q_-)^\dagger = Q_+/Q_-$ and $\tilde{g} = gQ_+/Q_-$ retains the reflection involution. Replace each evaluation matrix by its full Hermite matrix $P \mapsto j_{Q_\pm}(aUP)$, where $U = g/Q_-$. Multiplication by the two full unit jets and Hermite interpolation are invertible. The maps in PL33–34 then use the raw residue forms with denominators $g = UQ_-$ and $\tilde{g} = UQ_+$, and their respective Jacobians. The local antidiagonal and contraction calculation in PL24–25 proves the statement for every jet order; in particular no higher-jet coefficient is replaced by its value. The definition of $\Delta_{p,n}$ and identity PL52 apply unchanged with these full jet actions. For full packets whose source roots are critical and target roots are off-critical, the same proof of injectivity uses $q_p(X) = \prod_{\tau} (X - p^\tau)^{m_\tau}$ and the first $\deg Q_-$ defects. This statement defines and computes the defect on precisely those full packets; it makes no claim that all actual roots are critical.

26.7 Compact primitive representatives and the prime discrepancy

Define the exact logarithmic change of variables

$$(\mathcal{L}F)(u) = e^{-u/2}F(e^{-u}), \quad \widehat{f}(z) = \int_{\mathbb{R}} f(u)e^{-izu} du. \quad (\text{PL36})$$

Substitution gives $\widehat{\mathcal{L}F}(z) = \mathcal{M}F(\frac{1}{2} + iz)$ and $\mathcal{L}a(D) = (\partial_u^2 - \frac{1}{4})\mathcal{L}$. In particular the two Fourier conditions $\widehat{f}(\pm i/2) = 0$ are precisely the two Mellin endpoint conditions, with their signs kept.

Write $F_P = a(D)H_P$, where $H_P = S_{Q_1}\Theta(P(y(D))\phi_*) \in \mathcal{B}$, and set $h_P = \mathcal{L}H_P$. Choose a fixed smooth compactly supported χ equal to one on $[-1, 1]$, and put $\chi_R(u) = \chi(u/R)$. Then

$$f_{P,R} = (\partial_u^2 - \frac{1}{4})(\chi_R h_P) \in C_c^\infty(\mathbb{R}), \quad \widehat{f}_{P,R}(\pm i/2) = 0. \quad (\text{PL37})$$

The last equalities follow by integration by parts against $e^{\pm u/2}$, with no boundary term. Since $H_P \in \mathcal{B}$, all derivatives of h_P decay faster than every exponential at both ends. Thus $f_{P,R} \rightarrow \mathcal{L}F_P$ in every exponentially weighted smooth seminorm. In particular, for any integer N , integration by parts gives a sequence $\epsilon_R \rightarrow 0$ such that, uniformly for $0 \leq \sigma \leq 1$,

$$|\widehat{f}_{P,R}((\sigma - \frac{1}{2})/i + t) - G_P(\sigma + it)| \leq \epsilon_R(1 + |t|)^{-N}. \quad (\text{PL38})$$

Here the Fourier argument is $t + i(1/2 - \sigma)$; hence $1/2 + i[t + i(1/2 - \sigma)] = \sigma + it$, fixing its sign.

For either zero set, let $C_N^2 = \sum_{\rho} m_{\rho}(1 + |\Im \rho|)^{-2N}$, finite for $N \geq 2$, and use its weighted value norm $\|v\|^2 = \sum m_{\rho}|v_{\rho}|^2$. PL38 bounds the norm difference by $C_N \epsilon_R$. The reflection permutation has norm one, so

$$|\mathcal{W}(F_{P,R}, F_{P,R}) - \mathcal{W}(F_P, F_P)| \leq C_N \epsilon_R(2\|v(F_P)\| + C_N \epsilon_R). \quad (\text{PL39})$$

This is the entire zero sum, including every zero outside the selected packets. Apply it to both the actual and control zero sets. For the nonzero polynomial PL35, the actual value is positive and the control value is -2 . Taking the right side of PL39 less than half the positive actual value and less than one, respectively, proves that both opposite signs persist for all sufficiently large R within the compact primitive class. These approximants need not retain the original theta class; PL15 gives the exact same-class lift, while PL37 gives the separate compact approximation with its proved topology and error.

For completeness, the rational-prime arithmetic distinguishes the two forms exactly. Set

$$H(s) = \overline{G_P(1 - \bar{s})}G_Q(s), \quad k(x) = \frac{1}{2\pi i} \int_{\Re s=c} H(s)x^{-s} ds \quad (c > 1). \quad (\text{PL40})$$

Products of Mellin transforms from $\mathcal{B}$ are transforms of their multiplicative convolutions, so $k \in \mathcal{B}$ and $\mathcal{M}k = H$. The classical explicit formula of Guinand and Weil, in the convention written in the original author source [21], equations `bombieriexplicit-bombieriexplicit2`, is

$$\mathcal{W}_g(F_P, F_Q) = H(0) + H(1) - \sum_p (\log p) \sum_{r \geq 1} (k(p^r) + p^{-r}k(p^{-r})) - \mathcal{W}_{\infty}(k), \quad (\text{PL41})$$

$$\mathcal{W}_{\infty}(k) = (\log(4\pi) + \gamma_E)k(1) + \int_1^{\infty} \left(k(x) + x^{-1}k(x^{-1}) - \frac{2}{x}k(1) \right) \frac{dx}{x - x^{-1}}. \quad (\text{PL42})$$

The numerator vanishes at $x = 1$; all infinite-endpoint terms converge by $k \in \mathcal{B}$. Both pole terms in PL41 vanish for these primitive tests. This invocation is a cited human explicit-formula theorem, not a newly assumed positivity assertion.

The control has the additional exact rational term

$$d(s) = Q'_0(s)/Q_0(s) - Q'_1(s)/Q_1(s), \quad \mathcal{W}_{\tilde{g}}(F_P, F_Q) - \mathcal{W}_g(F_P, F_Q) = \sum_{z \in Z_+} H(z) - \sum_{\tau \in Z_-} H(\tau). \quad (\text{PL43})$$

The second equality follows directly by subtracting the two full zero multisets; every retained zero and multiplicity cancels. It is also exactly

$$\frac{1}{2\pi i} \int_{\Re s=c} d(s)(H(s) + H(1-s)) ds. \quad (\text{PL44})$$

Indeed $d(1-s) = -d(s)$. Integrate dH around a rectangle with vertical sides c and $1-c$, both outside the finite roots. The horizontal integrals tend to zero because d is rational and H decays rapidly in this fixed strip. Substitution $s \mapsto 1-s$ in the left integral produces PL44. The residues are $+H(z)$ at each inserted root and $-H(\tau)$ at each removed root, proving equality with PL43. Thus using the actual prime expression PL41 for the control without PL43 would change the arithmetic form. The paired defect PL34 records precisely this difference on the common primitive family. No universal positivity or RH conclusion follows.

27 The complete signed support and the absolute twistor charts

27.1 Human source and the exact categories

The source is Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1. The original author file `CC.tex` was read at lines 310–413 and 420–1207, including the entire relevant sections on the signed base, the Lydakis–Day product and proposed coproduct, the two charts, the geometric involution, complex conjugation, and odd Frobenius. Exact source locators are `F2`, `sect2`, `sect2.2`, `conv`, `opens`, `eq:restriction_maps`, `def:alpha_symmetry`, `thm:twistor_space_scheme`, and `prop:twistor_frobenius`.

The calculations below first take place in commutative pointed monoids. Their zero is absorbing and their identity is denoted 1. Put

$$B = \{0, 1, \epsilon\}, \quad \epsilon^2 = 1, \quad \epsilon \neq 1. \quad (\text{AT1})$$

A signed pointed monoid is supplied with a morphism from B . The target may specialize the sign, but all universal calculations retain $\epsilon \neq 1$. The source identifies spherical monoids’ tensor product with the relative smash product. We therefore write $\otimes_B$ for that pointed-monoid pushout: in particular $\epsilon \otimes 1 = 1 \otimes \epsilon$. No additive relation is introduced until an explicitly named ring or semiring receiver.

27.2 The actual full split base and its two spectra

Let L be a finite nontrivial bounded distributive lattice. Retain

$$G_L(\mathbb{Z}) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{a} = (a, 1_L) : a \in \mathbb{Z}\}. \quad (\text{AT2})$$

Its overlap is $e = z_{1_L} = \widehat{0}$, while $\tau = z_{0_L} \neq e$. Addition is coefficient addition and lattice join; multiplication is coefficient multiplication and lattice meet. Thus $\widehat{a}z_\lambda = z_\lambda$ and $z_\lambda z_\mu = z_{\lambda \wedge \mu}$. Let M_L be its underlying multiplicative pointed monoid with zero τ . There is an injective signed-base map

$$B \longrightarrow M_L, \quad 0 \mapsto \tau, \quad 1 \mapsto \widehat{1}, \quad \epsilon \mapsto \widehat{-1}. \quad (\text{AT3})$$

In particular $\epsilon z_\lambda = z_\lambda$ for every support label. This relation is retained below, including for e .

For precision a prime of a pointed monoid is a proper ideal whose complement is multiplicatively closed; additive closure is not part of this definition. The entire pointed-monoid spectrum of M_L can be computed:

$$\begin{aligned} P_c &= \{z_\lambda : c \not\leq \lambda\}, & 0_L < c \in L, \\ Q_E &= \{z_\lambda : \lambda \in L\} \cup \{\widehat{a} : a \neq 0, p \mid a \text{ for some } p \in E\}, & E \subset \{\text{rational primes}\}. \end{aligned} \quad (\text{AT4})$$

To prove exhaustion, an ideal containing any supported coefficient contains e by multiplication by e , and hence every support label. Its remaining prime condition is exactly that on the free commutative monoid of positive prime factors, with signs units; unique integer factorization gives Q_E . An ideal omitting e contains no supported coefficient. Its support labels form a downset whose complement is a nonempty meet-closed up-set. Because L is finite, that complement is $\uparrow c$ for the meet c of its elements, and $c \neq 0_L$. This gives P_c . Both displayed families directly satisfy the ideal and prime conditions, proving the classification.

The original semiring spectrum embeds topologically in this monoid spectrum by retaining exactly the additively closed primes. In the support family these are the P_c with c join prime, or equivalently join irreducible in a finite distributive lattice. Indeed join closure of its complement's complement is exactly $c \leq \lambda \vee \mu \Rightarrow c \leq \lambda$ or $c \leq \mu$. In the arithmetic family they are $Q_\emptyset$ and $Q_{\{p\}}$: two different primes in E generate 1 additively by their integer Bezout identity, and a singleton gives the usual prime ideal. Its subspace topology is the original one, since the basic opens in both cases are the same tests $a \notin P$. Thus the full semiring spectrum $\text{Spec}_{\text{lat}} L \triangleright \text{Spec } \mathbb{Z}$ has been preserved by an explicit embedding, not identified with the larger monoid spectrum.

27.3 Adjoining the imaginary generator without losing a support label

For any signed pointed monoid M , set

$$A_M = M[J]/(J^2 = \epsilon). \quad (\text{AT5})$$

There is an explicit normal form: its elements are (m, b) with $m \in M$ and $b \in \{0, 1\}$, identifying only $(0, 0)$ and $(0, 1)$. Write $(m, b) = mJ^b$. Multiplication is

$$(m, b)(n, c) = (mne^{bc}, b + c \bmod 2). \quad (\text{AT6})$$

The carry identity modulo two proves associativity, and the formula has $J = (1, 1)$ with $J^2 = \epsilon$. A map out of this monoid is uniquely determined by a signed map out of M and a root of $j^2 = \epsilon$; reduction of every power of j proves the universal property. This proves the normal form, including the injectivity of $M \rightarrow A_M$.

For $M = M_L$ it follows that above every nonbottom support label there are two distinct elements

$$z_\lambda, \quad z_\lambda J \quad (\lambda \neq 0_L), \quad (z_\lambda J)^2 = z_\lambda. \quad (\text{AT7})$$

The equality uses $\epsilon z_\lambda = z_\lambda$; it does not identify these two elements. There is a canonical map to an actual Gaussian coefficient receiver,

$$A_{M_L} \longrightarrow U(G_L(\mathbb{Z}[i])), \quad \widehat{a}J^b \longmapsto \widehat{ai^b}, \quad z_\lambda J^b \longmapsto z_\lambda, \quad (\text{AT8})$$

where U denotes forgetting addition. Its image is exactly the axial submonoid consisting of the support labels and nonzero real or purely imaginary integer coefficients. Its kernel congruence is exactly the identifications $z_\lambda J = z_\lambda$ for $\lambda \neq 0_L$: distinct nonzero axial coefficients remain distinct, and the normal form proves there are no other identifications. The additive closure of this image is the full $G_L(\mathbb{Z}[i])$, since $a + bi$ is a sum of its two displayed axes. This describes both the retained data and the precise additional quotient in passing to Gaussian coefficients.

There is no signed map $A_B \rightarrow M_L$: a root of $j^2 = \epsilon$ would be a unit, but the units of M_L are $\widehat{1}$ and $\widehat{-1}$, whose squares are 1. Thus (AT5) and (AT8) are the required root-adjoining maps, rather than an assumed root in the original integer coefficient monoid.

Adjoining this unit does not change the monoid prime topology. For every prime of A_M , membership of mJ^b is equivalent to membership of m , because J is a unit. Conversely the same rule extends every prime of M uniquely to one of A_M ; the normal form and the fact that ϵ is a unit prove well-definedness. Basic opens satisfy $D(mJ^b) = D(m)$. Hence

$$\text{Spec } A_M \xrightarrow{\sim} \text{Spec } M \quad (\text{AT9})$$

is a homeomorphism. The extra roots and the extra support tags therefore have an exact point-functor meaning even where they do not separate additional monoid primes.

27.4 The diagonal obstruction and the actual torsor coaction

For a free signed spatial monoid $B[N]$, its spatial diagonal $n \mapsto n \otimes n$ together with $\epsilon \mapsto \epsilon$ is defined over the fixed base. The relation $J^2 = \epsilon$ is not a relation internal to an unsigned spatial monoid independent of that base. The proposed diagonal must therefore be checked after imposing this relation.

Theorem 27.1 (Failure of the binary diagonal on the actual relation). *For $A = A_B = B[J]/(J^2 = \epsilon)$, no signed morphism $A \rightarrow A \otimes_B A$ can send J to $J_1 J_2$. Indeed*

$$(J_1 J_2)^2 = \epsilon^2 = 1 \neq \epsilon. \quad (\text{AT10})$$

The same obstruction occurs in A_M whenever the image of ϵ remains distinct from 1 in $A_M \otimes_M A_M$. The raw diagonal is defined after additionally imposing $\epsilon = 1$, but that is a different signed-base receiver.

Proof. A signed morphism must send J^2 to the same base element ϵ . Its proposed value has the square in (AT10). To verify that these elements remain different in the displayed target, map both J_1, J_2 to i in the multiplicative pointed monoid of $\mathbb{C}$, and send ϵ to -1 . This respects the defining tensor relations and distinguishes 1 from ϵ . Over M_L the original coefficient map to $\mathbb{C}$ also distinguishes them, so the same counterexample applies to the full base extension. After the specified sign collapse, the equality required for descent does hold, proving the last assertion. $\square$

The counit obstruction is equally exact: A_B has no B -point because neither 1 nor ϵ has square ϵ . Thus this imaginary-root object cannot have an identity section for a group law over B . This is a defect of applying the free spatial diagonal to this particular signed relation; it does not remove the valid coproduct on the free spatial core $B[T^{\mathbb{Z}}]$.

There is instead the genuine group object

$$H = B[U]/(U^2 = 1), \quad \Delta_H(U) = U_1 U_2, \quad e_H(U) = 1, \quad \iota_H(U) = U^{-1} = U. \quad (\text{AT11})$$

All base elements are fixed. Direct substitution proves the group identities. The actual imaginary-root coaction is

$$\rho : A_B \longrightarrow A_B \otimes_B H, \quad \rho(J) = J \otimes U. \quad (\text{AT12})$$

Its square is $\epsilon \otimes 1$, and applying either composite coaction gives $J \otimes U \otimes U$, proving coassociativity; the counit gives J back. Base change gives the same formulas over every original M_L .

Theorem 27.2 (The precise torsor identity). *The morphism of root functors $(j, u) \mapsto (j, ju)$ is represented by an isomorphism*

$$A_B \otimes_B A_B \xrightarrow{\sim} A_B \otimes_B H, \quad J_1 \mapsto J, \quad J_2 \mapsto JU, \quad (\text{AT13})$$

whose inverse sends $J \mapsto J_1$ and $U \mapsto J_1^{-1} J_2$. For every signed target the roots of $j^2 = \epsilon$, when nonempty, form a free transitive set under the units u with $u^2 = 1$.

Proof. All imaginary roots are units because $j^4 = 1$ and $j^{-1} = \epsilon j$. Therefore the inverse formula is defined, and its proposed U has square $\epsilon^{-1} \epsilon = 1$. Composing the two displayed formulas gives the identity on each generator, proving the isomorphism. For two roots j_1, j_2 the unique translating unit is $j_1^{-1} j_2$, with square one. This proves the assertion on every target and its base changes. $\square$

Here “torsor” means this explicit action and isomorphism; no unspecified topology on pointed monoids is needed. After passing to any ring with $\epsilon = -1$, the object is $\text{Spec } R[J]/(J^2 + 1)$, finite free of rank two and faithfully flat over R , with precisely the same μ_2 torsor identity. Faithful flatness follows because its module is R^2 . Thus it is also a torsor in the faithfully flat ring-scheme sense.

The canonical ternary operation is the torsor heap

$$h(j_1, j_2, j_3) = j_1 j_2^{-1} j_3, \quad h^*(J) = J_1 J_2^{-1} J_3. \quad (\text{AT14})$$

Its square is $\epsilon \epsilon^{-1} \epsilon = \epsilon$. It satisfies $h(x, y, y) = x = h(y, y, x)$ and $h(h(x, y, z), u, v) = h(x, y, h(z, u, v))$ by multiplication of units. Every odd raw product of roots also has square ϵ ; every even raw product has square one. These operations are different: $j_1 j_2 j_3 = \epsilon h(j_1, j_2, j_3)$. Its odd diagonal is the actual power $j^{2k+1} = \epsilon^k j$. Choosing a root j_0 after a specified base extension gives the binary law $j_1 j_0^{-1} j_2$ with identity j_0 , thereby identifying the torsor with μ_2 after that choice. No such choice was present over B .

27.5 The actual charts, overlap, and geometric involution

Use exactly the source charts, now in pointed-monoid notation:

$$A_+ = A_B[T], \quad A_- = A_B[S], \quad A_\circ = A_B[T, T^{-1}], \quad S = T^{-1}, \quad J_+ = J, \quad J_- = \epsilon J. \quad (\text{AT15})$$

The coaction (AT12) extends by fixing the spatial coordinate. It respects the overlap because $\rho(J_-) = \epsilon J U = \rho(\epsilon J)$. The source’s geometric involution is

$$\begin{aligned} \alpha^*(T) &= \epsilon T^{-1}, & \alpha^*(J) &= \epsilon J = J^{-1}; \\ \alpha_+^*(S) &= \epsilon T, & \alpha_+^*(J_-) &= J_+, \\ \alpha_-^*(T) &= \epsilon S, & \alpha_-^*(J_+) &= J_-. \end{aligned} \quad (\text{AT16})$$

It squares to the identity on every generator. For instance the overlap of $\alpha_+^*(J_-)$ is J , while $\alpha^*(\epsilon J) = \epsilon^2 J = J$. This verifies the nontrivial sign in the chart compatibility. The torsor coaction commutes with α : on J both give $\epsilon J U$, and on T both give ϵT^{-1} . The same formulas hold after the full base change from B to M_L , with α fixing every element of M_L .

For the pure spherical generic chart, its monomials have the form $m J^b T^r$, $m \in \{1, \epsilon\}$, $b = 0, 1$, $r \in \mathbb{Z}$. Invariance under (AT16) forces $r = 0$ and then $b = 0$. Thus its invariant pointed monoid is exactly B , as used in the source’s gluing argument. After the actual full base change its invariant pointed monoid is instead

$$(A_{M_L}[T, T^{-1}])^\alpha = M_L \cup \{z_\lambda J : 0_L < \lambda \in L\}. \quad (\text{AT17})$$

Indeed the exponent comparison still forces $r = 0$, and an odd J exponent is invariant precisely when $\epsilon m = m$. By (AT2), these nonzero m are exactly the support labels. This proves the exact changed invariant receiver while retaining the source’s original pure-base result.

For completeness the chart spectra also keep the entire original base. Every prime of $M[T]$ is determined by its contraction P and whether it contains T . If it omits T , then $m T^r \in P'$ precisely when $m \in P$; if it contains T , all positive exponents belong and the exponent-zero part is P . These rules give primes and prove exhaustion. Basic opens show the resulting spectrum is the product of $\text{Spec } M$ with the two-point affine monoid line. Gluing the two localizations in (AT15), and using (AT9), gives the product with the three-point projective monoid line. The injection of the full semiring spectrum from (AT4) is retained in this construction. This is a base change of the actual charts, not an identification with the source’s separate global amalgam.

27.6 Odd powers, fixed points, and the exact Frobenius bridge

For $n \geq 1$, power $x \mapsto x^n$ is a pointed-monoid endomorphism of any commutative pointed monoid. It fixes the base (AT1) precisely when n is odd. On the source charts the resulting maps are

$$F_n^*(T) = T^n, \quad F_n^*(J) = J^n = \epsilon^{(n-1)/2} J, \quad n \text{ odd.} \quad (\text{AT18})$$

They respect $J^2 = \epsilon$, every overlap, and α : for example $\alpha^*(T^n) = \epsilon^n T^{-n} = \epsilon T^{-n}$, and $F_n^*(\epsilon J) = \epsilon J^n$. They compose as $F_m F_n = F_{mn}$. On the torsor the equivariance identity is $\rho F_n = (F_n \otimes [n])\rho$, with $[n](U) = U^n = U$ for odd n . Thus these powers are defined without the invalid binary imaginary diagonal.

After adjoining the full original integer coefficient monoid there are two different exact maps. The relative map fixes M_L and uses (AT18) on T, J . The absolute pointed-monoid power also sends each $m \in M_L$ to m^n :

$$F_{n,\text{rel}}(mJ^b T^r) = mJ^{bn} T^{rn}, \quad F_{n,\text{abs}}(mJ^b T^r) = m^n J^{bn} T^{rn}. \quad (\text{AT19})$$

Both are signed for odd n . In the Gaussian polynomial receiver of (AT8), the relative map extends to the actual ring homomorphism fixing $\mathbb{Z}$ and sending $i \mapsto i^n$, $T \mapsto T^n$, and hence to the full split semiring while fixing every support label. The absolute map is the multiplicative power on that receiver. In characteristic zero it is not additive for $n > 1$, since $(1+1)^n = 2^n \neq 2 = 1^n + 1^n$. This is the exact comparison between the two coefficient operations.

Power induces the identity on every monoid prime spectrum: $x^n \in P$ if and only if $x \in P$, by primality and induction. For the relative chart map, the preceding chart-prime classification gives the same identity: it fixes the base prime and the condition whether T belongs. This does not imply triviality on field-valued points; those are calculated below. In particular the original semiring prime subspace of $G_L(\mathbb{Z})$ is fixed by absolute powers as a topological space, even though those maps are not semiring endomorphisms in characteristic zero.

The distinction is visible in the actual Gaussian ring spectrum. For the relative $n = 3$ substitution on $\mathbb{Z}[i][T]$, the inverse image of the prime $(T - 2)$ is $(T - 8)$: its quotient test is $f(i, T) \mapsto f(-i, 8)$, and Gaussian conjugation is an automorphism. For absolute cubing, inverse image of the same prime as a multiplicative ideal is $(T - 2)$ itself. Thus the identity statement about pointed-monoid powers has not been extended incorrectly to the additive substitution receiver.

The joint fixed set of all odd powers on the pure source chart is B : already the third power forces the spatial exponent to be zero and removes the odd imaginary exponent. On the full absolute base change, the exact fixed set is

$$\{\tau, \widehat{1}, \widehat{-1}\} \cup \{z_\lambda J^b : 0_L < \lambda \in L, b = 0, 1\}. \quad (\text{AT20})$$

The normal form proves this directly. A nonzero integer coefficient fixed by the third power is 1 or -1 . An odd imaginary exponent on such a coefficient would require $-a^3 = a$, impossible for a nonzero integer. By contrast every displayed support element is fixed by all odd powers because $z_\lambda^n = z_\lambda$ and $\epsilon z_\lambda = z_\lambda$. This includes both tags over the original supported zero e . For relative odd powers the integer coefficients are fixed as well, and the joint fixed set is exactly the right-hand side of (AT17). This gives both versions of the fixed-point comparison.

27.7 Complex points, the twistor quotient, and the action that descends

In a field-valued point the sign is evaluated as -1 . If the field has no root of -1 , the imaginary charts have no points. If it has characteristic different from two and roots $\pm i$, the chart overlap is exactly

$$(t, j)_+ \sim (t^{-1}, -j)_-. \quad (\text{AT21})$$

Thus there are two projective lines, with the first joining the $(+, i)$ and $(-, -i)$ charts. In characteristic two the unique field-valued root is 1 and the point sets form a single projective line. This is a statement about field-valued points, not a claim that the characteristic-two root scheme is reduced.

On the complex generic open the original involution is

$$\alpha(z, j) = (-z^{-1}, -j), \quad c(z, j) = (\bar{z}, -j). \quad (\text{AT22})$$

It interchanges the two spheres, and each orbit has a unique representative on the first sphere. Taking that representative after conjugation gives the exact real structure $\sigma(z) = -1/\bar{z}$, including $0 \leftrightarrow \infty$. It has no fixed point, since a nonzero finite fixed point would satisfy $|z|^2 = -1$ and the endpoints are exchanged.

There is a necessary descent check for the spatial action. Scaling both root sheets by $a_\lambda(z, j) = (\lambda z, j)$ satisfies $\alpha a_\lambda = a_{\lambda^{-1}} \alpha$; it does not preserve the α -orbits in general. Indeed the representatives (z, i) and $(-z^{-1}, -i)$ scale to $(\lambda z, i)$ and $(-\lambda z^{-1}, -i)$, which lie in one α -orbit exactly when $\lambda = \lambda^{-1}$. The exact action which does descend is

$$\tilde{a}_\lambda(z, i) = (\lambda z, i), \quad \tilde{a}_\lambda(z, -i) = (\lambda^{-1} z, -i). \quad (\text{AT23})$$

It is an action, extends across both poles, and commutes with α by direct substitution in (AT22). On the first representative sphere it gives the usual action $z \mapsto \lambda z$. Complex conjugation transforms its parameter by $\lambda \mapsto \bar{\lambda}^{-1}$, so

$$\sigma(\lambda z) = \bar{\lambda}^{-1} \sigma(z). \quad (\text{AT24})$$

This is the actual twistor equivariance, now with its descent map.

There is a real algebraic group recording exactly that parameter descent: the norm-one torus

$$\mathcal{T} = \text{Spec } \mathbb{R}[a, b]/(a^2 + b^2 - 1), \quad (a, b)(c, d) = (ac - bd, ad + bc). \quad (\text{AT25})$$

Over $\mathbb{C}$, the coordinate $\lambda = a + ib$ identifies it with $\mathbf{G}_m$, with inverse $a = (\lambda + \lambda^{-1})/2$, $b = (\lambda - \lambda^{-1})/(2i)$. The real structure on this complex parameter is $\lambda \mapsto \bar{\lambda}^{-1}$, as those formulas show. This realizes (AT23) and (AT24) after the specified real additive extension. Division by two is stated explicitly; no identification of μ_2 with a constant two-point group in characteristic two has been used.

For an odd n , let $\chi_4(n) = (-1)^{(n-1)/2}$. On the two sheets the actual map is $(z, j) \mapsto (z^n, j^n)$. Taking the first-sheet representative gives precisely

$$f_n(z) = \begin{cases} z^n, & \chi_4(n) = 1, \\ -z^{-n}, & \chi_4(n) = -1. \end{cases} \quad (\text{AT26})$$

Both formulas extend over zero and infinity. Put $a(z) = -1/z$. For odd n it commutes with $z \mapsto z^n$, and $a^2 = 1$. Since $\chi_4(mn) = \chi_4(m)\chi_4(n)$, these maps satisfy $f_m f_n = f_{mn}$. They commute with σ by conjugating their real coefficients and using the same odd sign. Their precise equivariance with the descended torus action is

$$f_n(\lambda z) = \lambda^{n\chi_4(n)} f_n(z). \quad (\text{AT27})$$

Thus the second case reverses the torus character and exchanges the poles. Its map of projective curves still has degree n : the pullback of the zero divisor is $n[0]$ in the first case and $n[\infty]$ in the second, with the opposite formula for the infinity divisor. These explicit divisor and character maps are the proved comparisons; no assertion about an unspecified Hodge bundle is needed.

27.8 The additive invariant conic and the prime two

There is also an exact additive receiver for the source's generic involution. In the ring $R_o = \mathbb{Z}[i][T, T^{-1}]$ put $\alpha(i) = -i$ and $\alpha(T) = -T^{-1}$. Define

$$u = T - T^{-1}, \quad v = i(T + T^{-1}). \quad (\text{AT28})$$

Both are invariant and $u^2 + v^2 = -4$.

Theorem 27.3 (The entire additive invariant receiver). *There are exact identifications*

$$R_o^\alpha = \mathbb{Z}[u, v]/(u^2 + v^2 + 4), \quad G_L(R_o)^\alpha = G_L(\mathbb{Z}[u, v]/(u^2 + v^2 + 4)). \quad (\text{AT29})$$

They retain every lattice label. The monoid invariant map from (AT17) lands in the original integer coefficient and support part; the new u, v are actual sums in this additive receiver.

Proof. An invariant Laurent polynomial has integer constant coefficient and, for every $r > 0$, a pair $a_r T^r + (-1)^r \bar{a}_r T^{-r}$, with $a_r \in \mathbb{Z}[i]$. Its real and imaginary integer parts are respectively multiples of

$$F_r = T^r + (-1)^r T^{-r}, \quad G_r = i(T^r - (-1)^r T^{-r}).$$

Both satisfy $X_{r+1} = uX_r + X_{r-1}$, with $F_0 = 2, F_1 = u, G_0 = 0, G_1 = v$. Thus all invariants are generated by u, v and integer constants. The indicated relation holds. It is the entire kernel: because it is monic in v^2 , any polynomial reduces to $A(u) + vB(u)$. In the Laurent receiver the first summand has integer coefficients and the second purely imaginary integer coefficients, so their sum vanishes only if both vanish. The element $u = T - T^{-1}$ is transcendental over $\mathbb{Z}$ (its powers have distinct highest T exponents), and $T + T^{-1}$ is nonzero in this domain. Hence $A = B = 0$. This proves the first identification. The involution fixes every z_λ and acts on a supported coefficient exactly by its ring involution, proving the second. Finally the normal-form fixed elements in (AT17) map either to their integer coefficient or to their original support label under (AT8), proving the stated comparison. $\square$

The conic in (AT29) has no real points: real squares cannot sum to -4 . It is an explicitly computed additive enlargement of the invariant receiver, not a claim that invariants commute with adding ring operations. Its proof also preserves the exact integer constant 4.

Over $\mathbb{C}$ the generic conic is the actual folded generic curve, with inverse coordinate

$$z = \frac{u - iv}{2}, \quad z^{-1} = -\frac{u + iv}{2}. \quad (\text{AT29a})$$

Their product is one by the conic equation. For each choice $j \in \{i, -i\}$ the corresponding unfolded coordinate is $T = (u - jv)/2$; the other choice gives $-T^{-1}$, so these are exactly one α -orbit. This proves the quotient statement directly on all generic complex points. Conjugating the real coordinates u, v induces $z \mapsto -1/\bar{z}$, as substitution in (AT29a) verifies. The torus (AT25) acts on this conic by

$$(u, v) \mapsto (au + bv, av - bu), \quad a^2 + b^2 = 1. \quad (\text{AT29b})$$

It preserves $u^2 + v^2 + 4$ and, in (AT29a), becomes $z \mapsto (a + ib)z$. Thus the repaired scalar descent has a concrete additive-coordinate action with the same signs. For odd n , the relative Frobenius acts on these invariant coordinates by $u \mapsto F_n$ and $v \mapsto \chi_4(n)G_n$, where the polynomials F_n, G_n were constructed by the exact recurrence in the proof above. This follows by substituting $T \mapsto T^n$ and $i \mapsto i^n$, and is the conic expression of (AT26).

Finally the root torsor's reduction at two is an actual scheme calculation. In $\mathbb{Z}[i]$, $(1+i)^2 = 2i$, so $(2) = (1+i)^2$ as ideals, and

$$\mathbb{Z}[i]/2\mathbb{Z}[i] \simeq \mathbb{F}_2[J]/((J+1)^2). \quad (\text{AT30})$$

The ring has a nonzero nilpotent $J + 1$ although it has only one field-valued root. At an odd prime the polynomial $J^2 + 1$ has no common root with its derivative $2J$, so its fibre is reduced and the degree-two algebra is etale there. The trace matrix in the basis $1, i$ is $\text{diag}(2, -2)$, with determinant -4 , in agreement with these exact fibres. This supplies the arithmetical relation to the prime two. The restriction of pointed-monoid powers to odd integers was separately proved by fixing ϵ ; neither calculation is substituted for the other.

$$\begin{array}{ccccc} M_L & \hookrightarrow & A_{M_L} & \xrightarrow{\rho} & A_{M_L} \otimes_B H \\ \downarrow & & \downarrow & & \\ U(G_L(\mathbb{Z}[i])) & \xlongequal{\quad} & U(G_L(\mathbb{Z}[i])) & & \end{array}$$

Figure 19: The actual signed bridge retains the full integer split base on the left. The upper inclusion adjoins $J^2 = \epsilon$. The middle downward map is (AT8): it identifies only the two imaginary tags above each nonbottom support label within its axial image. Its right-hand coaction is the valid torsor map (AT12), not the binary diagonal ruled out in Theorem 27.1. The full Gaussian receiver is obtained by taking the stated additive closure.

The free spatial coproduct, imaginary torsor, ternary operation, odd powers, full-support fixed points, Gaussian coefficient maps, and twistor quotient have all been related by their explicit morphisms. The conclusions use the author's signed chart relations and field hypotheses exactly. No Riemann hypothesis, smoothness of a global arithmetic curve, or spectral purity conclusion is asserted from these constructions.

28 The two original prime actions as a finite subscheme of the norm-one torus

The norm-one torus in (AT25) has an exact receiver in the original arithmetic zero classes. The construction below uses the actual primes $2, 3$, the original coordinate s , and every multiplicity. It does not replace a prime eigenvalue by its modulus. The torus comes from the explicit descent of the Connes–Consani twistor charts [25]; the full residue and prime actions used here are those proved in the full-jet chapter and in (PL23)–(PL28).

Let Z be a nonempty finite set of actual zeros of $g = 2\xi$, stable under $\iota\rho = 1 - \bar{\rho}$, with each full multiplicity m_ρ . Write

$$A_Z = \prod_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad t_\rho = s - \rho, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad A_{Z, \mathbb{R}} = \{f : f^\dagger = f\}. \quad (\text{TP1})$$

Every $f \in A_Z$ has the unique decomposition $f = (f + f^\dagger)/2 + i(f - f^\dagger)/(2i)$ into two elements of $A_{Z, \mathbb{R}}$. Consequently $A_{Z, \mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \simeq A_Z$, with the given conjugation. This identity keeps the actual reflection; it does not conjugate each off-critical factor separately.

For each original prime p put

$$\lambda_p = p^{-1/2} \Pi_p(1) = (p^{\rho-1/2} \sum_{j=0}^{m_\rho-1} (\log p)^j t_\rho^j / j!)_{\rho \in Z}, \quad a_p = \frac{\lambda_p + \lambda_p^{-1}}{2}, \quad b_p = \frac{\lambda_p - \lambda_p^{-1}}{2i}. \quad (\text{TP2})$$

Here Π_p is multiplication by the full p^s jet. The factor $p^{-1/2}$ is stated explicitly: the original action is recovered exactly as $p^{1/2} \lambda_p$. The exponential and its inverse are finite Taylor expressions in each original nilpotent coordinate. Reflection gives $\lambda_p^\dagger = \lambda_p^{-1}$, whence $a_p^\dagger = a_p$, $b_p^\dagger = b_p$, and $a_p^2 + b_p^2 = 1$. There is therefore a real algebra morphism

$$\Psi : \mathcal{B} := \mathbb{R}[a_2, b_2, a_3, b_3] / (a_2^2 + b_2^2 - 1, a_3^2 + b_3^2 - 1) \longrightarrow A_{Z, \mathbb{R}} \quad (\text{TP3})$$

with exactly the images in TP2. Its source is the coordinate algebra of the square of the real norm-one torus, not a freely assigned positivity condition.

Theorem 28.1 (The two primes recover every original jet). *The morphism Ψ is surjective. Thus the entire real packet is the finite closed subscheme of $\mathcal{T}^2$ defined by $I_Z = \ker \Psi$. Its complexification has coordinate maps $X_2 \mapsto \lambda_2$, $X_3 \mapsto \lambda_3$ from $\mathbb{C}[X_2^{\pm 1}, X_3^{\pm 1}]$. No positive-order jet is discarded.*

Proof. Two distinct roots ρ, σ cannot have equal values for both λ_2 and λ_3 . Otherwise $(\rho - \sigma) \log 2 = 2\pi in$ and $(\rho - \sigma) \log 3 = 2\pi im$ for integers m, n . A nonzero difference would make m, n nonzero and give $m \log 2 = n \log 3$, contradicting unique prime factorization after exponentiating (negative exponents can be cleared). Thus the two values separate the roots.

Choose a complex number c outside the following finite set. For every pair of distinct roots exclude a value solving $\lambda_2(\rho) + c\lambda_3(\rho) = \lambda_2(\sigma) + c\lambda_3(\sigma)$, if there is one. For every root exclude the unique value solving $(\log 2)\lambda_2(\rho) + c(\log 3)\lambda_3(\rho) = 0$. Then $h = \lambda_2 + c\lambda_3$ has distinct constant values h_ρ and nonzero first derivative at every root. In the local factor, $h - h_\rho = d_\rho t_\rho + O(t_\rho^2)$ with $d_\rho \neq 0$. Successively matching the coefficients of $t_\rho, t_\rho^2, \dots, t_\rho^{m_\rho-1}$ expresses t_ρ as a polynomial in $h - h_\rho$: at stage j the coefficient to be divided by is $d_\rho^j \neq 0$. This proves that the local map from $\mathbb{C}[X]$ has kernel $(X - h_\rho)^{m_\rho}$ and is onto.

For distinct roots these ideals are coprime. Bezout's identity for their powers gives polynomials equal to one modulo one power and zero modulo all others. Combining these with the local polynomial expressions proves that $\mathbb{C}[h] = A_Z$. In particular the image of $\mathbb{C}[X_2^{\pm 1}, X_3^{\pm 1}]$ is all of A_Z . Complexifying TP3 is that map, because $X_p = a_p + ib_p$, $X_p^{-1} = a_p - ib_p$. For a real fixed element choose a complex polynomial preimage and average it with its real conjugate. The average lies in $\mathcal{B}$ and still maps to that element. This proves surjectivity over $\mathbb{R}$. The auxiliary c proves generation; it changes neither original prime nor either of their actions. $\square$

There is also an explicit, exact description of the ideal, useful for retaining the full multiplicities:

$$I_Z \otimes_{\mathbb{R}} \mathbb{C} = \left\{ P : \frac{d^j}{ds^j} P(2^{s-1/2}, 3^{s-1/2}) \Big|_{s=\rho} = 0 \quad (\rho \in Z, 0 \leq j < m_\rho) \right\}. \quad (\text{TP4})$$

These are exactly the coefficients of the original Taylor map, so the formula follows by its kernel definition, without a radical replacement. For any other rational prime p , TP2 belongs to this same quotient. The constructive Bezout and local inversion proof above expresses it as a polynomial in the two original prime coordinates. Therefore Π_p on this packet is recovered by that polynomial followed by multiplication by $p^{1/2}$.

Proposition 28.2 (The real closed points and their full thickening). *As a real algebra, the packet has the exact factorization*

$$A_{Z, \mathbb{R}} \simeq \prod_{\rho=\iota\rho} \mathbb{R}[y_\rho]/(y_\rho^{m_\rho}) \times \prod_{\{\rho, \iota\rho\}, \rho \neq \iota\rho} (\mathbb{C}[t_\rho]/(t_\rho^{m_\rho}))_{\mathbb{R}}, \quad y_\rho = t_\rho/i. \quad (\text{TP5})$$

The real closed points of $\text{Spec } A_{Z, \mathbb{R}}$ are exactly the critical-line roots. Each off-critical reflection pair is a closed point with residue field $\mathbb{C}$ and the displayed thickening. At a real point $|\lambda_p| = 1$ for every prime. At a root with $\Re \rho \neq 1/2$ the modulus is exactly $p^{\Re \rho - 1/2}$.

Proof. At a fixed root $t_\rho^\dagger = -t_\rho$, so y_ρ is fixed. A polynomial in y_ρ is fixed exactly when all its coefficients are real. At a two-element orbit, one arbitrary complex truncated polynomial determines its partner uniquely by conjugating coefficients and replacing t by $-t$. Projection onto the chosen factor is a real algebra isomorphism. These computations prove TP5. The maximal ideal in each truncated factor is its positive-order ideal, giving the claimed residue fields. There is no unital real algebra map $\mathbb{C} \rightarrow \mathbb{R}$, since the square of the image of i would equal -1 .

Thus only the first factors give real-valued points. Finally the modulus follows directly from the original exponential in TP2. The real structure on the complex torus is $\lambda \mapsto \bar{\lambda}^{-1}$; its nonfixed points occur in precisely the pairs in TP5. $\square$

This calculation specifies the purity condition geometrically: it is the assertion that the relevant reduced arithmetic packet has only real closed points in this real torus. Merely supplying its real structure does not establish that assertion; TP5 retains and describes every nonreal point rather than deleting it. Its nilpotent part also remains explicit. With the original residue pairing and Jacobian, the exact pullback is

$$R_Z(f, J_g h) = \sum_{\rho \in Z} m_\rho \overline{f(\iota \rho)} h(\rho), \quad \ker J_g = \prod_{\rho} (t_\rho), \quad (\text{TP6})$$

as proved in PL24–27. Thus a fixed value contributes a positive one-dimensional block, while each two-element orbit contributes $\begin{pmatrix} 0 & m_\rho \\ m_\rho & 0 \end{pmatrix}$, of signs $+$, $-$. Every positive-order jet stays in the stated trace radical and in the perfect raw residue space. Surjectivity in TP3 means these are the signs and jets seen by the two actual prime-coordinate functions, not an unrelated choice of tests.

28.1 Its exact map over the full split-support spectrum

Let L be the original finite nontrivial distributive support lattice. Applying its coefficient operations to TP3 gives the surjective semiring morphism

$$G_L(\mathcal{B}) \longrightarrow G_L(A_{Z, \mathbb{R}}), \quad (b, 1_L) \longmapsto (\Psi b, 1_L), \quad z_\lambda \longmapsto z_\lambda. \quad (\text{TP7})$$

Every killed arithmetic coefficient goes to the supported zero e , and every original support label is fixed. On prime spectra the induced map is the topological embedding

$$\text{Spec}_{\text{lat}} L \triangleright \text{Spec } A_{Z, \mathbb{R}} \hookrightarrow \text{Spec}_{\text{lat}} L \triangleright \text{Spec } \mathcal{B}, \quad (\text{TP8})$$

equal to the identity on support primes and to the contraction of the closed packet $V(I_Z)$ on the arithmetic part. Indeed a support prime omits e , so a coefficient mapping to e cannot enter it. An arithmetic prime contains every support label, and its supported coefficients contract by Ψ . These observations give exactly TP8 using the prime classification of the foundational split-support paper [3]. Surjectivity of TP7 shows that the basic open sets in its domain spectrum are the restrictions of basic opens from the target, which proves the embedding assertion.

The topology of this image differs from that of the ring packet in a precise way: its image contains the whole support stratum, and its closure is the whole target spectrum. Each support point specializes to every arithmetic point in the ordinal-sum topology. Since L is finite and nontrivial, it has a support prime: choose a minimal nonzero join-irreducible by finite descent, and use its prime filter. Thus every arithmetic point lies in that closure, and every support point is already in the image. The image is proper. For example, the real torus has infinitely many pairs of real points, while the finite-dimensional algebra $A_{Z, \mathbb{R}}$ has only finitely many prime ideals by TP5. Evaluation at those torus points gives infinitely many distinct maximal ideals of $\mathcal{B}$, so they cannot all belong to $V(I_Z)$. Consequently TP8 is a proper dense embedding, with its finite arithmetic part closed inside $V(e)$. No ring closed-immersion statement has been transferred to the entire split spectrum.

29 All mixed finite amplitude covariances with the original two global conditions

The supplied continuation *Full-support reconstruction: the missing cohomology, the actual prime degree, and complete mixed-sign calculations* calculates a two-factor example over $\mathbb{Z}/3$ and $\mathbb{Z}/5$.

$$\begin{array}{ccc}
\mathrm{Spec}_{\mathrm{lat}} L \triangleright \mathrm{Spec} A_{Z, \mathbb{R}} & \xleftarrow{\mathrm{TP8}} & \mathrm{Spec}_{\mathrm{lat}} L \triangleright \mathrm{Spec} \mathcal{B} \\
\uparrow & & \uparrow \\
\mathrm{Spec} A_{Z, \mathbb{R}} & \xleftarrow{V(I_Z)} & \mathcal{T}^2
\end{array}$$

Figure 20: The original two primes determine the full finite jet packet by TP2–5. The bottom map is a closed immersion of real schemes. The top map retains every support prime and is a proper dense topological embedding. Both vertical maps identify the arithmetic parts inside $V(e)$. The change is proved directly from the foundational ordinal-sum spectrum; the diagram does not assign positivity to its nonreal closed points.

We retain its actual amplitude maps and counting inner products and calculate the entire spectrum for every finite number of factors. This is a finite receiver of the split-support maps. The formulas below do not identify it with an adelic completion or replace the original theta space by it.

Let R_i be finite nonzero rings, $N_i = |R_i| \geq 2$, and let L_i be finite nontrivial bounded distributive support lattices, $l_i = |L_i|$, for $1 \leq i \leq r$. All formal basis vectors in $\mathbb{C}[G_{L_i}(R_i)]$ and $H_i = \mathbb{C}[R_i]$ are orthonormal. Write $e_i = \delta_0$ and $c_i = N_i^{-1/2} \sum_{x \in R_i} \delta_x$. The original amplitude map is

$$\pi_i \delta_{(a,1)} = \delta_a, \quad \pi_i \delta_{z_\lambda} = \delta_0, \quad \pi_i \pi_i^* = I + (l_i - 1) e_i e_i^*. \quad (\text{MG1})$$

The overlapping notation $(0, 1) = z_1$ denotes one basis element. There are exactly l_i preimages of the amplitude zero and one preimage of every nonzero amplitude, proving the last identity.

Put $H = \bigotimes_i H_i$, $e = \bigotimes_i e_i$, $c = \bigotimes_i c_i$, $N = \prod_i N_i$, and $V = \{e, c\}^\perp$. Let P be the orthogonal projection onto V , $\pi = \bigotimes_i \pi_i$, and $T = P\pi$ with codomain V . Its exact covariance is

$$C = TT^* = PDP|_V, \quad D = \bigotimes_i (I + (l_i - 1) e_i e_i^*). \quad (\text{MG2})$$

Both global constraints are present once. The operator D is strictly positive; hence C is invertible on V and T is onto. The unique minimum-norm lift of $v \in V$ is $T^* C^{-1} v$, because it has image v and is orthogonal to $\ker T$. Its squared norm is $v^* C^{-1} v$. Thus C^{-1} is exactly the attained quotient metric.

For each proper subset $S \subsetneq \{1, \dots, r\}$ define

$$H_S = \bigotimes_{i \in S} \mathbb{C} e_i \otimes \bigotimes_{i \notin S} e_i^\perp, \quad q_S = \prod_{i \notin S} (N_i - 1), \quad \alpha_S = \prod_{i \in S} l_i. \quad (\text{MG3})$$

The empty product is one. These are all the original mixed blocks after removing the single all-zero vector e : $e^\perp = \bigoplus_{S \subsetneq [r]} H_S$, with $D|_{H_S} = \alpha_S I$ and $\dim H_S = q_S$. Moreover $c_S = \mathrm{proj}_{H_S} c$ has $\|c_S\|^2 = q_S/N$. This follows by using $\langle e_i, c_i \rangle = N_i^{-1/2}$ and multiplying all factor norms. In particular every c_S is nonzero and $\sum_S q_S = N - 1$.

Theorem 29.1 (Complete mixed covariance spectrum and determinant). *The characteristic polynomial on the original $(N - 2)$ -dimensional space V is exactly*

$$\det(xI_V - C) = \prod_{S \subsetneq [r]} (x - \alpha_S)^{q_S - 1} \frac{1}{N - 1} \sum_{S \subsetneq [r]} q_S \prod_{\substack{T \subsetneq [r] \\ T \neq S}} (x - \alpha_T). \quad (\text{MG4})$$

This formula remains valid when different α_S coincide. Its exact determinant and attained quotient determinant are

$$\det C = \left(\prod_{S \subsetneq [r]} \alpha_S^{q_S} \right) \frac{\sum_{S \subsetneq [r]} q_S / \alpha_S}{N - 1}, \quad \det G_{\text{quotient}} = (\det C)^{-1}. \quad (\text{MG5})$$

Proof. Inside H_S , the $(q_S - 1)$ -dimensional orthogonal complement of c_S belongs to V and retains eigenvalue α_S . What remains is the compression of the diagonal matrix $D_0 = \text{diag}(\alpha_S)$ to the hyperplane orthogonal to $d = (\sqrt{q_S})_S$. In an orthonormal basis whose final vector is $d/\sqrt{N-1}$, the determinant of the compressed block of $xI - D_0$ is the final principal cofactor. For x outside its finite diagonal spectrum, the cofactor identity gives

$$\det(xI - D_0) \frac{d^*(xI - D_0)^{-1}d}{N-1} = \frac{1}{N-1} \sum_S q_S \prod_{T \neq S} (x - \alpha_T).$$

The identity holds for all x because both expressions after cancellation are polynomials. Multiplying the already separated block factors proves MG4, including coincident eigenvalues. Evaluating the same cofactor identity at D_0 itself proves MG5, since every $\alpha_S > 0$. The empty hyperplane when $r = 1$ contributes determinant one, consistently with both formulas. $\square$

The remaining eigenvalues can be located without cancelling a mixed block. Between consecutive distinct values α_S , the secular function $\sum_S q_S/(x - \alpha_S)$ has strictly negative derivative, approaches $+\infty$ at the left endpoint from the right, and approaches $-\infty$ at the right endpoint from the left. It has exactly one root there. If a diagonal value occurs at h different subsets, its contribution in the compressed remaining block is $h - 1$. This follows either from MG4 or from the d -orthogonal subspace of that equal-eigenvalue block. All eigenvalues are positive and every multiplicity is specified.

For two factors, write $m = N_1$, $n = N_2$, $A = l_1$, $B = l_2$. The extra two-dimensional characteristic polynomial is

$$\frac{(n-1)(x-B)(x-1) + (m-1)(x-A)(x-1) + (m-1)(n-1)(x-A)(x-B)}{mn-1}. \quad (\text{MG6})$$

The other eigenvalues are A with multiplicity $n - 2$, B with multiplicity $m - 2$, and 1 with multiplicity $(m-1)(n-1) - 1$. Coincident values are combined using MG6. In the supplied example $(m, n, A, B) = (3, 5, 2, 2)$, MG6 equals $(x-2)(x-11/7)$. Hence

$$\text{spec } C = \{2^{[5]}, (11/7)^{[1]}, 1^{[7]}\}, \quad \det C = 352/7, \quad \det G_{\text{quotient}} = 7/352. \quad (\text{MG7})$$

This recovers the supplied result from the complete factor-wise amplitude map and determines all its higher-factor extensions.

There is an exact trace comparison for local versus global moment conditions as well. For units $u_i \in R_i^\times$, let U_i permute the basis by $x \mapsto u_i x$, and put $f_i = |\{x \in R_i : (u_i - 1)x = 0\}|$. Both e_i, c_i are fixed. The tensor product action preserves V and $V_{\text{loc}} = \bigotimes_i \{e_i, c_i\}^\perp \subseteq V$. Their orthogonal complement inside V is also invariant, and

$$\begin{aligned} \text{Tr}_V U &= \prod_i f_i - 2, \\ \text{Tr}_{V_{\text{loc}}} U &= \prod_i (f_i - 2), \\ \text{Tr}_{V \ominus V_{\text{loc}}} U &= \prod_i f_i - 2 - \prod_i (f_i - 2), \\ \dim(V \ominus V_{\text{loc}}) &= N - 2 - \prod_i (N_i - 2). \end{aligned} \quad (\text{MG8})$$

The trace of each permutation is its fixed-point count; subtracting its two fixed vectors gives the local formula. Tensoring and then subtracting the global two-dimensional fixed space proves all four identities. Multiplication by 2 on $\mathbb{Z}/3$ and $\mathbb{Z}/5$ has $f_1 = f_2 = 1$, giving traces $-1, +1, -2$ on the three respective spaces and dimension ten for the last. Because each U_i fixes e_i , it commutes with D, P, C and the exact minimum-norm section. Thus the trace comparison and the metric calculation refer to the same original receiver.

$$\begin{array}{ccccc}
\bigotimes_i \mathbb{C}[G_{L_i}(R_i)] & \xrightarrow{\pi} & H & \xrightarrow{P} & V \\
& & & & \parallel \\
& \nwarrow T^*C^{-1} & & & V
\end{array}$$

Figure 21: The original finite amplitude map retains every support label before the two global moment conditions are imposed. Equations MG1–5 calculate its complete covariance and the attained minimum-norm section. Proper mixed blocks in MG3 survive; replacing them by separate local two-condition spaces removes the trace and dimension computed in MG8. The supplied two-factor example is MG7; the all-factor formulas are derived here.

30 The actual relative-to-ordinary image on the full finite packet spectrum

The finite packet spectrum in TP8 has several arithmetic closed points. Its ordinary cohomology can therefore be calculated directly, rather than inferred from the irreducible arithmetic base. Let $P = \text{Spec}_{\text{lat}} L$ be the full finite nonempty support poset. Let $Z_{\mathbb{R}}$ be the finite discrete spectrum of the nonempty packet algebra $A_{Z, \mathbb{R}}$ of TP1, and put $c = |Z_{\mathbb{R}}| \geq 1$. A critical root counts once and an off-critical reflected pair counts once, by TP5. Its full split spectrum is exactly

$$Y_Z = P \triangleright Z_{\mathbb{R}}. \quad (\text{PI1})$$

We use the constant sheaf with value a specified complex vector space W and all its restriction maps equal to the identity. This is a sheaf of complex vector spaces on the actual prime topology. It is not silently made a module over the semiring structure sheaf. At every point its stalk is the stated W ; this choice is part of the construction.

Write $C^\bullet(P; W)$ for the strict-chain cochain complex, with its alternating face differential δ , and augment it by W in degree -1 with the constant-section map. Denote this augmented complex by $\tilde{C}^\bullet(P; W)$. The full finite-space resolution used in FL14–19 computes constant-sheaf cohomology by these cochains. The relative complex for the closed arithmetic part is the kernel of restriction from Y_Z to P . Every strict chain of Y_Z either lies entirely in P or ends at exactly one arithmetic point z . Thus, for $n \geq 1$,

$$C^n(Y_Z; W) = C^n(P; W) \oplus \bigoplus_{z \in Z_{\mathbb{R}}} C^{n-1}(P; W), \quad C^0(Y_Z; W) = C^0(P; W) \oplus W^{Z_{\mathbb{R}}}. \quad (\text{PI2})$$

For $(a, (b_z)_z)$ in degree n , its differential has components

$$\delta a, \quad \delta b_z + (-1)^{n+1} a. \quad (\text{PI3})$$

For $n = 0$, δb_z means the constant cochain with value b_z , so the second component is $b_z - a(x)$ on the edge (x, z) . For all other degrees the formula follows by deleting each vertex of $(x_0, \dots, x_n, z)$ with its actual alternating sign. This derives the differential, including the connecting sign.

Theorem 30.1 (Complete relative image and its kernel). *For every $n \geq 1$ there are canonical identifications*

$$\begin{aligned}
H_{Z_{\mathbb{R}}}^n(Y_Z; W) &= \mathbb{C}^{Z_{\mathbb{R}}} \otimes_{\mathbb{C}} \tilde{H}^{n-1}(P; W), \\
H^n(Y_Z; W) &= (\mathbb{C}^{Z_{\mathbb{R}}} / \Delta \mathbb{C}) \otimes_{\mathbb{C}} \tilde{H}^{n-1}(P; W).
\end{aligned} \quad (\text{PI4})$$

The relative-to-ordinary map is the displayed quotient on the first factor. It is onto, and its kernel is $\Delta \mathbb{C} \otimes \tilde{H}^{n-1}(P; W)$. Also $H_{Z_{\mathbb{R}}}^0(Y_Z; W) = 0$ and $H^0(Y_Z; W) = W$.

Proof. By PI2–3, the kernel of restriction has in degree n the direct sum of c copies of $\tilde{C}^{n-1}(P; W)$, with differential δ . It computes the first group of PI4. The constant augmentation is injective because $P \neq \emptyset$, so its degree-zero relative group vanishes. The connecting map from $H^n(P; W)$ sends $[a]$ to $((-1)^{n+1}[a])_{z \in Z_{\mathbb{R}}}$ in the relative degree $n+1$. For $n=0$ this is the negative diagonal map after quotienting constant sections, and for $n>0$ it is a signed diagonal injection. The long exact sequence of the surjective cochain restriction therefore gives the second group of PI4 and the exact quotient map, with no choice of an arithmetic point. In degree zero, every edge forces all values to coincide; the poset is connected because P and $Z_{\mathbb{R}}$ are nonempty and every pair (x, z) is an edge. Its global constant sections are precisely W . $\square$

The ordinary quotient also has the following explicit cochain receiver. Put $U = \mathbb{C}^{Z_{\mathbb{R}}}$, $V = U/\Delta\mathbb{C}$ and $A^\bullet = \tilde{C}^\bullet(P; W)$. The map

$$\rho_n(a, b) = (q \otimes 1)b : C^n(Y_Z; W) \longrightarrow V \otimes A^{n-1}, \quad q : U \twoheadrightarrow V \quad (\text{PI7})$$

commutes with the differential because $q\Delta = 0$. Its kernel is $K^n = C^n(P; W) \oplus A^{n-1}$, with $d_K(a, b) = (\delta a, \delta b + (-1)^{n+1}a)$. For $n \geq 1$, let $h_n(a, b) = ((-1)^n b, 0)$. Substitution gives $d_K h + h d_K = 1$ in positive degrees. In degree zero, with $\varepsilon(a, b) = b$ and $\iota(w) = ((w)_{x \in P}, w)$, it gives $h_1 d_K = 1 - \iota\varepsilon$. Thus the kernel retracts to $W[0]$. This proves the stated quotient on cohomology directly. Its restriction to the relative complex is the same $q \otimes 1$, without a choice of an arithmetic point.

For Boolean support with d atoms, P consists of d points. The calculation specializes to

$$\dim_{\mathbb{C}} H_{Z_{\mathbb{R}}}^1(Y_Z; \mathbb{C}) = c(d-1), \quad \dim_{\mathbb{C}} H^1(Y_Z; \mathbb{C}) = (c-1)(d-1), \quad H^{n \geq 2}(Y_Z; \mathbb{C}) = 0. \quad (\text{PI5})$$

For the six-point triangle-boundary support, $\tilde{H}^1(P; \mathbb{C}) = \mathbb{C}$ and all other reduced groups vanish. Thus its ordinary degree-two group is $\mathbb{C}^{Z_{\mathbb{R}}}/\Delta\mathbb{C}$, and its relative degree-two group is $\mathbb{C}^{Z_{\mathbb{R}}}$. In particular this packet has a nonzero relative-to-ordinary image when $c > 1$. The example comes from the same support complex in BP12, not an assigned cohomological degree.

30.1 Every original prime acts on these exact groups

Take $W = A_Z$, the entire original complex jet algebra, and let Π_p act by multiplication by its full p^s jet at every stalk. This is a sheaf endomorphism: all restriction maps are identities and commute with it. Formula PI3 proves chain compatibility and gives the action $I \otimes \Pi_p$ under PI4. If $h_{n-1} = \dim_{\mathbb{C}} \tilde{H}^{n-1}(P; \mathbb{C})$, then

$$\begin{aligned} \text{Tr}(\Pi_p \mid H_{Z_{\mathbb{R}}}^n(Y_Z; A_Z)) &= c h_{n-1} \sum_{\rho \in Z} m_{\rho} p^{\rho}, \\ \text{Tr}(\Pi_p \mid H^n(Y_Z; A_Z)) &= (c-1) h_{n-1} \sum_{\rho \in Z} m_{\rho} p^{\rho}. \end{aligned} \quad (\text{PI6})$$

Indeed the coefficient matrix on each full local jet factor is lower triangular in the original ascending basis $1, t_{\rho}, \dots, t_{\rho}^{m_{\rho}-1}$, with diagonal p^{ρ} repeated m_{ρ} times. Explicitly its row k , column j entry is $p^{\rho} (\log p)^{k-j} / (k-j)!$ for $k \geq j$ and zero otherwise. Tensoring with the explicitly counted topological factors gives the displayed traces. Its nilpotent coefficients are still the original Taylor coefficients; only the trace omits their strictly triangular entries. The ordinary dual of $\mathbb{C}^{Z_{\mathbb{R}}}/\Delta\mathbb{C}$ is the sum-zero subspace of $(\mathbb{C}^{Z_{\mathbb{R}}})^*$, so the dual receiving map is also specified without choosing a branch. Write $H = \tilde{H}^{n-1}(P; \mathbb{C})$, and let T be either $U \otimes H$ or $V \otimes H$. The exact perfect dual receiver, using FJ4–5, is

$$1_{T^*} \otimes D_R : T^* \otimes \overline{A_Z} \xrightarrow{\sim} (T \otimes A_Z)^*, \quad (\phi \otimes \bar{f}, t \otimes h) \longmapsto \phi(t) R(f, h). \quad (\text{PI8})$$

On the target the prime acts by $p(1 \otimes \Pi_p^{-1})^*$; on the source it acts by $1 \otimes \overline{\Pi_p}$. The equality follows by applying FJ6 to $R(\Pi_p f, h) = p R(f, \Pi_p^{-1} h)$, so every factor and inverse is specified.

The dual relative map is $q^* \otimes 1_{H^*} \otimes 1_{A_Z^*}$, where q^* embeds the sum-zero covectors into U^* . No topological self-pairing is assumed. Contracting by the original J_g gives exactly

$$\phi(t) \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} h(\rho). \quad (\text{PI9})$$

The two radicals are $T^* \otimes \overline{\ker J_g}$ and $T \otimes \ker J_g$, because the topological evaluation pairing is perfect and FJ4–5 identifies the coefficient radical. This retains all multiplicities through the exact dual maps.

Finally PI1 is the finite packet base, with the dense embedding TP8 into $P \triangleright \text{Spec } \mathcal{B}$. On the latter irreducible arithmetic base, the constant-coefficient sheaf has ordinary cohomology $\mathbb{C}[0]$: the sheaf-triple computation FL1–13 reduces it to the constant sheaf on $\text{Spec } \mathcal{B}$, whose restriction maps between nonempty opens are identities and which is flasque. The ring $\mathcal{B}$ is an integral domain, since after scalar extension it is the Laurent polynomial domain $\mathbb{C}[X_2^{\pm 1}, X_3^{\pm 1}]$. The same argument with A_Z coefficients gives $A_Z[0]$. Consequently the pullback from that ordinary positive-degree cohomology into PI4 is the zero map. The nonzero packet image calculated here has the exact base-change map just stated; it has not been identified with an already nonzero global interior group.

$$\begin{array}{ccc} \mathbb{C}^{Z_{\mathbb{R}}} \otimes \tilde{H}^{n-1}(P; W) & \longrightarrow & (\mathbb{C}^{Z_{\mathbb{R}}} / \Delta \mathbb{C}) \otimes \tilde{H}^{n-1}(P; W) \\ \parallel & & \parallel \\ H_{Z_{\mathbb{R}}}^n(Y_Z; W) & \longrightarrow & H^n(Y_Z; W) \end{array}$$

Figure 22: The actual relative-to-ordinary map on the full finite packet spectrum, PI2–4. Every support chain and every real arithmetic closed point occurs. With the original jet coefficient space, the prime action and trace on both sides are PI6. No extra prime-weight factor is inferred from the degree shift.

31 The complete Hermitian pulse matrix and its finite-prime Weil bound

The supplied reconstruction [27], Section 5, proposes a uniform coercivity constant for arbitrary complex combinations of its original prime-centred pulses. We prove its exact matrix and constant. The original widths, derivatives, amplitudes, Haar measure, prime factors, support faces, and complete Archimedean subtraction are retained. The earlier two-prime, four-sign proof is [26], FWP1–24, and its common Archimedean face construction is FWJ1–8. All estimates needed here are proved again.

31.1 The original local distributions and the exact test space

The original author TeX of Alain Connes [5], section `sectweilpos`, equations `wp` and `arch`, uses

$$\hat{h}(\xi) = \int_0^\infty h(x) x^{-i\xi} \frac{dx}{x}, \quad h^*(x) = \overline{h(x^{-1})}, \quad (h * k)(x) = \int_0^\infty h(y) k(x/y) \frac{dy}{y}. \quad (\text{FP1})$$

For $f = h * h^*$ its local distributions are

$$\begin{aligned} W_p(f) &= (\log p) \sum_{m \geq 1} p^{-m/2} (f(p^m) + f(p^{-m})), \\ W_{\text{arch}}(f) &= (\log(4\pi) + \gamma_E) f(1) \\ &\quad + \int_1^\infty (f(x) + f(x^{-1}) - 2x^{-1/2} f(1)) \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}. \end{aligned} \quad (\text{FP2})$$

On tests with $\widehat{h}(\pm i/2) = 0$, the complete Weil form in that source is

$$Q(h) = -W_{\text{arch}}(h * h^*) - \sum_p W_p(h * h^*). \quad (\text{FP3})$$

The positivity criterion for all admissible tests is attributed in that source to André Weil. No such universal conclusion is used or proved here. Equations FP1–3 fix the convention to which the following calculation applies.

Fix a set $I = \{p_1, \dots, p_s\}$ of distinct rational primes, where $s \geq 2$. Put $r_0 = 1$, $r_i = p_i$, and

$$M_I = \prod_{i=1}^s p_i, \quad \varepsilon = \frac{1}{16} \log \frac{M_I + 1}{M_I}, \quad a_j = \log r_j, \quad (\text{FP4})$$

$$w(t) = \begin{cases} \exp[-1/(1 - (t/\varepsilon)^2)], & |t| < \varepsilon, \\ 0, & |t| \geq \varepsilon, \end{cases} \quad z(t) = w''(t) - \frac{1}{4}w(t), \quad N = \int_{\mathbb{R}} z(t)^2 dt.$$

For every $c = (c_0, \dots, c_s) \in \mathbb{C}^{s+1}$, define the actual test

$$g_c(e^t) = G_c(t) = \sum_{j=0}^s c_j z(t - a_j). \quad (\text{FP5})$$

The coefficient N remains this literal integral; neither z nor g_c is divided by its norm.

Lemma 31.1 (Admissibility and the exact coefficient norm). *The functions w, z and g_c are smooth with compact support, $N > 0$, and*

$$\widehat{g}_c(i/2) = \widehat{g}_c(-i/2) = 0, \quad \|g_c\|_{L^2(dx/x)}^2 = N \sum_{j=0}^s |c_j|^2. \quad (\text{FP6})$$

Proof. Every derivative of the displayed bump inside its support is its exponential times a finite combination of powers of $(1 - (t/\varepsilon)^2)^{-1}$ and polynomials in t . The exponential tends to zero faster than every such power at both endpoints. Extension by zero is therefore smooth, with all endpoint derivatives zero. Integration by parts gives $\int z w = -\int |w'|^2 - \frac{1}{4} \int w^2 < 0$, so $z \neq 0$ and $N > 0$. For $b = \pm 1/2$, two integrations by parts give

$$\int_{\mathbb{R}} z(t) e^{bt} dt = (b^2 - \frac{1}{4}) \int_{\mathbb{R}} w(t) e^{bt} dt = 0. \quad (\text{FP7})$$

Translation by a_j multiplies this zero by e^{ba_j} . FP1 and $x = e^t$ now prove the two Fourier conditions.

For distinct positive integers A, D ,

$$|\log(A/D)| \geq \log(1 + 1/D). \quad (\text{FP8})$$

For $A > D$, use $A \geq D + 1$. For $A < D$, necessarily $D \geq 2$, and $D/A \geq D/(D - 1) > (D + 1)/D$. The case $D = 1$ is already covered by $A > D$. Every r_j is less than M_I , because $s \geq 2$. For $i \neq j$, apply FP8 to $A = r_i, D = r_j$:

$$|a_i - a_j| > \log(1 + 1/M_I) = 16\varepsilon. \quad (\text{FP9})$$

Thus the translates of z , each supported in an interval of radius ε , have disjoint supports. Expanding the squared norm in the original Haar measure $dt = dx/x$ proves FP6. $\square$

The map $c \mapsto g_c$ is therefore injective and has Gram matrix $N \text{Id}$. In particular every complex coefficient direction, including every phase and every vanishing coordinate, is retained. For $\widehat{f}_c = g_c * g_c^*$, compact support and Fubini give $\widehat{f}_c(\xi) = \widehat{g}_c(\xi) \widehat{g}_c(\overline{\xi})$; hence both pole moments $\widehat{f}_c(\pm i/2)$ vanish as well.

31.2 Every finite prime power and every complex coefficient

Keep the real autocorrelation

$$C(t) = \int_{\mathbb{R}} z(u)z(u-t) du. \quad (\text{FP10})$$

It is smooth, real and even, with support in $[-2\varepsilon, 2\varepsilon]$, $C(0) = N$, and $|C(t)| \leq N$ by Cauchy-Schwarz and translation invariance. Expanding the convolution in FP1 gives the exact formula

$$f_c(e^t) = \int_{\mathbb{R}} G_c(u)\overline{G_c(u-t)} du = \sum_{i,j=0}^s c_i \overline{c_j} C(t - a_i + a_j). \quad (\text{FP11})$$

This includes all $(s+1)^2$ ordered coefficient pairs. It gives $f_c(e^{-t}) = \overline{f_c(e^t)}$ and $f_c(1) = N\|c\|_2^2$.

Lemma 31.2 (Exact prime-lag elimination). *Put $\beta_i = (\log p_i)/\sqrt{p_i}$. Then*

$$W_p(f_c) = \begin{cases} 2N\beta_i \operatorname{Re}(\overline{c_0}c_i), & p = p_i \in I, \\ 0, & p \notin I. \end{cases} \quad (\text{FP12})$$

All higher powers and all other pulse-pair lags vanish exactly.

Proof. Let $n \geq 2$ be an integer, and consider an ordered summand of FP11 at $t = \log n$. Its argument is $\log(nr_j/r_i)$. If $nr_j \neq r_i$, FP8 with $A = nr_j, D = r_i < M_I$ gives

$$|\log(nr_j/r_i)| > 16\varepsilon > 2\varepsilon, \quad (\text{FP13})$$

so this summand is zero. If $nr_j = r_i$, then $r_j = 1$: otherwise nr_j is composite and cannot equal either 1 or one of the primes r_i . It follows that $j = 0, i \geq 1$, and $n = p_i$. This summand is $Nc_i\overline{c_0}$. The reciprocal evaluation is its conjugate by FP11. Finally, a prime power p^m equals p_i precisely when $p = p_i$ and $m = 1$. Substitution into the entire sum in FP2 proves FP12. This proof addresses all integers $n \geq 2$, so no finite-prime tail remains to be estimated. $\square$

31.3 The exact full Archimedean matrix on every support face

Write

$$K(t) = \frac{e^{-t/2}}{1 - e^{-2t}} \quad (t > 0), \quad c_{\text{arch}} = \log(4\pi) + \gamma_E. \quad (\text{FP14})$$

The substitution $x = e^t$ in FP2 gives exactly

$$W_{\text{arch}}(f_c) = c_{\text{arch}}f_c(1) + \int_0^\infty (f_c(e^t) + f_c(e^{-t}) - 2e^{-t/2}f_c(1))K(t) dt. \quad (\text{FP15})$$

Near zero the bracket is $O(t)$, because f_c is smooth, and $K(t) = O(t^{-1})$. Beyond the support of $f_c(e^t)$ the subtracted term remains and decays exponentially. The integral is thus convergent at both ends.

Define the original central value and each full cross integral:

$$\begin{aligned} A_\varepsilon &= c_{\text{arch}}N + 2 \int_0^\infty [C(t) - Ne^{-t/2}]K(t) dt, \\ J_\varepsilon(d) &= \int_{d-2\varepsilon}^{d+2\varepsilon} C(t-d)K(t) dt \quad (d > 16\varepsilon). \end{aligned} \quad (\text{FP16})$$

Both are real. Set $H_{ii} = 0$ and $H_{ij} = J_\varepsilon(|a_i - a_j|)$ for $i \neq j$, so H is a real symmetric $(s+1)$ -by- $(s+1)$ matrix. For every $A \subseteq I$, retain the corresponding face functional

$$Q_A(g_c) = -W_{\text{arch}}(f_c) - \sum_{p_i \in A} W_{p_i}(f_c). \quad (\text{FP17})$$

The face $A = I$ is the complete form Q , by FP12. Every proper face keeps exactly the same Archimedean functional.

Theorem 31.3 (The complete Hermitian coefficient matrix). *Writing E_{ij} for the matrix units on $\mathbb{C}^{s+1}$, one has*

$$Q_A(g_c) = c^* W_A c, \quad W_A = -A_\varepsilon \text{Id} - H - N \sum_{p_i \in A} \beta_i (E_{0i} + E_{i0}). \quad (\text{FP18})$$

This is valid simultaneously for every face and every complex coefficient vector.

Proof. The diagonal summands in FP11 contribute $\|c\|_2^2 A_\varepsilon$ to FP15. For each unordered pair i, j , put $d = |a_i - a_j|$. Pairing its two ordered terms in $f_c(e^t) + f_c(e^{-t})$ yields

$$2 \operatorname{Re}(\bar{c}_i c_j) (C(t - d) + C(t + d)). \quad (\text{FP19})$$

On $t > 0$, the second term is zero by $d > 16\varepsilon$. The first integrates to $J_\varepsilon(d)$; its integral has precisely the full interval in FP16. The subtraction involving $f_c(1)$ has no cross coefficient, since the pulse supports are disjoint. Consequently

$$W_{\text{arch}}(f_c) = A_\varepsilon \|c\|_2^2 + 2 \sum_{0 \leq i < j \leq s} J_\varepsilon(|a_i - a_j|) \operatorname{Re}(\bar{c}_i c_j) = c^* (A_\varepsilon \text{Id} + H) c. \quad (\text{FP20})$$

Together with FP12 this proves FP18, including its signs and all complex cross coefficients. Possible overlap between different cross-integral intervals does not change the sum: each comes from its own term of the finite convolution expansion. $\square$

31.4 A bound for the whole coefficient space

For $t > 0$,

$$K(t) \leq \frac{1}{1 - e^{-2t}} = 1 + \frac{1}{e^{2t} - 1} \leq 1 + \frac{1}{2t}, \quad 1 - e^{-t/2} \leq t/2. \quad (\text{FP21})$$

The first inequality uses $e^{-t/2} \leq 1$; the next uses $e^{2t} \geq 1 + 2t$. The second follows from $e^{-u} \geq 1 - u$. Since $C(t) \leq N$, these give

$$\begin{aligned} 2 \int_0^{2\varepsilon} [C(t) - N e^{-t/2}] K(t) dt &\leq 2N \int_0^{2\varepsilon} (1 - e^{-t/2}) K(t) dt \\ &\leq N \int_0^{2\varepsilon} (t + \tfrac{1}{2}) dt = N(2\varepsilon^2 + \varepsilon). \end{aligned} \quad (\text{FP22})$$

Because C vanishes outside this central interval, the rest of the central integral is evaluated exactly:

$$-2N \int_{2\varepsilon}^\infty e^{-t/2} K(t) dt = -2N \int_0^{e^{-2\varepsilon}} \frac{dy}{1 - y^2} = -N \log \coth \varepsilon. \quad (\text{FP23})$$

Here $y = e^{-t}$; the antiderivative is $\frac{1}{2} \log((1 + y)/(1 - y))$. Combining FP16, FP22 and FP23 gives

$$-\frac{A_\varepsilon}{N} \geq \log \coth \varepsilon - c_{\text{arch}} - \varepsilon - 2\varepsilon^2. \quad (\text{FP24})$$

No part of the infinite Archimedean subtraction has been dropped.

For any noncentral lag $d > 16\varepsilon$, the integral interval has length 4ε and lower endpoint $d - 2\varepsilon > 14\varepsilon$. Thus

$$|J_\varepsilon(d)| \leq N \int_{d-2\varepsilon}^{d+2\varepsilon} \left(1 + \frac{1}{2t}\right) dt \leq N \left(4\varepsilon + \frac{4\varepsilon}{2(d-2\varepsilon)}\right) \leq N(4\varepsilon + \tfrac{1}{7}). \quad (\text{FP25})$$

There are exactly s off-diagonal entries in each row of H . For an arbitrary complex c , the inequality $2|c_i||c_j| \leq |c_i|^2 + |c_j|^2$ gives

$$|c^* H c| \leq N(4\varepsilon + \frac{1}{7}) \sum_{i \neq j} |c_i||c_j| \leq Ns(4\varepsilon + \frac{1}{7}) \|c\|_2^2. \quad (\text{FP26})$$

This controls all phases at once, without imposing any sign condition on c .

The star matrix $S_A = \sum_{p_i \in A} \beta_i (E_{0i} + E_{i0})$ maps e_0 to $v_A = \sum_{p_i \in A} \beta_i e_i$, maps v_A to $\|v_A\|^2 e_0$, and vanishes on the orthogonal complement of their span. For $A = \emptyset$ it is zero. Therefore

$$\|S_A\| = \left(\sum_{p_i \in A} \beta_i^2 \right)^{1/2} \leq \frac{2}{e} \sqrt{|A|} \leq \frac{2}{e} \sqrt{s}. \quad (\text{FP27})$$

Indeed $x \mapsto (\log x)/\sqrt{x}$ has derivative $x^{-3/2}(1 - \frac{1}{2} \log x)$; its maximum on $x > 0$ is $2/e$, attained at $x = e^2$. Using FP18 and FP24–27 proves, for every nonzero c ,

$$\frac{Q_A(g_c)}{N\|c\|_2^2} \geq L_s(\varepsilon) := \log \coth \varepsilon - c_{\text{arch}} - (1 + 4s)\varepsilon - 2\varepsilon^2 - \frac{s}{7} - \frac{2\sqrt{s}}{e}. \quad (\text{FP28})$$

31.5 Every rational constant in the universal lower bound

We first record elementary strict inequalities with their proofs. For $x > 0$, polynomial division and integration give

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \int_0^x \frac{t^3}{1+t} dt < x - \frac{x^2}{2} + \frac{x^3}{3}. \quad (\text{FP29})$$

Also $\log(1+x) > x/(1+x)$ and $\log(1+x) < x$, by integrating the strict bounds $1/(1+x) < 1/(1+t) < 1$ for $0 < t < x$. Since $M_I \geq 6$, applying FP29 at $x = 1/6$ gives

$$0 < \varepsilon \leq \frac{1}{16} \log(7/6) < \frac{1}{16} \left(\frac{1}{6} - \frac{1}{72} + \frac{1}{648} \right) = \frac{25}{2592} < \frac{1}{100}. \quad (\text{FP30})$$

For $0 < u < 1$, integration of a finite geometric series gives

$$\log \frac{1+u}{1-u} > 2 \sum_{j=0}^m \frac{u^{2j+1}}{2j+1} \quad (m \geq 0), \quad (\text{FP31})$$

because the omitted integral of $2u^{2m+2}/(1-u^2)$ is positive. Use $u = 1/3, m = 2$ for $\log 2$, and $u = 1/9, m = 0$ for $\log(5/4)$. Since $100 = 2^6(5/4)^2$, this proves

$$\log 100 > 12 \left(\frac{1}{3} + \frac{1}{81} + \frac{1}{1215} \right) + \frac{4}{9} = \frac{1864}{405}. \quad (\text{FP32})$$

The derivative of $\tanh t$ is at most one, so $\tanh \varepsilon \leq \varepsilon$. FP30 and $2592/25 = 100(648/625)$ now give

$$\begin{aligned} \log \coth \varepsilon &\geq \log(1/\varepsilon) > \log(2592/25) \\ &= \log 100 + \log(648/625) > \frac{1864}{405} + \frac{23}{648} = \frac{5009}{1080}. \end{aligned} \quad (\text{FP33})$$

The bound on the second logarithm uses $\log(1 + 23/625) > (23/625)/(1 + 23/625) = 23/648$.

To bound the Archimedean constant, write $H_n = \sum_{k=1}^n 1/k$. The sequence $H_n - \log n$ decreases strictly to γ_E , since $\log(1 + 1/n) > 1/(n+1)$. Hence $\gamma_E < H_{10} - \log 10$. The inequality $\pi < 22/7$ has the following exact integral proof:

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \int_0^1 \left(x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2} \right) dx$$

$$= \frac{22}{7} - \pi. \quad (\text{FP34})$$

The displayed division is checked by multiplication; its last term integrates using $\arctan 1 = \pi/4$. Applying FP29 at $x = 9/35$ consequently gives

$$c_{\text{arch}} < H_{10} + \log(2\pi/5) < H_{10} + \log(44/35)$$

$$< \frac{7381}{2520} + \frac{9}{35} - \frac{1}{2} \left(\frac{9}{35} \right)^2 + \frac{1}{3} \left(\frac{9}{35} \right)^3 = \frac{9750961}{3087000}. \quad (\text{FP35})$$

Finally $99^2 = 9801 > 9800 = 2 \cdot 70^2$ and the positive exponential series $e > 1 + 1 + 1/2 + 1/6 = 8/3$ imply

$$\frac{2\sqrt{2}}{e} < 2 \frac{99}{70} \frac{3}{8} = \frac{297}{280}. \quad (\text{FP36})$$

Theorem 31.4 (Uniform bound on every face and every coefficient). *For every finite distinct-prime set I with $s \geq 2$, every face $A \subseteq I$, and every $c \in \mathbb{C}^{s+1}$, the original test FP5 satisfies*

$$Q_A(g_c) \geq \mu_* \|g_c\|_{L^2(dx/x)}^2, \quad \mu_* = \frac{6600077509}{144027072000} > \frac{1}{22}. \quad (\text{FP37})$$

For $c \neq 0$, the first inequality can be made strict.

Proof. First suppose $s = 2$. The function $9x + 2x^2$ is increasing for $x > 0$. Substituting FP30, FP33, FP35 and FP36 into FP28 gives the entirely rational calculation

$$L_2(\varepsilon) > \frac{5009}{1080} - \frac{9750961}{3087000} - 9 \frac{25}{2592} - 2 \left(\frac{25}{2592} \right)^2 - \frac{2}{7} - \frac{297}{280}$$

$$= \frac{6600077509}{144027072000} = \mu_*, \quad (\text{FP38})$$

$$\mu_* - \frac{1}{22} = \frac{587316599}{1584297792000} > 0.$$

Now let $s \geq 3$. After ordering the distinct primes, the first is at least 2, the second at least 3, and each remaining prime at least 5. Thus

$$M_I \geq 6 \cdot 5^{s-2}, \quad \varepsilon < \frac{1}{16M_I}, \quad \log \coth \varepsilon > \log 96 + (s-2) \log 5. \quad (\text{FP39})$$

The middle inequality uses $\log(1+x) < x$; the last uses $\coth \varepsilon \geq 1/\varepsilon$. FP32 and $\log(25/24) < 1/24$ give

$$\log 96 = \log 100 - \log(25/24) > \frac{1864}{405} - \frac{1}{24} = \frac{547}{120} + \frac{1}{405}. \quad (\text{FP40})$$

FP31 with $u = 2/3, m = 1$ also gives

$$\log 5 > 2 \left(\frac{2}{3} + \frac{8}{81} \right) = \frac{124}{81} = \frac{3}{2} + \frac{5}{162}. \quad (\text{FP41})$$

Consequently $\log \coth \varepsilon > \frac{547}{120} + \frac{3}{2}(s-2)$. Moreover $e > 8/3$ and $\sqrt{s} \leq s$ imply $2\sqrt{s}/e < 3s/4$. Insert these, $\varepsilon < 1/100$, and FP35 into FP28:

$$L_s(\varepsilon) > B_s := \frac{547}{120} + \frac{3}{2}(s-2) - \frac{9750961}{3087000}$$

$$- \frac{1+4s}{100} - \frac{1}{5000} - \frac{s}{7} - \frac{3s}{4}. \quad (\text{FP42})$$

Direct rational subtraction gives

$$\begin{aligned} B_3 &= \frac{1402183}{15435000}, & B_3 - \mu_* &= \frac{661629847}{14696640000} > 0, \\ B_{s+1} - B_s &= \frac{3}{2} - \frac{1}{25} - \frac{1}{7} - \frac{3}{4} = \frac{397}{700} > 0. \end{aligned} \quad (\text{FP43})$$

Thus $L_s(\varepsilon) > \mu_*$ for all $s \geq 3$, as well as for $s = 2$. FP6 and FP28 prove the strict bound for $c \neq 0$. For $c = 0$, both sides in FP37 are zero. $\square$

For $s = 2$ and the four original choices $c = (1, \pm 1, \pm 1)$, FP6 gives $\|g_c\|^2 = 3N$, so the new bound gives $Q(g_c) > 3\mu_*N$. In particular

$$3\mu_* - \frac{5981}{52500} = \frac{5653533017}{240045120000} > 0. \quad (\text{FP44})$$

This strengthens the original FWP bound without changing any part of the pulse construction. The calculation proves coercivity on the specified $(s+1)$ -dimensional complex test space for each finite prime set, and on every one of its retained support faces. It does not identify these spaces with the entire admissible test space in Weil's criterion.

32 Every mixed-prime spectral direction and its actual pulse receiver

This paper independently derives the spectrum and every finite pulse eigenspace proposed in Sections 6–8 of the supplied continuation *Full-support reconstruction: the missing cohomology, the actual prime degree, and complete mixed-sign calculations*. The calculation retains the original support diagonal, all $s!$ tensor summands, the original pulse amplitudes, and the actual map from a compression back to the full Hilbert space. It also derives every higher tensor evaluation on the same pulse tests. The finite positivity estimate used in the last section is the proved companion result [28]; no positivity or boundedness of a full Weil operator is assumed.

The analytic source is Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, original author source `rhready.tex`, subsection `sectweilpos`, equations `wp` and `arch` (source lines 841–868). Connes attributes the criterion there to André Weil. The exact programme antecedent is the pinned edition 033, PME6–10, with the original three-pulse calculation FWP2–23 and the shared-Archimedean support trace FWJ1–8. These antecedents and the supplied Sections 6–8 are identified separately from the new derivation below. No prime-spectrum foundation or classical Fourier theorem is claimed as a new discovery.

32.1 The original prime operator and the form domain

Use the Hilbert space $H = L^2(\mathbb{R}_+^\times, dx/x)$ with inner product linear in its first variable. The logarithmic coordinate is $x = e^u$. Our unitary Fourier transform and original Mellin transform are

$$(\mathcal{F}g)(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(e^u) e^{-itu} du, \quad \widehat{g}(t) = \sqrt{2\pi} \mathcal{F}g(t). \quad (\text{FMS1})$$

For a rational prime p put $\ell_p = \log p$ and $r_p = p^{-1/2}$. Let $(T_a F)(u) = F(u - a)$ on $L^2(\mathbb{R}, du)$. The norm-convergent series

$$B_p = (\log p) \sum_{n \geq 1} r_p^n (T_{n\ell_p} + T_{-n\ell_p}) \quad (\text{FMS2})$$

defines a bounded self-adjoint operator, since every translation has norm one and $\sum_{n \geq 1} 2r_p^n < \infty$. The two geometric series give

$$\mathcal{F}B_p\mathcal{F}^{-1} = M_{b_p}, \quad b_p(t) = (\log p) \left(\frac{1 - p^{-1}}{1 - 2p^{-1/2} \cos(t \log p) + p^{-1}} - 1 \right). \quad (\text{FMS3})$$

Here M_b denotes multiplication by b . If $g^*(x) = \overline{g(x^{-1})}$ and $f = g * g^*$, convolution is always with respect to dx/x . Direct substitution in the convolution gives

$$f(e^a) = \int_{\mathbb{R}} g(e^u) \overline{g(e^{u-a})} du.$$

The symmetric pair of translations in (FMS2) therefore gives the exact original finite-place functional:

$$\langle B_p g, g \rangle = W_p(f) = (\log p) \sum_{n \geq 1} p^{-n/2} \{f(p^n) + f(p^{-n})\}. \quad (\text{FMS4})$$

All terms, including $\log p$, both reciprocal arguments and the exponent $-n/2$, are retained.

For completeness, this operator is the receiver of the actual relative-prime energy map in PME6–10, not an assigned multiplier. Use additive Haar measure $\mu(\mathbb{Z}_p) = 1$ on $\mathbb{Q}_p$, and set

$$(\mathcal{U}_p \phi)(y) = \sqrt{p} \phi(y/p), \quad \xi_p = \mathbf{1}_{\mathbb{Z}_p} \in L^2(\mathbb{Q}_p).$$

Changing variables proves that $\mathcal{U}_p$ is unitary, and $\mathcal{U}_p^n \xi_p = p^{n/2} \mathbf{1}_{p^n \mathbb{Z}_p}$ for $n \geq 0$. Hence $\langle \mathcal{U}_p^n \xi_p, \xi_p \rangle = p^{-|n|/2}$ for every integer n . For compactly supported $F(u) = g(e^u)$ set

$$(\mathcal{E}_p g)(x) = \sum_{n \in \mathbb{Z}} F(x + n\ell_p) \mathcal{U}_p^n \xi_p, \quad 0 \leq x < \ell_p, \quad \mathcal{E}_p : H \longrightarrow L^2([0, \ell_p]; L^2(\mathbb{Q}_p)). \quad (\text{FMS5})$$

The sum is initially finite. Expanding its squared norm and partitioning $\mathbb{R}$ by the intervals $[n\ell_p, (n+1)\ell_p]$ gives

$$\|\mathcal{E}_p g\|^2 = \sum_{j \in \mathbb{Z}} p^{-|j|/2} \int_{\mathbb{R}} F(u) \overline{F(u - j\ell_p)} du.$$

Cauchy–Schwarz bounds the absolute sum by $(1 + r_p)(1 - r_p)^{-1} \|g\|^2$, so $\mathcal{E}_p$ extends boundedly. The $j = 0$ term is exactly $\|g\|^2$. Polarization now proves

$$B_p = (\log p)(\mathcal{E}_p^* \mathcal{E}_p - I_H). \quad (\text{FMS6})$$

Thus the subtracted identity in (FMS3) is required by the original energy receiver.

The denominator of the Poisson kernel in (FMS3) varies from $(1 - r_p)^2$ to $(1 + r_p)^2$. Consequently

$$b_p(\mathbb{R}) = [-\rho_p u_p, u_p], \quad u_p = \frac{2 \log p}{\sqrt{p} - 1}, \quad \rho_p = \frac{\sqrt{p} - 1}{\sqrt{p} + 1}. \quad (\text{FMS7})$$

The extrema occur at $\cos(t \log p) = -1$ and 1 , respectively.

The actual admissible domain is

$$\mathcal{D}_0 = \{g \in C_c^\infty(\mathbb{R}_+^\times) : \widehat{g}(i/2) = \widehat{g}(-i/2) = 0\}. \quad (\text{FMS8})$$

It is dense in H . Indeed the two moment functionals $F \mapsto \int F(u) e^{u/2} du$ and $F \mapsto \int F(u) e^{-u/2} du$ are linearly independent on $C_c^\infty(\mathbb{R})$: translate a nonnegative nonzero test by a nonzero real number to obtain an invertible two-by-two moment matrix. Any linear functional vanishing on their joint kernel therefore is a linear combination of those two moments. If an L^2 vector annihilates that kernel, its distribution equals a combination of $e^{u/2}$ and $e^{-u/2}$. No nonzero such combination is in L^2 : one coefficient produces exponential growth at $+\infty$, and the other at $-\infty$. Thus the orthogonal complement of $\mathcal{D}_0$ is zero. Algebraic tensor powers of $\mathcal{D}_0$ are consequently dense in the corresponding Hilbert tensor powers of H .

On $\mathcal{D}_0$ the complete Weil quadratic form has the convention

$$Q(g) = -W_{\text{arch}}(g * g^*) - \sum_p W_p(g * g^*). \quad (\text{FMS9})$$

Each sum on compact tests is finite: only prime powers in a fixed compact interval can occur in the convolution support. The Archimedean functional is the original one

$$W_{\text{arch}}(f) = (\log(4\pi) + \gamma_E)f(1) + \int_1^\infty \{f(x) + f(x^{-1}) - 2x^{-1/2}f(1)\} \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}. \quad (\text{FMS10})$$

For convolution squares the bracket is $O(\log x)$ near 1, so the integral is convergent. Polarization makes all these Hermitian quadratic forms sesquilinear on $\mathcal{D}_0$. No bounded representing operator for (FMS9) will be required.

32.2 The retained support diagonal and all permutation terms

Fix distinct primes $I = \{p_1, \dots, p_s\}$ with $s \geq 2$, and let Ω be the entire place set. The actual support faces in this calculation are the interval

$$A \mapsto (\Omega \setminus I) \cup A, \quad A \subseteq I.$$

This map preserves joins and meets and is a lattice isomorphism onto that interval. Its bottom contains the Archimedean place and every unselected place. In particular it is not the external absent support. The coefficient receiver used below is the complex vector space with basis these actual support states:

$$V_I = \mathbb{C}[\mathcal{P}(I)], \quad h_I = \sum_{A \subseteq I} (-1)^{|A|} \delta_A, \quad \Delta_r : V_I \longrightarrow V_I^{\otimes r}, \quad \Delta_r \delta_A = \delta_A^{\otimes r}. \quad (\text{FMS11})$$

The map Δ_r is extended linearly. In particular $\Delta_r h_I$ is the displayed signed sum of diagonals, and is not the tensor power $h_I^{\otimes r}$.

Let q_0 be the common sesquilinear form $-W_{\text{arch}} - \sum_{p \notin I} W_p$ on convolution pairs from $\mathcal{D}_0$, and set

$$q_A = q_0 - \sum_{p_i \in A} \langle B_{p_i} \cdot, \cdot \rangle.$$

The full support face $A = I$ is the complete form (FMS9); all other faces retain the very same q_0 . Let $\mathcal{Q} : V_I \rightarrow \{\text{sesquilinear forms on } \mathcal{D}_0\}$ send δ_A to q_A . The precise mixed map is $\mathcal{Q}^{\otimes r} \Delta_r$. Its value on h_I is the form

$$\mathcal{J}_r = \sum_{A \subseteq I} (-1)^{|A|} q_A^{\otimes r} \quad \text{on } \mathcal{D}_0^{\otimes_{\text{alg}} r}. \quad (\text{FMS12})$$

No product of unbounded operators is present in this definition.

Proposition 32.1 (Every tensor summand). *Write $C_0 = q_0$ and $C_i = \langle B_{p_i} \cdot, \cdot \rangle$ for $1 \leq i \leq s$. For a word $w \in \{0, 1, \dots, s\}^r$, let $n(w)$ be its number of nonzero entries. Then*

$$\mathcal{J}_r = \sum_{\substack{w \in \{0, \dots, s\}^r \\ \{1, \dots, s\} \subseteq \{w_1, \dots, w_r\}}} (-1)^{s+n(w)} C_{w_1} \otimes \dots \otimes C_{w_r}. \quad (\text{FMS13})$$

In particular $\mathcal{J}_r = 0$ for $r < s$. At $r = s$ its unique bounded Hilbert-space receiver is

$$\mathbb{M}_I = \sum_{\sigma \in S_s} B_{p_{\sigma(1)}} \otimes \dots \otimes B_{p_{\sigma(s)}}. \quad (\text{FMS14})$$

Proof. In the expansion of $q_A^{\otimes r}$ a fixed word w can occur exactly when A contains its nonzero labels $T(w)$; its coefficient there is $(-1)^{n(w)}$. The total coefficient is

$$(-1)^{n(w)} \sum_{A \supseteq T(w)} (-1)^{|A|} = (-1)^{n(w)+|T(w)|} (1-1)^{s-|T(w)|}.$$

It vanishes unless $T(w) = I$, in which case it is $(-1)^{s+n(w)}$. At $r = s$ each nonzero label must occur exactly once, there is no zero label, and the coefficient is 1. Thus every permutation occurs exactly once. Each factor B_p is bounded, so their finite tensor sum represents this form. Density proved after (FMS8) gives uniqueness of its bounded extension. No commutativity between q_0 and other forms is needed; the entries occupy separate tensor factors. $\square$

For $r > s$ formula (FMS13) generally retains the common form q_0 . It makes no bounded-extension assertion at those orders. The finite pulse restrictions and finite traces below exist at every order without such an assertion.

32.3 The entire mixed spectrum and its spectral projections

Theorem 32.2 (Exact permanent spectrum). *Put $u_i = u_{p_i}$, $\rho_i = \rho_{p_i}$ and $\rho_I = \max_i \rho_i$. The bounded self-adjoint operator $\mathbb{M}_I$ is unitarily equivalent to multiplication on $L^2(\mathbb{R}^s)$ by*

$$m_I(t_1, \dots, t_s) = \text{perm}[b_{p_i}(t_j)]_{i,j=1}^s = \sum_{\sigma \in S_s} \prod_{j=1}^s b_{p_{\sigma(j)}}(t_j). \quad (\text{FMS15})$$

Its spectrum and norm are

$$\sigma(\mathbb{M}_I) = [-s! \rho_I \prod_i u_i, s! \prod_i u_i], \quad \|\mathbb{M}_I\| = s! \prod_i u_i. \quad (\text{FMS16})$$

It has no eigenvalues. Its positive and negative spectral subspaces are both infinite-dimensional, and its kernel is zero.

Proof. The product Fourier transform sends each tensor summand of (FMS14) to its product multiplier, proving (FMS15). Every factor $b_{p_i}(t_j)/u_i$ lies in $[-\rho_i, 1]$. A nonnegative product of s such factors is at most 1. A negative product has at least one negative factor; its absolute value is at most ρ_I , since the other factors have absolute value at most 1. Termwise summation proves both displayed spectral bounds for m_I .

To prove their sharpness with all simultaneous prime frequencies, write $\mathbb{T}^s = (\mathbb{R}/2\pi\mathbb{Z})^s$. The orbit

$$t \mapsto (t \log p_1, \dots, t \log p_s) \pmod{2\pi}$$

is dense. Here is a proof of the exact density needed. For $n \in \mathbb{Z}^s \setminus \{0\}$ one has $\sum_i n_i \log p_i \neq 0$, since exponentiation and unique prime factorization forbid $\prod_i p_i^{n_i} = 1$. Thus

$$\frac{1}{2T} \int_{-T}^T e^{it \sum_i n_i \log p_i} dt = \frac{\sin(T \sum_i n_i \log p_i)}{T \sum_i n_i \log p_i} \rightarrow 0.$$

The average of the constant character is 1. Uniform approximation by trigonometric polynomials proves that orbit averages of any continuous function converge to its Haar integral. If a nonempty open set were missed, choose a nonnegative continuous function supported inside it, with strictly positive Haar integral. Its orbit averages would be zero, a contradiction.

The s variables $t_1, \dots, t_s$ are independent. Consequently the corresponding s columns have dense image in $(\mathbb{T}^s)^s$. On this compact connected torus the function given by the permanent of the Poisson expressions is continuous. Its maximum is attained by setting every phase equal to 0. Its minimum is attained by fixing an index i_0 with $\rho_{i_0} = \rho_I$, setting every phase in row i_0 equal to π , and every other phase equal to 0. Every permutation product is then exactly $-\rho_I \prod_i u_i$. These choices attain the two bounds. The image of a connected space in $\mathbb{R}$ is an interval. Hence the image on this torus is exactly the interval in (FMS16), and the closure of $m_I(\mathbb{R}^s)$ is that interval.

Every neighborhood of every point in this closed interval has preimage of positive Lebesgue measure in $\mathbb{R}^s$. Indeed density first gives a real tuple with its value in that neighborhood, and

continuity gives a nonempty Euclidean open neighborhood of that tuple with the same property. Thus the essential range is the entire interval. Outside it, multiplication by $m_I - z$ has bounded inverse. At any point of it, restrict a unit vector to a finite-measure part of the set $|m_I - z| < 1/n$ to obtain approximate eigenvectors. This proves the spectrum and norm assertions.

The denominator in each Poisson expression is strictly positive on the real axis. Thus m_I is real analytic on $\mathbb{R}^s$ and is nonconstant, since its range closure contains both endpoints of a nondegenerate interval. We recall the full measure argument: the zero set of a nonzero real-analytic function F on $\mathbb{R}^d$ has measure zero. For $d = 1$, its zeros are isolated unless it is identically zero. For the induction step expand in the last coordinate at zero. At least one coefficient $\partial_d^k F(x', 0)$ is a nonzero analytic function of x' ; otherwise analyticity on each vertical line forces F to vanish everywhere. By induction, the zero set of this one coefficient has measure zero. Outside it the vertical restriction is not identically zero and its zero set has one-dimensional measure zero. Fubini on bounded boxes proves the assertion. Apply this to $F = m_I - a$ for every real a . Therefore an L^2 eigenvector at a , which would be supported on $\{m_I = a\}$, is zero. Nonreal eigenvalues are excluded by self-adjointness. Both sign regions contain nonempty open sets, since the range closure has a strictly negative and a strictly positive value. Their L^2 spaces are infinite-dimensional, for example by using infinitely many disjoint measurable subsets of a bounded open ball in each region. $\square$

The proof also gives the full spectral measure, not merely its support. For every Borel set $E \subseteq \mathbb{R}$,

$$E_I(E) = (\mathcal{F}^{\otimes s})^{-1} M_{\mathbf{1}_{\{m_I \in E\}}} \mathcal{F}^{\otimes s}, \quad \langle E_I(E)G, G \rangle = \int_{\{m_I(t) \in E\}} |(\mathcal{F}^{\otimes s}G)(t)|^2 dt. \quad (\text{FMS17})$$

In particular the sign projections are obtained with $E = (0, \infty)$ and $E = (-\infty, 0)$, and their sum is the identity. Zero is in the continuous spectrum: M_I is injective with dense range, because its range orthogonal complement is its kernel, but its range is not closed. If it were closed, its inverse on that range would be bounded, contradicting the approximate zero-eigenvectors constructed in the proof.

32.4 The actual pulse map and its complete compression

Retain the exact pulse choices of the supplied continuation:

$$\begin{aligned} M_I &= \prod_{i=1}^s p_i, & \varepsilon &= \frac{1}{16} \log \frac{M_I + 1}{M_I}, \\ w(u) &= \begin{cases} \exp[-1/(1 - (u/\varepsilon)^2)], & |u| < \varepsilon, \\ 0, & |u| \geq \varepsilon, \end{cases} & z &= w'' - \frac{1}{4}w, & N &= \int_{\mathbb{R}} z(u)^2 du, & (\text{FMS18}) \\ a_0 &= 0, & a_i &= \log p_i, & g_c(e^u) &= \sum_{j=0}^s c_j z(u - a_j), & c &\in \mathbb{C}^{s+1}. \end{aligned}$$

The exponent in w dominates every inverse power of the distance to its endpoints, so all derivatives vanish at the endpoints. Thus w and z are smooth. Moreover $\int zw = -\int |w'|^2 - \frac{1}{4} \int w^2 < 0$, proving $N > 0$. Two integrations by parts with the zero boundary terms give

$$\int_{\mathbb{R}} z(u) e^{cu} du = (c^2 - \frac{1}{4}) \int_{\mathbb{R}} w(u) e^{cu} du = 0, \quad c = \pm \frac{1}{2}.$$

Translation multiplies each zero by e^{ca_j} . Therefore $g_c \in \mathcal{D}_0$ for every complex coefficient vector, and every tensor product of these functions retains both moment conditions in each separate variable.

Put $C(v) = \int z(u)z(u-v) du$. It is real and even, vanishes outside $[-2\varepsilon, 2\varepsilon]$, has $C(0) = N$, and satisfies $|C(v)| \leq N$. The pulse centers have pairwise distance greater than 16ε . Indeed, for distinct positive integers A, D with $D \geq 2$,

$$|\log(A/D)| \geq \log((D+1)/D). \quad (\text{FMS19})$$

For $A > D$ this follows from $A \geq D+1$; for $A < D$ use $D/(D-1) > (D+1)/D$. Apply this to ratios of distinct primes, using $D < M_I$; the distances from 0 are at least $\log 2$. It follows that the map

$$\mathcal{U} : \mathbb{C}^{s+1} \longrightarrow H, \quad (\mathcal{U}c)(e^u) = N^{-1/2} \sum_{j=0}^s c_j z(u - a_j), \quad g_c = \sqrt{N} \mathcal{U}c \quad (\text{FMS20})$$

is an isometry. This changes no original amplitude: g_c is exactly the unscaled function in (FMS18).

Proposition 32.3 (Lag calculation of every matrix entry). *For every rational prime q ,*

$$W_q(g_c * g_c^*) = \begin{cases} 2N\beta_i \Re(\overline{c_0} c_i), & q = p_i, \\ 0, & q \notin I, \end{cases} \quad \beta_i = \frac{\log p_i}{\sqrt{p_i}}. \quad (\text{FMS21})$$

Consequently, writing E_{ab} for the matrix taking e_b to e_a and killing all other basis vectors,

$$\mathcal{U}^* B_{p_i} \mathcal{U} = P_i = \beta_i (E_{0i} + E_{i0}), \quad \mathcal{U}^* B_q \mathcal{U} = 0 \quad (q \notin I). \quad (\text{FMS22})$$

Proof. Expand the convolution with every complex coefficient:

$$(g_c * g_c^*)(e^v) = \sum_{j,k=0}^s c_j \overline{c_k} C(v - a_j + a_k).$$

Take $v = \log n$ with $n = q^m \geq 2$. Terms $j = k$ vanish because $\log n \geq \log 2 > 2\varepsilon$. If $j = 0, k > 0$, the argument is $\log(np_k) > 2\varepsilon$. If $j > 0, k = 0$, the argument is $\log(n/p_j)$; it is zero only when $n = p_j$, and otherwise (FMS19) makes its absolute value greater than 16ε . Finally, when $j, k > 0$ and $j \neq k$, the argument is $\log(np_k/p_j)$. It cannot be zero: distinct primes cannot satisfy $np_k = p_j$ with $n \geq 2$. Again (FMS19), with denominator $p_j < M_I$, gives a gap greater than 16ε . Hence the only nonzero value at $v = \log n$ is $Nc_i \overline{c_0}$ when $n = p_i$. The value at $-\log n$ is its conjugate. Unique factorization gives $q^m = p_i$ precisely for $q = p_i, m = 1$. Substitution in (FMS4) proves (FMS21). The norm relation in (FMS20) and polarization of the Hermitian forms then prove every entry of (FMS22). $\square$

Set $U_s = \mathcal{U}^{\otimes s}$ and $b = \prod_i \beta_i > 0$. Tensoring the entrywise compression, without suppressing any term, gives the self-adjoint matrix

$$\mathfrak{M}_I = U_s^* \mathbb{M}_I U_s = \sum_{\sigma \in S_s} P_{\sigma(1)} \otimes \cdots \otimes P_{\sigma(s)} \quad \text{on } (\mathbb{C}^{s+1})^{\otimes s}. \quad (\text{FMS23})$$

32.5 Every pulse eigenspace, its inertia, and its leakage

The orthonormal tensor basis is indexed by words $w = (w_1, \dots, w_s) \in \{0, 1, \dots, s\}^s$. A word is called admissible here when its nonzero labels are pairwise distinct; this finite-word condition is separate from the moment condition on test functions. For an admissible word let $J \subseteq \{1, \dots, s\}$ be its occupied positions and $S \subseteq \{1, \dots, s\}$ its labels. Then $|J| = |S| = k$. Let $\mathcal{H}_{J,S}$ be the span of the $k!$ bijective assignments $J \rightarrow S$, with zeros in the other positions. Define

$$u_{J,S} = \frac{1}{\sqrt{k!}} \sum_{w \in \mathcal{H}_{J,S}} e_w. \quad (\text{FMS24})$$

The sum notation means the basis words spanning the indicated block.

Theorem 32.4 (Complete pulse spectrum). *For every complementary pair (J, S) and (J^c, S^c) the vectors*

$$v_{J,S}^\pm = \frac{u_{J,S} \pm u_{J^c,S^c}}{\sqrt{2}} \quad (\text{FMS25})$$

are eigenvectors of $\mathfrak{M}_I$ with respective eigenvalues

$$\pm b\sqrt{k!(s-k)!}.$$

Choose one member of each complementary pair to avoid duplication. All other directions are zero directions: the zero-sum subspace in each $\mathcal{H}_{J,S}$ and the span of words with a repeated nonzero label. These form its entire kernel.

For $0 \leq k \leq \lfloor s/2 \rfloor$ define

$$\nu_k = \begin{cases} \binom{s}{k}^2, & k < s/2, \\ \frac{1}{2}\binom{s}{s/2}^2, & k = s/2 \quad (s \text{ even}). \end{cases} \quad (\text{FMS26})$$

The two eigenvalues of magnitude $b\sqrt{k!(s-k)!}$ each have multiplicity ν_k . Writing $n_s = \frac{1}{2}\binom{2s}{s}$ and $d_s = (s+1)^s - \binom{2s}{s}$, the exact inertia and characteristic polynomial are

$$(n_+, n_-, n_0) = (n_s, n_s, d_s), \quad \det(\lambda I - \mathfrak{M}_I) = \lambda^{d_s} \prod_{k=0}^{\lfloor s/2 \rfloor} (\lambda^2 - b^2 k!(s-k)!)^{\nu_k}. \quad (\text{FMS27})$$

Its operator norm and product of all nonzero eigenvalues are

$$\|\mathfrak{M}_I\| = \sqrt{s!} b, \quad \text{pdet}(\mathfrak{M}_I) = (-1)^{n_s} b^{2n_s} \prod_{k=0}^{\lfloor s/2 \rfloor} [k!(s-k)!]^{\nu_k}. \quad (\text{FMS28})$$

Proof. The factor P_i kills e_j unless $j = 0$ or $j = i$, and sends e_0 to $\beta_i e_i$ and e_i to $\beta_i e_0$. For a permutation summand to act nontrivially on a word, the prime assigned to an occupied position must equal its label. Two equal nonzero labels therefore force every summand to vanish. For an admissible word in $\mathcal{H}_{J,S}$, the assigned prime is fixed at each occupied position, and the remaining primes may be assigned bijectively to J^c in exactly $(s-k)!$ ways. Each such assignment has coefficient b and gives one of the distinct complementary words. Thus

$$\mathfrak{M}_I e_w = b \sum_{v \in \mathcal{H}_{J^c,S^c}} e_v, \quad \mathfrak{M}_I u_{J,S} = b\sqrt{k!(s-k)!} u_{J^c,S^c}. \quad (\text{FMS29})$$

The matrix between the two word blocks is the constant-entry matrix $b\mathbf{1}$. Its only nonzero singular value is $b\sqrt{k!(s-k)!}$; the zero-sum subspaces are killed. Equivalently, summing the first equality proves the second, and applying the same argument to the complement gives the two-by-two matrix

$$\begin{array}{c|cc} & u_{J,S} & u_{J^c,S^c} \\ \hline u_{J,S} & 0 & b\sqrt{k!(s-k)!} \\ u_{J^c,S^c} & b\sqrt{k!(s-k)!} & 0 \end{array}.$$

Its two orthonormal eigenvectors are (FMS25). The word blocks and repeated-label words exhaust the tensor basis, so the listed kernel and eigenspaces are complete.

There are $\binom{s}{k}^2$ pairs (J, S) of size k . For $k < s/2$ each complementary pair has exactly one such member. At $k = s/2$ complementation has no fixed pair: no subset of a nonempty position set equals its complement. Hence divide the count by two in that case. This proves (FMS26). Counting s chosen elements from two disjoint sets of s elements gives Vandermonde's identity $\sum_k \binom{s}{k}^2 = \binom{2s}{s}$, so $\sum_{k \leq s/2} \nu_k = n_s$. The total dimension is $(s+1)^s$; this proves the inertia and characteristic polynomial. Finally $k!(s-k)! = s!/\binom{s}{k} \leq s!$, with equality at $k = 0$. Taking the largest magnitude and multiplying each positive-negative pair proves (FMS28). $\square$

For $s = 2$ the inertia is $(3, 3, 3)$; the nonzero magnitudes are $\sqrt{2}b$ once per sign and b twice per sign. For $s = 3$ the inertia is $(10, 10, 44)$; the magnitudes are $\sqrt{6}b$ once per sign and $\sqrt{2}b$ nine times per sign. These are evaluations of the general proof, not numerical evidence substituted for it.

The comparison with the full operator has an exact extra map. Put $\Pi_s = U_s U_s^*$, the orthogonal projection onto the pulse tensor space, and define

$$L_I = (I - \Pi_s) \mathbb{M}_I U_s : (\mathbb{C}^{s+1})^{\otimes s} \longrightarrow (\text{ran } U_s)^\perp. \quad (\text{FMS30})$$

Then

$$\begin{aligned} \mathbb{M}_I U_s &= U_s \mathfrak{M}_I + L_I, & U_s^* L_I &= 0, \\ L_I^* L_I &= U_s^* \mathbb{M}_I^2 U_s - \mathfrak{M}_I^2, & \|\mathbb{M}_I U_s v\|^2 &= \|\mathfrak{M}_I v\|^2 + \|L_I v\|^2. \end{aligned} \quad (\text{FMS31})$$

To prove the first line, split the identity into $\Pi_s + (I - \Pi_s)$ and use (FMS23). For the second line multiply $U_s^* \mathbb{M}_I (I - \Pi_s) \mathbb{M}_I U_s$ out, using $\mathbb{M}_I = \mathbb{M}_I^*$ and $U_s^* U_s = I$; orthogonality gives the norm equality.

The restriction of L_I to every eigenspace of $\mathfrak{M}_I$ is injective. Indeed if $\mathfrak{M}_I v = \lambda v$ and $L_I v = 0$, then the first line of (FMS31) gives $\mathbb{M}_I U_s v = \lambda U_s v$. The full operator has no eigenvectors, so $U_s v = 0$ and then $v = 0$. In particular

$$L_I : \ker \mathfrak{M}_I \hookrightarrow (\text{ran } U_s)^\perp, \quad L_I v = \mathbb{M}_I U_s v \quad (v \in \ker \mathfrak{M}_I). \quad (\text{FMS32})$$

Thus every nonzero radical vector in the specified compression survives under the full operator through this explicit injection. For an individual normalized compression eigenvector v , $\|(\mathbb{M}_I - \lambda)U_s v\| = \|L_I v\| > 0$. The finite and infinite spectral assertions are connected by (FMS31); neither is substituted for the other.

32.6 All higher tensor orders on the same tests

Here the common term on the pulse space is exactly the Archimedean term, since (FMS21) proves that every unselected finite prime is zero. To retain the exact coefficient receiver, put

$$K(t) = \frac{e^{-t/2}}{1 - e^{-2t}}, \quad c_{\text{arch}} = \log(4\pi) + \gamma_E,$$

and define the convergent integrals

$$\begin{aligned} A_\varepsilon &= c_{\text{arch}} N + 2 \int_0^\infty [C(t) - N e^{-t/2}] K(t) dt, \\ J_\varepsilon(d) &= \int_{d-2\varepsilon}^{d+2\varepsilon} C(t-d) K(t) dt \quad (d > 16\varepsilon). \end{aligned} \quad (\text{FMS33})$$

The integrand of the first line is bounded near zero, since $C(t) - N e^{-t/2} = O(t)$ and $K(t) = O(t^{-1})$. The original negative subtraction is exponentially integrable outside the support of C . Expansion of (FMS10), using the convolution in the proof of Proposition 32.3, gives

$$\begin{aligned} W_{\text{arch}}(g_c * g_c^*) &= A_\varepsilon \sum_{j=0}^s |c_j|^2 + 2 \sum_{0 \leq j < k \leq s} \Re(\overline{c_j} c_k) J_\varepsilon(|a_j - a_k|), \\ W_A &= -A_\varepsilon I - [J_\varepsilon(|a_j - a_k|)]_{j \neq k} - N \sum_{p_i \in A} \beta_i (E_{0i} + E_{i0}), \\ Q_A(g_c) &= c^* W_A c. \end{aligned} \quad (\text{FMS34})$$

The off-diagonal matrix in the second line has zero diagonal. For each pair of different centers, one of the two shifted correlations is supported on the positive half-axis and the other on the

negative half-axis. The two reciprocal values in (FMS10) therefore yield precisely the factor $2\Re(\overline{c_j}c_k)$ in the first line, with no assumption that c is real. The same formula follows when two positive lag intervals overlap, since it is a linear integral of their sum.

The companion finite-pulse theorem [28], FP37 with its complete proof FP38–43, with exactly (FMS18) and (FMS33), proves the quantitative matrix inequality

$$W_A \succeq \mu_* N I_{s+1} \quad (A \subseteq I), \quad \mu_* = \frac{6600077509}{144027072000} > \frac{1}{22}. \quad (\text{FMS35})$$

This is a proved theorem about the entire $(s+1)$ -dimensional complex pulse coefficient space and every finite support face. It is the only positivity input below. It makes no assertion on other compact tests or on a bounded global Weil operator. The companion proof evaluates the full subtraction in (FMS33); it is not an assumption that that integral has the desired sign.

For the original unscaled g_c put

$$A_c = -W_{\text{arch}}(g_c * g_c^*), \quad \alpha_i = 2N\beta_i \Re(\overline{c_0}c_i), \quad Q_A(g_c) = A_c - \sum_{p_i \in A} \alpha_i. \quad (\text{FMS36})$$

The symbol α_i is the quantity denoted a_i in the supplied continuation's Section 8; it is not the logarithmic pulse center a_i in (FMS18). This distinction changes no parameter or formula. For integers $r \geq 0$ define the literal support observation

$$J_r(c) = \sum_{A \subseteq I} (-1)^{|A|} \left(A_c - \sum_{p_i \in A} \alpha_i \right)^r. \quad (\text{FMS37})$$

At $r = 0$ each power is defined to be 1, as for a zeroth tensor power; the signed sum is zero since $s > 0$.

Theorem 32.5 (All orders, signs, bounds, and exact rate). *For every $c \in \mathbb{C}^{s+1}$, $J_r(c) = 0$ for $r < s$. For $r \geq s$,*

$$J_r(c) = \frac{r!}{(r-s)!} \left(\prod_{i=1}^s \alpha_i \right) \int_{[0,1]^s} \left(A_c - \sum_{i=1}^s t_i \alpha_i \right)^{r-s} dt. \quad (\text{FMS38})$$

For nonzero c , set

$$m_c = A_c - \sum_i (\alpha_i)_+, \quad M_c = A_c + \sum_i (-\alpha_i)_+. \quad (\text{FMS39})$$

These are the smallest and largest actual support-face values, and $m_c \geq \mu_* N \|c\|_2^2 > 0$. For every $r \geq s$ one has

$$\text{sgn } J_r(c) = \text{sgn} \prod_{i=1}^s \Re(\overline{c_0}c_i), \quad (\text{FMS40})$$

with $J_r(c) = 0$ exactly when at least one factor on the right vanishes. The complete quantitative bounds are

$$\frac{r!}{(r-s)!} \left| \prod_i \alpha_i \right| m_c^{r-s} \leq |J_r(c)| \leq \frac{r!}{(r-s)!} \left| \prod_i \alpha_i \right| M_c^{r-s}. \quad (\text{FMS41})$$

If every α_i is nonzero, then

$$\lim_{r \rightarrow \infty} |J_r(c)|^{1/r} = M_c, \quad \frac{|J_r(c)|}{M_c^r} \longrightarrow 1. \quad (\text{FMS42})$$

Proof. For a differentiable function F and a real α , including negative α , the fundamental theorem of calculus gives

$$F(x) - F(x - \alpha) = \alpha \int_0^1 F'(x - t\alpha) dt.$$

Apply this identity successively with $\alpha = \alpha_1, \dots, \alpha_s$ to $F(x) = x^r$ at $x = A_c$. The left side expands to exactly (FMS37), with its original sign $(-1)^{|A|}$. The s th derivative is zero for $r < s$, and is $r!(r-s)!^{-1}x^{r-s}$ otherwise. This proves both assertions through (FMS38), also when some $\alpha_i = 0$.

The affine function on the cube in (FMS38) is minimized by $t_i = 1$ for $\alpha_i > 0$ and $t_i = 0$ for $\alpha_i < 0$; its maximum uses the reverse choices. This proves (FMS39), and these choices are cube vertices, hence actual face values. The finite theorem (FMS35) gives the stated positive lower bound. The integral in (FMS38) is therefore strictly positive for every $r \geq s$. Since N and every β_i are strictly positive, the sign of its prefactor is exactly (FMS40). Bounding the integrand by its minimum and maximum, on a cube of volume one, proves (FMS41). At $c = 0$ all J_r with $r > 0$ are zero, so no separate positivity claim is required.

When every $\alpha_i \neq 0$, the unique maximizing face is $A_{\max} = \{p_i : \alpha_i < 0\}$, and its sign in (FMS37) is $\epsilon_c = (-1)^{|A_{\max}|} = \text{sgn} \prod_i \alpha_i$. All other corner values are strictly between 0 and M_c . Dividing (FMS37) by M_c^r proves convergence to ϵ_c and hence both limits. More explicitly, if $d_c = \min_i |\alpha_i| > 0$, every other corner is at most $M_c - d_c$, and all corners are positive. Thus the exact finite sum supplies the error bound

$$|J_r(c) - \epsilon_c M_c^r| \leq (2^s - 1)(M_c - d_c)^r. \quad (\text{FMS43})$$

□

For literal tests realizing all three possibilities, take $c_0 = 1$. The choice $c_i = 1$ for every i gives $J_r(c) > 0$ for every $r \geq s$. The choice $c_1 = -1$ and $c_i = 1$ for $i > 1$ gives $J_r(c) < 0$ at every such order. The choice $c_1 = i$ and $c_j = 1$ for $j > 1$ gives $J_r(c) = 0$ at every order, while $g_c \neq 0$ and every face form remains strictly positive by (FMS35). These are the original amplitude choices in (FMS18), including the complex phase in the third test.

These formulas retain all polynomial coefficients. An additional way to see them simultaneously is the exact exponential generating identity and its coefficient formula:

$$\begin{aligned} \sum_{r=0}^{\infty} \frac{J_r(c)}{r!} z^r &= e^{A_c z} \prod_{i=1}^s (1 - e^{-\alpha_i z}), \\ J_r(c) &= \sum_{\substack{n_0 \geq 0, n_i \geq 1 \\ n_0 + n_1 + \dots + n_s = r}} \frac{r!}{n_0! n_1! \dots n_s!} A_c^{n_0} \prod_{i=1}^s (-1)^{n_i+1} \alpha_i^{n_i}. \end{aligned} \quad (\text{FMS44})$$

The first identity follows by summing the finite subset exponential and factoring; expanding the $s+1$ entire power series proves the second. In particular $J_s(c) = s! \prod_i \alpha_i$ and $J_{s+1}(c) = (s+1)! \prod_i \alpha_i (A_c - \frac{1}{2} \sum_i \alpha_i)$. The first formula agrees exactly with the compression, including its original mass:

$$J_s(c) = N^s \langle \mathfrak{M}_I c^{\otimes s}, c^{\otimes s} \rangle = s! (2N)^s \left(\prod_i \beta_i \right) \prod_i \Re(\overline{c_0} c_i). \quad (\text{FMS45})$$

Indeed the matrix of q_A on $\mathcal{UC}^{s+1}$ is $N^{-1}W_A$. Proposition 32.1 at order s and (FMS23) give the first equality. The product of the individual star-matrix values gives the second.

Finally all orders have a finite trace receiver that preserves the support diagonal. Let Π_A be projection onto $\mathbb{C}\delta_A$ in V_I , let $D_{h_I}\delta_A = (-1)^{|A|}\delta_A$, and for a finite vector v put $R_v x = \langle x, v \rangle v$.

Then

$$\begin{aligned} \mathbf{O}_r &= \sum_{A \subseteq I} \Pi_A \otimes \mathbf{W}_A^{\otimes r}, \\ \text{Tr}[(D_{h_I} \otimes R_{c^{\otimes r}})\mathbf{O}_r] &= J_r(c). \end{aligned} \tag{FMS46}$$

For one support block, the rank-one trace is $(c^* \mathbf{W}_A c)^r$, because $\text{Tr}(TR_v) = \langle Tv, v \rangle$; an orthonormal basis expansion proves this identity directly. Summing the support blocks proves (FMS46). Each $\mathbf{W}_A$ is positive by the proved finite bound, so $\mathbf{O}_r$ is positive. The actual inserted D_{h_I} has both signs and is not a positive functional. Consequently both signs in (FMS40) are compatible with the positive face matrices through this exact trace map. This finite receiver does not take an unspecified infinite trace.

Precise scope of the result. The full Hilbert-space assertion is the evaluated bounded operator (FMS14), with every spectral projection in (FMS17). The finite tensor eigenspaces are its exact compression through (FMS20), with the complementary map (FMS30). The all-order trace keeps the same support diagonal and the exact common Archimedean number. These maps prove how the different sign calculations are related. They establish no global Weil positivity theorem, no zero off the critical line, and no new purity statement.

33 The complete prime-word graph and the fate of every packet class

This calculation uses the literal source family of the received *Constructive two-prime packets: exact theta lifts and bounded moment dynamics*, results SZ-20260923-901–903. Both of its complete TeX sources, `NOTE.tex` and `GENERATOR.tex`, were read. Their local source identities and original human reading coverage are retained in the private source ledger [29]. The mathematical context for the original theta quotient is Meyer [20]; the entire primitive section used here is proved in PL1–27 above. The present chapter computes the completions and their exact maps, using the new word family without changing its source metric.

33.1 The original source and its full finite observation

Retain the finite reflected packet Z , all multiplicities m_ρ , the original coordinates $t_\rho = s - \rho$, and $h(s) = c_h \prod_\rho (s - \rho)^{m_\rho}$. Put $d = \sum_\rho m_\rho$ and $A = \prod_\rho \mathbb{C}[t_\rho]/(t_\rho^{m_\rho})$. The primitive map $S : A \rightarrow L^2(\mathbb{R}_{>0}, dx)$ is exactly $Su = \text{Prim}_h(u)$ of PL; its Mellin transform is

$$\mathcal{M}(Su)(s) = a(s) \frac{g(s)}{h(s)} R_u(s), \quad a(s) = s(s-1), \quad g = 2\xi, \quad R_u = \text{rem}_h([a(g/h)]^{-1}u). \tag{GC1}$$

It has full selected jets u and vanishes at every unselected zero to its full order. Consequently S is injective. Its positive Gram $G = S^*S$ retains the original measure dx and all original local units; it is not the reflected Weil form. The latter's exact connecting formula is GC11 below.

Let $H_c = a_2 + cb_2 + c^2a_3 + c^3b_3$, where $\lambda_p = p^{s-1/2}$, $a_p = (\lambda_p + \lambda_p^{-1})/2$ and $b_p = (\lambda_p - \lambda_p^{-1})/(2i)$. Choose the finite selector c of CG3–5 in the received proof. To recall why it exists, the original prime characters for 2 and 3 separate every pair of distinct roots: equality would imply an impossible rational relation between $\log 2$ and $\log 3$. Each value collision or vanishing derivative is therefore the zero of a nonzero polynomial in c of degree at most three. Excluding the derivatives only at multiple roots leaves at most $3(\binom{r}{2} + r_2)$ bad integers, where $r = |Z|$ and r_2 counts the multiple roots. One of the integers from zero to that number avoids them all. Thus $z_\rho = H_c(\rho)$ are distinct, and the local linear term of $H_c - z_\rho$ is nonzero at each nonsimple factor. Substitution there is triangular with invertible diagonal; interpolation at the distinct nodes then gives the

$$\begin{array}{ccc}
(\mathbb{C}^{s+1})^{\otimes s} & \xrightarrow{U_s} & H^{\otimes s} \\
\mathfrak{M}_I \downarrow & & \downarrow \mathbb{M}_I \\
(\mathbb{C}^{s+1})^{\otimes s} & \xrightarrow[U_s^*]{U_s} & H^{\otimes s}
\end{array}$$

The square has the exact defect

$$\mathbb{M}_I U_s - U_s \mathfrak{M}_I = L_I, \quad U_s^* L_I = 0, \quad L_I^* L_I = U_s^* \mathbb{M}_I^2 U_s - \mathfrak{M}_I^2.$$

The full operator. Its spectrum is the entire interval

$$[-s! \rho_I \prod_i u_i, s! \prod_i u_i], \quad u_i = \frac{2 \log p_i}{\sqrt{p_i} - 1}, \quad \rho_I = \max_i \frac{\sqrt{p_i} - 1}{\sqrt{p_i} + 1}.$$

Both signs have infinite-dimensional spectral subspaces. There are no eigenvectors, including at zero.

The complete finite compression for two primes. Put $b = (\log p_1)(\log p_2)/\sqrt{p_1 p_2}$. Every one of its nine dimensions is represented in this table:

eigenvalue	$-\sqrt{2}b$	$-b$	0	b	$\sqrt{2}b$
multiplicity	1	2	3	2	1

The three-dimensional zero eigenspace maps injectively through L_I into the orthogonal complement of the pulse space. This is the calculated fate of its apparent zero directions under the original full operator.

The Weil face matrices on the same tests. Every face retains its original Archimedean integral and obeys $W_A \succeq \mu_* N I$, where $\mu_* = 6600077509/144027072000 > 1/22$. Their mixed tensor trace inserts D_{h_I} , whose diagonal entries are $(-1)^{|A|}$. The sign in the mixed trace comes from this specified insertion; the face matrices remain positive.

Figure 23: The exact maps from the finite tests to the full mixed-prime operator, including the square's nonzero defect. FMS14–17 prove the full spectrum, FMS20–32 give the compression and complementary map, and FP37–43 and FMS46 prove the positive face bound and signed trace. All original pulse amplitudes are recovered by $g_c = \sqrt{N} \mathcal{U} c$. The local analytic distributions are those of Connes [5], with the programme's prime-energy receiver in the complete source [26].

unital surjection

$$E : \mathbb{C}[X] \rightarrow A, \quad P \mapsto P(H_c), \quad \ker E = (\nu), \quad \nu(X) = \prod_{\rho} (X - z_{\rho})^{m_{\rho}} \in \mathbb{R}[X]. \quad (\text{GC2})$$

For completeness, the interpolation assertion follows by solving the finite inverse of each local substitution coefficient in increasing degree, then using the relatively prime powers $(X - z_{\rho})^{m_{\rho}}$. Their product divides exactly the polynomials whose full substituted jets vanish. The resulting map is onto and its kernel is the displayed degree- d ideal. The conjugation $H_c^{\dagger} = H_c$ makes the nodes conjugate in pairs with equal multiplicities, proving the real coefficients of ν . No nilpotent order has been removed.

Write H_c for the actual four-dilation operator:

$$H_c = \frac{1-ic}{2}V_2 + \frac{1+ic}{2}V_{1/2} + \frac{c^2-ic^3}{2}V_3 + \frac{c^2+ic^3}{2}V_{1/3}, \quad (V_q F)(x) = q^{-1/2}F(x/q). \quad (\text{GC3})$$

Each V_q is unitary for dx , with adjoint $V_{q^{-1}}$. Therefore H_c is bounded and self-adjoint. Its real Mellin-line multiplier and bound are

$$\begin{aligned} x_c(t) &= \cos(t \log 2) + c \sin(t \log 2) + c^2 \cos(t \log 3) + c^3 \sin(t \log 3), \\ |x_c(t)| &\leq K_c = (1 + c^2)^{3/2}. \end{aligned} \quad (\text{GC4})$$

Put $F_* = S1$ and $\Phi(P) = P(H_c)F_*$. Mellin Plancherel, with its original factor, gives the finite positive measure

$$\int f(x) d\mu(x) = \frac{1}{2\pi} \int_{\mathbb{R}} f(x_c(t)) |\mathcal{M}F_*(1/2 + it)|^2 dt, \quad \|\Phi(P)\|^2 = \int |P|^2 d\mu. \quad (\text{GC5})$$

Its mass is $\|F_*\|^2$, and its support lies in $[-K_c, K_c]$. The original source relation from PL and the received proof is $\Phi(P) = S(EP) + \Theta \mathcal{D}_P(1)$ with its explicitly integrated source $\mathcal{D}_P$. In particular $\Phi(\nu Q) = \Theta \mathcal{D}_{\nu Q}(1)$ is an actual theta relation, not merely a vanished finite observation. Conversely every theta relation has zero selected jets, so the kernel in GC2 is exactly the relation ideal in this word family.

33.2 Every native minimum tends to zero

The measure μ has no atoms. Indeed x_c is a nonconstant real-analytic function: its four distinct exponential frequencies are linearly independent, and the coefficient of the frequency $\log 2$ is nonzero. Their linear independence follows by differentiating a vanishing linear combination zero through order three; the resulting Vandermonde determinant is nonzero. Every level set of a nonconstant real-analytic function of one variable is discrete and therefore countable. Its inverse image under GC5 has Lebesgue measure zero, proving the assertion. Thus $\mu(\nu^{-1}(0)) = 0$.

Polynomials are dense in $\mathcal{H} = L^2(\mu)$. Here are the approximation details used in this statement. For a finite Borel measure on a compact interval, continuous functions are dense in L^2 : approximate the indicator of an open set O by $\min(1, n \operatorname{dist}(x, O^c))$ and use dominated convergence. Complements, finite intersections (take products of bounded approximants) and increasing countable unions show that indicators of all Borel sets are in the same closure; simple-function approximation finishes the claim. After an affine map to $[0, 1]$, the Bernstein polynomials of a continuous f approximate it uniformly. Indeed their error is bounded by its modulus of continuity at δ , plus $2\|f\|_{\infty}/(4n\delta^2)$, by the mean x and variance $x(1-x)/n$ of the binomial weights. First choose δ small and then n large. This proves the required polynomial density.

Multiplication by ν on $\mathcal{H}$ is bounded and self-adjoint, with zero kernel, because ν is real on the interval and vanishes on a set of zero measure. Its range is dense: the orthogonal complement of the range is its adjoint's kernel. Continuity of multiplication and polynomial density therefore imply

$$\overline{\nu \mathbb{C}[X]}^{L^2(\mu)} = \mathcal{H}. \quad (\text{GC6})$$

For $n \geq d - 1$, let G_n be the attained minimum Gram on A in its original ascending jet coordinates:

$$\langle G_n u, u \rangle = \min\{\|P\|_{L^2(\mu)}^2 : \deg P \leq n, EP = u\}. \quad (\text{GC7})$$

Each is positive definite by polynomial injectivity in $L^2(\mu)$; that injectivity follows because μ has infinitely many support points whereas a nonzero polynomial has finitely many zeros. The feasible space is a nonempty finite-dimensional affine space, so the minimum is attained. The Grams decrease in the positive-semidefinite order. For a fixed representative P_u , GC6 approximates $-P_u$ by νQ_j in $L^2(\mu)$. Then $P_u + \nu Q_j$ retains exactly u and its norm tends to zero. Consequently $G_n \downarrow 0$ as $n \rightarrow \infty$. This convergence is in operator norm in the finite-dimensional original coordinates: the positive diagonal entries tend to zero, their sum is the trace, and the norm is bounded by that sum.

Let E_0 be the full confluent jet matrix with columns $1, H_c, \dots, H_c^{d-1}$. If G_n^{word} denotes the received proof's quotient Gram in $1, X, \dots, X^{d-1}$, then

$$G_n = (E_0^{-1})^* G_n^{\text{word}} E_0^{-1}, \quad \det G_n = \frac{D_{n+1}}{|\det E_0|^2 B_{n-d+1}} \rightarrow 0. \quad (\text{GC8})$$

The numerator and denominator are the actual source and relation Hankel determinants of GC5, with the original measure and no rescaling. Explicitly D_{n+1} is the Gram determinant of $1, X, \dots, X^n$ and B_{n-d+1} is the Gram determinant of $\nu, X\nu, \dots, X^{n-d}\nu$, with $B_0 = 1$. The combined basis $1, \dots, X^{d-1}, \nu, \dots, X^{n-d}\nu$ has determinant one relative to the monomial basis because ν is monic. Subtracting its attained relation projections gives the Schur complement G_n^{word} and leaves the relation block. The determinant therefore factors as $D_{n+1} = B_{n-d+1} \det G_n^{\text{word}}$. The same quotient vector has original jet coordinates $E_0 v$ when its word coordinates are v , proving the displayed coordinate change and retaining $|\det E_0|^2$.

33.3 An explicit sequence retaining every prescribed jet

The convergence above admits a specified polynomial sequence. For $u \in A$ let P_u be its unique degree-below- d representative under E . Set $K = K_c$, $M_0 = \mu(\mathbb{R}) = \|F_*\|^2$, and use the actual suprema on $[-K, K]$:

$$f_\varepsilon(x) = -\frac{P_u(x)\nu(x)}{\nu(x)^2 + \varepsilon^2}, \quad L_\varepsilon = \frac{\|P'_u\|_\infty}{2\varepsilon} + \frac{\|P_u\|_\infty \|\nu'\|_\infty}{\varepsilon^2}. \quad (\text{GC13})$$

The derivative of f_ε has modulus at most L_ε : use $|\nu|/(\nu^2 + \varepsilon^2) \leq 1/(2\varepsilon)$ and $|\varepsilon^2 - \nu^2|/(\nu^2 + \varepsilon^2)^2 \leq 1/\varepsilon^2$. For $n \geq 1$, put $y = (x + K)/(2K)$ and define the explicit degree-at-most- n polynomial

$$q_{\varepsilon,n}(x) = \sum_{k=0}^n f_\varepsilon(-K + 2Kk/n) \binom{n}{k} y^k (1-y)^{n-k}, \quad P_{\varepsilon,n} = P_u + \nu q_{\varepsilon,n}. \quad (\text{GC14})$$

These exact coefficients use the original word variable; the actual source is $P_{\varepsilon,n}(H_c)F_*$. One has $EP_{\varepsilon,n} = u$, including every derivative order, and $\deg P_{\varepsilon,n} \leq d + n$. The variance of the Bernstein sampling point on $[-K, K]$ is at most K^2/n , so Cauchy-Schwarz and the derivative bound give $\|q_{\varepsilon,n} - f_\varepsilon\|_\infty \leq KL_\varepsilon/\sqrt{n}$. Since $P_u + \nu f_\varepsilon = P_u \varepsilon^2/(\nu^2 + \varepsilon^2)$, the triangle inequality yields the complete attained upper estimate

$$\begin{aligned} \langle G_{d+n} u, u \rangle^{1/2} &\leq \|P_{\varepsilon,n}\|_{L^2(\mu)} \\ &\leq \left(\int \frac{|P_u(x)|^2 \varepsilon^4}{(\nu(x)^2 + \varepsilon^2)^2} d\mu(x) \right)^{1/2} + \frac{\|\nu\|_\infty \sqrt{M_0} KL_\varepsilon}{\sqrt{n}}. \end{aligned} \quad (\text{GC15})$$

Taking $\varepsilon = 1/j$ and $n = j^6$ gives a predetermined sequence with original jets exactly u and source norm tending to zero. The second term is bounded by a fixed packet-dependent constant times j^{-1} ; the integral tends to zero by dominated convergence and the zero measure of $\nu^{-1}(0)$ proved above. No unproved quantitative rate for that integral is inserted. This supplies explicit polynomial representatives for all the vertical graph limits used next.

33.4 The completed two-observation space retains the entire packet

Consider now the actual two-observation map

$$P \longmapsto (\Phi(P), S(EP)), \quad \|P\|_{\text{graph}}^2 = \|P\|_{L^2(\mu)}^2 + \|S(EP)\|_{L^2(dx)}^2. \quad (\text{GC9})$$

Both terms use the original source observation dx ; the second uses its already defined primitive section. No new form on the packet is selected. Under GC5 its graph is (P, EP) inside $\mathcal{H} \oplus A_G$, where A_G is A with Gram $G = S^*S$. Its closure is the entire direct sum. To prove this, a vector (f, v) orthogonal to that graph is first tested on $P = \nu Q$. GC6 forces $f = 0$. The surjectivity of E then forces $v = 0$. The double orthogonal complement proves the claimed density.

The closure of the original relation ideal in this completed graph is exactly $\mathcal{H} \oplus 0$, by GC6 and GC2. Therefore the complete quotient, with its attained metric, is

$$\frac{\overline{\{(P, EP) : P \in \mathbb{C}[X]\}}}{\overline{\{(\nu Q, 0) : Q \in \mathbb{C}[X]\}}} = \frac{\mathcal{H} \oplus A_G}{\mathcal{H} \oplus 0} \xrightarrow{[(f, u)]} u \cong A_G. \quad (\text{GC10})$$

At finite cutoff its Gram in the original jets is exactly $G + G_n$, since the second observation is constant on the fibre $EP = u$. Hence its limit is G , and its determinant tends to $\det G > 0$. The forgetful projection onto the first source observation has kernel $0 \oplus A$. This explicitly identifies every class lost by that projection, including all nilpotent jets.

The extension of the original reflected Weil form is equally explicit:

$$\mathcal{W}((f, u), (h, v)) = R_Z(u, J_g v) = \sum_{\rho \in Z} m_\rho \overline{u(1 - \bar{\rho})} v(\rho). \quad (\text{GC11})$$

On polynomial graph vectors this is the original full Weil pairing, since GC1 makes all other zero terms vanish. It extends continuously by the finite projection to A_G ; this extension has radical $\mathcal{H} \oplus \prod_\rho (t_\rho)$. The exact map relating GC9's positive observation norm to GC11 is the second projection, followed by the residue map and J_g . For r_f reflection-fixed roots and r_p two-element reflection orbits, its finite nonzero signature is $(r_f + r_p, r_p)$, with $d - r_f - 2r_p$ zero jet directions. This follows from the value block m_ρ at a fixed root and the block $\begin{pmatrix} 0 & m_\rho \\ m_\rho & 0 \end{pmatrix}$ at each pair. Thus no identification with the positive graph norm is used to assign a sign to this arithmetic form.

Finally multiplication by X on the dense polynomial graph extends to (M_x, M_{H_c}) on $\mathcal{H} \oplus A_G$. Its bound is $\max(K_c, \|M_{H_c}\|_G)$, since the two factors have these bounds and $E(XP) = H_c EP$. The first component is self-adjoint. The second has the precise original defect

$$S^* E_c - E_c^* S = M_{H_c}^* G - G M_{H_c}, \quad E_c = H_c S - S M_{H_c} = \Theta \Psi_c. \quad (\text{GC12})$$

Expanding both sides and using $H_c^* = H_c$ proves the equality. Every discrepancy survives as the given theta cochain and the retained packet component. These are linear and Hilbert-space receivers of the already defined full-support maps; no algebra structure on all of $L^2(\mu)$ or equality of arithmetic forms is asserted.

33.5 Finite-cutoff errors in the retained coordinates

The convergence above also controls the complete finite Gram, without a choice of a unit-mass source. Let $e_1, \dots, e_d$ be the original ascending jet basis of A , and apply GC13–14 to each $u = e_i$, with $\varepsilon = j^{-1}$ and $n = j^6$. Denote the resulting polynomial by $P_{i,j}$. Define the actual matrix

$$T_j = \left(\int \overline{P_{i,j}(x)} P_{l,j}(x) d\mu(x) \right)_{i,l=1}^d, \quad \alpha_j = \|G^{-1/2} T_j G^{-1/2}\|. \quad (\text{GC16})$$

All entries use the original measure GC5 and the explicit coefficients GC14. The lift $u \mapsto \sum_i u_i P_{i,j}$ has observation u , so minimization proves

$$0 < G_n \preceq T_j \preceq \alpha_j G \quad (n \geq d + j^6), \quad \alpha_j \longrightarrow 0. \quad (\text{GC17})$$

Indeed GC15 sends each diagonal of T_j to zero, and its positive trace bounds its operator norm. The fixed positive matrix G then proves the last limit. No rate uniform in a growing packet is inferred from this fixed-packet statement.

Write $\tilde{G}_n = G + G_n$ for the exact graph quotient. In a fixed original subspace frame I with r independent columns, put $G^I = I^*GI$ and $\tilde{G}_n^I = I^*\tilde{G}_nI$. The complete restricted error satisfies

$$0 \leq \log \frac{\det \tilde{G}_n^I}{\det G^I} \leq r \log(1 + \alpha_j) \quad (n \geq d + j^6). \quad (\text{GC18})$$

Congruence by I preserves GC17; congruence by $(G^I)^{-1/2}$ then puts every eigenvalue between 1 and $1 + \alpha_j$. Taking their product proves GC18. In particular, at the four unchanged indices $q - 1, q, 2q - 1, 2q$, whenever $q - 1 \geq d + j^6$,

$$\begin{aligned} 0 &\leq \log \det \tilde{G}_{q-1}^I + \log \det \tilde{G}_q^I - \log \det \tilde{G}_{2q-1}^I - \log \det \tilde{G}_{2q}^I \\ &\leq 2r \log(1 + \alpha_j). \end{aligned} \quad (\text{GC19})$$

The lower bound uses the decreasing original minima separately in the pairs $(q - 1, 2q - 1)$ and $(q, 2q)$. The fixed $\log \det G^I$ cancels with signs $+, +, -, -$; GC18 bounds the two positive remainders and the negative remainders are nonnegative before their signs are applied. For $r = 0$ the empty determinant is one and the displayed return is zero.

The connection to the one-observation minimum is exact at *every* admissible cutoff, including before GC19 applies:

$$\begin{aligned} \log \det \tilde{G}_n^I &= \log \det(I^*G_nI) \\ &\quad + \log \det \left(I_r + (I^*G_nI)^{-1/2}G^I(I^*G_nI)^{-1/2} \right). \end{aligned} \quad (\text{GC20})$$

Factor out the positive I^*G_nI to prove this formula. Applying the four signed evaluations to both sides retains the whole second determinant. Thus the graph-completion calculation does not assign an asymptotic to the original one-observation return. Nor does it replace the original conductor source metric SCD41: SCD37–42 is the exact isometry and four-cutoff receiver for that source. Here GC20 identifies the metric change of the specific two-observation construction GC9, including its entire finite-cutoff correction.

33.6 The prime-action defect on every spectral direction

The theta cochain in GC12 carries quantitative information. Use the inner product conjugate-linear in the first variable. For any original eigenvector u of M_{H_c} , with eigenvalue z_ρ , self-adjointness of H_c gives

$$\text{Im} \langle Su, E_c u \rangle = -\text{Im}(z_\rho) \|Su\|^2, \quad \|E_c u\| \geq |\text{Im} z_\rho| \|Su\|. \quad (\text{GC21})$$

The first identity follows by substituting $E_c u = H_c Su - z_\rho Su$; the expectation of the self-adjoint operator is real. Cauchy–Schwarz proves the bound. Such a nonzero eigenvector is explicit even at a multiple root: take the local idempotent times $(H_c - z_\rho)^{m_\rho - 1}$. The nonzero local derivative from GC2 proves it is nonzero and that multiplication by H_c acts on it by z_ρ . For a nonfixed reflected pair, the two selected values are distinct conjugates, hence nonreal. Thus GC21 detects its nonzero discrepancy in the actual theta source, with the original primitive norm and without changing its eigenvector.

The complete critical-point jets also remain visible. At a reflection-fixed root with $m_\rho \geq 2$, z_ρ is real. Let e_ρ be its local idempotent and set $u_0 = e_\rho(H_c - z_\rho)^{m_\rho - 1}$, $u_1 = e_\rho(H_c - z_\rho)^{m_\rho - 2}$. They satisfy $(M_{H_c} - z_\rho)u_0 = 0$ and $(M_{H_c} - z_\rho)u_1 = u_0 \neq 0$. Direct substitution gives

$$\begin{aligned} \langle Su_0, E_c u_1 \rangle - \langle E_c u_0, Su_1 \rangle &= -\|Su_0\|^2, \\ \|Su_0\| \|E_c u_1\| + \|E_c u_0\| \|Su_1\| &\geq \|Su_0\|^2 > 0. \end{aligned} \quad (\text{GC22})$$

The two occurrences of $H_c - z_\rho$ cancel by self-adjointness, leaving exactly the displayed term and sign. The triangle inequality proves the second line. This calculates the retained higher-jet discrepancy even when the value of the prime word is real. The value pairing GC11 kills these higher jets, whereas GC22 proves that their source-action discrepancy is nonzero. The residue projection in GC11 and the theta cochain in GC12 are the exact maps connecting these statements.

33.7 The lost positive class gives an explicit form-domain defect

Choose a nonzero packet u with $w = T_Z(u, u) > 0$. Such a choice exists for every nonempty reflected packet: at a fixed root use its constant value vector, and at a nonfixed pair use the two constant values $(1, 1)$, putting all other coordinates zero. The corresponding values of w are m_ρ and $2m_\rho$. Let P_u be its degree-below- d representative and consider the actual word subspace

$$\mathcal{D}_u = \mathbb{C}P_u + \nu\mathbb{C}[X] \subset L^2(\mu), \quad Q_u(aP_u + \nu Q) = w|a|^2. \quad (\text{GC23})$$

The decomposition has unique scalar a , since applying E gives au . GC11 shows that Q_u is precisely the restriction of the original whole Weil form to this source subspace. Every other zero contribution vanishes on the word family. The domain is dense by GC6 and Q_u is nonnegative.

Nevertheless this nonnegative form is not closable in the original $L^2(\mu)$ source norm. The explicit polynomials $P_j = P_{1/j, j^6}$ from GC14 satisfy

$$P_j \longrightarrow 0 \text{ in } L^2(\mu), \quad Q_u(P_j - P_l) = 0, \quad Q_u(P_j) = w > 0. \quad (\text{GC24})$$

They all have observation u , so their differences belong to the relation ideal. They are therefore Cauchy for the form norm $\|P\|_{L^2(\mu)}^2 + Q_u(P)$. If a closed extension in the original source Hilbert space existed, completeness would give a form-norm limit. Its source component must be zero, whereas form-norm convergence would force its squared form value to be w . A quadratic form has value zero at zero, giving the contradiction. The completed graph GC10 supplies the precise retained object: this sequence converges there to $(0, u)$, whose second component and form value w survive. This establishes why the source projection alone cannot carry the packet form, even on a positive direction. It uses actual theta relations and the specified sequence; it makes no inference that the full Weil criterion has either sign.

$$\begin{array}{ccc} \mathbb{C}[X] & \xrightarrow{P \mapsto (P, EP)} & \mathcal{H} \oplus A_G \xrightarrow{\text{pr}_2} A_G \\ \Phi \downarrow & & \downarrow \text{pr}_1 \\ \overline{\Phi(\mathbb{C}[X])} & \xrightarrow{\cong} & \mathcal{H} \end{array}$$

Figure 24: The full graph closure is the direct sum, GC9–10. The original theta relations are dense in the first source component, GC6. The second observation retains every full jet with the original primitive Gram. GC11 evaluates the arithmetic form through that same second projection, and GC12 retains the source-operator discrepancy. The lower isometry is the original Mellin observation GC5.

34 The literal prime pulses observe every finite full zero packet

This note computes the exact map from the literal pulse family FP4–6 of [34] to the original nonreduced zero packet of [35]. It proves that an appropriate finite set of distinct primes makes this map onto, without assuming that the bump factor is a unit. Every local Taylor coefficient is retained. The final sections construct both attained quotient metrics and identify the whole complementary Weil contribution of their actual lifts. The result concerns finite-packet observation; it is not positivity of the original packet trace or of all Weil tests.

34.1 The original packet, pulse, and Mellin conversion

To distinguish the arithmetic completion from a pulse test, write $g_\zeta(s) = 2\xi(s)$ for the function denoted g in [35]. Thus

$$g_\zeta(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g_\zeta(s) = g_\zeta(1-s) = \overline{g_\zeta(\bar{s})}. \quad (\text{PO1})$$

Let Z be a finite nonempty set of actual nontrivial zeros, with each full multiplicity m_ρ retained. Fix the original nonzero leading coefficient c_h and define

$$\begin{aligned} h(s) &= c_h \prod_{\rho \in Z} (s - \rho)^{m_\rho}, & d &= \sum_{\rho \in Z} m_\rho, & A_h &= \mathbb{C}[s]/(h) \cong \prod_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \\ t_\rho &= s - \rho, & g_\zeta(\rho + t_\rho) &= u_\rho(t_\rho)t_\rho^{m_\rho}, & u_\rho(0) &= 2\lambda_\rho \neq 0. \end{aligned} \quad (\text{PO2})$$

The finite-dimensional coordinates throughout are the actual Taylor coefficients of t_ρ^k , $0 \leq k < m_\rho$. No unit u_ρ , multiplicity, or leading coefficient is replaced by a preferred value. For an entire function H ,

$$j_h H = \left(\sum_{k=0}^{m_\rho-1} \frac{H^{(k)}(\rho)}{k!} t_\rho^k \right)_{\rho \in Z}. \quad (\text{PO3})$$

The human source for the full finite-jet spectral description is Connes–Consani [33], original equation `evalmaprho`; our coordinate factorials in PO3 are stated explicitly.

Fix a finite set $I = \{p_1, \dots, p_n\}$ of distinct primes, where $n \geq 2$, and retain exactly the FP width and functions:

$$\begin{aligned} M_I &= \prod_{i=1}^n p_i, & \varepsilon_I &= \frac{1}{16} \log(1 + 1/M_I), & r_0 &= 1, & r_i &= p_i, \\ b(v) &= \begin{cases} e^{-1/(1-v^2)}, & |v| < 1, \\ 0, & |v| \geq 1, \end{cases} & w_\varepsilon(t) &= b(t/\varepsilon), & z_\varepsilon(t) &= w_\varepsilon''(t) - \frac{1}{4}w_\varepsilon(t), \\ g_{I,c}(e^t) &= \sum_{j=0}^n c_j z_{\varepsilon_I}(t - \log r_j), & N_I &= \int_{\mathbb{R}} z_{\varepsilon_I}(t)^2 dt > 0. \end{aligned} \quad (\text{PO4})$$

The smoothness, zero endpoint derivatives, and positivity of N_I follow directly from the bump: all derivatives are an exponential times a finite sum of rational powers of $1 - v^2$; $\int z_\varepsilon w_\varepsilon = -\int |w'_\varepsilon|^2 - \frac{1}{4} \int w_\varepsilon^2 < 0$. The original logarithmic norm is

$$\|g_{I,c}\|_{L^2(dx/x)}^2 = N_I \sum_{j=0}^n |c_j|^2. \quad (\text{PO5})$$

Indeed distinct centres have distance greater than $16\varepsilon_I$: for distinct positive integers a, b , $|\log(a/b)| \geq \log(1 + 1/b)$, and every $r_j < M_I$. Their supports, each of radius ε_I , are disjoint.

The exact conversion to the original Mellin space is

$$F_{I,c}(x) = x^{-1/2} g_{I,c}(x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{PO6})$$

Every $F_{I,c}$ is smooth with compact support in $\mathbb{R}_{>0}$, hence belongs to the original space

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} |x^b (-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}. \quad (\text{PO7})$$

This conversion does not preserve the unweighted $L^2(dx/x)$ norm of F . Its exact isometry is $\int_0^\infty x |F_{I,c}(x)|^2 dx/x = \|g_{I,c}\|^2$; PO5 always refers to the original pulse norm.

Define the entire bump transform, with its actual amplitude,

$$W_\varepsilon(z) = \int_{\mathbb{R}} w_\varepsilon(t) e^{zt} dt = \varepsilon \int_{-1}^1 b(v) e^{\varepsilon z v} dv, \quad a(s) = s(s-1), \quad B_\varepsilon(s) = a(s) W_\varepsilon(s - \tfrac{1}{2}). \quad (\text{PO8})$$

Two integrations by parts, with the literal zero endpoint terms, give $\int z_\varepsilon(t) e^{zt} dt = (z^2 - \frac{1}{4}) W_\varepsilon(z)$. Translation by $\log r_j$ multiplies this by r_j^z . Substitution $x = e^t$ in PO6 therefore proves

$$\boxed{\mathcal{M}F_{I,c}(s) = B_{\varepsilon_I}(s) \sum_{j=0}^n c_j r_j^{s-1/2}.} \quad (\text{PO9})$$

In particular $\mathcal{M}F_{I,c}(0) = \mathcal{M}F_{I,c}(1) = 0$. In the original convention $\widehat{g}(\xi) = \int g(e^t) e^{-i\xi t} dt$ from Connes [32], one has the exact signs

$$\mathcal{M}F(s) = \widehat{g}(i(s - \tfrac{1}{2})), \quad \widehat{g}(-i(s - \tfrac{1}{2})) = \mathcal{M}F(1-s). \quad (\text{PO10})$$

These two different arguments must not be interchanged without the displayed reflection.

34.2 All Taylor coefficients and the exact defect of a nonunit bump

Put $\alpha_\rho = \rho - \frac{1}{2}$. For $k \geq 0$ define

$$\begin{aligned} w_{\rho,k}(\varepsilon) &= \frac{W_\varepsilon^{(k)}(\alpha_\rho)}{k!} = \frac{\varepsilon^{k+1}}{k!} \int_{-1}^1 v^k b(v) e^{\varepsilon \alpha_\rho v} dv, \\ b_{\rho,k}(\varepsilon) &= \rho(\rho-1)w_{\rho,k}(\varepsilon) + (2\rho-1)w_{\rho,k-1}(\varepsilon) + w_{\rho,k-2}(\varepsilon), \end{aligned} \quad (\text{PO11})$$

where a coefficient with negative index is zero. These are the full Taylor coefficients of $W_\varepsilon(s - \frac{1}{2})$ and $B_\varepsilon(s)$ at the original point ρ .

Let $V_I : \mathbb{C}^{n+1} \rightarrow A_h$ have columns $v(r_j) = j_h(r_j^{s-1/2})$. In the actual coefficient basis, let D_ε be multiplication by $j_h B_\varepsilon$. Its block at ρ is lower triangular, with $(D_\varepsilon)_{\rho;k,\ell} = b_{\rho,k-\ell}(\varepsilon)$ for $k \geq \ell$ and zero otherwise. The full observation matrix is

$$\begin{aligned} E_I : \mathbb{C}^{n+1} &\longrightarrow A_h, & E_I c &= j_h \mathcal{M}F_{I,c}, & E_I &= D_{\varepsilon_I} V_I, \\ (V_I)_{\rho,k;j} &= r_j^{\alpha_\rho} \frac{(\log r_j)^k}{k!}, \\ (E_I)_{\rho,k;j} &= r_j^{\alpha_\rho} \sum_{\ell=0}^k b_{\rho,k-\ell}(\varepsilon_I) \frac{(\log r_j)^\ell}{\ell!}. \end{aligned} \quad (\text{PO12})$$

Here $\log r_0 = 0$; the zeroth power is one and positive powers are zero. Formula PO12 retains every bump derivative, the quadratic factor a , all prime scalars, and every nilpotent power.

No invertibility has yet been assumed. Define

$$\nu_\rho(\varepsilon) = \min\{m_\rho, \text{ord}_{z=\alpha_\rho} W_\varepsilon(z)\}. \quad (\text{PO13})$$

The function W_ε is not identically zero, since $W_\varepsilon(0) > 0$, so its order at every point is finite. Since $a(\rho) \neq 0$, multiplication by B_ε on the ρ -factor has the exact image, kernel, and quotient

$$\begin{aligned} \text{im } D_{\varepsilon,\rho} &= (t_\rho^{\nu_\rho}), & \ker D_{\varepsilon,\rho} &= (t_\rho^{m_\rho - \nu_\rho}), \\ A_\rho / \text{im } D_{\varepsilon,\rho} &\cong \mathbb{C}[t_\rho] / (t_\rho^{\nu_\rho}). \end{aligned} \quad (\text{PO14})$$

If $\nu_\rho = 0$, the displayed kernel is zero and quotient zero; if $\nu_\rho = m_\rho$, the image is zero and kernel all of A_ρ . To prove these statements, factor $B_\varepsilon(\rho + t) = t^{\nu_\rho} u(t)$ with $u(0) \neq 0$ when $\nu_\rho < m_\rho$.

Multiplication by u is invertible by finite recursive division, and multiplication by t^{ν_ρ} has exactly the displayed coefficient kernel and image. If $\nu_\rho = m_\rho$, the multiplication is zero directly.

For an arbitrary finite prime set, write $\mathcal{V}_I = \text{im } V_I$. The exact further coefficient and bump defects are

$$\begin{aligned} \text{im } E_I &= D_{\varepsilon_I} \mathcal{V}_I, \\ 0 &\longrightarrow \ker V_I \longrightarrow \ker E_I \xrightarrow{V_I} \mathcal{V}_I \cap \ker D_{\varepsilon_I} \longrightarrow 0, \\ 0 &\longrightarrow D_{\varepsilon_I} A_h / D_{\varepsilon_I} \mathcal{V}_I \longrightarrow A_h / D_{\varepsilon_I} \mathcal{V}_I \longrightarrow A_h / D_{\varepsilon_I} A_h \longrightarrow 0. \end{aligned} \quad (\text{PO15})$$

Each exactness assertion follows by taking the indicated kernel or image of its displayed linear map. If V_I is onto, the last quotient has dimension $\sum_\rho \nu_\rho(\varepsilon_I)$; thus PO14 is the precise missing-jet space of a nonunit bump.

34.3 Shrinking the prescribed width controls the whole bump

Write the unchanged bump moments

$$\mu_k = \int_{-1}^1 v^k b(v) dv, \quad \mu_0 > 0, \quad \mu_{2k+1} = 0, \quad 0 \leq \mu_{2k} \leq \mu_0. \quad (\text{PO16})$$

Uniform convergence of the exponential series on $[-1, 1]$ gives

$$W_\varepsilon(z) = \varepsilon \sum_{k=0}^{\infty} \frac{\mu_{2k}(\varepsilon z)^{2k}}{(2k)!}, \quad |W_\varepsilon(z) - \varepsilon \mu_0| \leq \varepsilon \mu_0 (\cosh(\varepsilon|z|) - 1) \leq \varepsilon \mu_0 (e^{\varepsilon|z|} - 1). \quad (\text{PO17})$$

The division by $\varepsilon \mu_0$ used to measure this estimate does not change the pulse or any entry of PO12. For all retained derivatives one also has

$$|w_{\rho,k}(\varepsilon)| \leq \frac{\varepsilon^{k+1} \mu_0}{k!} e^{\varepsilon|\alpha_\rho|}. \quad (\text{PO18})$$

This follows from PO11 and $|v| \leq 1$.

Let $R_Z = \max_{\rho \in Z} |\alpha_\rho|$. If $\varepsilon R_Z < \log 2$, then PO17 shows that

$$|W_\varepsilon(\alpha_\rho)| \geq \varepsilon \mu_0 (2 - e^{\varepsilon R_Z}) > 0 \quad (\rho \in Z). \quad (\text{PO19})$$

It follows that all $\nu_\rho = 0$ and D_ε is invertible. This proves invertibility; it does not replace the higher coefficients by zero. Its complete determinant and inverse recursion are

$$\begin{aligned} \det D_\varepsilon &= \prod_{\rho \in Z} [\rho(\rho-1)W_\varepsilon(\alpha_\rho)]^{m_\rho}, \\ d_{\rho,0} &= b_{\rho,0}^{-1}, \quad d_{\rho,k} = -b_{\rho,0}^{-1} \sum_{j=1}^k b_{\rho,j} d_{\rho,k-j} \quad (1 \leq k < m_\rho). \end{aligned} \quad (\text{PO20})$$

The polynomial $\sum d_{\rho,k} t_\rho^k$ is the exact inverse of B_ε in that factor, by multiplying the two finite series.

For the literal width of PO4, $\varepsilon_I < 1/(16M_I)$. Consequently the sufficient condition

$$M_I > R_Z/8 \implies \varepsilon_I R_Z < 1/2 < \log 2 \quad (\text{PO21})$$

proves the required unit property. The inequality $\log 2 = \int_0^1 (1+t)^{-1} dt > 1/2$ fixes the last constant. Adding primes increases M_I without changing the defining formula for the width. The actual new pulse has that smaller width; it is not identified with the old pulse.

More precisely, for $I \subseteq J$, let ι_{IJ} insert zero coefficients for the added primes. The exponential maps satisfy $V_J \iota_{IJ} = V_I$, and the full jet maps satisfy

$$D_{\varepsilon_I} E_J \iota_{IJ} = D_{\varepsilon_J} E_I. \quad (\text{PO22})$$

The multiplication operators commute, proving the equality. If D_{ε_I} is invertible, this is $E_J \iota_{IJ} = D_{\varepsilon_J} D_{\varepsilon_I}^{-1} E_I$. Thus PO22 is the exact comparison when the bump width changes; all unit Taylor data in the comparison are given by PO11 and PO20.

34.4 Prime logarithms determine every exponential polynomial

We give the uniqueness argument in full. It uses neither a prime-number theorem nor a spacing assumption on the primes. Its circle identity is the classical Jensen formula [30]; its arithmetic input is Euler's reciprocal-prime argument [31]. Both are derived completely below. For these historical works the cited archive metadata, rather than an unavailable original TeX source, was consulted.

Lemma 34.1 (An exponential polynomial has at most linearly many zeros in growing discs). *If*

$$f(z) = \sum_{\ell=1}^r P_\ell(z) e^{\alpha_\ell z} \quad (\text{PO23})$$

is a nonzero entire function, with polynomial P_ℓ , then there is a constant C_f such that its number of zeros in $|z| \leq T$, counted with multiplicity, is at most $C_f(T+1)$ for $T \geq 1$.

Proof. With $k = \max \deg P_\ell$ and $A = \max |\alpha_\ell|$, one has $|f(z)| \leq C(1+|z|)^k e^{A|z|}$. Factor its possible zero at the origin as $f(z) = z^q f_0(z)$, where $f_0(0) \neq 0$. Choose r with $2T < r < 3T$ whose circle contains no zero. Such radii exist because the zeros of a nonzero entire function are isolated. List the finitely many zeros ζ_j of f_0 inside that circle, with multiplicities. After dividing out $\prod_j (z - \zeta_j)$, the remaining holomorphic function has no zeros on the closed disc. Its logarithmic modulus is harmonic and satisfies the circle mean-value identity. For $|\zeta_j| < r$, the average of $\log |re^{i\theta} - \zeta_j|$ is $\log r$: expand $\log(1 - \zeta_j e^{-i\theta}/r)$ into its uniformly convergent power series and average its real part. Subtracting the central value therefore gives the exact formula

$$\sum_{|\zeta_j| < r} \log \frac{r}{|\zeta_j|} = \frac{1}{2\pi} \int_0^{2\pi} \log |f_0(re^{i\theta})| d\theta - \log |f_0(0)|. \quad (\text{PO24})$$

Each zero with $|\zeta_j| \leq T$ contributes at least $\log 2$. The right side is at most $\log C + k \log(1+3T) + 3AT - \log |f_0(0)|$; the factor r^{-q} can only decrease this bound for $T \geq 1$. This is at most a constant times $T+1$. Adding the q zeros at zero proves the lemma. $\square$

Lemma 34.2 (Prime reciprocals diverge). *For every finite set S of primes, $\sum_{p \notin S} p^{-1} = \infty$.*

Proof. It suffices to prove divergence over all primes. If that sum were finite, then

$$\log \prod_{p \leq X} (1 - p^{-1})^{-1} = \sum_{p \leq X} \int_0^{1/p} \frac{dt}{1-t} \leq 2 \sum_p p^{-1} \quad (\text{PO25})$$

would bound every finite product independently of X . Expanding its finitely many geometric factors, with nonnegative terms and unique factorization, shows that this product contains every term $1/m$ for $1 \leq m \leq X$, since each prime factor of such an m is at most X . Thus it is at least $\sum_{m \leq X} 1/m$, which tends to infinity by comparison with $\int_1^X dt/t$. This contradiction proves divergence. Removing finitely many terms leaves divergence unchanged. $\square$

Theorem 34.3 (Uniqueness on every cofinite prime-log set). *Let α_ℓ be pairwise distinct complex numbers, and let P_ℓ be polynomials. If the function PO23 vanishes at $\log p$ for all primes outside a finite set, then every P_ℓ is zero.*

Proof. First suppose f were nonzero. Lemma 34.1 would give at most $C(k+2)$ distinct prime logarithms in each interval $[k, k+1)$, since those are zeros in the disc of radius $k+1$. It would follow that

$$\sum_{p \notin S} \frac{1}{p} = \sum_{p \notin S} e^{-\log p} \leq C \sum_{k=0}^{\infty} (k+2) e^{-k} < \infty, \quad (\text{PO26})$$

contradicting Lemma 34.2. Hence $f \equiv 0$.

To exclude cancellation between distinct exponents, suppose $P_{\ell_0} \neq 0$ and set $m_\ell = \deg P_\ell + 1$ for each nonzero polynomial. Apply $\prod_{\ell \neq \ell_0} (\partial_z - \alpha_\ell)^{m_\ell}$ to $f \equiv 0$. Every term except the ℓ_0 term is annihilated. The latter becomes $e^{\alpha_{\ell_0} z}$ times $\prod_{\ell \neq \ell_0} (\partial_z + \alpha_{\ell_0} - \alpha_\ell)^{m_\ell} P_{\ell_0}(z)$. On the finite-dimensional polynomial space of degree at most $\deg P_{\ell_0}$, each factor has a nonzero constant diagonal and a strictly degree-lowering remainder. It is invertible by a finite geometric inverse. The resulting polynomial is therefore nonzero, a contradiction. Thus all P_ℓ vanish. $\square$

34.5 A finite literal pulse family is onto all original jets

Theorem 34.4 (Full-jet surjectivity with the original width). *For every finite packet PO2, there is a finite set I of distinct primes, $|I| \geq 2$, such that the literal observation E_I in PO12 is onto A_h . The set can be required to contain any prescribed finite prime set. It can also be chosen with $\varepsilon_I R_Z < \log 2$, so its bump factor is a proved unit with all its coefficients still given by PO11.*

Proof. For every prime p , consider the full vector

$$v(p) = \left(p^{\alpha_\rho} \frac{(\log p)^k}{k!} \right)_{\rho \in Z, 0 \leq k < m_\rho}. \quad (\text{PO27})$$

These vectors span $\mathbb{C}^d$, even after any finite set of primes is removed. If their span were proper, elementary finite-dimensional linear algebra would give a nonzero complex linear functional annihilating it. In coordinates, its prime values would be

$$\sum_{\rho \in Z} p^{\alpha_\rho} \sum_{k=0}^{m_\rho-1} \ell_{\rho,k} \frac{(\log p)^k}{k!} = 0. \quad (\text{PO28})$$

The exponents α_ρ are distinct. Theorem 34.3, applied to $z = \log p$, forces every coefficient $\ell_{\rho,k}$ to vanish, a contradiction.

Select d distinct prime vectors forming a basis. This selection can be performed successively: whenever the chosen span is proper, the spanning assertion supplies another prime outside it, also outside any prescribed finite exclusion. Take the union of these primes with the prescribed finite set. If needed, add primes so that the total cardinality is at least two and its product exceeds $R_Z/8$. There are arbitrarily large primes by Lemma 34.2, so finitely many additions suffice.

Call the resulting set I . The columns of V_I still contain the selected invertible d -by- d submatrix; enlarging I does not alter these exponential columns. Its width now satisfies PO21, so D_{ε_I} is invertible by PO19. Therefore $E_I = D_{\varepsilon_I} V_I$ is onto. Its retained invertible minor has determinant

$$\left(\prod_{\rho \in Z} [\rho(\rho-1)W_{\varepsilon_I}(\alpha_\rho)]^{m_\rho} \right) \det[v(p'_1) \cdots v(p'_d)] \neq 0, \quad (\text{PO29})$$

where $p'_1, \dots, p'_d$ are the selected basis primes. This proves surjectivity with the literal new width and every local factor included. $\square$

The proof provides existence of a finite spanning minor and an explicit width threshold. It does not assert a uniform bound for the size of those basis primes. The observation does not require a constant-column contribution: the d selected prime columns already form a basis; the original $r_0 = 1$ column remains present.

34.5.1 The actual cohomological map and prime convention

Retain the original source $V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even, } \phi(0) = 0, \int \phi = 0\}$, $\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx)$, and $Q = \mathcal{B}/\Theta V$. For $\Re s > 1$, substitution gives $\mathcal{M}\Theta\phi(s) = 2\zeta(s)M_+\phi(s)$, with $M_+\phi(s) =$

$\int_0^\infty \phi(x)x^s dx/x$. The latter is holomorphic for $\Re s > -2$: evenness and $\phi(0) = 0$ imply $\phi(x) = O(x^2)$ at zero, while the source is rapidly decreasing at infinity. The identity therefore continues to the critical strip and shows that all full actual zero jets of $\Theta\phi$ vanish. Consequently the exact observation factors as

$$\mathbb{C}^{n+1} \xrightarrow{c \rightarrow F_{I,c}} \mathcal{B}_{\text{pr}} \longrightarrow Q \xrightarrow{\bar{j}_h} A_h, \quad \bar{j}_h[F] = j_h \mathcal{M}F, \quad \bar{j}_h[F_{I,c}] = E_I c. \quad (\text{PO30})$$

Here $\mathcal{B}_{\text{pr}}$ imposes the two Mellin endpoint zeros, already verified in PO9. Surjectivity of the composite does not identify its kernel with the theta relation space: $\ker E_I$ means exactly vanishing at this finite packet.

The full original prime action on F is $U_p F(x) = F(x/p)$, with $\mathcal{M}U_p F(s) = p^s \mathcal{M}F(s)$. On a literal pulse translation $g(x) \mapsto g(x/p)$, conversion PO6 gives instead

$$x^{-1/2}g(x/p) = p^{-1/2}U_p F(x), \quad \bar{j}_h[p^{-1/2}U_p F] = p^{-1/2}\Pi_p \bar{j}_h[F], \quad \Pi_p = M_{p^s}. \quad (\text{PO31})$$

The local matrix is $p^{\rho-1/2} \sum_{k < m_\rho} (\log p)^k N_\rho^k / k!$, where N_ρ multiplies by t_ρ . This is the exact factor connecting the two original actions. PO30–31 are assertions on the original arithmetic coefficient spaces. Their finite observation has the following explicit support diagram, rather than a claim that every support stalk has constant coefficients. Let $\mathcal{J} = \{0, \dots, n\}$ be its source indices and $\mathcal{K} = \{(\rho, k) : \rho \in Z, 0 \leq k < m_\rho\}$ its target indices. Retain a declared incidence set $\mathcal{D} \subseteq \mathcal{K} \times \mathcal{J}$ containing every nonzero entry of the actual matrix E_I ; any additional declared zero entries remain in $\mathcal{D}$. The exact support trajectories and coefficient maps are

$$\begin{aligned} \mathcal{L}_{\mathcal{D}} &= \{(B, C) : B \subseteq \mathcal{J}, C \subseteq \mathcal{K}, \mathcal{D}(B) \subseteq C\}, \\ E_{B,C} &: \mathbb{C}^B \longrightarrow \mathbb{C}^C, \quad E_{B,C} = (E_I)_{C,B}, \\ j_{CC'} E_{B,C} &= E_{B',C'} j_{BB'} \quad \text{for } B \subseteq B', C \subseteq C'. \end{aligned} \quad (\text{PO54})$$

Here j means extension by zero. To prove the last square, apply it to a coordinate vector at $j \in B$. Every nonzero row coefficient $E_{\alpha j}$ has $\alpha \in \mathcal{D}(B) \subseteq C$; hence the two sides have exactly the same retained coefficients in C' . Declared zero entries change neither equality nor their support labels. Componentwise unions and intersections preserve the trajectory condition, so this is the original finite support-trajectory construction. Its top is $(\mathcal{J}, \mathcal{K})$, where the coefficient map is exactly $E_I : \mathbb{C}^{n+1} \rightarrow A_h$. Each smaller coefficient space is the indicated coordinate space, and need not have the dimension of its top. Thus PO12 is the terminal arithmetic fibre of this explicit diagram; it has not replaced the other support stalks. This support indexing is distinct from the prime-face index $A \subseteq I$ used for the Weil matrices below.

34.6 Two attained positive metrics, with their exact sections

Fix a prime set from Theorem 34.4, and write $E = E_I$, $N = N_I$. All adjoints in this section use the explicit coefficient bases $c_0, \dots, c_n$ and t_ρ^k . The standard coordinate inner product is used only to write these matrices; the source pulse norm remains the literal $N \text{Id}$ from PO5.

For completeness retain the exact Hermitian FP form. Let $C_\varepsilon(t) = \int z_\varepsilon(u) z_\varepsilon(u-t) du$, $K(t) = e^{-t/2}/(1 - e^{-2t})$, $c_{\text{arch}} = \log(4\pi) + \gamma_E$, and

$$\begin{aligned} A_\varepsilon &= c_{\text{arch}} N + 2 \int_0^\infty [C_\varepsilon(t) - N e^{-t/2}] K(t) dt, \\ J_\varepsilon(a) &= \int_{a-2\varepsilon}^{a+2\varepsilon} C_\varepsilon(t-a) K(t) dt, \\ (\mathbf{H}_\varepsilon)_{ij} &= \begin{cases} J_\varepsilon(|\log r_i - \log r_j|), & i \neq j, \\ 0, & i = j. \end{cases} \end{aligned} \quad (\text{PO32})$$

For each retained prime face $A \subseteq I$, the full matrix is

$$W_A = -A_\varepsilon \text{Id} - H_\varepsilon - N \sum_{p_i \in A} \frac{\log p_i}{\sqrt{p_i}} (e_{0i} + e_{i0}). \quad (\text{PO33})$$

In particular $W = W_I$ is the actual complete Weil matrix of these pulse functions. FP18–43 of [34] prove, including the entire Archimedean subtraction,

$$Q(g_{I,c}) = c^* W c, \quad W_A \geq \mu_* N \text{Id}, \quad \mu_* = \frac{6600077509}{144027072000} > \frac{1}{22}. \quad (\text{PO34})$$

For every nonzero c , the inequality proved there is strict. The complete companion source supplies that proof; it uses no positivity assumption on a zero packet. Thus W_A is positive definite on the whole coefficient space for every face, without suppressing any cross coefficient in PO33.

Theorem 34.5 (The two minimum sections). *The minimum section for the original pulse norm is*

$$L_2 = E^*(EE^*)^{-1}, \quad EL_2 = \text{Id}, \quad G_2 = N(EE^*)^{-1}. \quad (\text{PO35})$$

For every $u \in A_h$, $\min_{Ec=u} N\|c\|_2^2 = u^* G_2 u$, attained uniquely at $L_2 u$. Its pulled-back full Weil form is $G_{\text{lift}} = L_2^* W L_2$, which is positive definite.

The minimum section for the full Weil coefficient form is instead

$$G_W = (EW^{-1}E^*)^{-1}, \quad L_W = W^{-1}E^*G_W, \quad EL_W = \text{Id}, \quad L_W^* W L_W = G_W. \quad (\text{PO36})$$

It gives

$$\min_{Ec=u} c^* W c = u^* G_W u, \quad G_{\text{lift}} \geq G_W \geq \mu_* N(EE^*)^{-1}. \quad (\text{PO37})$$

The same constructions hold for each face after replacing W by its exact matrix W_A .

Proof. Since E is onto, EE^* is positive definite: $u^* EE^* u = \|E^* u\|^2 = 0$ implies that u annihilates the image of E , hence $u = 0$. Thus PO35 is defined and $EL_2 = \text{Id}$. Every solution is $c = L_2 u + k$, $k \in \ker E$; $\langle L_2 u, k \rangle = u^* (EE^*)^{-1} E k = 0$. Expanding the norm proves the asserted minimum and its uniqueness. It also gives $L_2^* L_2 = (EE^*)^{-1}$.

Positive definiteness of W follows from PO34. Therefore $EW^{-1}E^*$ is positive definite by the same argument, and PO36 is defined. For $k \in \ker E$, $(L_W u)^* W k = u^* G_W E k = 0$. Every solution now satisfies the exact decomposition

$$c^* W c = u^* G_W u + k^* W k, \quad c = L_W u + k, \quad E k = 0. \quad (\text{PO38})$$

This proves the unique attained minimum and PO36. Using $L_2 u$ as a competing solution proves $G_{\text{lift}} \geq G_W$. For every solution $Ec = u$, PO34 and PO35 give $c^* W c \geq \mu_* N\|c\|^2 \geq \mu_* N u^* (EE^*)^{-1} u$. Taking the attained minimum proves PO37. For $u \neq 0$, $L_W u \neq 0$, so the last inequality is strict by the strict version of PO34. Replacing W by W_A uses precisely the same proof. $\square$

These are different specified metrics. Their connecting identity is

$$G_{\text{lift}} - G_W = (L_2 - L_W)^* W (L_2 - L_W), \quad E(L_2 - L_W) = 0. \quad (\text{PO39})$$

It follows by applying the orthogonality in PO38 to $L_2 u = L_W u + (L_2 - L_W)u$ and polarizing. Thus neither minimum section is silently substituted for the other.

34.7 The exact complementary Weil contribution

Now require Z to be stable under $\iota\rho = 1 - \bar{\rho}$, as in the original reflected packet. For germs $u, v \in A_h$, retain the original raw residue form and its Jacobian contraction:

$$\begin{aligned} u^\dagger(s) &= \overline{u(1 - \bar{s})}, & R_Z(u, v) &= \sum_{\rho \in Z} \text{Res}_\rho \frac{u^\dagger(s)v(s)}{g_\zeta(s)} ds, & J_\zeta &= M_{g'_\zeta}, \\ T_Z(u, v) &= R_Z(u, J_\zeta v) = \sum_{\rho \in Z} m_\rho \overline{u(\iota\rho)} v(\rho). \end{aligned} \quad (\text{PO40})$$

The equality follows from $g'_\zeta/g_\zeta = m_\rho/t_\rho + u'_\rho/u_\rho$: only the simple pole contributes. The full residue block, with row $t_{\iota\rho}^j$ and column t_ρ^k , is

$$(-1)^j [t_\rho^{m_\rho-1-j-k}] u_\rho(t_\rho)^{-1}. \quad (\text{PO41})$$

Its antidiagonal is $(-1)^j/(2\lambda_\rho) \neq 0$; hence the raw pairing remains perfect. The Hermitian matrix of T_Z , denoted W_Z , has entries

$$(W_Z)_{\sigma,k;\rho,\ell} = m_\rho \mathbf{1}_{\sigma=\iota\rho} \mathbf{1}_{k=0} \mathbf{1}_{\ell=0}. \quad (\text{PO42})$$

All higher-jet coordinates remain in the packet even though this particular contraction vanishes on them. The multiplicities on a reflected pair agree by PO1.

We use the exact primitive packet section of [35], and recall its full construction to specify the complementary test. Let $U_h = g_\zeta/h$, let $u_h = j_h U_h$, an actual unit of A_h , and set

$$R_u = \text{rem}_h([a]^{-1} u_h^{-1} u), \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad D = -x\partial_x. \quad (\text{PO43})$$

The remainder has degree less than d ; both inverse units are taken with their full Taylor expansions. For $H \in \mathcal{B}$ with $\mathcal{M}H(\rho) = 0$, put

$$S_\rho H(x) = x^{-\rho} \int_x^\infty H(t) t^{\rho-1} dt, \quad S_h = c_h^{-1} \prod_{\rho \in Z} S_\rho^{m_\rho}, \quad \text{Prim}_h(u) = a(D) S_h \Theta(R_u(D) \phi_*). \quad (\text{PO44})$$

These expressions are well defined on the displayed arguments. For clarity, differentiating the integral gives $(D - \rho)S_\rho H = H$. The zero Mellin moment lets one replace the tail near zero by minus the integral from zero to x . Rapid decrease of H at both ends then bounds each weighted Euler derivative of $S_\rho H$; so $S_\rho H \in \mathcal{B}$. Its Mellin transform is $\mathcal{M}H(s)/(s - \rho)$, by integration by parts. Repeating removes the full zero orders, and uniqueness of the rapidly decreasing solution proves that the factors commute. Gaussian integration gives $\mathcal{M}\Theta\phi_* = g_\zeta$; multiplying by $R_u(s)$ supplies all required zeros of h . It follows that

$$\mathcal{M} \text{Prim}_h(u)(s) = a(s) U_h(s) R_u(s), \quad j_h \mathcal{M} \text{Prim}_h(u) = u. \quad (\text{PO45})$$

The section is primitive at 0, 1, and its transform has the full actual zero multiplicity at every zero outside Z . This verifies the exact properties used below, including the original factor c_h , units u_h , and coefficients of ϕ_* .

For $F, H \in \mathcal{B}$, retain the whole reflected form

$$\mathcal{W}(F, H) = \sum_{\rho \text{ actual}} m_\rho \overline{\mathcal{M}F(1 - \bar{\rho})} \mathcal{M}H(\rho). \quad (\text{PO46})$$

The series converges absolutely: integration by parts in $\log x$ gives arbitrary inverse powers of $|\Im\rho|$ uniformly in the critical strip. The zero-count bound $N(T) = O(T \log T)$ needed for

summation is proved directly from the unchanged Gaussian seed in [35], PL50. For the pulse conversion PO6, the original explicit formula of [32] gives

$$\mathcal{W}(F_{I,c}, F_{I,d}) = c^* W d. \quad (\text{PO47})$$

To check the sign conversion, its Fourier zero argument is $(\rho - \frac{1}{2})/i = -i(\rho - \frac{1}{2})$. For a convolution square, the value there is $\mathcal{M}F(1 - \rho)\overline{\mathcal{M}F(\bar{\rho})}$, by PO10 and the conjugate Fourier product. Reindex the entire zero multiset by $\rho \mapsto 1 - \rho$, which preserves multiplicity by PO1; the result is exactly PO46. Both pole terms vanish by PO9. The sesquilinear identity follows by polarization, with conjugate linearity in the first slot. Thus PO47 invokes the actual human explicit-formula convention, not a new choice of trace sign.

Define the original residual map on all pulse coefficients by

$$\mathcal{R}_I(c) = F_{I,c} - \text{Prim}_h(Ec) \in \mathcal{B}_{\text{pr}}. \quad (\text{PO48})$$

It has every full jet zero on Z , by PO12 and PO45. At every actual zero outside Z , the packet section vanishes to full order, so the residual retains exactly the original pulse's jets there. This is an actual function-space map, not a formal difference of assigned quadratic forms.

Theorem 34.6 (Exact paired residual and attained packet metrics). *For every coefficient pair c, d ,*

$$\begin{aligned} \mathcal{W}(\text{Prim}_h(Ec), \mathcal{R}_I(d)) &= \mathcal{W}(\mathcal{R}_I(c), \text{Prim}_h(Ed)) = 0, \\ \mathcal{W}(\mathcal{R}_I(c), \mathcal{R}_I(d)) &= c^*(W - E^*W_Z E)d. \end{aligned} \quad (\text{PO49})$$

For any right inverse L of E , put $\mathcal{R}_L(u) = F_{I,Lu} - \text{Prim}_h(u)$. Its exact whole complementary matrix is

$$\mathcal{W}(\mathcal{R}_L(u), \mathcal{R}_L(v)) = u^* D_L v, \quad \boxed{D_L = L^* W L - W_Z}. \quad (\text{PO50})$$

In particular $D_{L_2} = G_{\text{lift}} - W_Z$ and $D_{L_W} = G_W - W_Z$, with

$$D_{L_2} \geq D_{L_W} \geq \mu_* N (EE^*)^{-1} - W_Z. \quad (\text{PO51})$$

Proof. At a zero in Z , every value of the residual vanishes, including at its reflected partner because Z is stable. At any zero outside Z , every value of the primitive section vanishes, also at its reflected partner. Each summand of either cross form is therefore zero individually. The pure section term of PO46 reduces exactly to $T_Z(Ec, Ed)$ by PO45, with no omitted outside contribution. Expanding $F_{I,c} = \text{Prim}_h(Ec) + \mathcal{R}_I(c)$ in PO47 proves PO49. Substitution $c = Lu, d = Lv$ proves PO50. The matrix inequalities are precisely PO37 with W_Z subtracted; hence PO51 follows. $\square$

The residual class $[\mathcal{R}_L(u)]$ lies in $\ker(\bar{j}_h : Q \rightarrow A_h)$ by PO30. It is not asserted to be a theta relation. Thus PO48 is also the exact cohomological difference between a pulse lift and the same-jet primitive packet section.

Corollary 34.7 (Numerical lower bound in a negative packet direction). *For every nonzero u with $u^* W_Z u < 0$, the minimum-Weil lift satisfies*

$$\mathcal{W}(\mathcal{R}_{L_W}(u), \mathcal{R}_{L_W}(u)) > \mu_* N u^* (EE^*)^{-1} u - u^* W_Z u > 0. \quad (\text{PO52})$$

The same lower bound holds for the minimum-pulse-norm lift L_2 .

Proof. Use the strict version of PO37 for $u \neq 0$ in PO50, then the fact that $(EE^*)^{-1}$ is positive definite. For L_2 , use $D_{L_2} \geq D_{L_W}$. $\square$

More explicitly, on any nonfixed reflected pair $\{\rho, \iota\rho\}$ of common multiplicity m , retain the full coordinate vector $u = \eta$, whose two constant coefficients are $1, -1$ and whose other coordinates are zero. The original trace block is $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; hence $\eta^* W_Z \eta = -2m$. The exact numerical lower bound for its complementary whole-zero contribution is

$$\mathcal{W}(\mathcal{R}_{LW}(\eta), \mathcal{R}_{LW}(\eta)) > 2m + \frac{6600077509}{144027072000} N \eta^* (EE^*)^{-1} \eta. \quad (\text{PO53})$$

All entries of E in this bound are the full bump-and-prime Taylor factors of PO11–12; N is the unchanged integral PO4. For a scalar multiple $a\eta$, both sides scale by $|a|^2$. This computes the exact compensation for an indefinite packet direction whenever that direction occurs in the original packet; it does not assert existence of an off-critical zero. For a nonzero higher-jet vector with all constant coefficients zero, PO42 instead gives $T_Z(u, u) = 0$, and the same proof gives a strictly positive complementary lower bound $\mu_* N u^* (EE^*)^{-1} u$. The raw form PO41 continues to detect those jets.

The onto map and the positive pulse form therefore coexist through the explicit residual PO48. Neither surjectivity nor the attained positive metric G_W identifies it with the original Jacobian trace W_Z . Their exact difference is the matrix and full zero sum PO50.

35 The evaluated cost of retaining all packet jets in shrinking prime pulses

The preceding full-jet observation gives a concrete way to transport the positive pulse calculation into a complete finite packet. This section evaluates the original source cost of that transport, including its first matrix correction. The pulse, Haar measure and coefficient amplitude are precisely FP4–6 [28]; the conversion to the primitive Mellin coordinate is the one used in PL1–10. The prime set and all jet coordinates remain fixed where a matrix limit is taken.

Fix a finite reflected packet Z , with its actual multiplicities, and its algebra $A = \prod_\rho \mathbb{C}[t_\rho]/t_\rho^{m_\rho}$, $t_\rho = s - \rho$, $d = \sum_\rho m_\rho$. Fix a finite set J of prime centres, allowing the original centre 1, for which the complete exponential-column map is onto:

$$P_J : \mathbb{C}^J \longrightarrow A, \quad (c_r) \longmapsto \sum_{r \in J} c_r [r^{s-1/2}]. \quad (\text{PCO1})$$

The preceding prime-column spanning proof constructs such a set for every finite packet. This surjectivity is used only to invert its exact positive row Gram, not to remove any Taylor coefficient. Extend the ambient prime set I containing the prime centres in J , and retain its prescribed FP width $\varepsilon = \frac{1}{16} \log(1 + 1/\prod_{p \in I} p)$. All coefficients outside J are zero. Enlarging I gives $\varepsilon \rightarrow 0$ on this same coefficient face. No width or amplitude in FP is replaced by an independently chosen one.

Put

$$\begin{aligned} w_0(v) &= \begin{cases} e^{-1/(1-v^2)} & |v| < 1, \\ 0 & |v| \geq 1, \end{cases} & A_w &= \int_{-1}^1 (w_0'')^2 dv, & B_w &= \int_{-1}^1 (w_0')^2 dv, & C_w &= \int_{-1}^1 w_0^2 dv, \\ \mu_0 &= \int_{-1}^1 w_0 dv > 0, & \mu_2 &= \int_{-1}^1 v^2 w_0 dv, & \kappa_w &= \frac{\mu_2}{2\mu_0}, & 0 < \kappa_w &< \frac{1}{2}. \end{aligned} \quad (\text{PCO2})$$

Every constant is the indicated integral of the original bump. In particular $A_w > 0$: a smooth compactly supported function with zero second derivative would be affine and hence zero, whereas $w_0(0) = e^{-1}$. The literal FP source norm is

$$N_\varepsilon = A_w \varepsilon^{-3} + \frac{B_w}{2} \varepsilon^{-1} + \frac{C_w}{16} \varepsilon. \quad (\text{PCO3})$$

For proof, substitute $v = t/\varepsilon$ into the square of $z(t) = \varepsilon^{-2}w_0''(t/\varepsilon) - \frac{1}{4}w_0(t/\varepsilon)$. Its cross term is $-\frac{1}{2}\varepsilon^{-1} \int w_0''w_0$. Integration by parts, with both endpoint values and derivatives zero, turns that integral into $-B_w$. This proves all three terms and their signs.

Let M be multiplication by $s - \frac{1}{2}$ in the original jet basis, and M_a multiplication by $a(s) = s(s-1)$. The latter is invertible since the selected nontrivial zeros avoid 0, 1. Define the complete, commuting matrix

$$B_\varepsilon = \mu_0^{-1} \int_{-1}^1 w_0(v) e^{\varepsilon v M} dv. \quad (\text{PCO4})$$

The integral is entrywise in the finite original coordinates; the matrix exponential retains every nilpotent power. For the literal pulse g_c set $F_c(x) = x^{-1/2}g_c(x)$. The two norms agree, since $\int |F_c|^2 dx = \int |g_c|^2 dx/x$. Its full Mellin jet matrix is exactly

$$E_\varepsilon = \varepsilon \mu_0 M_a B_\varepsilon P_J. \quad (\text{PCO5})$$

Indeed two integrations by parts give $\int z(t) e^{\beta t} dt = (\beta^2 - \frac{1}{4})\varepsilon \int w_0(v) e^{\varepsilon \beta v} dv$. The centre $\log r$ contributes r^β , with $\beta = s - \frac{1}{2}$, so $\beta^2 - \frac{1}{4} = a(s)$. Applying the full Taylor map to this entire identity gives PCO5, including every derivative of both factors.

The inverse and its error can be controlled in these same coordinates. Write $m = \|M\|$ for the Euclidean operator norm and $\varepsilon_0 = (2 \max(1, m))^{-1}$. Evenness of w_0 and the matrix exponential series give, for $0 < \varepsilon \leq \varepsilon_0$,

$$B_\varepsilon = I + \kappa_w \varepsilon^2 M^2 + R_\varepsilon, \quad \|R_\varepsilon\| \leq \frac{\varepsilon^4 m^4 e^{1/2}}{24}. \quad (\text{PCO6})$$

To prove the bound, retain terms through degree three in the exponential series. The integrated odd terms vanish and the remaining tail is bounded by $|\varepsilon v|^4 m^4 e^{|\varepsilon v| m}/24$. Use $|v| \leq 1$ and divide the integral by its mass μ_0 . In particular $\|B_\varepsilon - I\| < 1/2$, since the quadratic bound is at most $1/8$ and the fourth-order bound is at most $e^{1/2}/384 < 1/128$ (use $e^{1/2} < 3$). The convergent geometric series proves invertibility, and

$$B_\varepsilon^{-1} = I - \kappa_w \varepsilon^2 M^2 + O_Z(\varepsilon^4). \quad (\text{PCO7})$$

Here the error is effective in the displayed coordinates. For example set $r = m^4 e^{1/2}/24$ and $b = \kappa_w m^2 + \varepsilon_0^2 r$; its norm is at most $\varepsilon^4(r + 2b^2)$, by $(I + Y)^{-1} = I - Y + Y^2(I + Y)^{-1}$ and $\|(I + Y)^{-1}\| \leq 2$.

Define the fixed full packet metric

$$K_J = (M_a P_J P_J^* M_a^*)^{-1}. \quad (\text{PCO8})$$

All cross entries and every confluent row are included. The minimum of the actual Haar norm over the fibre $E_\varepsilon c = u$ has Gram

$$H_\varepsilon = N_\varepsilon (E_\varepsilon E_\varepsilon^*)^{-1} = \frac{N_\varepsilon}{\varepsilon^2 \mu_0^2} B_\varepsilon^{-*} K_J B_\varepsilon^{-1}. \quad (\text{PCO9})$$

The first equality follows by writing the minimum solution as $c = E_\varepsilon^* (E_\varepsilon E_\varepsilon^*)^{-1} u$: all other solutions differ by an orthogonal kernel vector. For the second equality, use PCO5 and the fact that M_a commutes with B_ε , then invert by congruence.

Combining PCO3, PCO7 and PCO9 evaluates both leading orders:

$$\begin{aligned} H_\varepsilon &= \frac{A_w}{\mu_0^2 \varepsilon^5} K_J \\ &\quad + \frac{1}{\mu_0^2 \varepsilon^3} \left[\frac{B_w}{2} K_J - A_w \kappa_w ((M^2)^* K_J + K_J M^2) \right] + O_{Z,J}(\varepsilon^{-1}). \end{aligned} \quad (\text{PCO10})$$

This is a matrix-norm expansion in the original jets. Products of the explicit PCO7 error bounds with the three terms of PCO3 give its stated remainder. It retains the off-diagonal confluent correction, which is lost by keeping only root values.

Its full determinant, including the multiplicities, is

$$\begin{aligned} \log \det H_\varepsilon &= -5d \log \varepsilon + d \log(A_w/\mu_0^2) + \log \det K_J \\ &\quad + \varepsilon^2 \left[\frac{dB_w}{2A_w} - 2\kappa_w \operatorname{Re} \operatorname{tr}(M^2) \right] + O_{Z,J}(\varepsilon^4), \\ \operatorname{tr}(M^2) &= \sum_{\rho \in Z} m_\rho (\rho - \tfrac{1}{2})^2. \end{aligned} \tag{PCO11}$$

Indeed PCO9 gives $\det H_\varepsilon = (N_\varepsilon/(\varepsilon^2 \mu_0^2))^d \det K_J / |\det B_\varepsilon|^2$. For a matrix Y of norm below one, $\log \det(I + Y) = \operatorname{tr}(Y) + O(\|Y\|^2)$ follows by taking traces of the absolutely convergent logarithm series. Apply this to PCO6, taking twice the real part for the squared modulus. The scalar logarithm of PCO3 supplies the other quadratic term. The displayed trace follows directly from each triangular local block of M , with all m_ρ diagonal entries retained.

Finally let W_I be the complete positive FP matrix on the fixed coefficient face J , at its ambient width. Its minimum under the same observation is $H_\varepsilon^{\text{Weil}} = (E_\varepsilon W_I^{-1} E_\varepsilon^*)^{-1}$. FP37 gives, with its proved original constant $\mu_* = 6600077509/144027072000$,

$$H_\varepsilon^{\text{Weil}} \succeq \mu_* H_\varepsilon. \tag{PCO12}$$

This follows by minimizing $c^* W_I c \geq \mu_* N_\varepsilon \|c\|^2$ on the same fibre, so no change of section or observation is hidden. Let W_Z be the reflected full-packet value form and use the preceding exact primitive splitting of the pulse. The complementary whole-zero contribution for its minimum-Weil lift is therefore

$$u^*(H_\varepsilon^{\text{Weil}} - W_Z)u \geq \mu_* u^* H_\varepsilon u - u^* W_Z u. \tag{PCO13}$$

For every fixed nonzero u , PCO10 makes its lower bound grow as a positive constant times ε^{-5} . For a reflected nonfixed pair of multiplicity m_ρ and the value vector $(1, -1)$ with all higher jets and other factors zero, $u^* W_Z u = -2m_\rho$. Its complete complementary contribution is at least $\mu_* u^* H_\varepsilon u + 2m_\rho$. This is the actual cost and complementary term of the surjective positive pulse receiver. Every factor belongs to the specified original test; no sign of an omitted root packet is inferred from finite-pulse positivity alone.

$$\begin{array}{ccc} (\mathbb{C}^J, N_\varepsilon I) & \xrightarrow{c \mapsto x^{-1/2} g_c} & \mathcal{B}_{\text{pr}} \\ & \searrow E_\varepsilon & \downarrow j_Z \mathcal{M} \\ & & (A, H_\varepsilon) \end{array} \quad \begin{array}{c} \curvearrowright S \end{array}$$

Figure 25: Every arrow retains the full original Taylor coefficients. The pulse observation is PCO5, and the attained norm is PCO9. The primitive section S has a different representative in the same packet; subtracting it leaves exactly the complementary Weil contribution PCO13. Along the actual shrinking ambient widths, the quotient metric has the complete ε^{-5} and ε^{-3} terms PCO10. The fixed column set J remains in the original support face.

36 Complete support cohomology of the original maps

The construction starts with the distinct zeros of $G(R)$ and retains both of them. It takes finite matrices with their original coefficients and declared supports to a diagram of cohomology

modules over every compatible support trajectory. This entire diagram carries a canonical action of $G_{\mathcal{L}}(R)$. Its full-support fibre is exactly the original kernel with its original minimum metric. Nonzero composites are computed before a sequence is called a complex.

The foundational input is the original semimodule classification [2], lines 458–945, especially the source labels `thm:fiber-modules`, `thm:reconstruct-semimodule`, `thm:equivalence-semimodules-diagrams`, and `thm:free-semimodule-boolean-cube`. All reconstruction assertions needed here are proved below. The original source and conductor matrices are those of OPG5, CEP2–5, PCL17–21, and GIP1–6 in the pinned source collection [6]. No coordinate support is identified across a basis change without its explicit connecting maps.

36.1 The two zeros and cancellation paths

Let R be a nonzero commutative ring with identity, and let

$$S = G(R) = R \sqcup \{\tau\}, \quad e = 0_R \in R \subset S. \quad (\text{SCD1})$$

On the supported copy of R , the operations are the ring operations. The new element τ is the additive identity and multiplicative absorber. Thus $r + (-r) = e$, not τ . The ring projection $p : S \rightarrow R$ sends τ to 0_R and fixes R . The support character $\chi : S \rightarrow \{0, 1\}$ sends τ to zero and every supported scalar to one. Checking whether the factors or summands are supported proves that both are semiring morphisms. These are the same maps as in the baseline note [1].

For finite sets I, J , a declared support for an R -matrix $d : R^I \rightarrow R^J$ is a relation $D \subseteq J \times I$ containing every pair with $d_{ji} \neq 0_R$. It may also contain supported zeros. Define

$$\tilde{d}_{ji} = \begin{cases} d_{ji} \in R \subset S & (j, i) \in D, \\ \tau & (j, i) \notin D, \end{cases} \quad D(A) = \{j : \exists i \in A, (j, i) \in D\}. \quad (\text{SCD2})$$

The sparse declaration contains exactly the nonzero entries; the fully entrywise declaration has $D = J \times I$. All results below retain whichever support is declared.

A vector of S^I is a pair (A, x) with $A \subseteq I$, $x \in R^A$. Zero entries of x are still supported; outside A its entries are τ . Direct matrix multiplication gives

$$\tilde{d}(A, x) = (D(A), d_{D(A), Ax}). \quad (\text{SCD3})$$

Every row with a declared incident edge receives a supported sum, including a sum that cancels to e .

Proposition 36.1. *For $d : R^I \rightarrow R^J$ and $f : R^J \rightarrow R^K$ with supports D, F ,*

$$p(\tilde{f}\tilde{d}) = fd, \quad \chi(\tilde{f}\tilde{d})_{ki} = \bigvee_{j \in J} (\mathbf{1}_F(k, j) \wedge \mathbf{1}_D(j, i)). \quad (\text{SCD4})$$

If $fd = 0_R$, the lifted composite has entry e at every declared length-two path and τ elsewhere. It is the global zero matrix over S exactly when there are no such paths.

Proof. A product in the sum is supported exactly when both entries are supported. Its ring value is $f_{kj}d_{ji}$, even when that product is zero. A nonempty sum of supported terms is supported and has the ring sum as its value. Only an empty supported sum is τ . This proves every assertion, including rings with zero divisors. The cancellation certificate $\sum_j f_{kj}d_{ji} = 0_R$ therefore retains all of its individual paths. $\square$

In particular,

$$d_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad d_1 = \begin{pmatrix} 1 & -1 \end{pmatrix}, \quad d_1 d_0 = 0_R, \quad \tilde{d}_1 \tilde{d}_0 = (e). \quad (\text{SCD5})$$

The exact morphism p carries this e to 0_R . It does not identify e and τ in S .

36.2 Every compatible support trajectory

Fix a finite cochain complex with its original based matrices:

$$C^n = R^{I_n}, \quad d_n : C^n \rightarrow C^{n+1}, \quad d_{n+1}d_n = 0_R. \quad (\text{SCD6})$$

Only finitely many I_n are nonempty. Fix every declared support D_n . A support trajectory is a sequence $A = (A_n)_n$, $A_n \subseteq I_n$, with $D_n(A_n) \subseteq A_{n+1}$. Let $\mathcal{L}_D$ be their componentwise ordered set.

Theorem 36.2. *The lattice $\mathcal{L}_D$ is complete and distributive. Its meet and join are componentwise intersection and union, its bottom is the empty trajectory, and its top is $\mathbf{I} = (I_n)_n$. For arbitrary seed sets $U_n \subseteq I_n$, their smallest containing trajectory is*

$$\text{cl}_D(U)_n = U_n \cup \bigcup_{m < n} D_{n-1} \cdots D_m(U_m). \quad (\text{SCD7})$$

This closure is monotone, idempotent, and preserves unions.

Proof. An edge starting in a union starts in one member, whose next component contains its endpoint. An edge starting in an intersection has its endpoint in every next component. Thus arbitrary unions and intersections are trajectories, with the stated empty conventions. Set-theoretic distributivity proves the lattice assertion. Every path endpoint in SCD7 is forced in any containing trajectory. Appending one edge proves that SCD7 itself is a trajectory. This proves minimality and all closure laws. $\square$

Equivalently the graded vertices (n, i) carry directed edges $(n, i) \rightarrow (n+1, j)$ from D_n . Degrees strictly increase, so reachability defines a finite partial order. The trajectories are exactly its upward-closed subsets. Every path remains visible.

Theorem 36.3. *For $A \in \mathcal{L}_D$ put*

$$C_A^n = R^{A_n}, \quad d_{n,A} = (d_n)_{A_{n+1}, A_n}. \quad (\text{SCD8})$$

These are cochain complexes. Extension by zero $j_{AB}^n : C_A^n \rightarrow C_B^n$ for $A \subseteq B$ is an injective cochain map. Consequently

$$Z_A^n = \ker d_{n,A}, \quad B_A^n = \text{im } d_{n-1,A}, \quad H_A^n = Z_A^n / B_A^n, \quad h_{AB}^n[x] = [j_{AB}^n x] \quad (\text{SCD9})$$

defines a diagram of R -modules, with exact composition of its maps.

Proof. If $i \in A_n$ and $d_{n,ji} \neq 0$, then $j \in A_{n+1}$. Thus the restricted intermediate sum removes no nonzero term:

$$(d_{n+1,A}d_{n,A})_{ki} = \sum_{j \in I_{n+1}} d_{n+1,kj}d_{n,ji} = 0_R.$$

Every declared intermediate path is also included in A_{n+1} , so its cancellation is retained. The same support condition says that applying d_n to a vector supported in A_n produces a vector in $R^{A_{n+1}}$. Therefore extension by zero commutes with the differential. It maps cycles and boundaries to cycles and boundaries; this proves the quotient map is well defined. Linearity, identity, and composition follow from zero extension. $\square$

For $A \subseteq B$, identify the smaller coordinate modules with their zero-extended copies. The exact kernels and cokernels are

$$\ker h_{AB}^n = (Z_A^n \cap B_B^n) / B_A^n, \quad \text{coker } h_{AB}^n = Z_B^n / (Z_A^n + B_B^n). \quad (\text{SCD10})$$

Indeed a smaller cycle becomes zero precisely when it becomes a larger boundary, and the image is $(Z_A^n + B_B^n) / B_B^n$. For SCD5 there are exactly six trajectories: the empty one; the last vertex; either middle vertex with the last; both middle vertices with the last; and the full set. The penultimate one has $H^1 = R(1, 1)$, which becomes a boundary in the full exact complex. The last-vertex fibre has $H^2 = R$. Thus exactness at full support does not discard the nontrivial cohomology at smaller supports.

36.3 Reconstruction and the full $G_{\mathcal{L}_D}(R)$ base

Define the disjoint union, retaining its index even when a module is zero,

$$\mathcal{H}_D^n = \bigsqcup_{A \in \mathcal{L}_D} H_A^n. \quad (\text{SCD11})$$

For elements in two fibres use

$$(A, [x]) + (B, [y]) = (A \cup B, h_{A, A \cup B}[x] + h_{B, A \cup B}[y]). \quad (\text{SCD12})$$

A supported $r \in R$ acts within its fibre, and τ acts by the bottom-fibre zero. Define $\mathcal{C}_D^n, \mathcal{Z}_D^n, \mathcal{B}_D^n$ from C_A^n, Z_A^n, B_A^n in the same way.

Theorem 36.4. *These operations make $\mathcal{H}_D^n$ a $G(R)$ -semimodule. Its support lattice is exactly $\mathcal{L}_D$, its fibres are exactly H_A^n , and its transports are h_{AB}^n . It is the quotient of $\mathcal{Z}_D^n$ by the congruence*

$$(A, x) \sim (B, y) \iff A = B \text{ and } x - y \in B_A^n. \quad (\text{SCD13})$$

Thus different support indices are never identified.

Proof. Three summands may all be transported to $A \cup B \cup C$. Functoriality of h makes either parenthesization equal to the same sum in that module. Commutativity and the bottom identity follow in the same way. Linearity of transport proves both scalar distributive laws. In particular $(r + s)x = rx + sx$ occurs in the same fibre, also when $r + s = 0_R$. The rules involving τ follow from its action. The supported zero acts by $e(A, [x]) = (A, 0)$, and $(A, 0) + (B, 0) = (A \cup B, 0)$, identifying the support lattice. Boundaries transport to boundaries, so SCD13 is preserved under addition and scalars. Its fibre quotient is precisely H_A^n . $\square$

There is an exact distinction between this construction and a falsely asserted global square-zero lift. Put

$$\mathcal{D}_n(A, x) = (A, d_{n,A}x), \quad \mathbf{z}_{n,n+2}(A, x) = (A, 0). \quad (\text{SCD14})$$

They are S -linear and $\mathcal{D}_{n+1}\mathcal{D}_n = \mathbf{z}_{n,n+2}$. For nontrivial $\mathcal{L}_D$, this is not the global zero map, whose output has index $\emptyset$. The usual preimage of that global zero is not the union of fibre kernels. Cohomology is taken in the additive category of R -module diagrams on the fixed lattice, as explicitly specified in SCD9, then reconstructed by SCD11–13.

The exact return map to the original free semimodule is

$$E_n : \mathcal{C}_D^n \rightarrow S^{I_n}, \quad E_n(A, x) = (A_n, x). \quad (\text{SCD15})$$

It is S -linear because supports of sums are unions and zero extensions commute with addition. It can fail to be injective: two distinct trajectories may have the same n -th support. The original lifted matrix is related to it by

$$E_{n+1}\mathcal{D}_n(A, x) = T_{D_n(A_n), A_{n+1}}(\tilde{d}_n E_n(A, x)). \quad (\text{SCD16})$$

Here T inserts supported zeros at the additional coordinates, exactly as in the Boolean-cube diagram of $S^{I_{n+1}}$. SCD3 and SCD8 prove this identity at every coordinate.

We now include all support scalars. For the bounded distributive lattice $\mathcal{L} = \mathcal{L}_D$, define

$$G_{\mathcal{L}}(R) = \{(r, \mathbf{I}) : r \in R\} \cup \{z_A = (0_R, A) : A \in \mathcal{L}\} \subset R \times \mathcal{L}, \quad (\text{SCD17})$$

with coordinatewise operations

$$(r, A) + (s, B) = (r + s, A \cup B), \quad (r, A)(s, B) = (rs, A \cap B).$$

It is a subsemiring of the product of the ring R and the lattice semiring $(\mathcal{L}, \cup, \cap)$: closure follows because a nonzero ring coordinate requires full support. The zero is $z_{\emptyset}$, the unit is $(1, \mathbf{I})$, and $(0_R, \mathbf{I}) = z_{\mathbf{I}}$ is the supported e .

Theorem 36.5 (Canonical action of every support scalar). *Every module diagram on $\mathcal{L}$, and in particular the cochain, cycle, boundary and cohomology diagrams above, has the action*

$$(r, \mathbf{I})(A, x) = (A, rx), \quad z_\lambda(A, x) = (\lambda \cap A, 0). \quad (\text{SCD18})$$

This defines a $G_{\mathcal{L}}(R)$ -semimodule. Restricting along

$$G(R) \longrightarrow G_{\mathcal{L}}(R), \quad \tau \longmapsto z_\emptyset, \quad r \longmapsto (r, \mathbf{I}) \quad (\text{SCD19})$$

recovers exactly the preceding $G(R)$ -semimodule. The scalar map is injective whenever $\emptyset \neq \mathbf{I}$.

Proof. The two rules agree at the shared element $(0_R, \mathbf{I}) = z_{\mathbf{I}}$: both give $(A, 0)$. A supported scalar preserves the label and acts linearly. For a zero-support scalar, additivity on vectors is exactly

$$\lambda \cap (A \cup B) = (\lambda \cap A) \cup (\lambda \cap B),$$

because linear transport sends a zero to a zero. Scalar addition of two such elements uses the other distributive identity, $(\lambda \cup \mu) \cap A = (\lambda \cap A) \cup (\mu \cap A)$. If one scalar is $(r, \mathbf{I})$ and the other is z_λ , their sum is $(r, \mathbf{I})$; the vector sum is $(A, rx) + (\lambda \cap A, 0) = (A, rx)$, as required. For two supported scalars, scalar additivity is the ordinary module law, even if their sum is the supported zero.

For scalar multiplication, two zero-support scalars give $(\lambda \cap \mu) \cap A$. A supported scalar followed or preceded by z_λ gives the same zero in $\lambda \cap A$. Two supported scalars satisfy the ordinary module multiplication law, also when their product is zero in R . The unit acts as identity. The global zero maps to the zero of the bottom fibre, hence is the required absorber. All semimodule axioms have now been checked. The homomorphism SCD19 follows directly from SCD17, and its restriction gives the declared action of both τ and every supported scalar. Injectivity follows from the distinct support coordinates. $\square$

The maps in SCD14 and the cohomology quotient are linear for this larger semiring: each takes the zero in a fibre to the zero in the same fibre. Coordinate extraction SCD15 is equivariant along the scalar morphism

$$G_{\mathcal{L}}(R) \rightarrow G_{\mathcal{P}(I_n)}(R), \quad (r, \mathbf{I}) \mapsto (r, I_n), \quad z_A \mapsto z_{A_n},$$

because the coordinate map $A \mapsto A_n$ preserves union, intersection, bottom and top.

Theorem 36.6 (Exact additive group completion and arithmetic receiver). *The map*

$$\epsilon_{\text{full}} : \mathcal{H}_D^n \rightarrow H_{\mathbf{I}}^n, \quad (A, [x]) \longmapsto h_{A, \mathbf{I}}^n[x] \quad (\text{SCD20})$$

is the universal additive group completion. It is also the universal linear map to an R -module, for the scalar projection $\pi : G_{\mathcal{L}}(R) \rightarrow R$, $(r, A) \mapsto r$. Thus the original cohomology is recovered exactly on the arithmetic quotient $G_{\mathcal{L}}(R)/(e) \cong R$, while every other fibre remains in $\mathcal{H}_D^n$.

Proof. The displayed map preserves addition and is onto, using the top fibre. Any additive map from $\mathcal{H}_D^n$ to an abelian group kills every $(A, 0)$, since it is additively idempotent and the target permits cancellation. But

$$(A, [x]) + (\mathbf{I}, 0) = (\mathbf{I}, h_{A, \mathbf{I}}[x]).$$

Applying the map forces its value on $(A, [x])$ to be its value on that transported top-fibre element. The restriction to the top fibre is a group homomorphism. This proves unique factorization through SCD20 for every abelian target, hence the universal group-completion assertion. For R -linear targets the same proof retains scalar linearity.

The scalar projection is a semiring morphism. Its kernel ideal is the full zero-amplitude fibre $\{z_A : A \in \mathcal{L}\}$, which equals the principal ideal (e) : $ez_A = z_A$, while multiplying e by any scalar

and adding such products always has zero amplitude. Moreover addition by $e = z_{\mathbf{I}}$ sends every zero-amplitude scalar to e . The congruence identifying e with the global zero therefore identifies every z_A with zero, and leaves exactly the ring operations on R . Conversely π separates all different ring amplitudes. Thus this quotient is exactly R . Finally SCD20 sends $z_\lambda(A, [x])$, a fibre zero, to zero and sends supported scalars to their usual action, proving its π -linearity. $\square$

36.4 Gluing, matrix changes, and the space defined by curvature

For $A, B \in \mathcal{L}_D$, the original coordinate inclusions give a degreewise exact sequence of complexes

$$0 \rightarrow C_{A \cap B} \xrightarrow{x \mapsto (x, -x)} C_A \oplus C_B \xrightarrow{(x, y) \mapsto x + y} C_{A \cup B} \rightarrow 0. \quad (\text{SCD21})$$

Surjectivity follows by assigning coordinates in A to the first summand and those in $B \setminus A$ to the second. A pair summing to zero has equal and opposite entries supported in the intersection, which proves exactness. SCD8 proves compatibility with the differentials.

If z is a cycle in the union, choose $z = x + y$ in the two summands. Then $d_A x = -d_B y = w$ is an intersection cycle. Set

$$\partial[z] = [w], \quad \cdots \rightarrow H_{A \cap B}^n \rightarrow H_A^n \oplus H_B^n \rightarrow H_{A \cup B}^n \xrightarrow{\partial} H_{A \cap B}^{n+1} \rightarrow \cdots \quad (\text{SCD22})$$

A new splitting changes (x, y) by $(t, -t)$ with t in the intersection, so w changes by dt . Splitting a primitive of a boundary shows independence of the representative of $[z]$. For exactness at the middle term, if a pair of cycles sums to a boundary, subtract a split primitive to make its sum zero. It then comes from an intersection cycle. If $\partial[z] = 0$, subtract an intersection primitive from one splitting term and add it to the other; both become cycles, proving exactness at the union. Finally if an intersection cycle maps to boundaries in both larger spaces, choose primitives with opposite signs. Their sum is a union cycle mapping to that intersection class. These arguments verify all three repeating positions and also that each composite is zero.

For an actual cochain map $F_n : C^n \rightarrow C'^n$, let P_n be a declared support containing its nonzero coefficients, and let D' be the target differential supports. Then

$$F_{\#} A = \text{cl}_{D'}((P_n(A_n))_n) \quad (\text{SCD23})$$

preserves joins and bottom, and the same matrices restrict to cochain maps $C_A \rightarrow C'_{F_{\#} A}$. This follows from $d'F = Fd$ and the defining support closure; hence it induces cohomology maps. If G is another map with declared supports Q_n , and the declared support of GF is contained in $Q_n P_n$ while retaining every nonzero coefficient, then

$$(GF)_{\#} A \subseteq G_{\#} F_{\#} A. \quad (\text{SCD24})$$

Indeed the trajectory on the right contains every composite coordinate before cancellation. Its minimal closed subtrajectory contains the direct product support. The composite cohomology map equals the direct one followed by this inclusion, since their underlying matrices are equal. This is the exact comparison when a cancellation changes the minimal support. An invertible basis change gives an isomorphism at full support; it is not asserted to give a bijection of coordinate support lattices.

The reconstructed map is always $G(R)$ -linear. Its exact compatibility with the additional lattice scalars is the transport of a zero along

$$F_{\#}(\lambda \cap A) \subseteq F_{\#} \lambda \cap F_{\#} A.$$

The left side is the support of the image of $z_\lambda(A, x)$; the right side is the support obtained by applying $z_{F_{\#} \lambda}$ to the image of (A, x) . Both amplitudes are zero, and the stated inclusion gives their exact diagram map. It follows from monotonicity of $F_{\#}$. If $F_{\#}$ also preserves intersections

and top, it induces a semiring morphism on the two $G_{\mathcal{L}}(R)$ bases, and this comparison is equality. Thus an arbitrary matrix change has the proved support transport, without an unproved scalar-base isomorphism.

Now allow arbitrary consecutive matrices with

$$\Omega_n = d_{n+1}d_n \quad (\text{SCD25})$$

possibly nonzero, and retain the same trajectory lattice. For a trajectory A , the restricted composite is exactly $(\Omega_n)_{A_{n+2}, A_n}$, by the intermediate-index proof of SCD8. Define the forbidden vertices to be those (n, i) whose i -th column in Ω_n is not the zero vector. Include all ancestors of these vertices in the forbidden set, and let T be its complement.

Theorem 36.7 (The curvature defines a maximal exact support space). *The set T is a trajectory. The trajectories whose restricted matrices are complexes are exactly*

$$\{A \in \mathcal{L}_D : A \subseteq T\}. \quad (\text{SCD26})$$

They form a complete distributive lattice with top T .

Proof. An edge from a non-forbidden vertex cannot reach a forbidden vertex, since that would make its starting vertex an ancestor of a forbidden vertex. Hence T is forward closed. If column i of Ω_n is nonzero, choose k with $(\Omega_n)_{ki} \neq 0$. At least one summand $d_{n+1,kj}d_{n,ji}$ is nonzero, so there is a declared path from (n, i) to $(n+2, k)$. Every trajectory containing the first contains the last, and its restricted curvature therefore has that nonzero entry. It cannot be a complex. It also cannot contain an ancestor of the first vertex. Conversely, a trajectory contained in T contains no initial column on which Ω_n is nonzero. All restricted composites are therefore zero. Closure under unions and intersections and the top assertion follow directly. $\square$

This retains the full original curvature rather than declaring it zero. There is also an exact correction using submodules beyond coordinate supports. For consecutive maps d, f ,

$$d(\ker(fd)) = \text{im } d \cap \ker f. \quad (\text{SCD27})$$

Both inclusions follow from $f(dx) = 0$. Restricting the first domain to $\ker(fd)$ gives a genuine complex, while retaining the explicit defect fd and the exact connecting inclusion. Thus the obstruction specifies both a coordinate-support lattice and its potentially larger linear submodule receiver.

36.5 The original source, conductor and invariant observation

At every literal original cutoff retain

$$N \in \{q-1, q, 2q-1, 2q\}, \quad \mathcal{P}_N = \mathbb{C}[S]_{\leq N}, \quad E = \mathbb{C}[S]/(\chi_k), \quad U_N : \mathcal{P}_N \rightarrow E, \quad p \mapsto [p]. \quad (\text{SCD28})$$

The polynomial χ_k is the complete original monic root polynomial of degree q . The source inner product is the original one, with its complete mass. Polynomial and primary coordinates are connected by the actual Vandermonde matrix.

The observation is exactly the original OPG5 map:

$$P_k = \frac{1}{n+1} \sum_{a=0}^n (C^{-a})^{\otimes k}, \quad L = P_k \Pi^{\otimes k} U^{\otimes k} \iota, \quad B = \text{im } L, \quad j\Lambda = L, \quad K = \ker \Lambda. \quad (\text{SCD29})$$

The U in this tensor formula is the original period-unit map, distinct from the remainder U_N . Every original period and invariant column in P_k, Π, U, ι is kept. See the exact OPG4–5 definition and original cycle action.

Let $Z_{\chi,N}$ be multiplication by the complete $\chi = \chi_k$, including its conversion to the original source coordinates. Use $\mathcal{P}_{N-q} = 0$ when $N = q - 1$. Then

$$\mathcal{P}_{N-q} \xrightarrow{Z_{\chi,N}} \mathcal{P}_N \xrightarrow{\Lambda U_N} B \quad (\text{SCD30})$$

is a complex, with degrees zero, one and two. Its cohomology is

$$H^0 = 0, \quad H^1 \xrightarrow{\sim} K, \quad [p] \mapsto U_N p, \quad H^2 = 0. \quad (\text{SCD31})$$

Multiplication by a monic nonzero polynomial is injective. Division by χ proves $\ker U_N = \chi \mathcal{P}_{N-q}$; U_N is onto since $N \geq q - 1$. A cycle therefore maps to K , every member of K has a lift, and two lifts differ by exactly a relation. The map to B is onto, proving the last assertion. Applying SCD7–20 to these complete matrices supplies the full $G_{\mathcal{L}_D}(\mathbb{C})$ -cohomology object, with precisely K as its terminal arithmetic fibre.

For the conductor graph, retain its original simple-quartet domain $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$. No conclusion about a more general multiplicity is imported from this graph presentation. Retain all coefficients of $A = \sum_{a,b=0}^8 a_{ab} e_{ab}^{(8)}$, all its shifts b_{ab} , and the original lower polynomial $\chi' = \chi_{k-8}$. The prime in χ' denotes that polynomial, not a derivative. The maps are

$$(\mathcal{T}_A p)(S') = \sum_{a,b=0}^8 a_{ab} p(S' + b_{ab}), \quad C_{(i,j),(u,w)} = \begin{cases} a_{u-i,w-j} & 0 \leq u-i, w-j \leq 8, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{SCD32})$$

The root identity $\beta'_{ij} + b_{ab} = \beta_{i+a,j+b}$ gives $\chi' \mid \mathcal{T}_A(\chi h)$ for every h . Thus $\mathcal{T}_A$ induces the map on quotient spaces with matrix $V_{\beta'}^{-1} C V_{\beta}$. The transpose C^T is multiplication by the nonzero biform A , hence injective in the polynomial domain. Therefore C is onto. Its kernel $V_A \subset E$ contains the actual K , by OCF17 and CEP5.

The complete pivot and remaining invariant-image maps are retained:

$$C N_k^T = 0, \quad S_k^T N_k^T = I, \quad \text{im } N_k^T = V_A, \quad J_R = S_k Z_k. \quad (\text{SCD33})$$

These identities hold in the original primary coordinates. Here Z_k contains all additional invariant columns, so after imposing the conductor equation the complete residual equations still include $J_R^T V_{\beta} p = 0$, as in CG29. The actual residual observation is

$$\Xi = \Lambda|_{V_A} : V_A \rightarrow B_A = \Lambda(V_A), \quad \ker \Xi = K.$$

In the original fixed residual frame it is Ξ_k^{fix} ; changing to that frame retains its full invertible target map. No formula below substitutes a historical rank numeral for $\dim K$ or $\dim B_A$.

Let v be the first nonzero translation moment order. The actual invertible source map S_N takes a polynomial to its first v original source coordinates and its $\mathcal{T}_A$ image. To see invertibility, Taylor's finite expansion says $\mathcal{T}_A$ lowers each degree at least v by exactly v and kills exactly $\mathcal{P}_{v-1}$. The first v source coordinates are an isomorphism on that kernel. The source and target dimensions are equal, so the displayed injective map is invertible. Define the original graph maps

$$\begin{aligned} X_N &= \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq N-v-q'}, & D_N &= \text{diag}(I_v, M_{\chi'}), \\ E_N &= U_N S_N^{-1} D_N : X_N \rightarrow V_A, & B_N &= \Xi E_N : X_N \rightarrow B_A. \end{aligned} \quad (\text{SCD34})$$

Here D_N includes the exact conversion to the lower original source coordinates. The equality $S_N^{-1} D_N X_N = U_N^{-1}(V_A)$ follows from divisibility by χ' . Consequently E_N is onto V_A .

The full original relation columns are

$$\begin{aligned} \mathcal{Z}_N h &= \begin{pmatrix} P_{<v} Z_{\chi,N} h \\ \mathcal{U}_{A,N-q} h \end{pmatrix}, & D_N \mathcal{Z}_N &= S_N Z_{\chi,N}, \\ (\mathcal{U}_{A,N-q} h)(S') &= \sum_{a,b=0}^8 a_{ab} d_{ab}(S') h(S' + b_{ab}), & d_{ab}(S') &= \chi(S' + b_{ab}) / \chi'(S'). \end{aligned} \quad (\text{SCD35})$$

Each d_{ab} is the original complete product of root differences. Multiplication by χ' proves the lower equality using SCD32; the upper block is exactly the retained lower source coordinates. Thus $\mathcal{Z}_N$ is injective and $\ker E_N = \text{im } \mathcal{Z}_N$.

The correct graph complex and its full-support cohomology are

$$\mathcal{P}_{N-q} \xrightarrow{\mathcal{Z}_N} X_N \xrightarrow{B_N} B_A, \quad H^1 \xrightarrow{\sim} K, \quad [x] \mapsto E_N x, \quad H^0 = H^2 = 0. \quad (\text{SCD36})$$

Injectivity, surjectivity and $\ker \Xi = K$ prove these statements exactly as in SCD31. The chain map to SCD30 is

$$(I, S_N^{-1} D_N, B_A \hookrightarrow B). \quad (\text{SCD37})$$

Its squares are SCD35 and $\Lambda U_N S_N^{-1} D_N = \Xi E_N$. Its induced middle cohomology map is the identity of K . On support trajectories the exact map is SCD23 with the actual coefficient supports of these three maps. Hence the support comparison is proved as well.

Choose the full original kernel frame I_K with actual rank $r = \dim K$. In original orthonormal source coordinates let $G_N = (U_N U_N^*)^{-1}$, and set

$$W_{K,N} = D_N^{-1} S_N U_N^* G_N I_K, \quad E_N W_{K,N} = I_K.$$

The inverse of D_N is used only on its indicated image; the minimum lift is in $U_N^{-1}(V_A)$. Then the exact kernel is

$$\ker B_N = \text{im}[\mathcal{Z}_N, W_{K,N}], \quad \text{rank}[\mathcal{Z}_N, W_{K,N}] = (N - q + 1) + r. \quad (\text{SCD38})$$

Applying E_N to a relation among those columns first kills the coefficients of the independent I_K . Injectivity of $\mathcal{Z}_N$ then kills the remainder. Conversely subtract the corresponding $W_{K,N}$ lift from any vector in $\ker B_N$; what remains lies in $\ker E_N$. This proves every column and rank assertion, including the empty relation list at $N = q - 1$.

The longer sequence $X_N \xrightarrow{E_N} V_A \xrightarrow{\Xi} B_A$ has the exact curvature

$$\Xi E_N = B_N : X_N \twoheadrightarrow B_A. \quad (\text{SCD39})$$

It is a complex only when the actual B_A is zero. In all cases its zero-curvature submodule is precisely SCD38. Thus the nonzero defect of that proposed sequence is retained, while SCD36 gives the genuine original graph complex.

36.6 Exact return to the original metric and arithmetic action

The full-support identification preserves the actual minimum metric. For a fixed $x \in K$, all representatives in SCD30 differ by χh . Minimizing their complete source norm is precisely the definition of the original quotient norm on E . Thus in the unchanged full kernel frame it is

$$H_N^K = I_K^* G_N I_K. \quad (\text{SCD40})$$

For the graph complex, its actual source metric is

$$H_N^{\text{actual}} = D_N^* S_N^{-1} D_N. \quad (\text{SCD41})$$

The map $S_N^{-1} D_N$ in SCD37 is then an isometry onto its image. Every representative of $x \in K$ lies in $U_N^{-1}(V_A)$, so the same minimum gives exactly SCD40. This proof does not substitute the alternative graph product metric $D_N^* D_N$ for SCD41. For all four cutoffs, the full cohomology determinant is therefore the original determinant, including the original signed return

$$\begin{aligned} & \log \det H_{q-1, \mathbf{I}}^{\text{coh}} + \log \det H_{q, \mathbf{I}}^{\text{coh}} - \log \det H_{2q-1, \mathbf{I}}^{\text{coh}} - \log \det H_{2q, \mathbf{I}}^{\text{coh}} \\ &= \log \det H_{q-1}^K + \log \det H_q^K - \log \det H_{2q-1}^K - \log \det H_{2q}^K. \end{aligned} \quad (\text{SCD42})$$

Use the same I_K at every endpoint and the original empty determinant convention. This exact identity assigns no missing asymptotic coefficient.

The original multiplication $M : E \rightarrow E$ need not preserve K . Its failure to descend through observation is explicitly computable. Keep the full minimum section and observed action

$$Q_N = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_N, \quad P_N = L_N \Lambda, \quad M_{B,N} = \Lambda M L_N.$$

Then

$$\Lambda M - M_{B,N} \Lambda = \Lambda M (I - P_N) = \Lambda M I_K (H_N^K)^{-1} I_K^* G_N. \quad (\text{SCD43})$$

For proof, direct multiplication gives the first equality. The minimum section is orthogonal to $\ker \Lambda = K$; hence $I - P_N$ is the G_N -orthogonal projector onto K . The displayed expression $I_K (H_N^K)^{-1} I_K^* G_N$ is the same projector: multiply to verify idempotence, image, identity on K , and weighted self-adjointness. This proves the final equality without deleting its kernel block.

The exact action-compatible subspace and defect space are

$$0 \rightarrow K \cap M^{-1} K \rightarrow K \xrightarrow{\Lambda M} \text{im}(\Lambda M|_K) \rightarrow 0. \quad (\text{SCD44})$$

The kernel assertion is its definition; the last target is the actual image. Thus this is the exact obstruction to obtaining an action on the original middle cohomology by M .

Retain every higher observation as well:

$$K^{(j)} = \bigcap_{a=0}^j \ker(\Lambda M^a), \quad j \geq 0. \quad (\text{SCD45})$$

The connecting maps are inclusions, and $K^{(j+1)} = K \cap M^{-1} K^{(j)}$. If two successive spaces are equal, the common space is M -invariant: membership in $K^{(j+1)}$ implies that the M -image belongs to $K^{(j)}$. It is then killed by every further observation, so the sequence stabilizes. It has at most $\dim K$ strict drops. Its stable value is the largest M -invariant subspace of K , since every such subspace is killed by every ΛM^a , while the stable value itself is invariant. This is a specified space with exact connecting maps, rather than an assumed descent.

The original metrics, periods and roots are unchanged throughout. The construction provides a complete support object over the absolute base, its exact arithmetic receiver, and the spaces defined by the two original curvature defects SCD39 and SCD43. No support trajectory or additional invariant column is discarded.

36.7 One support base for the four original cutoffs

The four support bases need not be identified by choosing bases again. There is a single exact common construction. For a fixed original source order $s \in \{1, k\}$, retain the original source inclusions $i_{NM} : \mathcal{P}_N \hookrightarrow \mathcal{P}_M$, $N < M$, at the four cutoffs. On relation polynomials use $j_{NM} : \mathcal{P}_{N-q} \hookrightarrow \mathcal{P}_{M-q}$ and on B use the identity. Their original matrices satisfy

$$i_{NM} Z_{\chi,N} = Z_{\chi,M} j_{NM}, \quad U_M i_{NM} = U_N, \quad \Lambda U_M i_{NM} = \Lambda U_N. \quad (\text{SCD46})$$

These are identities on polynomials before passing to any coordinates. Their matrices include whatever original basis conversions are required. They form cochain maps of SCD30 and induce precisely the identity of K in SCD31.

For the graph presentations, the corresponding map is

$$F_{NM}^X = D_M^{-1} S_M i_{NM} S_N^{-1} D_N, \quad E_M F_{NM}^X = E_N, \quad F_{NM}^X Z_N = Z_M j_{NM}. \quad (\text{SCD47})$$

The inverse of D_M is taken only on its actual image: a polynomial whose class lies in V_A still has that property after source inclusion, so S_M of that polynomial has its second component

divisible by χ' . This proves that the map is defined. Cancellation of the displayed inverse maps, followed by SCD35 and SCD46, proves the two identities. Composition for three source cutoffs follows by the same cancellation. The induced cohomology map is again the identity of K .

Now form a finite coefficient graph whose vertices record the cutoff, cochain degree and coefficient index. Include every declared differential edge, every edge of these original horizontal source maps, and, if both presentations are retained together, every edge of the graph-to-polynomial map SCD37. Let $\mathcal{L}_{\text{all}}$ be all subsets closed under these edges. One can include both source orders as separately labelled copies, without identifying their metrics. The proof of SCD7 applies to this finite diagram: unions and intersections remain closed, and forward closure is the least containing closed set. Thus $\mathcal{L}_{\text{all}}$ is a finite bounded distributive lattice with all the original coefficient supports retained.

The coordinate projection to any one of the complexes preserves union, intersection, bottom and top, and takes a master trajectory to a valid local trajectory. Restricting every original matrix at a master trajectory therefore gives a diagram of complexes in which all original squares still commute. To verify a square entry, note that every intermediate index with a nonzero contribution is forced into the master trajectory by its first edge; its second edge is retained as well. Thus the restricted entry is the original full entry, with all its cancellation terms.

For each cutoff reconstruct

$$\mathcal{H}_{\text{all},N}^1 = \bigsqcup_{A \in \mathcal{L}_{\text{all}}} H_{N,A|N}^1. \quad (\text{SCD48})$$

These are modules over the *same* $G_{\mathcal{L}_{\text{all}}}(\mathbb{C})$ by SCD18. The horizontal cohomology maps preserve the master label A , commute with union transports, and take a zero in a fibre to the zero in the same fibre. They are therefore $G_{\mathcal{L}_{\text{all}}}(\mathbb{C})$ -linear: for z_λ both composites give the zero at $\lambda \cap A$; for a supported scalar the assertion is ordinary linearity. Their additive group completions all equal the same K , with identity connecting maps and the four separate original metrics SCD40. This proves a common absolute-base receiver for the complete four-cutoff diagram, including the graph maps, without merging its source metrics.

36.8 Sources and verification scope

The accompanying exact checker verifies the complete six-trajectory cancellation example and its cohomology changes; the $G_{\mathcal{L}}(R)$ scalar laws; and the full source/graph/minimum identities on a specified positive four-root auxiliary example at all four cutoffs. This example tests the proved algebra. It is not a replacement for the original period matrices or an evaluation of a hypothetical zeta zero. No floating-point test proves a native sign or coefficient.

The classification source is read as original LaTeX, not a PDF or a summary. The accompanying source-use ledger records exact hashes, reading coverage and receiving formulas. The elementary group-completion, gluing and cohomology constructions above have complete proofs here; the human semiring reference below is contextual provenance, not a claim that an unread external theorem supplies a missing calculation.

37 The attained metric on every support cohomology fibre

This section works on the fibre complexes constructed from the original coefficient matrices. Their support labels retain which relation columns and ambient coordinates are present. For two nested trajectories $A \subseteq B$, write $K_A \subseteq K_B$ for the cycle spaces and $I_A \subseteq I_B$ for the relation images, all embedded by zero extension in the same original middle coefficient space V . In particular $I_A \subseteq K_A$ and $I_B \subseteq K_B$. No replacement inner product is introduced: let G be the original positive-definite Hermitian form on V .

For a subspace $I \subseteq V$, denote its G -orthogonal projection by P_I . With any full-column-rank frame Z for I ,

$$P_I = Z(Z^*GZ)^{-1}Z^*G. \quad (39)$$

The empty frame defines the zero projection. To verify the formula, its square equals itself, its image is I , and $Z^*G(x - P_I x) = 0$ for every x ; these three properties characterize the orthogonal projection independently of the chosen frame.

Theorem 37.1 (Exact transport and its new boundary kernel). *The quotient $\mathbf{H}_A = K_A/I_A$ has attained norm*

$$\|[x]\|_A^2 = \inf_{z \in I_A} \|x + z\|_G^2 = \|(1 - P_{I_A})x\|_G^2.$$

The inclusion of trajectories induces the well-defined linear map

$$t_{AB} : \mathbf{H}_A \rightarrow \mathbf{H}_B, \quad [x] \mapsto [x], \quad \ker t_{AB} = (K_A \cap I_B)/I_A.$$

For every $x \in K_A$ its entire norm loss is

$$\|[x]\|_A^2 - \|t_{AB}[x]\|_B^2 = x^*G(P_{I_B} - P_{I_A})x = \|(P_{I_B} - P_{I_A})x\|_G^2. \quad (40)$$

The map is a contraction, and its defect has rank at most $\dim I_B - \dim I_A$.

Proof. Write $x = P_{I_A}x + (1 - P_{I_A})x$. For $z \in I_A$, orthogonality gives $\|x + z\|_G^2 = \|P_{I_A}x + z\|_G^2 + \|(1 - P_{I_A})x\|_G^2$. The first term attains zero at $z = -P_{I_A}x$, proving the quotient formula. Because $I_A \subseteq I_B$, two representatives defining the same class at A also define the same class at B . The image vanishes precisely for representatives in $K_A \cap I_B$, giving the exact kernel.

Nested projections obey $P_{I_A}P_{I_B} = P_{I_A} = P_{I_B}P_{I_A}$. The second equality follows because $P_{I_A}x \in I_B$; the first follows by taking adjoints in the G inner product. Thus $P_{I_B} - P_{I_A}$ is the orthogonal projection onto $I_B \cap I_A^\perp$. Subtract the two attained norm formulas to obtain (40); its last expression and its rank follow from this projection description. This proof retains every actual relation vector and its position in the original metric. $\square$

Theorem 37.2 (The complete determinant correction). *Let the columns of U_A be a basis of $K_A \cap I_A^\perp$, so they represent a basis of $\mathbf{H}_A$. Put*

$$\begin{aligned} H_A &= U_A^*GU_A, \\ E_{AB} &= U_A^*G(P_{I_B} - P_{I_A})U_A, \\ H_{B \leftarrow A} &= U_A^*G(1 - P_{I_B})U_A = H_A - E_{AB}. \end{aligned} \quad (41)$$

Here $0 \preceq E_{AB} \preceq H_A$, and $H_{B \leftarrow A}$ is exactly the pullback of the attained quotient metric, including its zero directions. Its kernel is the coordinate representation of $(K_A \cap I_B)/I_A$. When that kernel is zero,

$$\log \det H_{B \leftarrow A} - \log \det H_A = \log \det \left(1 - H_A^{-1/2} E_{AB} H_A^{-1/2} \right). \quad (42)$$

Every eigenvalue of the matrix subtracted from 1 lies in $[0, 1)$; at most $\dim I_B - \dim I_A$ are nonzero. If it has an eigenvalue 1, the corresponding cohomology class becomes a boundary at B and both sides must instead be restricted to a specified surviving subspace before taking an ordinary determinant.

Proof. Theorem 37.1 evaluated on $U_A c$ gives every displayed form identity and inequality. Its zero norm at B is equivalent to membership in I_B , proving the kernel assertion. Congruence by $H_A^{-1/2}$ transforms the metric difference into the matrix in (42); multiplicativity of determinant yields that formula, and positivity proves the eigenvalue claims. Rank cannot increase under congruence or compression. Thus the term records the original relative position of the relations rather than their number alone. $\square$

37.1 Return through all four original cutoffs

Use precisely the four indices and signs

$$(N_1, N_2, N_3, N_4) = (q - 1, q, 2q - 1, 2q), \quad (\sigma_1, \sigma_2, \sigma_3, \sigma_4) = (+1, +1, -1, -1).$$

For each index use its original form G_N , actual relation spaces and attained quotient basis. Whenever the transported spaces have no killed class, define $\Lambda_N = H_{A,N}^{-1/2} E_{AB,N} H_{A,N}^{-1/2}$. Applying the proved identity separately at each index gives the exact signed return

$$\sum_{j=1}^4 \sigma_j \log \det H_{B \leftarrow A, N_j} = \sum_{j=1}^4 \sigma_j \log \det H_{A, N_j} + \sum_{j=1}^4 \sigma_j \log \det(1 - \Lambda_{N_j}). \quad (43)$$

No sign is assigned to the last sum by the separate contraction inequalities: the four signs in the return are retained. If a class is killed, Theorem 37.2 identifies exactly its kernel before a surviving determinant is formed.

Proposition 37.3 (An evaluated example of the full support correction). *Take the complex $\mathbb{C} \xrightarrow{(1,1)^\top} \mathbb{C}^2 \rightarrow 0$. The trajectory with no relation column and both middle coordinates has $I_A = 0$. Activating the relation gives $I_B = \mathbb{C}(1, 1)^\top$. On the original class represented by $x = (1, 0)^\top$ and metric $G_N = \text{diag}(e^N, 1)$,*

$$\|x\|_A^2 = e^N, \quad \|x\|_B^2 = \frac{e^N}{1 + e^N}, \quad \frac{\|x\|_B^2}{\|x\|_A^2} = \frac{1}{1 + e^N}.$$

For m independent copies, the four-cutoff logarithmic correction is exactly

$$2mq + m(\log(1 + e^{-(2q-1)}) + \log(1 + e^{-2q}) - \log(1 + e^{-(q-1)}) - \log(1 + e^{-q})). \quad (44)$$

Its difference from $2mq$ has absolute value at most $m(e^{-(2q-1)} + e^{-2q} + e^{-(q-1)} + e^{-q})$.

Proof. The attained norm is the minimum over $t \in \mathbb{C}$ of $e^N|1 + t|^2 + |t|^2$. Completing the square gives the minimizer $t = -e^N/(1 + e^N)$ and the stated value. The class is not killed because $(1, 0)$ is not a multiple of $(1, 1)$. Orthogonal direct sums multiply the ratios. Take logarithms at all four cutoffs and use $\log(1 + e^N) = N + \log(1 + e^{-N})$; their signed linear combination is $2q$. Finally $0 \leq \log(1 + t) \leq t$ for $t \geq 0$ follows by integrating $(1 + t)^{-1} \leq 1$, proving the error bound. $\square$

This example is completely specified and is not substituted for a native arithmetic coefficient. It demonstrates with an attained minimum that a support transition can contribute at scale kq when m has scale k . Equation (43), evaluated with the original programme spaces, is the receiver required for that programme calculation.

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- [28] *The complete Hermitian pulse matrix and a uniform finite-prime Weil bound*, companion independent mathematical derivation, `FINITE_PRIME_PULSE_BOUND.tex`, 23 September 2026. Theorem at FP37, proof FP38–43; the entire 579-line source was read for this calculation. The complete proof accompanies this cumulative derivation. Its finite complex-coefficient and all-support-face lower bound, with the exact original width and Archimedean subtraction, is the input (FMS35). This is a programme proof, not a substitute human-source attribution.
- [29] Programme derivation, *Constructive two-prime packets: exact theta lifts and bounded moment dynamics*, 23 September 2026, results SZ-20260923-901–903. Complete local unpublished sources `NOTE.tex` and `GENERATOR.tex`, CG1–24 and `eq:measure`, `eq:volume`, `eq:matrix-defect`. Both sources are retained without alteration in the working source collection; their complete proofs were read before the GC receiving calculation. No public source link is claimed while publication is held. Its exact finite-check receipt and numerical example remain separate: the latter is not interval-certified. GC1–24 supplies the full closure and graph-receiver proofs used here.
- [30] Johan Ludwig William Valdemar Jensen, *Sur un nouvel et important théorème de la théorie des fonctions*, Acta Mathematica **22** (1899), 359–364, doi:10.1007/BF02417878. The journal archive metadata was checked; no original TeX or PDF was read. PO24 is derived in full here from the circle mean-value identity.
- [31] Leonhard Euler, *Variae observationes circa series infinitas*, Eneström E72, Commentarii academiae scientiarum Petropolitanae **9** (1744), 160–188; written 1737. The Euler archive record was checked for attribution and metadata; no original TeX or PDF was read. The reciprocal-prime divergence needed here is proved completely in PO25.
- [32] Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, original author source `rhready.tex`, section `sectweilpos`, lines 850–864, equations `wp` and `arch`. The positivity criterion is attributed there to André Weil. PO10 and PO47 retain its Fourier and Haar conventions explicitly.
- [33] Alain Connes and Caterina Consani, *Hochschild homology, trace map and zeta cycles*, arXiv:2207.10419, original author source `Hochschild-zeta.tex`, lines 458–464, equation `evalmaprho` in the proof of Proposition `laplacian`; full multiplicity jets and the triangular action. PO2–3 state all coordinates used here.

- [34] Companion derivation, *The complete Hermitian pulse matrix and a uniform finite-prime Weil bound*, `FINITE_PRIME_PULSE_BOUND.tex`, complete source retained alongside this note, FP4–6, FP18–28 and FP29–43. It proves the literal norm, all face matrices and the exact constant used in PO34. Its original two-prime programme input is the complete pinned edition 033, FWP1–24 and FWJ1–8.
- [35] Companion derivation, *Primitive representatives of the complete arithmetic jet packets*, `PRIMITIVE_PACKET_LIFT.tex`, complete source retained alongside this note, PL10–18, PL23–28 and PL50. PO43–45 repeat the full packet section and its relevant proof. Its original programme convention is the pinned tau-base source, Sections 2–6.